



# **Uganda National Roads Authority**

## **Asset Management System**

### **Pavement Management System User Manual**

## *Asset Management System Documentation Map*

This document contains the Pavement Management System User Manual for the Uganda National Roads Authority (UNRA). It forms part of the Asset Management System documentation as shown on the previous page.

### **Support**

This software application was implemented under the Uganda Roads Data Collection Study (Credit No. 3976-UG), by Roughton International and Aurecon South Africa (Pty) Ltd, on behalf of the Uganda National Roads Authority (UNRA).

Technical support may be available according to agreement in force between Aurecon and UNRA.

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# Uganda National Roads Authority

## Asset Management System

### Pavement Management System User Manual

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## 1. Introduction

This Pavement Management System User Manual describes the use of software tools by the Uganda National Roads Authority (UNRA) to generate forward works programs to develop and maintain the national road network. These PMS tools can also be used to predict Asset Value.

The tools covered in this manual are:

- dTIMS<sup>TM</sup> CT (hereafter referred to as **dTIMS**) Life Cycle Cost Analyser (LCCA),
- **dTIMS MD** (Management Dashboard to provide access to managers to dTIMS),
- **dTIMS WF** (Workflow solution to document procedures in dTIMS),
- Highway Development and Management system (**HDM-4**) and
- Roads Economic Decision Model (**RED**)

Together these tools form the Pavement Management System (PMS) for UNRA.

dTIMS is the tool on which UNRA's PMS is stored, maintained and analysed. UNRA's road network definition and data is stored in a dTIMS database, also referred to as UNRA's dTIMS database. dTIMS MD and dTIMS WF are add-on solutions (tools) to the dTIMS software with capabilities as listed below.

dTIMS MD:

- provides access to the data of UNRA's dTIMS database,
- provides access to the database management capabilities in dTIMS and it
- provides access to selected routines and information of any life cycle cost analyses in UNRA's dTIMS database.

dTIMS WF automates, standardize and documents dTIMS processes such as:

- importing/exporting of data in UNRA's dTIMS database,
- preparation of data and information for a life cycle cost analysis,
- transformation (manipulation) of data in UNRA's dTIMS database and
- performing a life cycle costs analysis.

The *Asset Management System Policies and Procedures Manual* describe the circumstances and frequencies with which these tools should be used.

## 2. Roles and Responsibilities

The Asset Management Unit of the Directorate of Planning is responsible for using the Pavement Management System to generate forward works programs to develop and maintain the National Road Network. Specifically, the Asset Management Unit is responsible for the activities listed in Figure 1 below.

**Figure 1 - Asset Management Unit Responsibilities Relating to Pavement Management System**

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Setup of the various Pavement Management System models</li> <li>• Setup of budgets</li> <li>• Setup of default data</li> <li>• Preparation of data in dTIMS prior to export to the Pavement Management System tools</li> <li>• Running the LCCA</li> <li>• Running HDM-4</li> <li>• Running RED</li> <li>• Calculating Asset Value</li> <li>• Presentation of Analyses</li> <li>• Presentation of Results in the Geographic Information System (GIS).</li> </ul> | <ul style="list-style-type: none"> <li>• Chapter 7</li> <li>• Chapter 9</li> <li>• Chapter 8</li> <li>• Chapter 10</li> <li>• Chapter 11</li> <li>• Chapter 12</li> <li>• Chapter 13</li> <li>• Chapter 14</li> <li>• Chapter 15</li> <li>• Chapter 16</li> </ul> |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 3. System Overview

Figure 2 below gives an overview of the Pavement Management System.

Basic models for Vehicle Operating Costs (VOCs), Road Deterioration, and Road Treatment Effects have been established during the customization of this system, based on information and data supplied by UNRA. These models and their parameters may be refined by UNRA in future.

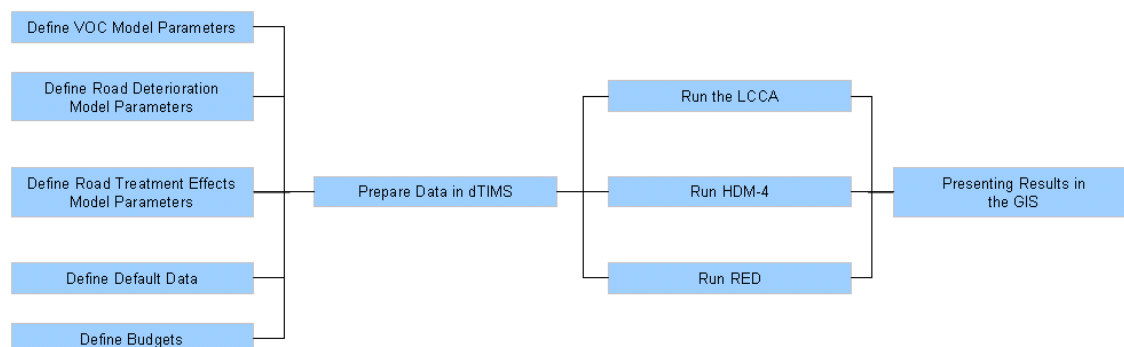
The PMS allows the user to define various default data to be used in cases where actual data is not provided. These might include items such as visual condition data, roughness data, climatic indicators, projected traffic growth rates, etc.

Budget data should be provided to allow generation of forward works programs, optimized under budget constraints.

A series of procedures must be followed to prepare data for the PMS tools. These PMS tools (RED, HDM-4 and dTIMS LCCA) are exact with regards to the format and content of data and input parameters. After the data has been prepared, then the dTIMS LCCA, HDM-4 and/or RED may be run to generate forward works programs.

The resultant programs can be presented in chart and/or graph form, and may also be presented in UNRA's Geographical Information System (GIS).

**Figure 2 - PMS System Overview**



### 3.1. RED system

#### 3.1.1. Introduction

RED is the Roads Economic Decision Model for Economic Evaluation of Low Volume Roads.

According to the Africa Technical Transport Note<sup>1</sup>:

- *HDM-4 is not particularly customized for low-volume roads (less than 200 vehicles per day),*
- *HDM-4 does not capture all the benefits associated with rural investments and*
- *HDM-4 requires a series of inputs which are impractical to collect for low traffic levels.*

*RED is presented as a simplified economic evaluation model to fulfill the planning and programming need of highway agencies in charge of low-volume roads, without demanding input parameters that may be unrealistic and costly to collect while presenting the results in a practical and effective manner.*

<sup>1</sup> According to the Africa Technical Transport Note, Road Maintenance Initiative. SSATP Note No 18. April 1999.

The data exported from the dTIMS database include all roads and should thus be filtered by the PMS user to include selected low-volume roads only. It will be up to the user to decide which low-volume road segments should be analysed; the software makes provision for 285 segments to be analyzed at a time.

### 3.1.2. Software

The RED software is a set of Excel files with spread sheets programmed with macros and formulas for the purpose of analysing maintenance and/or rehabilitation alternatives. Whenever reference is made to RED or the RED software, it implicates the Excel files. The main files that form the RED software are a VOC file, a MAIN file, a PROGRAM file and a RISK file. An additional Excel file (DATA FROM dTIMS) was added to customise the road network data that was exported from dTIMS to be used in the RED program file.

All files customised for use by UNRA, reside with documentation and related files on the directory **C:\UNRA\_PMS\RED**. Along with these files, the executable file **EBM-32.EXE** should reside on the directory **C:\UNRA\_PMS\RED**, otherwise the budget optimisation feature of the RED software will not work. There are also backups files under **C:\UNRA\_PMS\RED\UNRA Masterfiles** in case the working files become corrupted or invalid. Never change the information in the backup directory and files, but copy them to the working directory; **C:\UNRA\_PMS\RED** before proceeding to the preparation of data stage or analysis stage.

The files customised for UNRA are:

- **UNRA RED - HDM-4 VOC models.xls** – file containing the parameters necessary to develop VOC models.
- **UNRA RED - Program.xls** – file to be used when a road network/sub-network/list of roads are analysed.
- **UNRA RED - Main.xls** – file to be used for project level analysis of a specific road or road section.
- **UNRA RED - Data from dTIMS.xls** – file that is exported from UNRA's dTIMS database for Program analysis. This file contains formulae to customise UNRA's road network/sub-network into the correct format before the network is pasted into UNRA RED – Program.xls.
- **UNRA RED – Risk.xls** – file to be used for risk analysis.
- File **EBM-32.EXE** should always accompany these files to enable optimisation of budgets.

The advanced user can make copies of the above files to any other directory or computer in order to analyse several data sets/networks. Care should be taken to ensure the referencing between the RED files stays intact and always keep the set of files in the same directory.

The pdf files under the **C:\UNRA\_PMS\RED\RED MANUALS** directory are manuals developed by the software developer; these should be used in combination with this manual whenever the RED software is used.

**Remember that the user is only allowed to change the yellow cells in the RED Excel spread sheets!**

## 3.2. dTIMS software

The dTIMS software has been customized with UNRA's data, vehicle operating cost models, pavements performance curves, life cycle cost analyses, queries, data transformations, etc. Certain items are tabulated in this document, but a copy of any of the objects in UNRA's dTIMS system can be printed or exported. A soft copy UNRA's dTIMS system is available on the documentation directory (C:\UNRA\_PMS\Documentation). The extent of UNRA's dTIMS system is too vast to include all items as tables in this report.

The Help item in the dTIMS software should be used in combination with this manual whenever the dTIMS software is used. The basic operations of the dTIMS software are not repeated in this manual, but it is assumed that the user has the basic knowledge to operate the software.

All concepts and operations of the dTIMS software are explained in detail in the dTIMS CT Users Guide on C:\UNRA\_PMS\dTIMS\_Manuals. This guide was compiled by the developers of dTIMS namely Deighton Associates. Users are advised to study this guide before continuing and to use the guide throughout the operation of UNRA's dTIMS system.

### 3.2.1. dTIMS WF add-on tool

dTIMS Workflow was acquired to assist the personnel of UNRA in the execution of the tasks and processes in this manual. The ultimate goal is to complete a life cycle cost analysis based on the HDM-4 pavement performance models. Consult Appendix A of this manual with detailed instruction on the installation and operation of dTIMS WF for UNRA. The majority of the LCCA operations and tasks, detailed in this manual, are included in Workflow tasks. However, the user should always use this manual in combination with the Workflows in dTIMS WF to make sure all background and information are used.

### 3.2.2. dTIMS MD add-on tool

dTIMS Workflow was acquired to provide easy access to UNRA's pavement management system. dTIMS MD provides management access to network analyses and provide the ability to adjust and investigate budget scenarios for their impact on the future predicted performance of UNRA's road network. Consult Appendix B of this manual with detailed instruction on the installation and operation of dTIMS MD for UNRA. Only analyses and budgets that were executed in dTIMS (CT) can be viewed in dTIMS MD; the operators of dTIMS MD are therefore dependent on the PMS operators who should complete the LCCA's in dTIMS before any user can use the MD tool for forward planning.

The following are of importance for dTIMS MD:

***Remember, any changes to your password will require changes to the "web.config" file usually on C:\Inetpub\wwwroot\dTIMSmd, otherwise MD will not work.***

Open the "web.config" file located in the dTIMS MD application folder using notepad. Locate the following entry "<identity impersonate". Edit this line to include the login information for your own username. This line should look like this.

```
<identity impersonate="true " userName="your username" password="your password" />
```

Save changes to Web.Config file.

***Any changes to the road network in GIS, requires a new shape file to be created to define the roads. Follow the instructions in Appendix B: Execution of dTIMS MD: Configuration, every time a new set of shape files are created.***

## 3.3. HDM-4 software

The HDM4 version 2.5 workspace for UNRA, was installed on the directory C:\UNRA\_PMS\HDM4 under the objects.dat file. The HDM4 software has manuals to be found on C:\UNRA\_PMS\HDM4\ENG. These manuals contain Applications Guides, User Guides, Analytical Framework and Model Descriptions, and more to be used when conducting any analysis in HDM4.



The HDM4 software has built-in pavement performance models and vehicle operating cost models. It is the responsibility of the user to update costs, calibration factors for pavement performance models and works standards regularly, to ensure the HDM4 software is pertinent to the UNRA road network.

### 3.4. Physical Implementation

Figure 3 below shows the physical implementation of the Road Management System in UNRA.

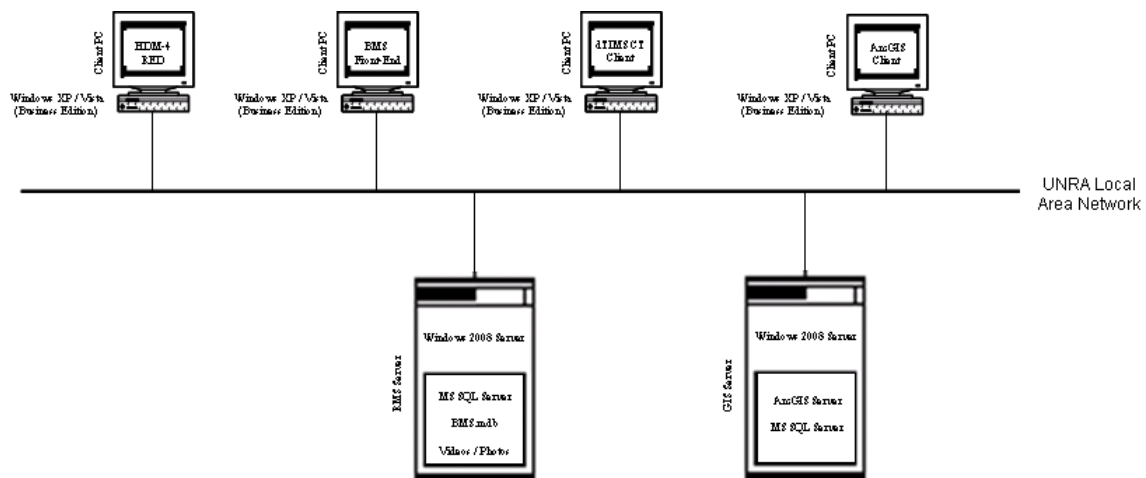
Servers:

- A RMS Server running Windows 2008 Server runs the dTIMS CT database (Uganda\_CT). This is implemented on MS SQL Server 2005. This server also holds the back-end of the Bridge Management System (BMS) MS Access database file. It also holds video files and photographs that are referenced by the asset management database.
- A GIS Server runs ESRI ArcGIS Server, which operates on MS SQL Server 2005. This is housed on a separate physical server running Windows 2008 Server.

PCs:

- Figure 3 depicts client software running on 4 separate PCs, however all client software may be installed on a single PC if UNRA so wishes. Also, the number of client installations depends only on the number of licenses held by UNRA for the respective client software.
- dTIMS CT, WF and MD Client software are installed on client PC's running Windows XP (Business Edition) or Windows Vista (Business Edition) or Windows 7 Enterprise. The Life Cycle Cost Analyser (LCCA) is a separate module of dTIMS CT that is installed on one or two of the client machines.
- ArcGIS client software is also installed on a PC running Windows XP (Business Edition) or Windows Vista (Business Edition) or Windows 7 Enterprise.
- The Bridge Management System (BMS) front-end database also resides on a client PC.
- HDM-4 and RED are stand-alone packages that run on a single client PC.

**Figure 3 - Road Management System Physical Implementation**



## 4. Operation of PMS Tools Software (dTIMS, RED & HDM-4)

### 4.1. dTIMS

#### 4.1.1. Execute dTIMS

How to execute dTIMS and open the dTIMS database of UNRA:

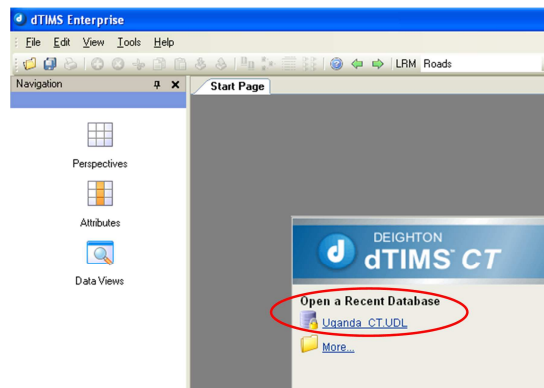
- Open the dTIMS (**dTIMS CT v8 Enterprise**) software from the desktop or via the Start/programs button.
- Select the UNRA database file (Uganda\_CT.UDL) from the folder C:\Databases according to one of the two methods below.

##### Method 1:

Select the Uganda\_CT.UDL database if it is already in the “Recent Databases list”.

The previous username and password will automatically be used.

If a different username and password are to be used, first enter these before clicking on Uganda\_CT\_UDL.



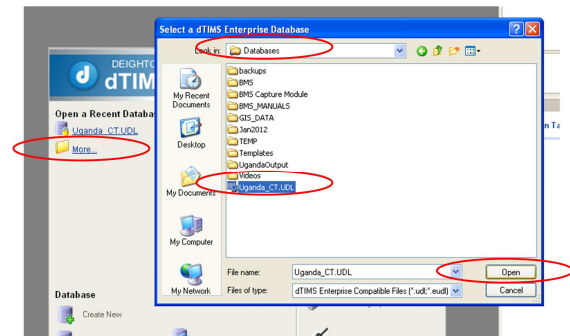
##### or Method 2:

Click on **More....**

Select path : C:\DATABASES

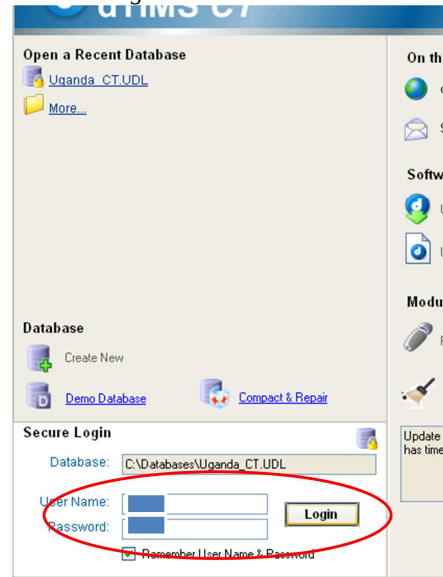
Select file: *Uganda\_CT.UDL*

Click on **Open**



Then enter User Name and Password according to the security levels below.

Click on **Login**



The following security level is assigned for CT users:

- Level 7 (Username = **lcca**, Password = **lcca**): For the dTIMS LCCA Manager to “manage and run the LCCA”. USE THIS SECURITY LEVEL FOR ALL DTIMS OPERATIONS DESCRIBED IN THIS MANUAL.
- Level 6 (Username = **data**, Password = **data**): For the dTIMS Manager to “manage all perspectives, attributes, views and queries relating to the road network, including bridge data from the BMS”.
- Level 8 (Username = **admin**, Password = **unraadmin**): For the dTIMS LCCA administrator. DO NOT USE THIS SECURITY, IT PROTECTS SOME OF THE CUSTOMISED OBJECTS.

These security levels allow the users access to the functionalities in dTIMS according to the responsibilities of the user. The security codes serves as a safety system to protect the data and integrity of UNRA’s database. Care should always be taken when any changes are made to the customised system as this could jeopardise the functioning of the system and accuracy of the results. Any changes to expressions, queries, transformations, perspectives, attributes should be done preferably by experienced users after a backup of the system has been done.

#### 4.1.2. Compact and repair dTIMS

The dTIMS software has the option to **Compact & Repair**. This functionality is for general system maintenance to clean-up temporary files, to remove temporary DRP’s and to delete unused space in the database. It is recommended that this option is run regularly; especially after large data files were imported, a dFRAG object was run or large volumes of transformations were executed. Select this option from the Start Page after dTIMS was opened but before UNRA’s database is opened.

#### 4.1.3. Backup

Consult the RMS Systems Administration Manual for details on how to back up the dTIMS system. It is recommended that a backup of UNRA’s dTIMS system is made at least once a week, but a backup is mandatory whenever new data sets are imported or whenever any changes are made to either the data or system objects.

#### 4.1.4. Execute dTIMS WF (Workflow)

To execute dTIMS WF: In Internet Explorer, navigate to:  
[http://www.dtimes.net/wf/v8.6/dTIMS\\_WF.xbap](http://www.dtimes.net/wf/v8.6/dTIMS_WF.xbap).

Use the same security usernames and password as for dTIMS CT.

#### 4.1.5. Execute dTIMS MD (Management Dashboard)

To configure dTIMS MD for the first time to connect to your database: In Internet Explorer, navigate to:  
<http://localhost:8080/ContentConfig.aspx>

To execute dTIMS MD: In Internet Explorer, navigate to:  
<http://localhost:8080/Login.aspx?Connect=DBPath>.

Use the same security usernames and password as for dTIMS CT.

### 4.2. HDM-4

#### 4.2.1. Execute HDM-4

How to execute HDM-4 and open the HDM-4 database (objects.dat file) of UNRA:

- Open the HDM-4 software from the desktop or via the Start/programs button.
- Open the HDM-4 workspace for UNRA (objects.dat file) on c:\UNRA\_PMS\HDM4.

#### 4.2.2. Backup

UNRA's HDM4 system resides on a directory C:\UNRA\_PMS\HDM4. It is recommended that this directory with all files or a selection of sub-directories are backed-up before any HDM-4 analysis is run and also at frequent intervals during analyses.

Backups should be stored on separate hardware and media.

Backups should also be moved off-site if possible.

The HDM-4 system consists of the following:

- The principle file containing the customised HDM-4 system for UNRA is "objects.dat".
- Road network master file is "UNRA Network.mdb". This file has been customized with queries to prepare the output data from dTIMS for HDM-4.
- Directory "UNRA masterfiles" contains master files in case the working files become corrupt.
- Directories such as "work", "rundata" and "ENG", containing work files, analyses output file and manuals.

#### 4.3. Execute RED

RED is a set of Excel files on C:\UNRA\_PMS\RED customised with formulae and macros. It is recommended that this directory with all files or a selection of sub-directories are backed-up before any RED analysis is run and also at frequent intervals during analyses.

Backups should be stored on separate hardware and media.

Backups should also be moved off-site if possible.

The RED system consists of the following:

- A set of Excel files (preceded with "UNRA RE") on C:\UNRA\_PMS\RED.
- The RED network file is UNRA RED - Data from dTIMS.xls. This file has been customized with data from dTIMS for RED.
- Directory "UNRA masterfiles" contains master files in case the working files become corrupt.
- Directory "RED original files" contains the original RED software/file as it was downloaded from the internet.

## 5. Important concepts in UNRA's life cycle cost analysis tools

### 5.1. Basic concepts RED, HDM-4 and dTIMS

The concept of "an analysis" in the three PMS software tools is different. The table below gives an overview of the PMS software tools with regards to their life cycle analysis methodologies. The customised life cycle cost analysis in dTIMS is based on the pavement performance and road user models of HDM-4, thus the comparison below is given for HDM-4 versus RED.

| HDM-4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | RED                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Background                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Background                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <p>The Highway Development and Management System - HDM-4 - is a software system for investigating choices in investing in road transport infrastructure; it involves the strategic planning of road works showing the consequences of various investment policies and funding levels.</p> <p>The HDM-4 decision support models are aimed in supporting decision-makers to develop road maintenance, rehabilitation and upgrading programmes by evaluating long-term road investment alternatives and consequences.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <p>According to the Transport Note, SSATP Note No 18, April 1999; the Roads Economic Decision model (RED) performs an economic evaluation of road investments and maintenance options customized to the characteristics of low-volume roads such as:</p> <ul style="list-style-type: none"> <li>• high uncertainty of the assessment of traffic, road condition, and future maintenance of unpaved roads;</li> <li>• periods during a year with disrupted passability;</li> <li>• levels of service and corresponding road user costs defined not only through roughness;</li> <li>• high potential to influence economic development; and</li> <li>• beneficiaries other than motorized road users.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| The life cycle cost analysis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | The life cycle cost analysis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <p>A network level life cycle cost analysis in HDM-4 (or dTIMS based on the HDM-4 models) is done for long term budget planning and for optimising the maintenance strategies.</p> <p>HDM-4 consist of the analytical models to calculate the deterioration of the road structure and surface due to the particular traffic load, weathering and existing distresses on each road. These models are typically roughness, potholes, cracking, rutting, ravelling, etc. Works effects are then triggered when certain intervention criteria is reached, e.g. when roughness is high (&gt; 6.6 IRI) the system triggers the need for heavy rehabilitation. In HDM-4 (and dTIMS), a variety of intervention alternatives are triggered over an analysis period (10-20 years) on each road segment of the road network investigated. The road user and investment costs of the works are calculated, as well as the effects of the works on the pavement structure and surface. These are then translated into a benefit indicator that can either be economical or condition-based allowing the optimisation of funding levels to achieve maximum returns in terms of the benefit indicator.</p> <p>An important aspect of this analysis is to acquire condition assessment data on the road network to be investigated. The existing condition on each road plays an important role in the further deterioration of the pavement, influencing both the timing of intervention measures and the costs. The traffic volumes per vehicle class should be provided for each road to be evaluated.</p> | <p>RED is a simplified economic evaluation for low-volume rural roads that measures the benefits to road users and consumers of reduced transport costs.</p> <p><i>The main simplification of RED with relation to the HDM-4 models is that it considers a constant level of service, during the analysis period, for the with and without project cases, while the HDM-4 models include road deterioration equations.</i></p> <p><i>RED reduces the input requirements for low-volume roads, and the existing condition levels such as gravel thickness and roughness area not required for the evaluation. The traffic volumes per vehicle class are however crucial information that should be provided for each road to be evaluated.</i></p> <p>The road deterioration equations of the HDM-4 models, which vary over time the roughness of a given road as function of condition, traffic, environment, and maintenance characteristics, are <u>not</u> implemented on RED. Rather, RED uses the concept of average levels of service, which is considered reasonable for low volume unpaved roads for the following reasons:</p> <ul style="list-style-type: none"> <li>• convenience in defining levels of service for low-volume roads with parameters other than average roughness and gravel thickness,</li> <li>• difficulty in measuring or estimating the roughness of unpaved roads and determining the grading frequency to be applied to unpaved roads,</li> <li>• seasonal change in road condition and passability and</li> <li>• cyclical nature of the road deterioration under a proper maintenance policy.</li> </ul> |

## 5.2. Intervention measures

Intervention measures are the road works modeled in the life cycle cost analysis tools; dTIMS and HDM-4. These intervention measures model the following:

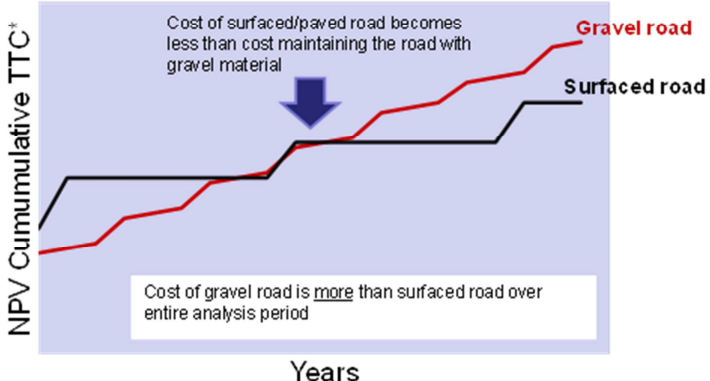
- The quantity and costs of works.
- Timing of works.
- The effect of the road works on the pavement performance models (e.g. cracking, roughness, etc.)

### 5.2.1. dTIMS

In dTIMS, the following road works are entered as TREATMENTS:

| Treatment  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HeavyRehab | <p>Heavy Rehabilitation is when the entire surfacing and one or two structural layers (base and sub-base) are removed and replaced with new material. Heavy rehabilitation or reconstruction is necessary when the road exhibits severe deformation – depressions and failures – caused by instability in the base structure of the road.</p> <p>The cost of this treatment is expensive and the unit cost is required per square metres which will be applied to the entire carriage way area of the road. Heavy rehabilitation triggers when:</p> <ul style="list-style-type: none"> <li>• roughness is severe ( &gt; 6.6 IRI) or</li> <li>• rutting is severe (&gt; 20 mm) or</li> <li>• roughness is in warning category (&gt; 5.5 IRI) and area with wide cracking exceeds 10% of the carriage way area or condition is in the very poor category.</li> </ul>                                                                                                                                                                                                                                                                                                                                                               |
| LightRehab | <p>Light Rehabilitation is triggered when the some deterioration of the pavement structure has occurred, but a heavy rehabilitation is not required of the entire structural layer. This treatment involves one of the following typical works: the removal/replacement of the surfacing and the removal/replacement of some sections of the base layer, or the placement of a new pavement layer (overlay) that could be proceeded by a mill.</p> <p>The cost of this treatment is considerably lower (approx. 50%) than a heavy rehabilitation and the unit cost is required per square metres which will be applied to the entire carriage way area of the road. Light rehabilitation is triggered when:</p> <ul style="list-style-type: none"> <li>• roughness is less than 5.5 IRI and area of wide cracking exceeds 10% of the carriage way area or</li> <li>• roughness is more than 5.5 IRI but area of wide cracking does not exceed 10% of the carriage way area or</li> <li>• area of wide cracking exceeds 10% of the carriage way area but the overall condition of the road segment is above 45%.</li> </ul> <p>In all cases here the roughness should not exceed 6.6 IRI and rutting should not exceed 20 mm.</p> |
| Reseal     | <p>Reseal/Seal extend the serviceable life of pavements by providing an all-weather surfacing to prevent the ingress of water. Therefore once cracking has initiated on the surface, a new reseal treatment will be triggered in the life cycle cost analysis. Reseals are cost-effective</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

| Treatment    | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              | <p>preventive maintenance, with costs small in comparison to full depth heavy rehabilitation. Reseals will not be triggered when roughness is poor (<math>&gt; 4.2</math> IRI) or when rutting is poor (<math>&gt; 20\text{mm}</math>) because pavement deformation cannot be repaired by resealing a road surface. The cost to enter should be an average unit cost of the road agency spent on a variety of reseals that could include convectional binders, modified binders and emulsions.</p> <p>The cost of this treatment is significantly lower than the rehabilitation treatments and the unit cost is required per square metres which will be applied to the entire carriage way area of the road. Reseals are triggered when:</p> <ul style="list-style-type: none"> <li>• All cracking exceeds 2%, but wide cracking is less than 15% on road carrying more than 4000 vehicles a day or</li> <li>• Wide cracking ranges between 5% and 15% on roads carrying less than 4000 vehicles per day or</li> <li>• All cracking exceeds 2%, wide cracking do NOT exceed 10% and roughness is in the moderate category (4.2 – 5.5 IRI).</li> </ul> |
| Reseal_Light | <p>Reseal-light is a light chip seal on low volume roads. The characteristics of this treatment is the same as a Reseal (above), except the cost is lower because conventional binders are used and not modified binders.</p> <p>Reseal-light is triggered when:</p> <ul style="list-style-type: none"> <li>• All cracking exceeds 1%, but wide cracking is less than 5% on road carrying less than 4000 vehicles a day.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Regravel     | <p>The reduction in gravel wearing course material on gravel roads occurs due to weathering and traffic. Regravelling is the process of replacing the gravel wearing course with new suitable material, then spreading and compacting it.</p> <p>The alternative to regravelling is to upgrade the road to surfaced standards, but this option will only be feasible on high trafficked roads where the NPV cost of the upgrade is less than the NPV of savings in road user costs.</p> <p>The unit cost is required per kilometre and regravelling is triggered when the thickness of the gravel wearing course is less than 30mm.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| UpgradePave  | <p>This treatment provides the alternative to upgrade a gravel road to a surfaced road thereby providing an all-weather, waterproof and skid resistance surfacing. The initial cost of an upgrade is relatively high, but the life cycle cost analysis tool will only consider this treatment when the capital cost, regular maintenance cost and road user cost of the surfaced road are lower than that of a gravel road. This treatment is therefore beneficial on high trafficked gravel roads where the savings in road user costs exceeds the high capital/maintenance investment needed for the upgrade.</p> <p>Upgrade to paved standard is considered on all roads carrying more than 200 vehicles per day, provided economic benefits prevail for the treatment.</p>                                                                                                                                                                                                                                                                                                                                                                         |

| Treatment                                          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                    | <p><b>TTC*: Surfaced versus gravel pavement type – High traffic volume</b></p>  <p>NPV Cumulative TTC*</p> <p>Years</p> <p>Gravel road</p> <p>Surfaced road</p> <p>Cost of surfaced/paved road becomes less than cost maintaining the road with gravel material</p> <p>Cost of gravel road is <u>more</u> than surfaced road over entire analysis period</p> <p>* TTC = Totals transport costs, inclusive of road user costs and initial/maintenance costs over analysis period</p><br><p><b>TTC*: Surfaced versus gravel pavement type – Low traffic volume</b></p>  <p>NPV Cumulative TTC*</p> <p>Years</p> <p>Surfaced road</p> <p>Gravel road</p> <p>Upgrade to surfaced/paved standard will NOT be considered here.</p> <p>Cost of gravel road is less than surfaced road over entire analysis period</p> <p>* TTC = Totals transport costs, inclusive of road user costs and initial/maintenance costs over analysis period</p> |
| Crack_seal<br><br>(Not used by UNRA:<br>July 2012) | Crack sealing occurs in the first analysis year, based on the most recent visual assessment information, when all cracking has initiated, but no wide cracking exists.<br><br>The unit cost is required per metre length of cracks. The area cracking is converted to this unit for cost calculations.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Patching<br><br>(Not used by UNRA:<br>July 2012)   | Patching occurs when the number of potholes exceeds 5 potholes per km.<br><br>The unit cost is required per square metre that will be applied to the area affected.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| FogSpray<br><br>(Not used by UNRA:                 | Fogspray involves a light application of diluted bitumen emulsion that is applied to a bituminous surfacing. It is a maintenance procedure on an existing bituminous surfacing to assist in retaining the aggregates in place, sealing the layer against water ingress or to “rejuvenate” the surfacing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |



| Treatment  | Description                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| July 2012) | <p>when it is showing signs of “dryness” resulting from ageing/hardening of the bitumen.</p> <p>Fogspray triggers on sealed roads with no wide cracking, where all crack has initiated but the area affected is less than 2% of carriage way area. Also the roughness is less than 4.2 IRI, rut depth is less than 12 mm and there are less than 5 potholes per km.</p> <p>The unit cost is required per square metres.</p> |

### 5.2.2. HDM-4

In HDM-4, the following road works are entered as WORK STANDARDS:

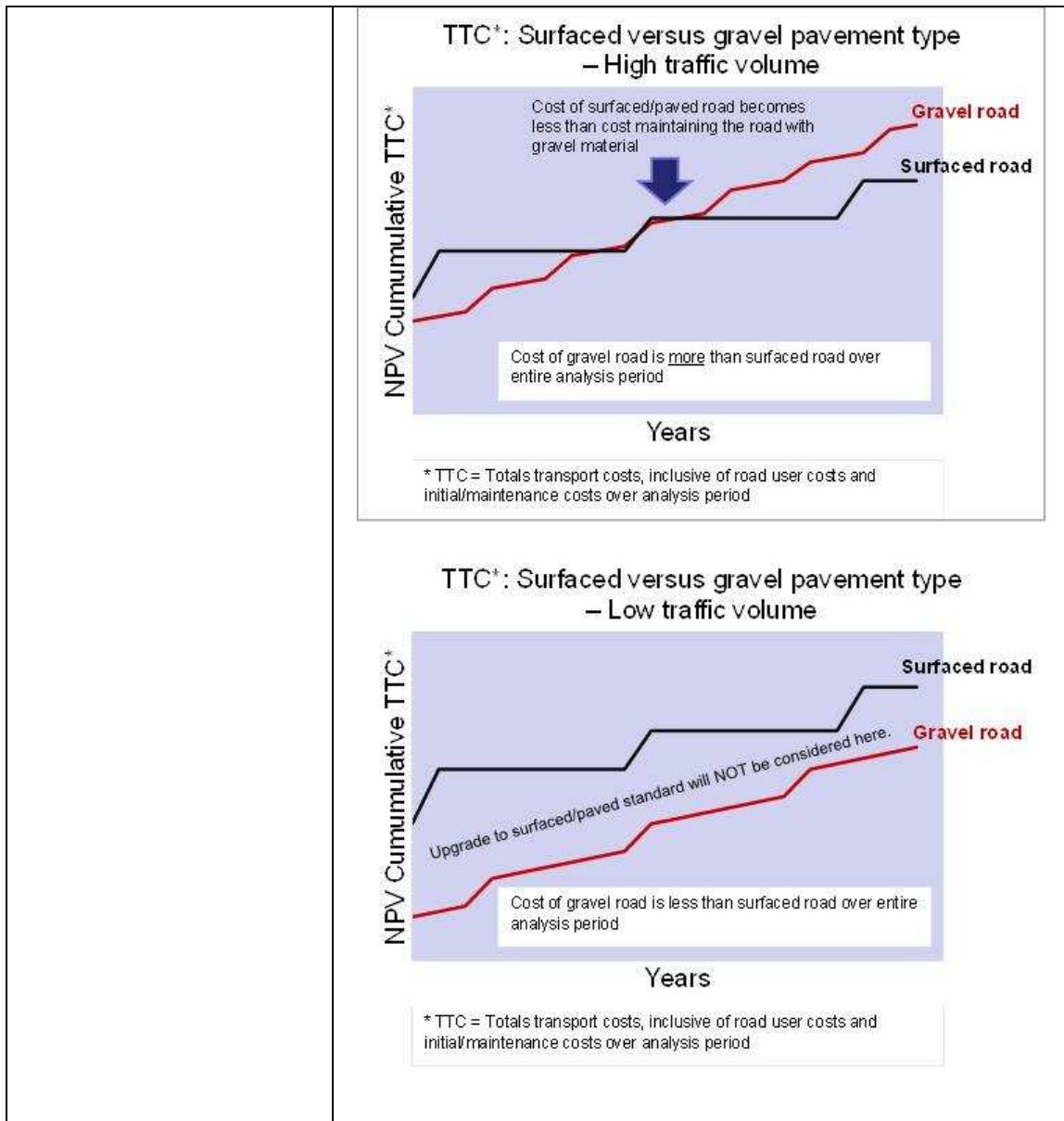
| Maintenance Standard                                                    | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Maint&amp;Rehab Bituminous:</b><br><br>Recon/HeavyRehab<br>(RECON)   | <p>Heavy Rehabilitation is when the entire surfacing and one or two structural layers (base and sub-base) are removed and replaced with new material. Heavy rehabilitation or reconstruction is necessary when the road exhibits severe deformation – depressions and failures – caused by instability in the base structure of the road.</p> <p>The cost of this treatment is expensive and the unit cost is required per square metres which will be applied to the entire carriage way area of the road. Heavy rehabilitation triggers when:</p> <ul style="list-style-type: none"> <li>• roughness is severe ( &gt; 6.6 IRI) or</li> <li>• rutting is severe (&gt; 20 mm) or</li> <li>• roughness is in warning category (&gt; 5.5 IRI) and area with wide cracking exceeds 10% of the carriage way area or condition is in the very poor category.</li> </ul>                                                                                                                                                                                                                           |
| <b>Maint&amp;Rehab Bituminous:</b><br><br>Overlay/LightRehab<br>(OVERL) | <p>Light Rehabilitation is triggered when the some deterioration of the pavement structure has occurred, but a heavy rehabilitation is not required of the entire structural layer. This treatment involves one of the following typical works: the removal/replacement of the surfacing and the removal/replacement of some sections of the base layer, or the placement of a new pavement layer (overlay) that could be proceeded by a mill.</p> <p>The cost of this treatment is considerably lower (approx. 50%) than a heavy rehabilitation and the unit cost is required per square metres which will be applied to the entire carriage way area of the road. Light rehabilitation is triggered when:</p> <ul style="list-style-type: none"> <li>• roughness is less than 5.5 IRI and area of wide cracking exceeds 10% of the carriage way area or</li> <li>• roughness is more than 5.5 IRI but area of wide cracking does not exceed 10% of the carriage way area.</li> </ul> <p>In all cases here the roughness should not exceed 6.6 IRI and rutting should not exceed 20 mm.</p> |
| <b>Maint&amp;Rehab Bituminous:</b><br><br>Reseal<br>(RSEAL)             | <p>Reseal/Seal extend the serviceable life of pavements by providing an all-weather surfacing to prevent the ingress of water. Therefore once cracking has initiated on the surface, a new reseal treatment will be triggered in the life cycle cost analysis. Reseals are cost-effective</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

| Maintenance Standard                                                              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                   | <p>preventive maintenance, with costs small in comparison to full depth heavy rehabilitation. Reseals will not be triggered when roughness is poor (<math>&gt; 4.2</math> IRI) or when rutting is poor (<math>&gt; 20\text{mm}</math>) because pavement deformation cannot be repaired by resealing a road surface. The cost to enter should be an average unit cost of the road agency spent on a variety of reseals that could include convectional binders, modified binders and emulsions.</p> <p>The cost of this treatment is significantly lower than the rehabilitation treatments and the unit cost is required per square metres which will be applied to the entire carriage way area of the road. Reseals are triggered when:</p> <ul style="list-style-type: none"> <li>All cracking exceeds 2%, but wide cracking is less than 15% on road carrying more than 4000 vehicles a day or</li> <li>Wide cracking ranges between 5% and 15% on roads carrying less than 4000 vehicles per day or</li> <li>All cracking exceeds 2%, wide cracking do NOT exceed 10% and roughness is in the moderate category (4.2 – 5.5 IRI).</li> </ul> |
| <b>Maint&amp;Rehab Bituminous:</b><br><br>Reseal light (LSEAL)                    | <p>Reseal-light is a light chip seal on low volume roads. The characteristics of this treatment is the same as a Reseal (above), except the cost is lower because conventional binders are used and not modified binders.</p> <p>Reseal-light is triggered when:</p> <ul style="list-style-type: none"> <li>All cracking exceeds 1%, but wide cracking is less than 5% on road carrying less than 4000 vehicles a day.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Maint&amp;Rehab Bituminous AND Routine - Bituminous:</b><br><br>Routine (ROUT) | <p>Routine is triggered as cost per kilometre per year by routine works not affected by the intervention of the Maintenance and Rehabilitation works. It should therefore include items such as cleaning of culverts, bush clearing, etc.</p> <p>A cost per km is required which will be applied for each year in the analysis period.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Maint&amp;Rehab Bituminous:</b><br><br>Patching (PATCH)                        | <p>Patching occurs when the number of potholes exceeds 5 potholes per km.</p> <p>The unit cost is required per square metre that will be applied to the area affected.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

| Maintenance Standard                                                                                                                          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Maint&amp;Rehab Unsealed:</b><br><br>Regravel with selected material (GRAV1)<br><br>And/or<br><br>Regravel with crushed aggregates (GRAV2) | <p>The reduction in gravel wearing course material on gravel roads occurs due to weathering and traffic. Regravelling is the process of replacing the gravel wearing course with new suitable material, then spreading and compacting it.</p> <p>The alternative to regravelling is to upgrade the road to surfaced standards, but this option will only be feasible on high trafficked roads where the NPV cost of the upgrade is less than the NPV of savings in road user costs.</p> <p>The unit cost is required per kilometre and regravelling is triggered</p> |

| Maintenance Standard                                                                 | Description                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                      | when the thickness of the gravel wearing course is less than 30mm.                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Maint&amp;Rehab Unsealed:</b><br><br>Grading<br>(GRAD)                            | <p>Light motorised grading, little or no water, no roller compaction. Grading is assumed for frequencies of once per year, however this can extended to a matrix of frequencies dependant on traffic volumes should such information become available.</p> <p>The unit cost is required per kilometre that is graded and it is therefore dependant on the frequency of the intervention, e.g. 30 day cycle, 60 day cycle, 260 day cycle, etc.</p> |
| <b>Maint&amp;Rehab Unsealed:</b><br><br>Spot regravell<br>(SPREGR)                   | <p>Spot regravelling is done on gravel roads with gravel material less than 50mm. It is therefore a holding action to postpone the regravelling of the entire surface. It adds 40m<sup>3</sup>/km/year, therefore requiring a unit cost per cubic metres.</p> <p>The unit cost is required per kilometre and regravelling is triggered when the thickness of the gravel wearing course is less than 30mm.</p>                                     |
| <b>Maint&amp;Rehab Unsealed AND Routine - Unsurfaced:</b><br><br>Routine<br>(ROUT_U) | <p>Routine is triggered as cost per kilometre per year by routine works not affected by the intervention of the Maintenance and Rehabilitation works. It should therefore include items such as cleaning of culverts, bush clearing, etc.</p> <p>A cost per km is required which will be applied for each year in the analysis period.</p>                                                                                                        |

| Improvement Standard        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Upgrade to Pave<br>(UPGRAD) | <p>This treatment provides the alternative to upgrade a gravel road to a surfaced road thereby providing an all-weather, waterproof and skid resistance surfacing. The initial cost of an upgrade is relatively high, but the life cycle cost analysis tool will only consider this treatment when the capital cost, regular maintenance cost and road user cost of the surfaced road are lower than that of a gravel road. This treatment is therefore beneficial on high trafficked gravel roads where the savings in road user costs exceeds the high capital/maintenance investment needed for the upgrade.</p> <p>Upgrade to paved standard is considered on all roads carrying more than 200 vehicles per day, provided economic benefits prevail for the treatment.</p> |



## 6. Process to conduct a life cycle cost analysis on UNRA's road network

The chapters hereafter contain the sequential flow of processes to be followed to conduct a life cycle cost analysis (LCCA) on UNRA's road network. It is a detailed set of instruction describing the following processes:

- Verify the integrity of UNRA's road network to be analysed.
- Input of all parameters and costs influencing the LCCA.
- Establish the data (file, perspectives, tables, etc) in the prescribed format to enable LCCA in HDM4, RED and dTIMS software.
- Conduct the analyses.
- Review the results.

**Workflow:** The majority of steps/tasks to be completed in dTIMS have been included in a set of Workflow objects in dTIMS WF. All chapters hereafter will indicate the Workflow step that was customised in dTIMS WF.

**Workflow:** Before you continue, perform **Step 00: Verify Data** in UNRA's dTIMS WF.

## 7. Pavement Management System Models

Use level 7 (Username = **lcca**, Password = **lcca**) security for all dTIMS operations in this chapter.

### 7.1. Introduction

The following models are used in the Pavement Management System:

- Vehicle Operating Cost Model
- Road Treatment Effects Model
- Pavement Deterioration Model

This chapter describes how the parameters for each of the above models are established and/or stored in the relevant system; RED, HDM-4 and dTIMS. The collection and processing of the data used to generate these parameters is described in the accompanying *PMS Models* documentation.

### 7.2. Vehicle Operating Cost Model (For RED and dTIMS)

**Workflow:** Included in **Step 01: Update VOC** of UNRA's dTIMS WF.

The three life cycle cost analysis (LCCA) tools described in this manual require vehicle operating cost models to calculate the economic benefit and economic consequence of maintenance policies and budget constraints. The customization of the vehicle operating costs models for dTIMS and RED are described herewith, whilst the VOC models for HDM-4 are described Chapter 11.3.

The establishment of the VOC models, described in this paragraph, should be completed before the user can continue to perform any life cycle cost analyses in RED and/or dTIMS.

**Methodology:** The VOC models are completed using the RED software, namely the Excel file; **UNRA RED - HDM4 VOC models.xls**. The VOC results are then copied and pasted into a *Cross Tab Transformation* in dTIMS. For the RED software the results are copied via an instruction programmed within the VOC RED Excel file to the Main RED Excel File.

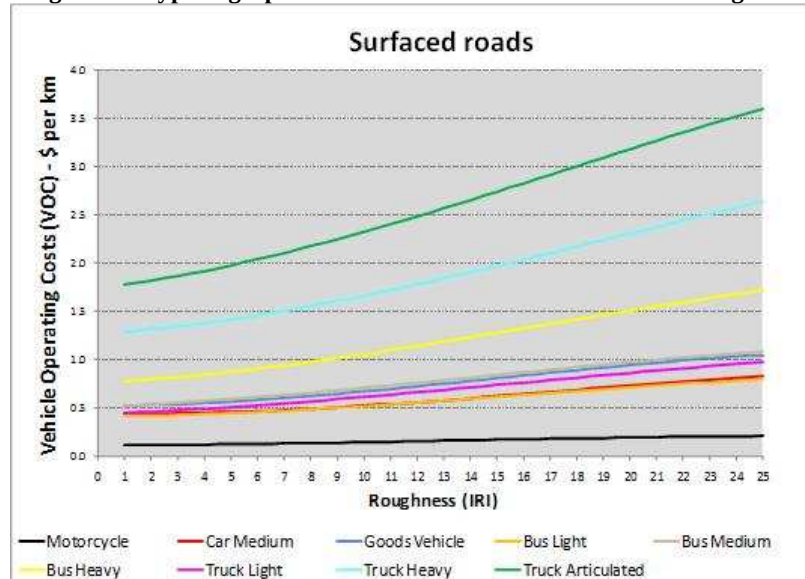
The steps hereafter establish the latest VOC models for use in RED and dTIMS:

1. Open the file **UNRA RED - HDM-4 VOC models.xls** residing on **C:\UNRA\_PMS\RED**.
2. Open the sheet **Basic Info Data**.
3. Enter the required data in the **Basic Info Data** sheet: The data in this sheet has already been set-up to correspond to the data and characteristics of the UNRA road network. There are however data to be updated on an annual basis and/or whenever the software is used. Economic unit costs are dynamic and it is up to the user to ensure these costs are updated before each analysis. Life cycle cost analyses are driven by the economic benefits calculated from the costs and the

parameters of the **Basic Info Data** sheet; the integrity of the data is therefore important. **Remember to change yellow cells only.** Cells to be updated regularly are:

- a. Year (D4)
  - b. Economic unit costs of the Vehicle Fleet (C39:K45)
  - c. All other yellow cells can be changed by an advanced user experienced in HDM-4's VOC models.
4. Open the sheet **Calibration Data** sheet if changes are necessary to the calibration data of the Vehicle fleet used for UNRA. This is for advanced users experienced in HDM-4's VOC models only. The system has been customised according to HDM-4's default values and no changes are thus needed here.
  5. Open the sheet **Compute Results** and click on the 1<sup>st</sup> button **Compute Results**. Wait for the processing to complete. The comparisons between vehicle operating costs and road roughness are now calculated.
  6. Make sure the filename in cell F23 in sheet **Compute Results** (shaded in yellow), contain the correct filename of the main file. UNRA's customised system requires the filename: UNRA RED - Main.XLS. (This file resides on the same directory as the file: **UNRA RED - HDM-4 VOC models.xls**)
  7. Click on the button; *Copy and Paste results into Main Module*. Wait for the processing to complete. (Click the OK button when copy and paste have completed)
  8. The user can now explore any of the worksheets from **Coefficients** onwards for the results of the VOC calculations.
  9. For UNRA, an extra sheet was added at the end of the list namely; **For\_dTIMS**. This sheet contains the relationship between roughness and VOC. The data is shown in graphs and tables that can be copied and pasted to any reports or other electronic media. It is the responsibility of the user to ensure the calculated VOC models are satisfactory. The graph below illustrates a typical relationship between VOC and roughness as customised in the sheet; **For\_dTIMS**.

**Figure 4 – Typical graph customized for UNRA: VOC versus Roughness**



10. Save the file; **UNRA RED - HDM-4 VOC models.xls**, but do not close the file.

11. The VOC models are now ready for use in the RED software.
12. Open the dTIMS software and open the UNRA database. Log in as “lcca” user. See paragraph 4.1.
13. In Work With Data, click on Cross Tab Transformations. Click on the item **CT\_VOC**, and open the Values under the Cross Tab Cells.
14. Use the mouse, shift-key and arrow buttons make all 4 columns and 18 rows are visible on the screen. High-light all the cells with values, i.e. cells from row2col2 to row19col5, see area shaded in blue in figure below.

**Figure 5 – The cross tab transformation cells for the coefficients of the VOC model**

| Row\AnalMaster->Trat_AADT | -1 | 0-1      | 1-2       | 2-3      | 3-4      |
|---------------------------|----|----------|-----------|----------|----------|
| -1                        | 0  | 0        | 0         | 0        | 0        |
| 1-2                       | 0  | 0.071996 | 8.8E-05   | 5.5E-05  | 0        |
| 2-3                       | 0  | 0.311507 | -0.005883 | 0.001026 | -1.7E-05 |
| 3-4                       | 0  | 0.375668 | -0.006532 | 0.001179 | -2.2E-05 |
| 4-5                       | 0  | 0.344888 | -0.001363 | 0.000818 | -1.4E-05 |
| 5-6                       | 0  | 0.434076 | 0.004764  | 0.000824 | -1.6E-05 |
| 6-7                       | 0  | 0.696219 | 0.009242  | 0.001647 | -3.1E-05 |
| 7-8                       | 0  | 0.374159 | 0.005709  | 0.000703 | -1.3E-05 |
| 8-9                       | 0  | 1.104126 | 0.006713  | 0.002011 | -2.9E-05 |
| 9-10                      | 0  | 1.583896 | 0.023756  | 0.002541 | -4.4E-05 |
| 10-11                     | 0  | 0.064869 | 0.002311  | -3.9E-05 | 1E-06    |
| 11-12                     | 0  | 0.285733 | 0.001926  | 0.000672 | -1E-05   |
| 12-13                     | 0  | 0.323324 | 0.00672   | 0.000628 | -1.2E-05 |
| 13-14                     | 0  | 0.315917 | 0.006567  | 0.000523 | -9E-06   |
| 14-15                     | 0  | 0.401572 | 0.014125  | 0.00052  | -1E-05   |
| 15-16                     | 0  | 0.673106 | 0.020939  | 0.001366 | -2.5E-05 |
| 16-17                     | 0  | 0.347168 | 0.013936  | 0.000406 | -7E-06   |
| 17-18                     | 0  | 1.059226 | 0.026103  | 0.001479 | -1.7E-05 |
| 18-19                     | 0  | 1.538056 | 0.051059  | 0.002237 | -3.6E-05 |

15. Use ALT & TAB to go back to the **For\_dTIMS** sheet in UNRA RED - HDM-4 VOC models.xls. Highlight and COPY the cells D9 through to G26 as indicated in the Figure 6. The matrix copied here should exactly match the number of rows and columns highlighted in the previous step.

**Figure 6 – Cells to highlight and copy in UNRA RED - HDM-4 VOC models.xls**

**Vehicle Operating Costs Function of Roughness Coefficients**

VOC (\$/veh-km) as a Function of Roughness (IRI)  
 $VOC = a_0 + a_1 \cdot IRI + a_2 \cdot IRI^2 + a_3 \cdot IRI^3$

|        |              |                   | a0       | a1        | a2       | a3        |
|--------|--------------|-------------------|----------|-----------|----------|-----------|
| 9 AX1  | Terrain: ALL | Motorcycle        | 0.107701 | 0.001639  | 0.000259 | -0.000007 |
| 10 AX2 |              | Car Medium        | 0.443449 | -0.004138 | 0.001469 | -0.000027 |
| 11 AX3 |              | Goods Vehicle     | 0.524415 | -0.000479 | 0.001938 | -0.000043 |
| 12 AX4 |              | Bus Light         | 0.409469 | 0.002476  | 0.001054 | -0.000021 |
| 13 AX5 | Paved        | Bus Medium        | 0.509305 | 0.010912  | 0.001145 | -0.000027 |
| 14 AX6 |              | Bus Heavy         | 0.757688 | 0.014722  | 0.001901 | -0.000038 |
| 15 AX7 |              | Truck Light       | 0.435102 | 0.010200  | 0.000965 | -0.000020 |
| 16 AX8 |              | Truck Heavy       | 1.267252 | 0.017889  | 0.002611 | -0.000045 |
| 17 AX9 |              | Truck Articulated | 1.738779 | 0.033978  | 0.003014 | -0.000056 |
| 18 AY1 | Terrain: ALL | Motorcycle        | 0.101006 | 0.003839  | 0.000164 | -0.000005 |
| 19 AY2 |              | Car Medium        | 0.422841 | 0.003019  | 0.001140 | -0.000021 |
| 20 AY3 |              | Goods Vehicle     | 0.477372 | 0.012210  | 0.001403 | -0.000033 |
| 21 AY4 |              | Bus Light         | 0.381389 | 0.010329  | 0.000759 | -0.000016 |
| 22 AY5 | Gravel       | Bus Medium        | 0.478007 | 0.020212  | 0.000839 | -0.000020 |
| 23 AY6 |              | Bus Heavy         | 0.735053 | 0.026401  | 0.001618 | -0.000032 |
| 24 AY7 |              | Truck Light       | 0.409571 | 0.018351  | 0.000666 | -0.000014 |
| 25 AY8 |              | Truck Heavy       | 1.223850 | 0.037143  | 0.002081 | -0.000033 |
| 26 AY9 |              | Truck Articulated | 1.693230 | 0.061562  | 0.002685 | -0.000047 |

16. Use ALT & TAB to go back to the open dTIMS software. The matrix of values should still be highlighted. Press CTRL & V to paste the roughness coefficients into the cross tab window. Make sure the data has pasted correctly. The columns correspond to the columns; a0 through a3, shown in Figure 6. Click OK to accept the changes. Make sure CT\_VOC is still selected and click the Save button to save the changes made to the Cross Tab Transformations.

**Figure 7 - Road Management System Physical Implementation**

**Cross Tab Transformation Cells**

Row: AY1 to AY19  
 Column: a0 to a3

| Row\Col | a0       | a1        | a2       | a3        |
|---------|----------|-----------|----------|-----------|
| AY1     | 0.107701 | 0.001639  | 0.000259 | -0.000007 |
| AY2     | 0.443449 | -0.004138 | 0.001469 | -0.000027 |
| AY3     | 0.524415 | -0.000479 | 0.001938 | -0.000043 |
| AY4     | 0.409469 | 0.002476  | 0.001054 | -0.000021 |
| AY5     | 0.509305 | 0.010912  | 0.001145 | -0.000027 |
| AY6     | 0.757688 | 0.014722  | 0.001901 | -0.000038 |
| AY7     | 0.435102 | 0.010200  | 0.000965 | -0.000020 |
| AY8     | 1.267252 | 0.017889  | 0.002611 | -0.000045 |
| AY9     | 1.738779 | 0.033978  | 0.003014 | -0.000056 |
| AY10    | 0.101006 | 0.003839  | 0.000164 | -0.000005 |
| AY11    | 0.422841 | 0.003019  | 0.001140 | -0.000021 |
| AY12    | 0.477372 | 0.012210  | 0.001403 | -0.000033 |
| AY13    | 0.381389 | 0.010329  | 0.000759 | -0.000016 |
| AY14    | 0.478007 | 0.020212  | 0.000839 | -0.000020 |
| AY15    | 0.735053 | 0.026401  | 0.001618 | -0.000032 |
| AY16    | 0.409571 | 0.018351  | 0.000666 | -0.000014 |
| AY17    | 1.223850 | 0.037143  | 0.002081 | -0.000033 |
| AY18    | 1.693230 | 0.061562  | 0.002685 | -0.000047 |
| AY19    |          |           |          |           |

17. The updated VOC models are now ready for use in both software packages: dTIMS and RED.
18. Close both the dTIMS and RED files/software.



### 7.3. Pavement Deterioration Model

**Workflow:** *No steps were customised in UNRA's dTIMS WF.*

Both RED and HDM-4 software have built-in pavement performance models and no customization of the software is therefore needed. However, the user does have the ability to manipulate the performance models with calibration factors to influence the rate of deterioration in HDM-4. This is detailed in the chapters describing the operation of the LCCA for each system.

The dTIMS software has no built-in pavement performance models, and is the only system (of the three LCCA tools) that was customized with pavement performance models. The HDM-4 pavement performance models were customized within dTIMS for use by UNRA. The pavement performance models in dTIMS should only be changed by advanced users experienced in the technical detail of HDM-4 modeling and experienced in the customization of life cycle costs analyses in dTIMS.

The pavement performance models are detailed in Table 10 through Table 12 of Appendix A. All items concerning the pavement performance models are detailed in Appendix A.

### 7.4. Road Treatment Effects Model

**Workflow:** *No steps were customised in UNRA's dTIMS WF.*

The treatment models customized in dTIMS are detailed in Table 13 through Table 14 of Appendix A. The advanced user can change the treatment parameters at his/her own risk, but usually the user will only need to change the unit cost per treatment, see next chapter.

## 8. Update Default Data

**Workflow:** *Included in Step02: Update Unit Costs of UNRA's dTIMS WF,*

**Workflow:** *Included in Step 03: Update Asset Value Costs of UNRA's dTIMS WF and*

**Workflow:** *Included in Step 04: Update Default Traffic Info of UNRA's dTIMS WF.*

Use level 7 (Username = **lcca**, Password = **lcca**) security for all dTIMS operations in this chapter.

The operations in this chapter are necessary before an analysis can be performed in RED, dTIMS and HDM-4.

The PMS database in dTIMS consists of road network inventory data, condition data, traffic data, etc. The data is collected via visual surveys, high speed surveys and other methods. In certain cases, if specific data items are not available in dTIMS, the PMS has to rely on default data values to ensure the pavement performance models continue to produce feasible answers.

Apart from the data in UNRA's dTIMS database, the life cycle cost analyses also require information such as unit costs, traffic growth factors, calibration factors, etc. needed in the pavements performance models. ***These default values have been programmed in dTIMS as expressions with names prefixed with "a\_".*** It is therefore easy to identify the default expressions. Note that these expressions are only displayed for the "lcca" user.

This chapter lists the default values required for the data preparation for analyses in RED, HDM-4 and/or dTIMS. The items are categorized as follows:

| Expression prefix | Description                                                                                                                                                                                                                                                       |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| a_AgencyName      | Name of road agency – this is not used anywhere in the system.                                                                                                                                                                                                    |
| a_Asset           | All items related to asset calculations of Visual assessment data and for forward prediction of asset value in dTIMS LCCA. These expressions can also be accessed by the “data” user. Make sure the cost items are Present Value with no allowance for inflation. |
| a_Cost            | All unit cost items and related cost items. Make sure the cost items are Present Value with no allowance for inflation.                                                                                                                                           |
| a_HDM4            | All items related to the HDM-4 pavement performance curves customized into dTIMS and HDM-4                                                                                                                                                                        |
| a_Traf            | All default values related to traffic. To be used in cases where no traffic data is available per vehicle class.                                                                                                                                                  |

Steps to check and update the default values in dTIMS:

1. Run the dTIMS software as “lcca” user and open the UNRA database.
2. In Analyze&Report / Analysis Expressions: Inspect all expressions prefixed with “**a\_Cost**” and “**a\_Asset**”. The cost and asset expressions should be checked before every LCCA and changed if the cost is outdated. These default values are shaded in blue in Table 1.
3. The default traffic input data is stored under the expressions prefixed with “**a\_Traf**”. Change only when new information becomes available from the TIS.
4. **For advanced users only:** In Analyze&Report / Analysis Expressions: Inspect the remainder of the expressions starting with “**a\_HDM4**”. These expressions influence the operation of the pavement performance models in all three LCCA tools. The values should only be changed if the user is experienced with HDM-4 performance modelling. In certain cases the values should correspond to predefined item lists (tables) in the HDM-4 database of UNRA and the integrity of analyses to follow could be jeopardized if invalid data is entered. These default values are also shown in Table 1. *The cost items shaded in blue refer to the unit cost values of the treatments, these items should be updated at least once a year!*
5. When all cost expressions and other expressions have been updated: Save all changes to the expressions and close the dTIMS software or continue to entering budget scenarios in dTIMS.

| Table 1 – Default values         |                                                                                       |               |
|----------------------------------|---------------------------------------------------------------------------------------|---------------|
| Default Expression Name          | Description                                                                           | TheExpression |
| a_AgencyName                     | Name of Agency                                                                        | 'UNRA'        |
| a_AssetValue_Pav_ReconstructCost | Enter the cost of a reconstruction on paved roads, per sq meters                      | 167           |
| a_AssetValue_Unp_GravelCost      | Enter the cost to upgrade an earth road to a gravel road, per km                      | 45000.0       |
| a_AssetValue_Unp_RegravelCost    | Enter the cost to regravel an existing gravel roads, per km                           | 30000.0       |
| a_AssetValue_Year_to_Calc        | Enter the year for which asset values should be calculated on the Visuals perspective | 2009.0        |
| a_Cost_EconomicFactor            | Economic cost as factor of Financial cost                                             | 0.68          |
| a_CostBlade                      | Default Cost to blade a gravel road (per km)                                          | 3500.0        |
| a_CostCrackSeal                  | Default Crack Seal Cost ( per metre)                                                  | 3.0           |
| a_CostFogSpray                   | Default cost for a fog spray (per square meteres)                                     | 1.6           |

| Table 1 – Default values |                                                                            |                                                                                                                                                                                                                                                 |
|--------------------------|----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Default Expression Name  | Description                                                                | TheExpression                                                                                                                                                                                                                                   |
| a_CostGravel             | Default Cost to upgrade an earth road to gravel standards (per km)         | 45000.0                                                                                                                                                                                                                                         |
| a_CostHeavyRehab         | Default Cost of a Heavy rehab (per square meteres)                         | 100.0                                                                                                                                                                                                                                           |
| a_CostLightRehab         | Default of a Light Rehab (per square meteres)                              | 60.0                                                                                                                                                                                                                                            |
| a_CostMSeal              | Default Cost of a Modified Seal (per square meteres)                       | 15.5                                                                                                                                                                                                                                            |
| a_CostPatching           | Default cost for Patching (per square meteres)                             | 100.0                                                                                                                                                                                                                                           |
| a_CostPaveUpgradeHigh    | Default Cost to upgrade a gravel road to high paved standards (per km)     | 1000000.0                                                                                                                                                                                                                                       |
| a_CostPaveUpgradeLow     | Default Cost to upgrade a gravel road to low paved standards (per km)      | 1000000.0                                                                                                                                                                                                                                       |
| a_CostRegravel           | Default Cost to regravel a gravel road (per km)                            | 30000.0                                                                                                                                                                                                                                         |
| a_CostReseal_Light       | Light Reseal Default cost (per square meteres)                             | 10.0                                                                                                                                                                                                                                            |
| a_CostSeal               | Default Cost of a Reseal (per square meteres)                              | 13.5                                                                                                                                                                                                                                            |
| a_HDM4_Altitude          | Default altitude in metres                                                 | 1200.0                                                                                                                                                                                                                                          |
| a_HDM4_CALIB_ITEM        | Name of calibration item in HDM4. Name to match items defined in HDM4.     | IF(HDM4_Temp->SURF_CLASS_Temp = 'AC','UNRA Seal on Granular',IF(HDM4_Temp->SURF_CLASS_Temp = 'SD','UNRA AC on Granular',IF(HDM4_Temp->SURF_CLASS_Temp = 'G','UNRA Gravel Type A',IF(HDM4_Temp->SURF_CLASS_Temp = 'E','UNRA Earth Type A','')))) |
| a_HDM4_CBR               | Default CBR where no data available                                        | 4.0                                                                                                                                                                                                                                             |
| a_HDM4_CDB               | Default construction defects indicator of base.                            | 1.0                                                                                                                                                                                                                                             |
| a_HDM4_Climate_Zone      | Default climate zone in HDM4. Name to match climate items defined in HDM4. | 'Humid Tropical'                                                                                                                                                                                                                                |
| a_HDM4_ConstructionAge   | Default age since construction, if no data available.                      | 19.0                                                                                                                                                                                                                                            |
| a_HDM4_CRK_CRT           | Crack retardation time due to maintenance, in years.                       | 0.0                                                                                                                                                                                                                                             |
| a_HDM4_CURVATURE         | Default HDM4 curvature, deg / km.                                          | 15.0                                                                                                                                                                                                                                            |
| a_HDM4_CWmax             | Maximum carriageway width for occurrence of edge break                     | 7.2                                                                                                                                                                                                                                             |
| a_HDM4_DRAIN_COND        | Default condition of drainage according to HDM4's ratings.                 | 2.0                                                                                                                                                                                                                                             |
| a_HDM4_DRAIN_TYPE        | Default drain type for HDM4 - 7 = no drainage effects.                     | 7.0                                                                                                                                                                                                                                             |
| a_HDM4_ELANES            | Number of effective lanes.                                                 | 2.0                                                                                                                                                                                                                                             |
| a_HDM4_ENFORCEMNT        | Default speed limit enforce factor.                                        | 1.1                                                                                                                                                                                                                                             |
| a_HDM4_GravelAge         | Default age since previous regravel, if no data available.                 | 5.0                                                                                                                                                                                                                                             |
| a_HDM4_GravThickMax      | Gravel thickness after regravelling.                                       | 150.0                                                                                                                                                                                                                                           |
| a_HDM4_HBASE             | Default thickness of base layer, mm.                                       | 150.0                                                                                                                                                                                                                                           |
| a_HDM4_HSNEW             | Default thickness of most recent surfacing, mm.                            | 20.0                                                                                                                                                                                                                                            |
| a_HDM4_HSOLD             | Default thickness of underlying previous surface layers, mm.               | 20.0                                                                                                                                                                                                                                            |
| a_HDM4_HSUBBASE          | Default thickness of subbase, mm.                                          | 150.0                                                                                                                                                                                                                                           |

| Table 1 – Default values |                                                                                                    |                 |
|--------------------------|----------------------------------------------------------------------------------------------------|-----------------|
| Default Expression Name  | Description                                                                                        | TheExpression   |
| a_HDM4_KCI               | Crack initiation calibration factor                                                                | 1.1             |
| a_HDM4_KCP               | Crack progression calibration factor                                                               | 0.2             |
| a_HDM4_KEB               | Calibration factor for edge break progression                                                      | 1.0             |
| a_HDM4_KGM               | Roughness environmental calibration factor                                                         | 1.0             |
| a_HDM4_KGP               | Roughness progression calibration factor                                                           | 1.0             |
| a_HDM4_KPP               | Pothole progression calibration factor                                                             | 1.0             |
| a_HDM4_KRP               | Rut depth progression calibration factor                                                           | 0.5             |
| a_HDM4_KVI               | Ravelling initiation calibration factor                                                            | 1.0             |
| a_HDM4_KVP               | Ravelling progression calibration factor                                                           | 1.0             |
| a_HDM4_m_ENVIRONCOEFF    | Environmental coefficient for roughness progression.                                               | 0.015           |
| a_HDM4_MMP               | Default value for Mean Monthly Precipitation, metres/month.                                        | 0.042           |
| a_HDM4_NUM_RFS           | Default number of rise and falls, per km.                                                          | 2.0             |
| a_HDM4_PPT               | Percentage potholes to patch for HDM4 modelling.                                                   | 90.0            |
| a_HDM4_Ravel_RRF         | Ravelling retardation factor.                                                                      | 1.0             |
| a_HDM4_REL_COMPCT        | Default resilient modulus of soil cement in GPa.                                                   | 91.0            |
| a_HDM4_RES_MODULU        | Default compaction in base, subbase, etc, %                                                        | 15.0            |
| a_HDM4_RF                | Default rise and fall, m/km.                                                                       | 10.0            |
| a_HDM4_SA_For_Gravel     | NA for UNRA                                                                                        | FALSE           |
| a_HDM4_SIGM_ADRAL        | Default acceleration noise, m/s <sup>2</sup> .                                                     | 0.1             |
| a_HDM4_SN                | Default structural number .                                                                        | 2.5             |
| a_HDM4_SNP_DRY           | Default SN measured in dry season?                                                                 | TRUE            |
| a_HDM4_SPEED_LIM         | Default speed limit, km/h.                                                                         | 100.0           |
| a_HDM4_Speedflow         | Default Speed flow in HDM4. To match item names in HDM4 exactly.                                   | 'Two Lane Road' |
| a_HDM4_SUGB_MATRL        | Default subgrade material code.                                                                    | 1.0             |
| a_HDM4_SUPERELEV         | Default superelevation , %                                                                         | 2.5             |
| a_HDM4_SurfaceAge        | Default age since previous surfacing, if no data available.                                        | 9.0             |
| a_HDM4_TEXT_DEPTH        | Default texture depth (mm).                                                                        | 0.7             |
| a_HDM4_TLF_CrackRavl     | Effects of pothole patching frequency for cracking and ravelling.                                  | 0.3             |
| a_HDM4_TLF_Enlargement   | Effects of pothole patching frequency - enlargement.                                               | 0.0             |
| a_HDM4_TRAF_FLOW         | Default traffic flow model in HDM4. To match item names in HDM4 exactly.                           | 'Free-Flow'     |
| a_HDM4_XFRI              | Default speed reduction of Motorised traffic, due to road side activities.                         | 1.0             |
| a_HDM4_XMT               | Default speed reduction of Motorised traffic.                                                      | 1.0             |
| a_HDM4_XNMT              | Default speed reduction of NON-Motorised traffic.                                                  | 1.0             |
| a_Traf_Growth            | Default traffic annual growth percentage (%) to be used where growth is negative or not available. | 4.0             |
| a_Traf_Veh1_Axles        | Default no of axles for Vehicle type 1. (Motorcycle)                                               | 2.0             |

| Table 1 – Default values |                                                                                                                |               |
|--------------------------|----------------------------------------------------------------------------------------------------------------|---------------|
| Default Expression Name  | Description                                                                                                    | TheExpression |
| a_Traf_Veh1_E80Fact      | Default E80 Factor for Vehicle type 1. (Motorcycle)                                                            | 0.0           |
| a_Traf_Veh1_Split        | Default split of traffic for Vehicle type 1 % of total daily traffic. (Motorcycle)                             | 24.0          |
| a_Traf_Veh2_Axles        | Default no of axles for Vehicle type 2. (Car)                                                                  | 2.0           |
| a_Traf_Veh2_E80Fact      | Default E80 Factor for Vehicle type 2. (Car)                                                                   | 0.0           |
| a_Traf_Veh2_Split        | Default split of traffic for Vehicle type 2. % of total daily traffic. (Car) – NOT USED FOR UNRA               | 21.0          |
| a_Traf_Veh3_Axles        | Default no of axles for Vehicle type 3. (LGV)                                                                  | 2.0           |
| a_Traf_Veh3_E80Fact      | Default E80 Factor for Vehicle type 3. (LGV)                                                                   | 0.7           |
| a_Traf_Veh3_Split        | Default split of traffic for Vehicle type 3. % of total daily traffic. (LGV) – NOT USED FOR UNRA               | 20.0          |
| a_Traf_Veh4_Axles        | Default no of axles for Vehicle type 4. (Minibus)                                                              | 2.0           |
| a_Traf_Veh4_E80Fact      | Default E80 Factor for Vehicle type 4. (Minibus)                                                               | 0.8           |
| a_Traf_Veh4_Split        | Default split of traffic for Vehicle type 4. % of total daily traffic. (Minibus) – NOT USED FOR UNRA           | 19.0          |
| a_Traf_Veh5_Axles        | Default no of axles for Vehicle type 5. (Bus medium)                                                           | 2.0           |
| a_Traf_Veh5_E80Fact      | Default E80 Factor for Vehicle type 5. (Bus medium)                                                            | 0.9           |
| a_Traf_Veh5_Split        | Default split of traffic for Vehicle type 5. % of total daily traffic. (Bus medium) – NOT USED FOR UNRA        | 1.0           |
| a_Traf_Veh6_Axles        | Default no of axles for Vehicle type 6. (Bus large)                                                            | 2.0           |
| a_Traf_Veh6_E80Fact      | Default E80 Factor for Vehicle type 6. (Bus large)                                                             | 1.8           |
| a_Traf_Veh6_Split        | Default split of traffic for Vehicle type 6. % of total daily traffic. (Bus large) – NOT USED FOR UNRA         | 1.0           |
| a_Traf_Veh7_Axles        | Default no of axles for Vehicle type 7. (Truck light)                                                          | 2.0           |
| a_Traf_Veh7_E80Fact      | Default E80 Factor for Vehicle type 7. (Truck light)                                                           | 1.0           |
| a_Traf_Veh7_Split        | Default split of traffic for Vehicle type 7. % of total daily traffic. (Truck light) – NOT USED FOR UNRA       | 5.0           |
| a_Traf_Veh8_Axles        | Default no of axles for Vehicle type 8. (Truck heavy)                                                          | 3.0           |
| a_Traf_Veh8_E80Fact      | Default E80 Factor for Vehicle type 8. (Truck heavy)                                                           | 2.0           |
| a_Traf_Veh8_Split        | Default split of traffic for Vehicle type 8. % of total daily traffic. (Truck heavy) – NOT USED FOR UNRA       | 5.0           |
| a_Traf_Veh9_Axles        | Default no of axles for Vehicle type 9. (Truck articulated)                                                    | 6.0           |
| a_Traf_Veh9_E80Fact      | Default E80 Factor for Vehicle type 9. (Truck articulated)                                                     | 3.5           |
| a_Traf_Veh9_Split        | Default split of traffic for Vehicle type 9. % of total daily traffic. (Truck articulated) – NOT USED FOR UNRA | 4.0           |
| a_Traf_Year              | Year of projected AADT data                                                                                    | 2009.0        |

## 9. Budgets

**Workflow:** No steps were customised in UNRA's dTIMS WF. (This chapter acquires the budget amounts to be analysed from the road agency's sources; the budget amounts will be inputted in dTIMS in chapter 11).

Use level 7 (Username = **lcca**, Password = **lcca**) security for all dTIMS operations in this chapter.

The operations in this chapter are necessary before an analysis can be performed in dTIMS. However, for both RED and HDM-4 network or sub-network analyses, the level of funding to be investigated (e.g. current budget) should be obtained at this stage.

Various budgets may be analyzed in the LCCA tools. Steps to follow:

- Determine the annual funding level available for maintenance, rehabilitation and upgrading projects over the next 3 to 5 years. It is important to note that the funding levels (budgets) analyzed in the LCCA's refers only to that portion of UNRA's total budget earmarked for the intervention activities (treatments) defined in the relevant systems. The activities are:
  - Crack seal and pothole patching.
  - Reseal, normal and light.
  - Rehabilitation, light (such as overlays) and heavy.
  - Regravelling.
  - Upgrade unsealed roads to surfaced standards.
- Any life cycle cost analysis runs over a period of 10 to 20 years. The expected future budgets are therefore projected from the given budgets of the first three to five years.
- Tabulate the funding levels to be analyzed, see example in Table 2. For the life cycle cost analyses in dTIMS, the budgets can be split between surfaced (Paved) and unsealed (Unpaved) roads if it is required to analyze the fixed allocation of budget per pavement type.
- The budgets determined here will later be entered into the LCCA systems.

### Important notes:

- Only a portion of the UNRA's budget is included in any of the life cycle cost analyses. This budget portion is relevant to the intervention activities investigated in the analysis, it is those activities that can be associated with road deterioration. Budget amounts for items such as personnel, routine maintenance, over heads, structures, new roads, etc., should be excluded from the analysis and the need for these items are not modeled in the LCCA..
- All current and future budget amounts are entered in Present Value, no allowance is made for inflation.

**Table 2 – Example of a Budget Scenario to be analyzed**

| Year                                                                              | Total budget of UNRA | Portion of budget earmarked for treatments analyzed in dTIMS, RED and HDM-4 | Portion of budget earmarked for PAVED treatments analyzed in dTIMS, RED and HDM-4 | Portion of budget earmarked for UNPAVED treatments analyzed in dTIMS, RED and HDM-4 |
|-----------------------------------------------------------------------------------|----------------------|-----------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 2010                                                                              | USD 450 million      | USD 200 million                                                             | USD150 million                                                                    | USD50 million                                                                       |
| 2011                                                                              | USD 450 million      | USD 200 million                                                             | USD150 million                                                                    | USD50 million                                                                       |
| 2012                                                                              | USD 450 million      | USD 200 million                                                             | USD150 million                                                                    | USD50 million                                                                       |
| 2013                                                                              | USD 450 million      | USD 200 million                                                             | USD150 million                                                                    | USD50 million                                                                       |
| 2014                                                                              | USD 450 million      | USD 200 million                                                             | USD150 million                                                                    | USD50 million                                                                       |
| 2015 onwards                                                                      | USD 450 million      | USD 200 million                                                             | USD150 million                                                                    | USD50 million                                                                       |
| In this example an annual amount of USD250 million will not be included the LCCA. |                      |                                                                             |                                                                                   |                                                                                     |

## 10. Preparation of Data

Use level 7 (Username = **lcca**, Password = **lcca**) security for all dTIMS operations in this chapter.

This chapter describes the necessary steps to be completed before a life cycle cost analysis (LCCA) can be executed in HDM-4, dTIMS<sup>TM</sup> CT (also referred to as dTIMS) and RED.

Figure 9 and describe the data flow and processes that have been implemented to prepare data for any analysis in either HDM-4, RED, or dTIMS. Each step of this “data preparation” process is described in this chapter, whilst subsequent chapters describe the running of each of these tools.

Appendix A contains all the items in the customized system for UNRA. Any item described in the steps below is detailed in Appendix A.

**Figure 8 – Overview: data flow from dTIMS perspective to REWD, HDM-4 and DTIMS LCCA**

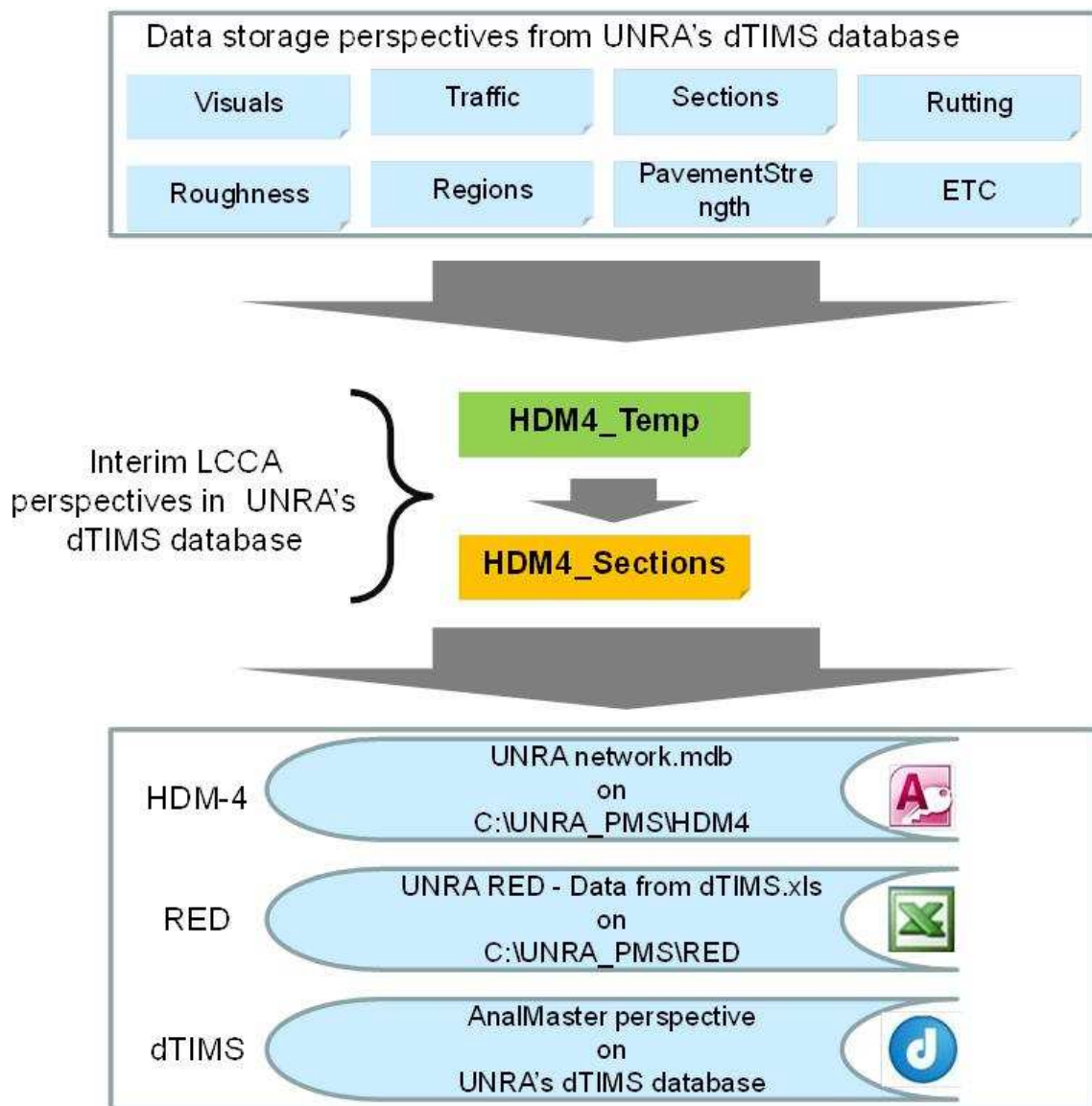
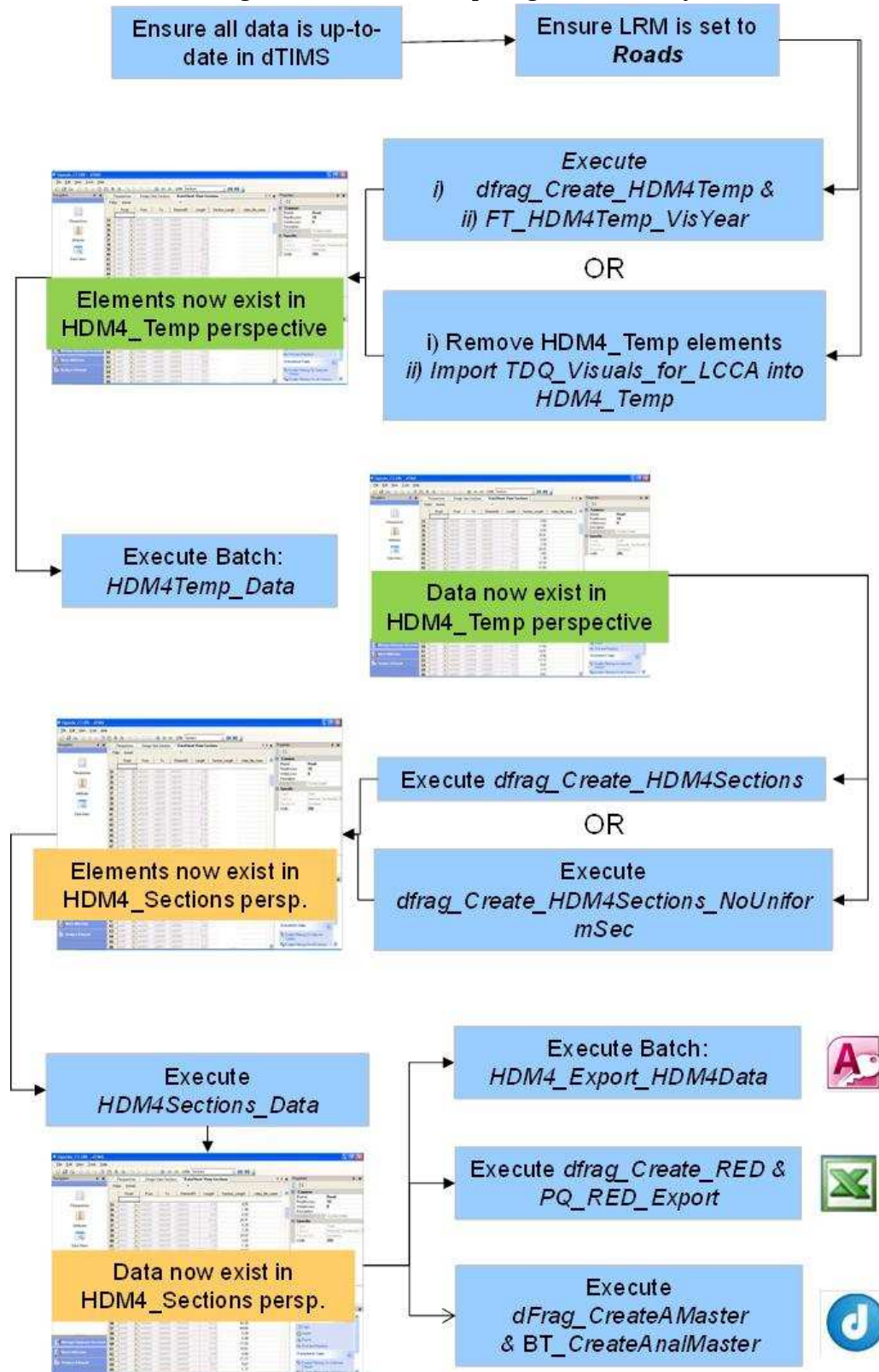




Figure 9 – Overview: Preparing Data for Analysis





The steps hereafter can be summarised as follows:

- Elements and data are created on the COMMITTED\_PROJECTS perspective, if necessary.
- Elements of the Visuals perspective are duplicated onto the HDM4\_Temp perspective. This is done by a) manually removing all elements from the HDM4\_Temp perspective and importing the elements from a Time Dependent Query on the Visuals perspective, or b) executing a dFRAG object that automatically extracts the elements of one year's data from the Visuals perspective and creating the elements on the HDM4\_Temp perspective.
- The attribute data of the data storage perspectives; Visuals, Roughness, Sections, Traffic, etc. are then transformed onto the HDM4\_Temp perspective.
- The HDM4\_Temp perspective is then used to create the HDM4\_Sections perspective containing the final set of elements. The elements in HDM4\_Sections perspective will either be a) uniform sections where “alike” adjacent sections from HDM4\_Temp perspective are combined or b) the elements will be an exact duplication of the elements of the HDM4\_Temp perspective if no uniform sectioning is required.
- The data of the HDM4\_Sections perspective is then transformed from the HDM4\_Temp perspective.
- The road elements and data of the HDM4\_Sections perspective are then used to create the perspectives and data files that will be used for life cycle cost analyses in RED, HDM4 and dTIMS.

## 10.1. Find missing data

### 10.1.1. Traffic data

**Workflow:** Included in **Step 05: Find Missing Traffic Data** of UNRA's dTIMS WF.

For the analysis of road networks, traffic data is elemental in determining the future performance of roads and it is also used during the optimization process where road maintenance and rehabilitation are selected under constrained funding scenarios. Road segments/pieces without traffic data or where MT\_AADT <= 0, cannot be analysed and should therefore be established prior to the analysis and updated with appropriate default values.

- In dTIMS, click on “Tools” in the menu bar and select “Database Reports”.
- Select the “Traffic” perspective from the list of perspectives in the drop down box on the “Select Perspective” pop up window.
- Click on the “Export” button and save the “Missing data report” for the Traffic perspective to an Excel file.
- Obtain the missing elements and data and update the Traffic perspective.
- Run the “Database Report” again for the Traffic perspective until all roads and road segments have data.

### 10.1.2. Other perspectives

**Workflow:** These can be/ were investigated in **Step 00: Verify Data** in UNRA's dTIMS WF.

The “Database Report” can be executed for any of the other perspectives, to investigate and fix missing road elements in comparison to the Roads perspective.

## 10.2. Enter the Committed Projects

**Workflow:** Included in **Step 06a: Remove Committed Projects** of UNRA's dTIMS WF and

**Step 06b: Import Committed Projects** of UNRA's dTIMS WF.

Committed projects are those works the road agency is committed to. Committed projects are optional and can be specified in the perspective: COMMITTED\_PROJECTS. All projects entered in this perspective will automatically be committed in any analysis in dTIMS. All funding scenarios that are investigated in dTIMS should have sufficient budget to cover the costs of these committed projects, otherwise the system will return an error when trying to optimise budget scenarios. Committed projects are typically one or more of the following:

- projects that are already in progress,
- projects that have a 100% certainty that they will commence within the next few (1-3) years, and
- any other projects the Road Agency are committed to and that should be included in the Maintenance, Rehabilitation and Upgrading Program generated from the LCCA.

Note:

- Committed treatments can only be entered for a LCCA in dTIMS.
- Works can only be committed if there are visual assessment data for the road or road segment. Verify that elements exist in the Visuals perspective for all works treatments to be committed for road segments. The elements should be for the year of the most recent visual assessments. ***If no assessment data is present for such a road segment, create the element/s and data on the Visuals perspective manually.***

If there are no works projects to be committed for this analysis:

- In dTIMS: Open the Data Sheet View of the COMMITTED\_PROJECTS perspective.
- Remove all elements (if there are any)
- Proceed to the next step “Generating Analyses”.

If there are committed projects:

- Verify that elements and data exist for all the projects on the Visuals perspective,
- Then enter the committed projects according to the details hereafter;
- *COM\_TRT* is the treatment name to match the treatment names in dTIMS exactly. Valid entries are:
  - HeavyRehab
  - LightRehab
  - Regravel
  - Reseal
  - Reseal\_Light
  - UpgradePave
- *COM\_Year* is the treatment year as a sequential number. E.g. enter 2 if the treatment is scheduled for the second year in the analysis period.
- *COM\_Cost* is the treatment cost. Entries are:
  - Nnnn (Numeric). E.g. 100000 to indicate a total cost of \$1000000 for the treatment.
  - -1 to indicate the formulae in the Analysis Expressions of dTIMS will be used to calculate the financial and economic cost.
  - 0 to indicate no cost. This can be used when the budget does not include the funding for committed treatments.
- *COM\_USE\_SUB*. Enter TRUE to indicate subsequent treatments should be triggered.
- *COM\_USE\_ANC*. Enter FALSE, because no ancillary treatments have been specified for UNRA.

- *COM\_ExBen* is the attribute for exogenous benefits to ensure the treatments has adequate benefits in terms of its objective function. Always enter an amount greater or equal to the total cost of the treatment, even if zero cost was entered. (This will not affect the cost of the treatment).

### 10.3. Create the elements and data of HDM4\_Temp perspective

**Workflow:** Included in **Step 07: Create Elements and Data in HDM4\_Temp perspective** of UNRA's dTIMS WF.

1. Before proceeding; all data relevant to the LCCA should have been imported into dTIMS. This includes the Road Network Definition (Roads and Sections); various Inventory Data (including Number of Lanes and Terrain Type); automated survey data (including Roughness and Rutting); Visual Condition Data; Traffic Data; and Pavement Strength and History Data. Refer to the RMS User Manual for details on UNRA's perspectives and for instructions on how to import such data into dTIMS.

Also all steps preceding this chapter should have been completed before continuing.

2. Run the dTIMS software but *do not open the database*. Instead, run the **Compact & Repair** option. Wait until the operation completes. (If you receive an "insufficient disk space" error during any of steps to follow, close the UNRA database, run the **Compact & Repair** option and restart the step that gave the error)
3. Now open the UNRA database as "lcca" user.
4. **Set the LRM to Roads.**
5. Two methods are available to duplicate the elements from the Visuals perspective for a life cycle cost analysis. Follow one of the steps in the table below.
- 6.

| <b>Option a</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Option b</b>                                                                                                                                                                                                                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Select road sections for multiple years of visual assessment data. <b>Choose this option when visual assessment data was collected over a number of years.</b> The user is prompted for a minimum visual assessment year and no data older than this year would be selected. A Time Dependant Query is then used to select elements according to the most recent visual assessment data. Visual assessment data that spreads over more than year would be extracted with this method.</p> | <p>Select road section for a single year's data. <b>Choose this option when visual assessment data was collected over a single year and the set of data in the Visuals perspective has the same value in Vis_YEAR.</b> The user is prompted for a visual assessment year and a dFRAG object is used to extract exactly one year's data</p> |
| <p>The elements are selected according to the Analysis Expression:<br/> <b>"bF_TDQ_Create_HDM4Temp_fromVisuals":</b><br/> Visuals-&gt;Vis_YEAR &gt;=<br/> PROMPT_NUMBER('MINIMUM YEAR OF VISUALS: ') And NOT IFDEFAULT(Visuals-&gt;ROAD_TYPE) and Visuals-&gt;ROAD_TYPE &gt; "<br/> And Traffic-&gt;traffic_year &gt; 0.0 and NOT IFDEFAULT(Traffic-&gt;traffic_year) and</p>                                                                                                                | <p>The elements are selected according to the Analysis Expression:<br/> <b>"bF_dFRAG_Create_HDM4Temp":</b><br/> Visuals-&gt;Vis_YEAR =<br/> PROMPT_NUMBER('Year of visuals: ')<br/> And NOT IFDEFAULT(Visuals-&gt;ROAD_TYPE) and Visuals-&gt;ROAD_TYPE &gt; " And Traffic-</p>                                                             |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>IF(Visuals-&gt;ROAD_TYPE = 'SD' or Visuals-&gt;ROAD_TYPE = 'AC',NOT IFDEFAULT(Visuals-&gt;CO_ALLCRCKS) ,NOT IFDEFAULT(Visuals-&gt;C_GRAVEL_THICK_UNP) )</p> <ul style="list-style-type: none"> <li>• Open the Data Sheet View of the HDM4_Temp perspective and delete all elements (Edit / Remove all elements).</li> <li>• Close the Data Sheet View when done.</li> <li>• In dTIMS/Work With Data / Time Dependent Query: Execute <b>TDQ_Visuals for_LCCA</b>. Enter the minimum year of visuals to include in the analysis, e.g. 2009. This step creates the file TDQ_Visuals.mdb on c:\UNRA_PMS\Exports.</li> <li>• In Access: Open the file that was just created in dTIMS by the TDQ. Sort the data on the ElementID and verify that the elements are valid. The TDQField is a duplication of the Vis-Year attribute storing the year for which the element was selected from the Visuals perspective. Usually no action is required here. Close and exit Access.</li> <li>• In Manage Database Structure /Perspective: Select (click on) HDM4_Temp perspective and select Data Sheet View. There should be zero elements. Import now the <b>Elements</b> and <b>Data</b> from the Access file that was created by the Time Dependent Query in the previous step. DO NOT import New Attributes!</li> <li>• Close and open the Data Sheet View of the HDM4_Temp perspective to refresh the data. The elements on HDM4_Temp should be identical to the elements of the Access file that was created by the Time Dependent Query.</li> <li>• The data could also be exported from the open Data Sheet View to check the elements. The only attribute to verify is Vis_YEAR that should have values not less than the minimum year entered at the beginning of this step.</li> <li>• This step creates the road elements only, the data attributes are completed during the next step. Close the Data View when done.</li> </ul> | <p>&gt;traffic_year &gt; 0.0 and NOT IFDEFAULT(Traffic-&gt;traffic_year) and IF(Visuals-&gt;ROAD_TYPE = 'SD' or Visuals-&gt;ROAD_TYPE = 'AC',NOT IFDEFAULT(Visuals-&gt;CO_ALLCRCKS) ,NOT IFDEFAULT(Visuals-&gt;C_GRAVEL_THICK_UNP) )</p> <ul style="list-style-type: none"> <li>• In dTIMS/Analyse &amp; Report/dFRAG Objects: Execute <b>dfrag_Create_HDM4Temp</b>. Enter the year of the most recent visuals in dTIMS, e.g. 2009. You will be prompted to enter this value more than once! This creates the elements of the HDM4_Temp perspective. This step could take a few hours (4 hours or more) to complete.</li> <li>• In dTIMS/Work With Data/ Formula Transformations: Execute <b>FT_HDM4Temp_VisYear</b>. Enter the year of the most recent visuals in dTIMS, e.g. 2009. Make sure to enter the same YEAR as entered during the previous step.</li> <li>• The Data Sheet View of HDM4_Temp could also be exported to check the elements. All elements should be verified and also verify whether the elements of the COMMITTED_PROJECTS perspective are included in the HDM4_Temp perspective. The only attribute to verify is Vis_YEAR that should have the same value as the year that was entered by the user.</li> <li>• This step creates the road elements only, the data attributes are completed during the next step. Close the Data View when done.</li> </ul> |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

7. In dTIMS / Work With Data / Batch Operations: Execute **HDM4Temp\_Data**. This step transforms data from the Visuals, Traffic, Sections and other perspectives to the elements of the HDM4\_Temp perspective. The file, **Road Network Export File Format.pdf**, under the documentation of the HDM4 software describes each data attribute (field) in the HDM4\_Temp perspective.

The user can inspect all the Perspective and Formula Transformations in the Batch Operation **HDM4Temp\_Data** to familiar him/herself with:

- the source of each data attribute (Perspective Transformations),
- values assumed if no data is available (by means of Formula Transformations and default values listed as “a\_” in Database Expressions),
- conversion calculations (Formula Transformations) that are done to make the data compatible with the input requirements of HDM-4.

In Manage Database Structure /Perspective: Select (click on) HDM4\_Temp perspective and select Data Sheet View. Check the data in the HDM4\_Temp perspective to ensure all data is valid. No “-1” values are allowed in any of the numeric fields. The data view can either be exported to be checked in Access and Excel, or check the data in the data view by scrolling and using the filters. Any obvious inaccuracies with regards to the elements and/or data should be fixed in the source perspectives, thereafter the previous steps of this chapter should be completed again. The process to validate the elements and data in HDM4\_Temp perspective will continue in the next paragraph.

Close the Data View when done.

The elements on Perspective HDM4\_Temp have now been created.

## 10.4. Validate HDM4\_Temp elements

**Workflow:** Included in **Step 08: Validate HDM4\_Temp Elements and Data** of UNRA’s dTIMS WF.

The purpose of this query is to list all road segments, obtained from the HDM4\_Temp perspective against the Road perspective. The user can then inspect the “road segments without visual assessment data” as these are the road segments with no elements on the HDM4\_Temp perspective that will NOT be analysed. The road segments (elements) of the HDM4\_Temp perspective are obtained from the Visuals perspective. To fix any outstanding or incorrect elements, modify the Visuals perspective and repeat all previous steps in this chapter.

8. In dTIMS / Work With Data / SCD Queries: Execute **SCDQ\_LCCACheck\_RoadSegments**. This query creates the file; **C:\UNRA\_PMS\Exports\SCDQ\_LCCA\_CHECK\_ROADSEGMENTS.xls**.
9. Open the Excel file created in the previous step.
10. In some cases dTIMS produces elements (rows) with zero length; FROM and TO columns have the same value. These rows can be deleted to be excluded from the verification process.
11. The output of this query is investigated for road segments that have no visual assessment data. These are the roads that have no information in the attributes Vis\_YEAR, COND\_YEAR, CRACKS\_ACA\_Temp, etc. See example of yellow shaded elements below. These road segments will subsequently not be included in any life cycle cost analysis, except if elements and

data for these road segments are entered in the VISUALS perspective and all previous steps in this chapter are repeated.

**Figure 10 – Verify road segments on UNRA’s database with no visual assessment data**

| Road | From   | To     | Length | road_class | MT_AADT | traffic_year | ROAD_TYPE | Vis_YEAR | CO_ALLCRACKS | CO_WIDECRACKS |
|------|--------|--------|--------|------------|---------|--------------|-----------|----------|--------------|---------------|
| A001 | 0.000  | 1.000  | 1      | A          | 11956   | 2009         | AC        | 2009     | 4            | 5             |
| A001 | 1.000  | 2.000  | 1      | A          | 11956   | 2009         | AC        | 2009     | 4            | 5             |
| A001 | 2.000  | 2.338  | 0.338  | A          | 11956   | 2009         | AC        | 2009     | 4            | 5             |
| A001 | 2.338  | 3.338  | 1      | A          | 8641    | 2009         | AC        | 2011     | 4            | 5             |
| A001 | 3.338  | 4.308  | 0.97   | A          | 8641    | 2009         | AC        | 2011     | 4            | 5             |
| A001 | 4.308  | 4.951  | 0.643  | A          | 7837    | 2009         | AC        | 2011     | 4            | 5             |
| A001 | 4.951  | 5.000  | 0.049  | A          | 17630   | 2009         | AC        | 2011     | 4            | 5             |
| A001 | 5.000  | 5.081  | 0.081  | A          | 17630   | 2009         |           |          |              |               |
| A001 | 5.081  | 5.206  | 0.125  | A          | 5582    | 2009         | AC        | 2009     | 4            | 5             |
| A001 | 5.206  | 6.206  | 1      | A          | 4281    | 2009         | AC        | 2009     | 3            | 4             |
| A001 | 6.206  | 6.883  | 0.677  | A          | 4281    | 2009         | AC        | 2009     | 3            | 4             |
| A001 | 6.883  | 7.883  | 1      | A          | 8685    | 2009         | AC        | 2009     | 3            | 4             |
| A001 | 7.883  | 8.883  | 1      | A          | 8685    | 2009         | AC        | 2009     | 3            | 4             |
| A001 | 8.883  | 9.391  | 0.508  | A          | 8685    | 2009         | AC        | 2009     | 3            | 4             |
| A001 | 9.391  | 10.000 | 0.609  | A          | 7624    | 2009         |           |          |              |               |
| A001 | 10.000 | 17.737 | 7.737  | A          | 7624    | 2009         | AC        | 2011     | 3            | 4             |
| A001 | 17.737 | 19.146 | 1.409  | A          | 5111    | 2009         | AC        | 2011     | 3            | 4             |
| A001 | 19.146 | 20.000 | 0.854  | A          | 4045    | 2009         | AC        | 2011     | 3            | 4             |
| A001 | 20.000 | 20.146 | 0.146  | A          | 4045    | 2009         |           |          |              |               |
| A001 | 20.146 | 21.146 | 1      | A          | 4045    | 2009         | AC        | 2009     | 4            | 5             |
| A001 | 21.146 | 22.146 | 1      | A          | 4045    | 2009         | AC        | 2009     | 4            | 5             |
| A001 | 22.146 | 23.146 | 1      | A          | 4045    | 2009         | AC        | 2009     | 4            | 5             |
| A001 | 23.146 | 24.146 | 1      | A          | 4045    | 2009         | AC        | 2009     | 4            | 5             |
| A001 | 24.146 | 25.146 | 1      | A          | 4045    | 2009         | AC        | 2009     | 5            | 5             |
| A001 | 25.146 | 26.146 | 1      | A          | 4045    | 2009         | AC        | 2009     | 4            | 5             |
| A001 | 26.146 | 27.146 | 1      | A          | 4045    | 2009         | AC        | 2009     | 3            | 4             |
| A001 | 27.146 | 28.146 | 1      | A          | 4045    | 2009         | AC        | 2009     | 4            | 5             |
| A001 | 28.146 | 29.146 | 1      | A          | 4045    | 2009         | AC        | 2009     | 3            | 5             |
| A001 | 29.146 | 30.146 | 1      | A          | 4045    | 2009         | AC        | 2009     | 3            | 4             |
| A001 | 30.146 | 31.146 | 1      | A          | 4045    | 2009         | AC        | 2009     | 3            | 4             |
| A001 | 31.146 | 32.146 | 1      | A          | 4045    | 2009         | AC        | 2009     | 3            | 4             |
| A001 | 32.146 | 33.146 | 1      | A          | 4045    | 2009         | AC        | 2009     | 3            | 4             |
| A001 | 33.146 | 34.146 | 1      | A          | 4045    | 2009         | AC        | 2009     | 3            | 4             |
| A001 | 34.146 | 35.146 | 1      | A          | 4045    | 2009         | AC        | 2009     | 3            | 4             |
| A001 | 35.146 | 36.146 | 1      | A          | 4045    | 2009         | AC        | 2009     | 3            | 4             |
| A001 | 36.146 | 37.146 | 1      | A          | 4045    | 2009         | AC        | 2009     | 3            | 4             |
| A001 | 37.146 | 38.146 | 1      | A          | 4045    | 2009         | AC        | 2009     | 3            | 4             |
| A001 | 38.146 | 38.446 | 0.3    | A          | 4045    | 2009         |           |          |              |               |
| A001 | 38.446 | 39.146 | 0.7    | A          | 4045    | 2009         | AC        | 2009     | 3            | 4             |
| A001 | 39.146 | 40.146 | 1      | A          | 4045    | 2009         | AC        | 2009     | 3            | 4             |
| A001 | 40.146 | 41.146 | 1      | A          | 4045    | 2009         | AC        | 2009     | 2            | 3             |

12. The output of this query is also investigated for road segments that have no traffic data. These are the roads that have no information in the attributes Traffic\_Year and MT\_AADT. See example of blue shaded elements below. These road segments will subsequently not be included in any life cycle cost analysis, except if elements and data for these road segments are entered in the TRAFFIC perspective and all previous steps in this chapter are repeated.

**Figure 11 – Verify road segments on UNRA’s database with no traffic data**

| Road | From   | To     | Length | road_class | MT_AADT | traffic_year | ROAD_TYPE | Vis_YEAR |
|------|--------|--------|--------|------------|---------|--------------|-----------|----------|
| C961 | 17.000 | 18.000 | 1      | C          | 111     | 2009         | G         | 2009     |
| C961 | 18.000 | 19.000 | 1      | C          | 111     | 2009         | G         | 2009     |
| C961 | 19.000 | 20.000 | 1      | C          | 111     | 2009         | G         | 2009     |
| C961 | 20.000 | 21.000 | 1      | C          | 111     | 2009         | G         | 2009     |
| C961 | 21.000 | 22.000 | 1      | C          | 111     | 2009         | G         | 2009     |
| C961 | 22.000 | 23.000 | 1      | C          | 111     | 2009         | G         | 2009     |
| C961 | 23.000 | 24.000 | 1      | C          | 111     | 2009         | G         | 2009     |
| C961 | 24.000 | 25.000 | 1      | C          | 111     | 2009         | G         | 2009     |
| C961 | 25.000 | 26.000 | 1      | C          | 111     | 2009         | G         | 2009     |
| C961 | 26.000 | 27.000 | 1      | C          | 111     | 2009         | G         | 2009     |
| C961 | 27.000 | 28.000 | 1      | C          | 111     | 2009         | G         | 2009     |
| C961 | 28.000 | 29.000 | 1      | C          | 111     | 2009         | G         | 2009     |
| C961 | 29.000 | 30.000 | 1      | C          | 111     | 2009         | G         | 2009     |
| C961 | 30.000 | 30.731 | 0.731  | C          | 111     | 2009         | G         | 2009     |
| C042 | 0.000  | 0.860  | 0.86   | C          |         |              |           |          |
| C043 | 0.000  | 5.411  | 5.411  | C          |         |              |           |          |
| C044 | 0.000  | 6.952  | 6.952  | C          |         |              |           |          |
| C045 | 0.000  | 38.552 | 38.552 | C          |         |              |           |          |
| C046 | 0.000  | 22.975 | 22.975 | C          |         |              |           |          |
| C150 | 0.000  | 11.119 | 11.119 | C          |         |              |           |          |
| C151 | 0.000  | 0.560  | 0.56   | C          |         |              |           |          |
| C152 | 0.000  | 14.338 | 14.338 | C          |         |              |           |          |
| C176 | 0.000  | 4.048  | 4.048  | C          |         |              |           |          |
| C177 | 0.000  | 4.048  | 4.048  | C          |         |              |           |          |
| C178 | 0.000  | 28.130 | 28.13  | C          |         |              |           |          |
| C179 | 0.000  | 1.598  | 1.598  | C          |         |              |           |          |
| C180 | 0.000  | 35.430 | 35.43  | C          |         |              |           |          |
| C181 | 0.000  | 1.989  | 1.989  | C          |         |              |           |          |
| C182 | 0.000  | 8.875  | 8.875  | C          |         |              |           |          |
| C183 | 0.000  | 0.785  | 0.785  | C          |         |              |           |          |
| C191 | 0.000  | 11.124 | 11.124 | C          |         |              |           |          |
| C193 | 0.000  | 9.919  | 9.919  | C          |         |              |           |          |
| C194 | 0.000  | 17.943 | 17.943 | C          |         |              |           |          |
| C195 | 0.000  | 9.452  | 9.452  | C          |         |              |           |          |
| C215 | 0.000  | 48.368 | 48.368 | C          |         |              |           |          |
| C216 | 0.000  | 25.963 | 25.963 | C          |         |              |           |          |
| C217 | 0.000  | 35.583 | 35.583 | C          |         |              |           |          |
| C218 | 0.000  | 31.956 | 31.956 | C          |         |              |           |          |
| C924 | 59.000 | 60.000 | 1      | C          | 27      | 2009         | G         | 2009     |

## 10.5. Validate HDM4\_Temp data

**Workflow:** Included in **Step 08: Validate HDM4\_Temp Elements and Data** of UNRA’s dTIMS WF.

The purpose of this query is to list the road segments of the HDM4\_Temp perspective, with all the data attributes that were obtained from the Visuals, Traffic and other perspectives. The user can then inspect the data attributes and verify the data that will be analysed. To fix any data, modify the data in the Visuals perspective and repeat all previous steps in this chapter.

13. In dTIMS / Work With Data / Perspective Queries: Execute **PQ\_LCCACheck\_Data**. This query creates the file; C:\UNRA\_PMS\Exports\ PQ\_LCCA\_CHECK\_DATA.XLS.
14. The output of this query should be investigated to ensure all data attributes have valid data.
  - All road segments (elements) without data in the Traffic\_Year or MT\_AADT columns will NOT be analysed. Fix the data in the Traffic perspective and re-run all the previous steps in this chapter.
  - Road\_Type= “SD” or “AC”: All attributes except GRAV\_THICK, should have valid data.

- Road\_Type = “G” or ”E”: The attributes Vis\_YEAR, GRAV\_THICKUNP and ROUGHNESS should have valid data.
  - Verify the data against the guidelines in the RMS User Manual. If any of the attributes listed in the query have no data or invalid data, the life cycle cost analysis will receive incorrect information.
  - To fix any missing data, create the elements and/or data on the relevant perspective and re-run all the previous steps in this chapter.
15. Execute the Data Sheet View of HDM4\_Temp perspective. Validate the data in all the attributes. The data can be inspected in dTIMS or it can be exported to Excel or Access. Make sure the source perspectives, e.g. traffic, visuals, etc. have valid values, if necessary fix the data and re-run all the previous steps in this chapter.
  16. To inspect the transformations from the sources perspectives to HDM4\_Temp: Open the batch operation HDM4Temp\_Data and validate each object that is included in the process to transfer data to the HDM4\_Temp perspective.
  17. DO NOT PROCEED IF THE ELEMENTS AND DATA OF HDM4\_TEMP ARE NOT 100% CORRECT.

The elements and data of perspective HDM4\_temp have now been validated.

## 10.6. Create the elements and data of HDM4\_Sections perspective

**Workflow:** Included in **Step 09: Create Elements and Data in HDM4\_Sections perspective** of UNRA's dTIMS WF.

18. Close the UNRA dTIMS database *but do not exit dTIMS*. Run the **Compact & Repair** option again. Wait until the operation completes. Open the database for UNRA again as “lcca” user.
19. Make sure the location referencing method (LRM) in dTIMS is still set to *Roads*. In Analyze&Report / dFRAG Objects select an appropriate dFRAG object according to the two options below. : Execute either
  - a) **dFrag\_Create\_HDM4Sections** or
  - b) **dFRAG\_Create\_HDM4Sections\_NoUniformSec**
 These steps create the elements of the HDM4\_Sections perspective.

| Option a                                                                                                                                                                                                                                                                     | Option b                                                                                                                                                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Combines the elements of the HDM4_Temp perspective (with elements identical to the Visual elements) into uniform sections, based on the condition and characteristics of the pavements. It reduces the number of records to analyse and compiles practical project elements. | Does not involve any combination of sections and the road elements from the visual survey are used for the LCCA. The typical section length analyzed is therefore 1km. |
| In Analyze&Report / dFRAG Objects:<br>Execute <b>dFrag_Create_HDM4Sections</b>                                                                                                                                                                                               | In Analyze&Report / dFRAG Objects: Execute <b>dFRAG_Create_HDM4Sections_NoUniformSec</b>                                                                               |

20. In Manage Database Structure /Perspective: Select (click on) HDM4\_Sections perspective and select Data Sheet View. For option a) elements should typically be longer than 1 km and the number of elements should be less than number of elements in the **HDM4\_Temp** perspective. For option b) the number of elements should match the number of elements in the **HDM4\_Temp** perspective. The data should also be exported from the open Data Sheet View to validate the



elements. This step creates the road elements only, the data attributes are filled during the next step. Close the Data View when done.

21. Close the UNRA dTIMS database **but do not exit dTIMS**. Run the **Compact & Repair** option again. Wait until the operation completes. Open the database for UNRA again as “lcca” user..

Ensure that the location referencing method (LRM) in dTIMS is still set to Roads.

In Work With Data / Batch Operations: Execute **HDM4Sections\_Data**. This transforms data from the **HDM4\_Temp** perspective to the elements of the **HDM4\_Sections** perspective. The HDM4\_Sections perspective now has all the data of the network as required by the HDM-4 software. The format of this perspective is exactly the same as the input required by HDM-4. The file, **Road Network Export File Format.pdf**, under the documentation of the HDM4 software describes each data attribute (field) in the HDM4\_Sections perspective. This perspective’s data is also transformed to the analysis master (AnalMaster) in dTIMS in the steps (following hereafter) for use in the dTIMS LCCA.

The user can inspect all the Perspective and Formula Transformations in the Batch Operation **HDM4Sections\_Data** to familiar him/herself with:

- the source of each data attribute (Perspective Transformations),
- values assumed if no data is available (by means of Formula Transformations and default values listed as “a\_” in Database Expressions),
- conversion calculations (Formula Transformations) that are done to make the data compatible with the input requirements of HDM-4.

In Manage Database Structure /Perspective: Select (click on) HDM4\_Sections perspective and select Data Sheet View. Check the data in the HDM4\_Sections perspective to ensure all data is valid. The data view can either be exported to be checked in Access and Excel, or check the data in the data view by scrolling and using the filters. Basic checks should include the total length of network, the validity of data (e.g. make sure there are no blank values, and/or that default data has been used where appropriate). Note that all default data is described in chapter 8 of this manual. Close the Data View when done.

The elements on Perspective HDM4\_Sections have now been created and validated.

## 10.7. Export the analysis file for HDM4

**Workflow:** Included in **Step 10: Export File for Strategic/Programme Analyses in HDM4** of UNRA’s dTIMS WF.

22. Close the UNRA dTIMS database **but do not exit dTIMS**. Run the **Compact & Repair** option again. Wait until the operation completes. Open the database for UNRA again as “lcca” user.

Ensure the location referencing method (LRM) in dTIMS is still set to Roads.

In Work With Data / Batch Operations: Execute **HDM4\_Export\_HDM4DATA**. This overwrites the file **UNRA\_Network.mdb** in the directory C:\UNRA\_PMS\HDM4. A copy of this file is stored on directory **C:\UNRA\_PMS\HDM4\UNRA\_Masterfiles** as a backup. Never overwrite any files on the UNRA Masterfiles directory, make a copy if necessary.

The analysis file for the HDM4 software has now been exported from dTIMS!

Chapter 11.3 has further instructions to manipulate this file for import into HDM-4.

## 10.8. Create and export the analysis file for RED

**Workflow:** Included in **Step 11: Create and Export File for RED Analyses** of UNRA's dTIMS WF.

23. Ensure the location referencing method (LRM) in dTIMS is still set to *Roads*.

In Analyze&Report / dFRAG Objects: Execute **dFrag\_Create\_RED**. These steps create the elements of the RED perspective. Elements are combined where the surface types of adjoining elements are the same, and the link id's of the adjoining elements are the same.

In Manage Database Structure /Perspective: Select (click on) **RED** perspective and select Data Sheet View. The elements in the RED perspective should correspond to the HDM4 elements, with combined elements where the surface type is unchanged on adjacent sections. The data should also be exported from the open Data Sheet View to check the elements. This step creates the road elements which will be used as the master perspective during the next step when data for the RED software is exported. Close the Data View when done.

24. In **Explorer**: Open the file **UNRA RED – Data from dTIMS.xls** in the directory C:\UNRA\_PMS\RED. Go to the sheet **From\_dTIMS1** and clear this sheet of all data. This sheet (**From\_dTIMS1**) will be populated with data during the step hereafter. Leave all other sheets as is.
25. In **dTIMS**: In Work With Data / Perspective Queries: Execute **PQ\_RED\_Export**. This updates the file **UNRA RED – Data from dTIMS.xls**<sup>2</sup> in C:\UNRA\_PMS\RED. Select **Yes** to overwrite the existing table. A copy of this file is stored on directory **C:\UNRA\_PMS\RED\UNRA MasterFiles** as a backup. Never overwrite any files on the UNRA Masterfiles directory, always use copy and paste to create a new working file.

The analysis file for the RED software has now been exported from dTIMS!

This file will be manipulated further in Chapter 13 for input into the RED software. This step overwrites the cells of sheet **From\_dTIMS1** in the file **UNRA RED – Data from dTIMS.xls** in the directory C:\UNRA\_PMS\RED.

## 10.9. Create the analysis perspective for an analysis in DTIMS

**Workflow:** Included in **Step 12: Create AnalMaster perspective for dTIMS Analyses** of UNRA's dTIMS WF.

26. Close the UNRA dTIMS database *but do not exit dTIMS*. Run the **Compact & Repair** option again. Wait until the operation completes. Open the database for UNRA again as "lcca" user.
27. Make sure the location referencing method (LRM) in dTIMS is still set to *Roads*.
28. In Analyze&Report / dFRAG Objects: Execute **dFrag\_Create\_AMaster**. This creates the elements on the AnalMaster perspective. Elements are identical to the elements of the HDM4\_Sections perspective. This step prepares the data for a life cycle cost analysis in dTIMS.

<sup>2</sup> This file has been customized by Aurecon to be used a life cycle cost analysis in the RED software. The output of the perspective query should always be stored in this file to enable the user to make the necessary manipulations necessary for a RED analysis.

29. Close the UNRA dTIMS database *but do not exit dTIMS*. Run the **Compact & Repair** option again. Wait until the operation completes. Open the database for UNRA again as “lcca” user.
30. In Work With Data / Batch Operations: Execute **BT\_Ceate\_AnalMaster**. This transforms data from the HDM4\_Sections perspective to the elements of the AnalMaster perspective. The AnalMaster perspective has now all the data of the network as required by dTIMS for a LCCA.

The user can inspect all the Perspective and Formula Transformations in the Batch Operation **BT\_Ceate\_AnalMaster** to familiar him/herself with:

- the source of each data attribute (Perspective Transformations),
- values assumed if no data is available (by means of Formula Transformations and default values listed as “a\_” in Database Expressions),
- conversion calculations (Formula Transformations) that are done to make the data compatible with the input requirements of HDM-4.

In Manage Database Structure /Perspective: Select (click on) AnalMaster perspective and select Data Sheet View. Check the data in the AnalMaster perspective to ensure all data is valid. The data view can either be exported and checked in Access and Excel, or check the data in the Data Sheet View by scrolling and using the filters. Basic checks should include the total length of network, the validity of data (e.g. make sure there are no blank values, and/or that default data has been used where appropriate). Note that all default data is described in chapter 8 of this manual. Note that only elements with a TRUE value in the AllFieldsValid attribute will be analysed. Refer to Chapter 11 for running the LCCA. Close the Data View when done.

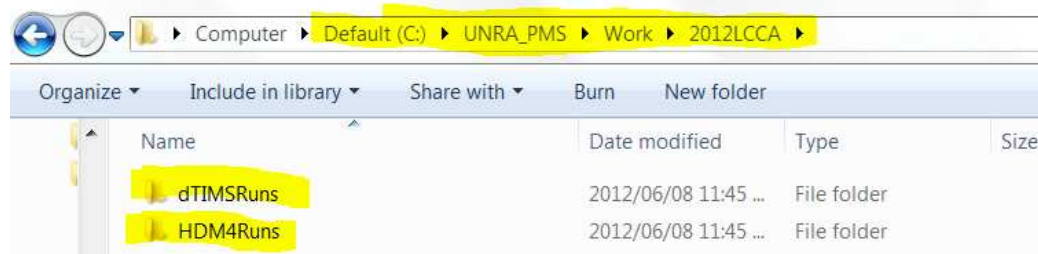
The elements for the analysis perspective in dTIMS have now been created.

31. Close the UNRA dTIMS database and exit dTIMS.

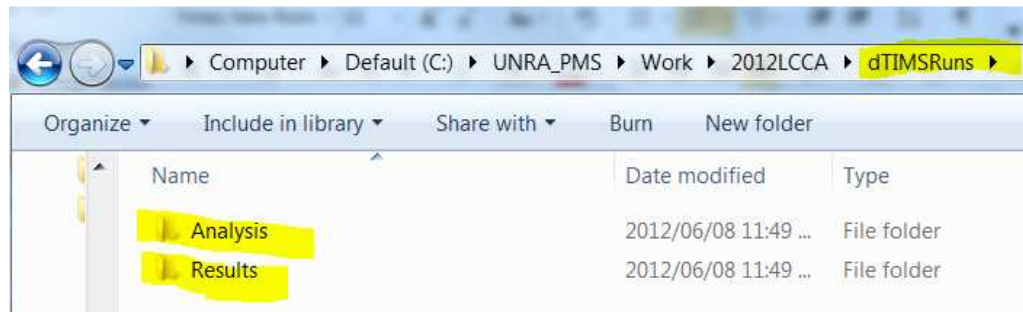
## 10.10. Create a folder structure to store the analysis results

**Workflow:** No steps were customised in UNRA’s dTIMS WF.

32. At this stage it is recommended that a folder structure is created on the directory C:\UNRA\_PMS\WORK to store the results and analysis files. The user can create a sub-folder for each analysis version that is run to make sure analysis results are not accidentally overwritten. E.g. C:\UNRA\_PMS\WORK\2012LCCA can be created with subfolders dTIMSRuns and HDM4Runs if both HDM4 and dTIMS analyses are to be complete in 2012. Use sub-folders within sub-folder is multiple versions are run.



Separate folders can then be created for each system e.g. Analysis and Results.



33. Proceed now to one of the analyses methods; namely RED, or dTIMS or HDM-4.

## 11. Running a LCCA in dTIMS software

A network level life cycle cost analysis has been customized in dTIMS based on the pavement and road user performance models of HDM-4. Several budget scenarios can be added to each analysis to investigate the consequences of future funding scenarios.

Chapter 17 has tables containing the customized lcca system for UNRA. The items in the Analysis & Reports object can also be investigated to obtain an understanding of the lcca in dTIMS.

Complete all LCCA steps with **LRM** set to **Roads**.

### 11.1. Generating Analyses

**Workflow:** Included in **Step 13:Generate strategies in dTIMS** and **Step 14:Optimise Budgets in dTIMS** of UNRA's dTIMS WF.

1. Run the dTIMS software but *do not open the database*. Instead, run the **Compact & Repair** option. Wait until the operation completes, then open the UNRA dTIMS database as "lcca" user.
2. During the steps in the previous chapter the elements and data of the AnalMaster perspective were created. The AnalMaster perspective defines the road network that will be analysed during the LCCA in dTIMS. In Manage Database Structure /Perspective: Select (click on) AnalMaster perspective and select Data Sheet View. All elements with a TRUE value in the AllFieldsValid attribute will be analysed. Repeat the steps of the previous chapter if the data is not valid in this perspective. Close the Data View when done.
3. In Analyse & Report/Analysis Variables check whether all the cost expressions starting with **a\_Cost** and **a\_Asset** are valid. These values have already been checked in chapter 6, but the unit costs can still be changed here prior to the LCCA in dTIMS. All Asset Value calculations and predictions will be based on the unit costs of the expressions:
  - a\_AssetValue\_Pav\_ReconstructCost (enter unit cost per square meters),
  - a\_AssetValue\_Unp\_GravelCost (enter unit cost per kilometer) and
  - a\_AssetValue\_Unp\_RegravelCost (enter unit cost per kilometer).

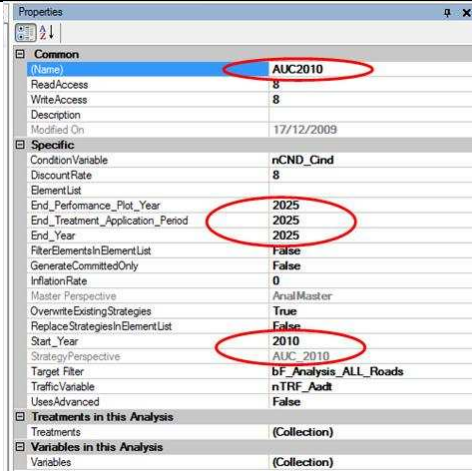
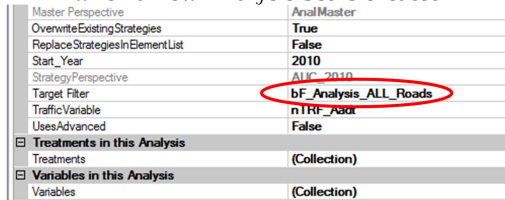
In Work with Data/ Cross Tab Transformations; click on CT\_VOC and open the Values tab in the properties window. The VOC constants should have values for all rows and columns; except for row = -1 and column = -1 where the zero-value is used. These constants have already been populated in chapter 5, but the VOC constants can still be changed here prior to the LCCA in dTIMS.

4. This step is to either duplicate one of the master analysis sets to create a new analysis set or to adjust the analysis set UNRA\_LCCA.

**Workflow:** The analysis set **UNRA\_LCCA** was customised for the Workflow of UNRA's dTIM WF. However any other analyses, created in dTIMS CT can be analysed and optimised in dTIMS WF (following the steps hereafter).

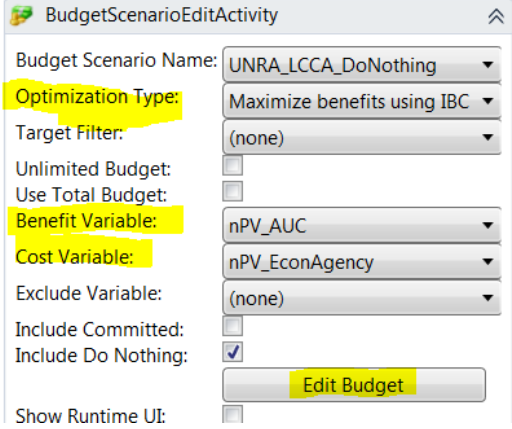
**Management Dashboard:** UNRA's dTIMS MD is used to navigate to any LCCA that was already analysed and optimised in dTIMS (or using dTIMS WF). These steps hereafter should have completed successfully before the management dashboard can be used.

| Edit/Create Analysis Set                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Option a (using dTIMS WF)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Option b (using dTIMS)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <i>Workflow: Included in Step 13: Generate strategies in dTIMS of UNRA's dTIMS WF.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Use one of the two master analysis sets to create new analysis set and complete the LCCA process with dTIMS WF.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <p>The WF will guide you to analysis set <b>UNRA_LCCA</b> under Go to Analyze &amp; Report/Analysis Sets.</p> <p>Make changes the filter, start year, end years and discount rate if necessary.</p> <p>Make sure the start year is at least equal to, or greater than the year of the most recent visual assessment data.</p> <p>The analysis set; UNRA_LCCA was customised for the objective function area-under-the-condition-curve (AUC). This benefit methodology has the objective to maximise road condition, thereby preserving the road network as an asset for the road agency.</p> | <p>Go to Analyze &amp; Report/Analysis Sets.</p> <p>Decide which of the two master analysis sets will be used; Master_AUC or Master_TTC. The two master sets are identical except for the benefit functions that are used to selected strategies. The Master_AUC analysis has the objective to maximise road condition. The Master_TTC has the objective to minimise the total transport cost. For the TTC objective, lower trafficked roads could result in no intervention strategy being selected because the road user cost savings are too low in comparison to the cost of the intervention strategy. Click on one of the analysis sets; Master_AUC or Master_TTC. Use the COPY and PASTE features to create a new analysis set.</p> |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Change the security to "Level 7" for the ReadAccess, WriteAccess and ExecuteWith.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <p>The only properties mandatory to change are; the Name, the three end-year properties, Start_Year and StrategyPerspective. Use the same descriptive name for the Name and StrategyPerspective. E.g. if an analysis is required in 2010, using the AUC benefit, it is recommended that the name AUC2010 is used for both the Name and StrategyPerspective. Make sure difference between the StartYear and the three End years are equal to your analysis period (10 or 15 year analysis is recommended). See Figure 12 for mandatory items to change in the properties window.</p>                                                                                                                                                        |

| Edit/Create Analysis Set                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Option a (using dTIMS WF)                                                                                   | Option b (using dTIMS)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Workflow: Included in Step 13: Generate strategies in dTIMS of UNRA's dTIMS WF.                             | Use one of the two master analysis sets to create new analysis set and complete the LCCA process with dTIMS WF.                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                                                                                             |  <p><b>Figure 12 – Mandatory property items to change when a new Analysis Set is created</b></p>                                                                                                                                                                                                                                                                                                                                                            |
|                                                                                                             | <p>Change the target filter if necessary: The three options are bF_Analysis_ALL_Roads, bF_Analysis_Flexible_Roads and bF_Analysis_Unsealed_Roads. The three expressions select either all roads, or surfaced flexible roads only, or unsealed roads only respectively. The two Master Analysis Sets were customised to run an analysis for all roads.</p> <p><b>Figure 13 – Optional property items to change when a new Analysis Set is created</b></p>  |
| Save the analysis set you just created.                                                                     | Save the analysis set you just created.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <i>dTIMS WF will automatically generate strategies, in other words EXECUTE the analysis set; UNRA_LCCA.</i> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

5. This step will create/change budget scenarios to optimise for the analysis set that was created during the previous step.

| Edit/Create Budget Scenarios |                        |
|------------------------------|------------------------|
| Option a (using dTIMS WF)    | Option b (using dTIMS) |

|                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><i>Workflow: Included in Step 14:Optimise Budgets in dTIMS of UNRA's dTIMS WF.</i></p> <p>(The steps are duplicated below)</p>                                                                                                                                                                                                                                                                 | <p>Create any number of budget scenarios for the analysis set that was just created.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <p>The WF will guide you to the budget scenarios associated with the analysis set <b>UNRA_LCCA</b>.</p> <p>Make changes where applicable, but most importantly EDIT THE BUDGET amounts to be optimised.</p> <p>Make sure the optimisation type, benefit variable and cost variable are as indicated below:</p>  | <p>Click on Analyze &amp; Report/Budget Scenarios to create budget scenarios for the analysis set just created. Click on New Item under Tasks to create a new budget item. Enter new property values for the following property items:</p> <ul style="list-style-type: none"> <li>Name: Enter a descriptive name that contains the name of the analysis too. E.g. <b>AUC2009_USD200m</b>, indicating a budget scenario with a USD200 million annual budget, or <b>AUC2009_Unconstr</b>, indicating an unconstrained budget scenario is investigated.</li> <li>Description can be entered as required.</li> <li>Select the appropriate analysis set. This name should <u>match exactly</u> the name of the analysis set you created in the previous step. E.g. AUC2010.</li> <li>Change the security to <b>“Level 7”</b> for the ReadAccess, WriteAccess and ExecuteWith.</li> <li>Chart color: Select a color from the list. The color selected is used in the network graphs showing average network condition for different budget scenarios..</li> </ul> |
|                                                                                                                                                                                                                                                                                                                                                                                                   | <ul style="list-style-type: none"> <li>UnlimitedBudgets: <ul style="list-style-type: none"> <li>FALSE when a constrained budget is used, or</li> <li>TRUE when unlimited funding is investigated.</li> <li>If an annualised funding level is investigated, this value should be FALSE.</li> </ul> </li> <li>Range of the Efficiency Frontier: 40%.</li> <li>Start_IBC_At_Do_Nothing: TRUE.</li> <li>Multi_Pass: TRUE.</li> <li>Number_Of_Minutes: 60</li> <li>Use Total Budget: <ul style="list-style-type: none"> <li>TRUE if a single value per year is entered for all roads (paved and unpaved) to be analysed,</li> <li>FALSE if the budget allocations for paved (flexible) and unpaved (unsealed) roads are entered separately.</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                               |
|                                                                                                                                                                                                                                                                                                                                                                                                   | <ul style="list-style-type: none"> <li>Committed under Strategies to include: TRUE, else FALSE when no committed treatments were entered during the previous steps.</li> <li>Do Nothing under Strategies to include: TRUE</li> <li>Benefit under Variables: <ul style="list-style-type: none"> <li>Select nPV_AUC if the analysis set was copied from</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |



| Edit/Create Budget Scenarios                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Option a (using dTIMS WF)                                                                                                                                                                                                          | Option b (using dTIMS)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Workflow: Included in Step 14:Optimise Budgets in dTIMS of UNRA's dTIMS WF.<br><br>(The steps are duplicated below)                                                                                                                | Create any number of budget scenarios for the analysis set that was just created.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                                                                                                                                                                                                                    | <p>Master_AUC (<b><i>This is the recommended benefit function to use.</i></b>)</p> <ul style="list-style-type: none"> <li>o Select nPV_TTC_Benefit if the analysis set was copied from Master_TTC.</li> <li>• Cost under variables: nPV_EconAgency</li> <li>• Budget under yearly Budgets: Can only be entered if UnlimitedBudget = FALSE. Enter the maximum available funding per year in US Dollars.</li> <li>• Save the budget scenario when done.</li> </ul> <p>Normally more than one budget scenario is created to investigate the influence of increased/reduced funding levels. Repeat the previous steps to create more scenarios or use the copy and paste facility to copy an existing scenario</p> <p>More budget scenarios can also be added later after an analysis has run. It is not always clear during this stage which funding levels are appropriate to maintain and improve road networks.</p> <p>Also investigate the budgets scenarios that were created for the analysis set UNRA_LCCA, it gives an indication of possible budget scenario to create and investigate.</p> |
| Save the budget scenarios after changes were made<br><br><b><i>dTIMS WF will automatically optimise the budget scenarios, in other words EXECUTE each of the budget scenarios associated with the analysis set; UNRA_LCCA.</i></b> | Save the budgets scenarios you just created.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

6. The analysis set with budget/s is now ready and the next step is to generate strategies and optimize budget scenarios.

| Generate Strategies and Optimise Budgets                             |                                                                                                                                                                                                                                                                                                                                        |
|----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Option a (using dTIMS WF)                                            | Option b (using dTIMS)                                                                                                                                                                                                                                                                                                                 |
| dTIMS WF automatically completes these tasks during steps 13 and 14. | <p>Close all tabs and go to Analyze &amp; Report/Analysis Sets.</p> <p>Click on the analysis set name you have just created, e.g. AUC2010 and select <b><i>Generate strategies and Optimize for AUC2010</i></b> from the tasks pane. This step takes a while to complete.</p> <p>The two steps to generate strategies and optimize</p> |



| Generate Strategies and Optimise Budgets |                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Option a (using dTIMS WF)                | Option b (using dTIMS)                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                          | <p>can also be done separately, see the Manual for the dTIMS software for more information.</p> <p>After an analysis set has been run (generate strategies has completed), budget scenarios can be added/changed and re/optimized. Go to Analyze &amp; Report/Budget Scenarios, add/change budget scenarios and use the Optimize task in the task pane to complete the optimization for one or more budget scenarios.</p> |

7. The results of the analysis can now be viewed and investigated in dTIMS or dTIMS MD.

**Workflow:** *Step 13:Generate strategies in dTIMS and Step 14:Optimise Budgets in dTIMS are the last steps in the workflow sequence for UNRA's life cycle cost analysis. The viewing of results can be done either in dTIMS MD or dTIMS, there are no workflow steps for these.*

**Budget scenarios can be re-optimised if changes are necessary to the funding levels and/or fund allocations of budget scenarios. Use the steps above or alternatively use Step 14 of the UNRA's workflow.**

## 11.2. Viewing Results in dTIMS

All results are viewed from the **Tasks Pane** under the **Analyze & Report/Budget Scenarios** tab. This section handles the checking of analysis results as well as retrieving data for input into maintenance and rehabilitation plans and reports.

Use the description of the performance models (analysis variables) in paragraph 17.1 when viewing the results.

### 11.2.1. Viewing the programme/treatment costs:

The first step is usually to check the costs of the budget scenarios that were optimised. The costs can be displayed per budget category (Programme Costs) and per treatment type (Treatment Costs/Lengths). The costs and lengths are given as a total for the road network analysed.

1. Select one budget scenario at a time, by clicking on the name. Select **Programme Costs** or **Treatment Costs/Lengths** from the task menu. To display treatment lengths the **Show treatment length** value in the properties menu should be changed to TRUE.
2. Both the grid data and the chart image can be copied from here and pasted into Word, Excel or other software.

### 11.2.2. Viewing the strategies per road element:

The results can also be checked and reviewed per road element. This allows the user to inspect the different alternatives (strategies) for maintaining the road element and it also shows the alternative (strategy) that was selected for each budget scenario.

1. Select one budget scenario at a time, by clicking on the name. Select **Review and Adjust Strategies** from the task menu. Use the Default Strategy Review Layout Under the tasks menu to display the LCCA results per road element. More information (chart and tables) can be dragged onto the display window from the toolbar.

### 11.2.3. Viewing charts that compare budget scenarios:

Multiple budget scenarios can also be compared at network level.

1. Select multiple budget scenarios. Select any of the items under **Charts that Compare Budget Scenarios** from the task menu to display the network level comparison for the selected scenarios.
2. Both the grid data and the chart image can be copied from here and pasted into Word, Excel or other software.

### 11.2.4. Generate a construction programme for a specific budget scenario:

This procedure exports the selected strategy on each road section for a specific budget scenario to a file (Excel, Access, etc.) for manipulation into a maintenance and rehabilitation programme.

1. Select one budget scenario at a time, by clicking on the name. Select **Construction Programmes** from the task menu.
2. The options can be changed as needed but it is recommended that the report is done by section, all treatments are included, at least 5 years of treatments are included and the values of the DataView **V\_AnalMaster\_For\_ConstructionProgramme** are included.
3. The data can then be further manipulated into a Maintenance and Rehabilitation Plan in Excel or Access.
4. The table below describes the output fields/columns of the Construction Programme.

| Column/Field Heading | Description of content                                                                                                   |
|----------------------|--------------------------------------------------------------------------------------------------------------------------|
| Analysis Set         | Name of analysis set that was executed in dTIMS.                                                                         |
| BudScen              | Name of budget scenario that was optimised in dTIMS.                                                                     |
| Year                 | Year of the first treatment that was selected for the road segment under the funding constraints of the budget scenario. |
| ElementID            | Element id of the road segment in the perspective: AnalMaster.                                                           |
| Length               | Length in kilometres of the road segment in the perspective: AnalMaster.                                                 |
| Road                 | Road name of the road segment in the perspective: AnalMaster.                                                            |
| From_Add             | Start km of the road segment in the perspective: AnalMaster.                                                             |
| From_Offset          | Same as previous, ignore this.                                                                                           |
| From_Desc            | Not used.                                                                                                                |
| To_Add               | End km of the road segment in the perspective: AnalMaster.                                                               |
| To_Offset            | Same as previous, ignore this.                                                                                           |
| To_Desc              | Not used.                                                                                                                |
| ElementKey           | Not used.                                                                                                                |
| Crk_Crka             | For paved roads: Percentage carriageway area with all cracks.                                                            |
| Crk_Crkw             | For paved roads: Percentage carriageway area with wide cracks.                                                           |
| Inv_Region           | Region name.                                                                                                             |
| Inv_Station          | Station name.                                                                                                            |
| Inv_FromDesc         | Description at start of road section in AnalMaster.                                                                      |
| Inv_ToDesc           | Description at end of road section in AnalMaster.                                                                        |
| Inv_Link             | Link name.                                                                                                               |
| Inv_Terrain          | Terrain code for Flat, Rolling or Mountainous .                                                                          |
| Inv_Width            | Carriageway width of road in metres.                                                                                     |
| Pot_PotNo            | For paved roads: No of 0.1m <sup>2</sup> potholes per km.                                                                |
| Pot_Ptch             | For paved roads: No data are recorded for this (Percentage area patched).                                                |
| Rgh_Irir             | For paved roads: Average roughness on the road segment from measured values, in IRI (m/km).                              |
| Rgh_Year             | For paved roads: Year in which roughness was measured.                                                                   |
| Rut_Rutm             | For paved roads: Average rut depth of the road segment in mm.                                                            |
| Rvl_Ravl             | For paved roads: Percentage area of carriage way with ravelling.                                                         |
| Struc_Pavetype       | Pavement type: F = Flexible (paved), U = Unsealed (unpaved).                                                             |

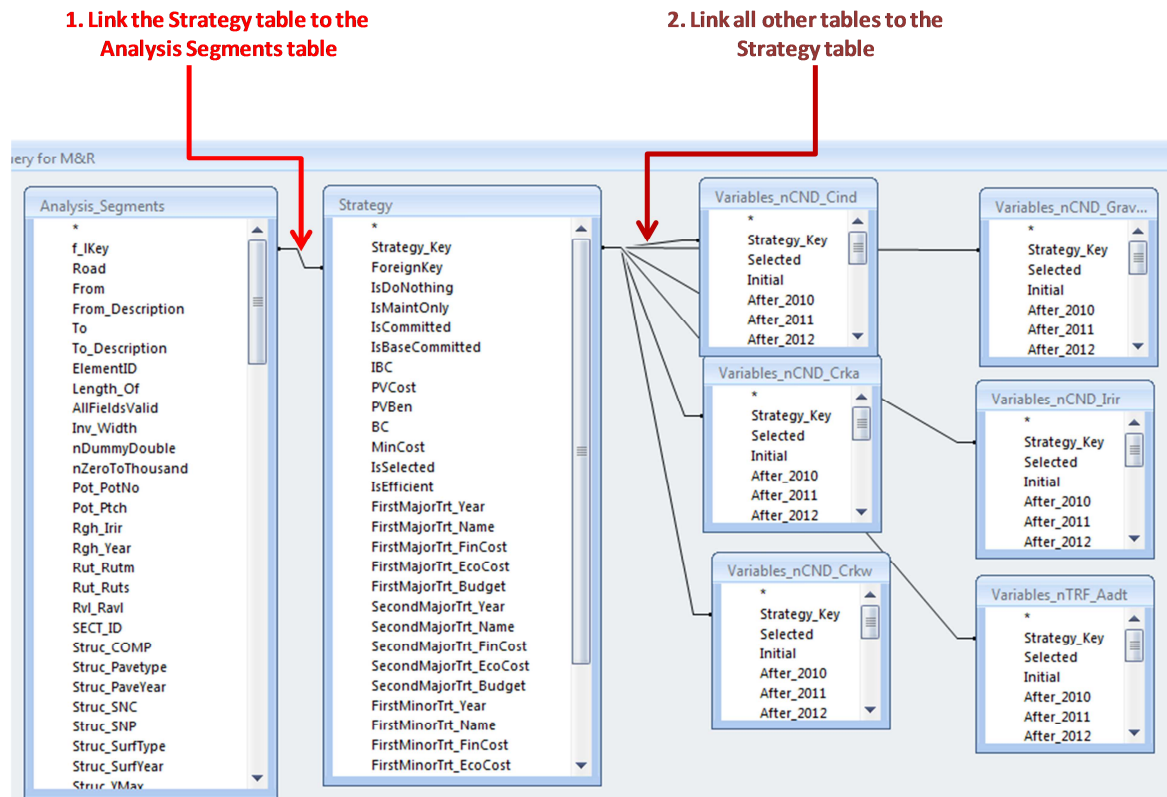
| Column/Field Heading | Description of content                                                                                                                                                                              |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Struc_PaveYear       | For paved roads: Year of construction or rehabilitation of structural pavement layers. 1900 if no data is available.                                                                                |
| Struc_SNC            | For paved roads: Modified structural number from falling weight deflections, for HDM-4 modelling.                                                                                                   |
| Struc_SNP            | For paved roads: Adjusted structural number from falling weight deflections, for HDM-4 modelling. (This will be the same as the previous column).                                                   |
| Struc_SurfType       | Surface type:<br>FA = Asphalt on paved road<br>FC = Chip and spray on paved road<br>FS = Slurry on paved road<br>UG = Unsealed road with gravel wearing course<br>UE = Earth road<br>US = Sand road |
| Struc_SurfYear       | For paved roads: Year of most recent surfacing. 1900 if no data is available.                                                                                                                       |
| Traf_AADT            | Average annual daily traffic in both directions for the road segment.                                                                                                                               |
| Traf_AADH_Split      | Percentage of heavy vehicle on the road segment.                                                                                                                                                    |
| Traf_AADT_Year       | Year of traffic data.                                                                                                                                                                               |
| Unp_GravThick        | For unpaved roads: Average gravel thickness of road segment, in mm.                                                                                                                                 |
| VCI                  | Average Visual Condition Index (%) for the road segment, derived from the visual assessment distresses. Ranges from 100% (very good), to 0% (very poor).                                            |
| Treatment            | Name of the first treatment that was selected for the road segment under the funding constraints of the budget scenario.<br><br>This is not necessarily the optimum treatment.                      |
| Cost                 | Cost of the first treatment that was selected for the road segment under the funding constraints of the budget scenario.                                                                            |

5. Examples of Construction programs (also called Maintenance, Rehabilitation and Upgrading Plans) can be found on “C:\UNRA\_PMS\Sample report”.

#### 11.2.5. Export strategies for a specific budget scenario:

The option is recommended for users with Access experience. This procedure exports the selected strategy on each road section for a specific budget scenario to a file (Excel, Access, etc.) for further manipulation. The predicted future performance of the difference distresses (models) are exported enabling the user further calculations on the prediction of roughness, rutting, cracking, gravel thickness, etc.

1. Select one budget scenario at a time, by clicking on the name. Select *Export Strategies* from the task menu.
2. Select an output database name (Access database is recommended) and change the options as needed. The default export options are recommended. Include the analysis variables (performance models) as required and start the export.
3. After the export has completed, the database can be opened in Access. The tables should be linked according to Figure 14 when creating a query.

**Figure 14 – How to link exported tables**

### 11.3. Manipulating the Results for Importing into dTIMS

The output from the Construction Programme of a budget scenario in dTIMS can be manipulated for importing back into UNRA's dTIMS database.

Investigate the attributes of the perspective: **Results\_dTIMS** in UNRA's dTIMS database to comprehend the data needs of the HDM4 results. The attributes are also detailed below in a table.

Many methods can be used to populate the Results\_dTIMS perspective in dTIMS with the Construction Programme of a particular budget scenario. The method below is one of many methods the user can adopt to prepare the Results\_dTIMS perspective with data. Alternatively the strategies can be exported for a budget scenario and the data can be manipulated further in Access and/or Excel.

This procedure exports the selected strategy on each road section for a specific budget scenario to a file (Excel, Access, etc.) for manipulation into an import file for dTIMS.

1. Select one budget scenario at a time, by clicking on the name. Select **Construction Programmes** from the task menu.
2. The options can be changed as needed but it is recommended that the report is done by section, all treatments are included, at least 5 years of treatments are included and the values of the DataView **V\_AnalMaster\_For\_ConstructionProgramme** are included.
3. The table below describes the output fields/columns of the Construction Programme.

| Column/Field Heading | Description of content                               |
|----------------------|------------------------------------------------------|
| Analysis Set         | Name of analysis set that was executed in dTIMS.     |
| BudScen              | Name of budget scenario that was optimised in dTIMS. |

| Column/Field Heading | Description of content                                                                                                                                                                              |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Year                 | Year of the first treatment that was selected for the road segment under the funding constraints of the budget scenario.                                                                            |
| ElementID            | Element id of the road segment in the perspective: AnalMaster.                                                                                                                                      |
| Length               | Length in kilometres of the road segment in the perspective: AnalMaster.                                                                                                                            |
| Road                 | Road name of the road segment in the perspective: AnalMaster.                                                                                                                                       |
| From_Add             | Start km of the road segment in the perspective: AnalMaster.                                                                                                                                        |
| From_Offset          | Same as previous, ignore this.                                                                                                                                                                      |
| From_Desc            | Not used.                                                                                                                                                                                           |
| To_Add               | End km of the road segment in the perspective: AnalMaster.                                                                                                                                          |
| To_Offset            | Same as previous, ignore this.                                                                                                                                                                      |
| To_Desc              | Not used.                                                                                                                                                                                           |
| ElementKey           | Not used.                                                                                                                                                                                           |
| Crk_Crka             | For paved roads: Percentage carriageway area with all cracks.                                                                                                                                       |
| Crk_Crkw             | For paved roads: Percentage carriageway area with wide cracks.                                                                                                                                      |
| Inv_Region           | Region name.                                                                                                                                                                                        |
| Inv_Station          | Station name.                                                                                                                                                                                       |
| Inv_FromDesc         | Description at start of road section in AnalMaster.                                                                                                                                                 |
| Inv_ToDesc           | Description at end of road section in AnalMaster.                                                                                                                                                   |
| Inv_Link             | Link name.                                                                                                                                                                                          |
| Inv_Terrain          | Terrain code for Flat, Rolling or Mountainous .                                                                                                                                                     |
| Inv_Width            | Carriageway width of road in metres.                                                                                                                                                                |
| Pot_PotNo            | For paved roads: No of 0.1m <sup>2</sup> potholes per km.                                                                                                                                           |
| Pot_Ptch             | For paved roads: No data are recorded for this (Percentage area patched).                                                                                                                           |
| Rgh_Irir             | For paved roads: Average roughness on the road segment from measured values, in IRI (m/km).                                                                                                         |
| Rgh_Year             | For paved roads: Year in which roughness was measured.                                                                                                                                              |
| Rut_Rutm             | For paved roads: Average rut depth of the road segment in mm.                                                                                                                                       |
| Rvl_Ravl             | For paved roads: Percentage area of carriage way with ravelling.                                                                                                                                    |
| Struc_Pavetype       | Pavement type: F = Flexible (paved), U = Unsealed (unpaved).                                                                                                                                        |
| Struc_PaveYear       | For paved roads: Year of construction or rehabilitation of structural pavement layers. 1900 if no data is available.                                                                                |
| Struc_SNC            | For paved roads: Modified structural number from falling weight deflections, for HDM-4 modelling.                                                                                                   |
| Struc_SNP            | For paved roads: Adjusted structural number from falling weight deflections, for HDM-4 modelling. (This will be the same as the previous column).                                                   |
| Struc_SurfType       | Surface type:<br>FA = Asphalt on paved road<br>FC = Chip and spray on paved road<br>FS = Slurry on paved road<br>UG = Unsealed road with gravel wearing course<br>UE = Earth road<br>US = Sand road |
| Struc_SurfYear       | For paved roads: Year of most recent surfacing. 1900 if no data is available.                                                                                                                       |
| Traf_AADT            | Average annual daily traffic in both directions for the road segment.                                                                                                                               |
| Traf_AADH_Split      | Percentage of heavy vehicle on the road segment.                                                                                                                                                    |
| Traf_AADT_Year       | Year of traffic data.                                                                                                                                                                               |
| Unp_GravThick        | For unpaved roads: Average gravel thickness of road segment, in mm.                                                                                                                                 |
| VCI                  | Average Visual Condition Index (%) for the road segment, derived from the visual assessment distresses. Ranges from 100% (very good), to 0% (very poor).                                            |

| Column/Field Heading | Description of content                                                                                                                                                         |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Treatment            | Name of the first treatment that was selected for the road segment under the funding constraints of the budget scenario.<br><br>This is not necessarily the optimum treatment. |
| Cost                 | Cost of the first treatment that was selected for the road segment under the funding constraints of the budget scenario.                                                       |

4. Export the Construction Programme to an Excel file; recommended Table name is “CP”. Then open the Excel file.
5. Refer to the sample file **“Sample for HDM4 Results Import.xls”** on **“C:\UNRA\_PMS\dtims\dtims Masterfiles”** for more details.
  - a. In this example the Worksheet **“CP”** was exported from the Construction Programme in dtims.
  - b. The steps that follow refer to this sample.
6. Work sheet “CP”: Insert a column at the beginning (column A). Heading is “Key”, enter the formula as follows: =E2&"\_"&D2. The content should be “Road\_year-of-treatment”, e.g. “1003\_2012”.
7. Work Sheet “CP”: Change the heading of column H to “From” and change the heading of column K to “To”.
8. Make a copy of the work sheet “CP” and call it “CP\_Unique”.
9. Go to work sheet “CP\_Unique” and delete all duplicates. Use Data/Remove Duplicates functionality and select ElementID as the only attribute to select on.
10. Now sort the work sheet “CP\_Unique” on Road (ascending) and From\_Add (ascending).
11. Add ONE empty rows at the top.
12. Note that the first four columns (Key, Analysis Set, BudScen and Year) will not be imported.
13. Enter the following formula in the “ElementID” column: =LEFT(G3&"\_"&RIGHT("000000"&VALUE(H3)\*1000,6)&"\_"&AO3&"\_"&AP3,40). The results should be a value consisting of roadname\_from-km\_analysis-start-year\_budget-name. E.g A001\_000000\_2011\_AUC2011\_UNC.
14. Delete the last two columns “Treatment” and “Cost”, these will be populated from the original work sheet “CP”.
15. Add the following 22 columns at the end (After VCI). (The Data Sheet View of Results\_dtims in UNRA’s dtims database can be exported to get the accurate field names). See sample file for formulae to populate these columns with data.

| Attribute/Column name | Description of content                                                               |
|-----------------------|--------------------------------------------------------------------------------------|
| Analysis_StartYR      | Enter the start year of the analysis period in dtims.                                |
| BudgetName            | Enter a name to identify the budget scenario (max 19 characters)                     |
| MRCostYr01            | Cost of the Treatment (Maintenance, Rehabilitation or Upgrading intervention. Year 1 |
| MRCostYr02            | Cost of the Treatment (Maintenance, Rehabilitation or Upgrading intervention. Year 2 |
| MRCostYr03            | Cost of the Treatment (Maintenance, Rehabilitation or Upgrading intervention. Year 3 |
| MRCostYr04            | Cost of the Treatment (Maintenance, Rehabilitation or Upgrading intervention. Year 4 |
| MRCostYr05            | Cost of the Treatment (Maintenance, Rehabilitation or Upgrading intervention. Year 5 |
| MRCostYr06            | Cost of the Treatment (Maintenance, Rehabilitation or Upgrading intervention. Year 6 |
| MRCostYr07            | Cost of the Treatment (Maintenance, Rehabilitation or                                |

| Attribute/Column name | Description of content                                                                |
|-----------------------|---------------------------------------------------------------------------------------|
|                       | Upgrading intervention. Year 7                                                        |
| MRCostYr08            | Cost of the Treatment (Maintenance, Rehabilitation or Upgrading intervention. Year 8  |
| MRCostYr09            | Cost of the Treatment (Maintenance, Rehabilitation or Upgrading intervention. Year 9  |
| MRCostYr10            | Cost of the Treatment (Maintenance, Rehabilitation or Upgrading intervention. Year 10 |
| MRTreatmentYr01       | Name of the Treatment (Maintenance, Rehabilitation or Upgrading intervention. Year 1  |
| MRTreatmentYr02       | Name of the Treatment (Maintenance, Rehabilitation or Upgrading intervention. Year 2  |
| MRTreatmentYr03       | Name of the Treatment (Maintenance, Rehabilitation or Upgrading intervention. Year 3  |
| MRTreatmentYr04       | Name of the Treatment (Maintenance, Rehabilitation or Upgrading intervention. Year 4  |
| MRTreatmentYr05       | Name of the Treatment (Maintenance, Rehabilitation or Upgrading intervention. Year 5  |
| MRTreatmentYr06       | Name of the Treatment (Maintenance, Rehabilitation or Upgrading intervention. Year 6  |
| MRTreatmentYr07       | Name of the Treatment (Maintenance, Rehabilitation or Upgrading intervention. Year 7  |
| MRTreatmentYr08       | Name of the Treatment (Maintenance, Rehabilitation or Upgrading intervention. Year 8  |
| MRTreatmentYr09       | Name of the Treatment (Maintenance, Rehabilitation or Upgrading intervention. Year 9  |
| MRTreatmentYr10       | Name of the Treatment (Maintenance, Rehabilitation or Upgrading intervention. Year 10 |

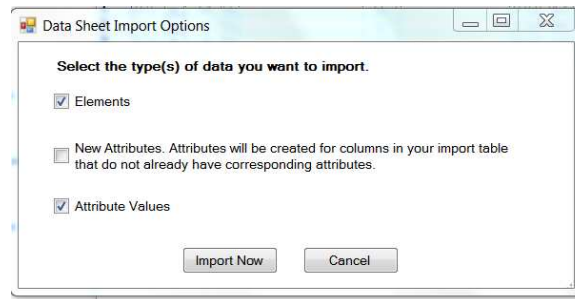
16. Enter the values 0 to 9 cells “AQ1:AZ1”, and again in cells “BA1:BJ1”. See BLUE cells below.

|       | Col AQ     | Col AR     | Col AS     | Col AT     | Col AU     | Col AV     | Col AW     | Col AX     | Col AY     | Col AZ     |
|-------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|
| Row 1 | 0          | 1          | 2          | 3          | 4          | 5          |            |            | 8          | 9          |
| Row 2 | MRCostYr01 | MRCostYr02 | MRCostYr03 | MRCostYr04 | MRCostYr05 | MRCostYr06 | MRCostYr07 | MRCostYr08 | MRCostYr09 | MRCostYr10 |

|       | Col BA          | Col BB          | Col BC          | Col BD          | Col BE          | Col BF          | Col BG          |
|-------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|
| Row 1 | 0               | 1               | 2               | 3               | 4               | 5               | ETC             |
| Row 2 | MRTreatmentYr01 | MRTreatmentYr02 | MRTreatmentYr03 | MRTreatmentYr04 | MRTreatmentYr05 | MRTreatmentYr06 | MRTreatmentYr07 |

- Use the sample file and copy the formulae from sheet “CP\_Unique”, Cells “AQ3 to BJ3you’re your worksheet. Copy the formula down up to the end of your worksheet.
- Make a copy of the work sheet “CP\_Unique” and call it “CP\_Unique\_VALUES”.
- Open work sheet “CP\_Unique\_VALUES”.
- Select all cells. Then copy all cells and PASTE AS VALUES.
- Now delete the first row.
- Select all cells and REPLACE (CTRL & H) “#N/A” with blank.
- Sheet “**CP\_Unique\_VALUES**” are now ready for importing, but first move the data to an Access file.
- Save and close the Excel sheet.
- Open Access.
- Create a new Microsoft Access database (2002-2003 format). It must have extension MDB!
- Then use “External Data” / “Import & Link” to access the Excel file you have just created.
- Make sure to link the work sheet “CP\_Unique\_VALUES” according the example above. (No primary key is needed).
- Close the Access file when done.
- In UNRA’s dTIMS database: Open the data Sheet View of Results\_dTIMS perspective.
- Make sure the LRM is set to “Roads”.

32. Now import the data from the Access table into perspective Results\_dTIMS of UNRA's dTIMS database. Import both Elements and Attribute values.



## 12. Running a LCCA in HDM-4 software

Volumes One through Four of the HDM-4 Manuals contain detailed explanations and guidance on the use of the HDM-4 software and methodology behind analyses and performance models. The running of analyses in HDM-4 should always be supported with the HDM-4 manuals. The step-by-step user instructions of HDM-4 are not repeated in this manual, but should be obtained from the HDM-4 manuals.

**Important:** The steps in this chapter will import the new road network created in UNRA's dTIMS database. Whenever a new road network is imported in HDM-4, the following important points should be considered:

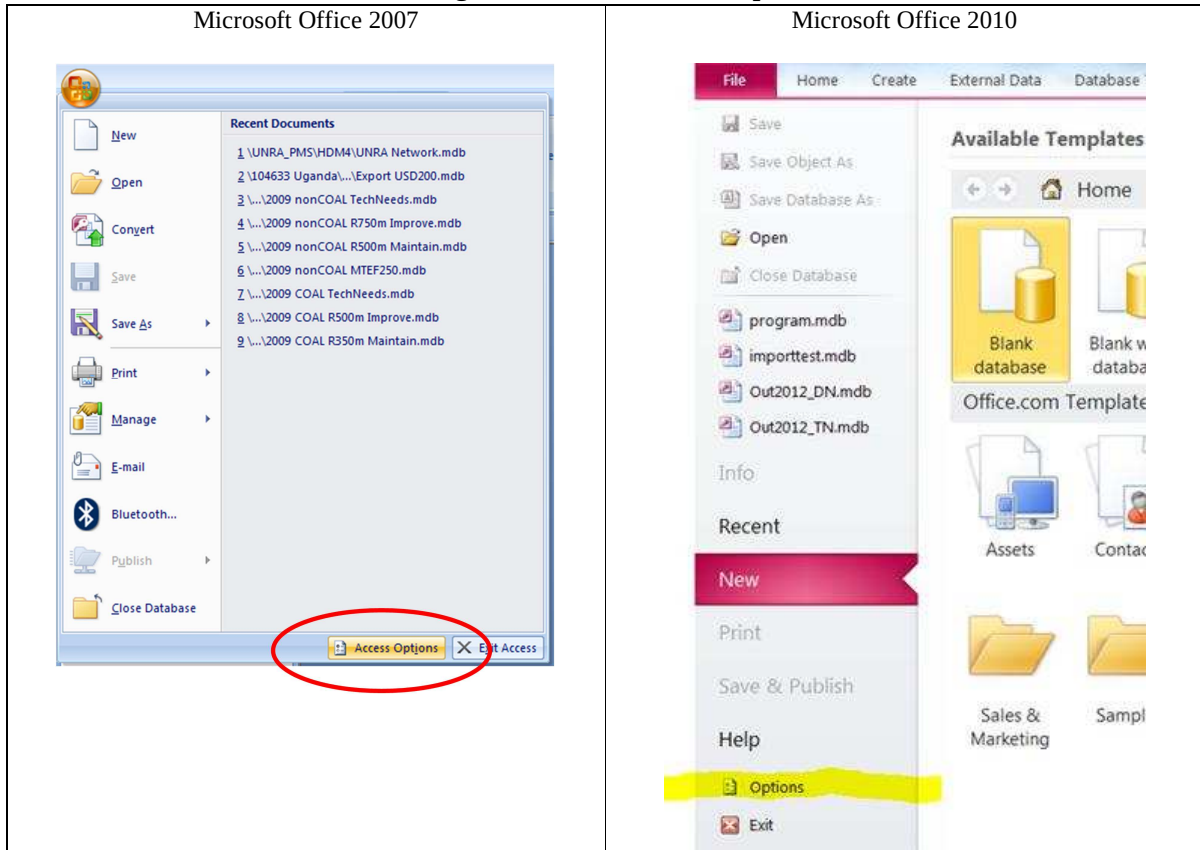
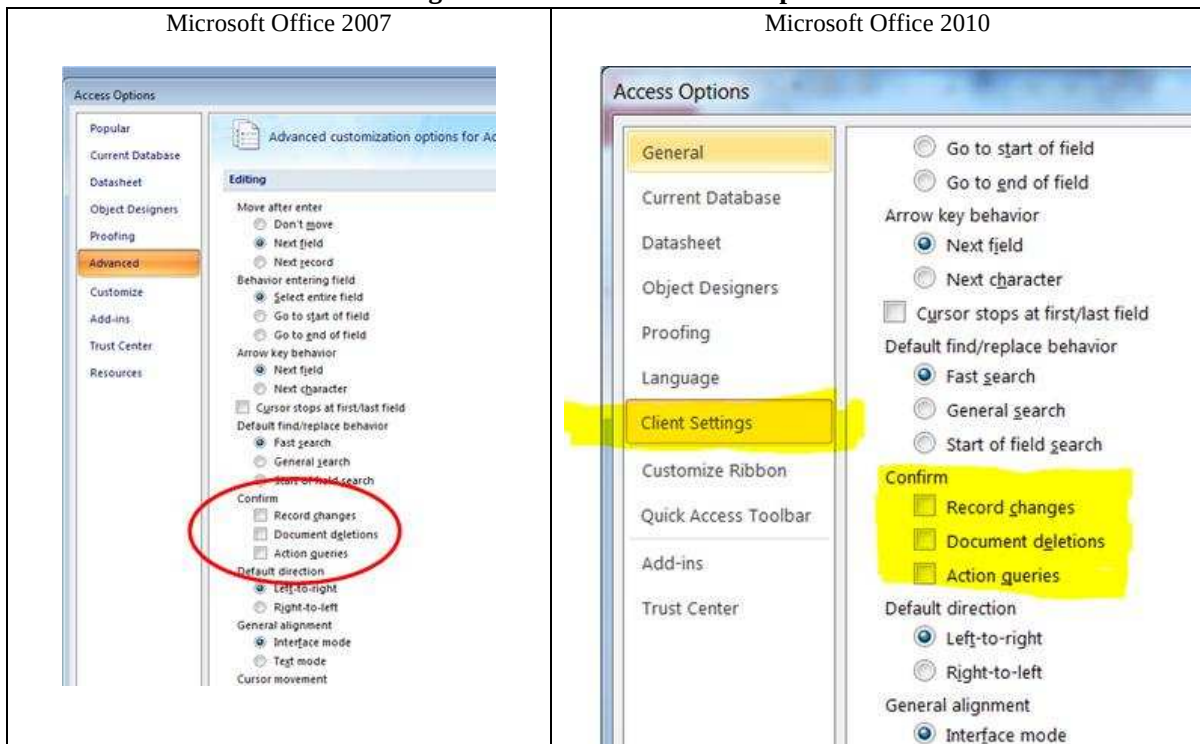
- Make a backup of the objects.dat file under c:\UNRA\_PMS\HDM4.
- The old road network should ALWAYS be deleted or renamed, because in HDM-4 records are appended to the existing road network if the road id's are not the same.
- Whenever a new road network has been imported, existing Project, Programme and Strategy analyses cannot be used. Any analysis object that used the old/deleted road network cannot be used for the newly imported road network.
- Create new Project, Programme and Strategy analyses from scratch after a new road network has been imported/created.

Follow the steps hereafter to prepare the data and complete an economic evaluation of roads in the HDM-4 software.

### 12.1. Prepare the road network file generated by dTIMS in Access

1. Using Access: Open the file *UNRA Network.mdb* in the directory C:\UNRA\_PMS\HDM4. This file was created and exported from the dTIMS software; see Chapter 10 of this report.
2. Deactivate the setting to confirm deletions and changes: By selecting Access Options (MSOffice 2007) / Options (MSOffice 2010), see Figure 15. Then unselect the three Confirm options as shown in Figure 16. If these settings are not changed, the user will have to answer "YES" to all confirmations during the next step.



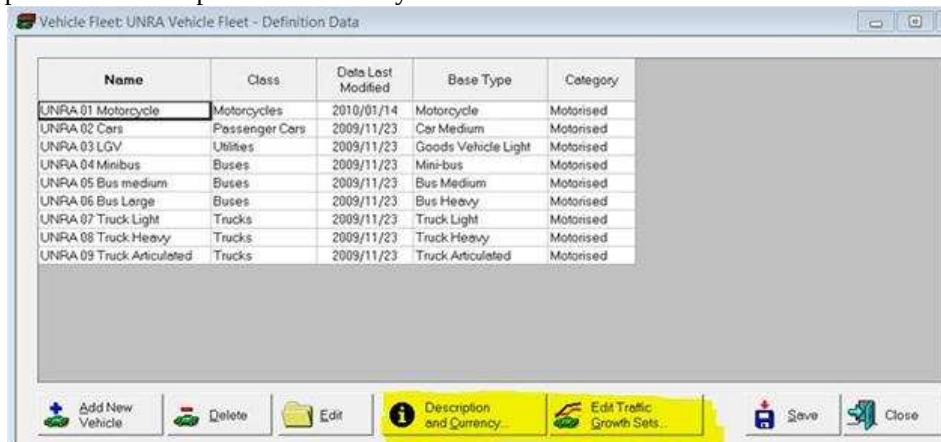
**Figure 15 – Select Access Options****Figure 16 – Unselect Confirm Options**

3. Open and run the Form: frm\_RunAllQueries.
4. Close the form and exit Access when done.

## 12.2. Update the Vehicle Fleet costs and growths.

Run the HDM4 software and open the HDM4 workspace on c:/UNRA\_PMS.

1. Open **UNRA Vehicle Fleet** under Vehicle Fleets.
2. Edit each of the nine vehicle classes and update the Economic Unit Costs, in US Dollars, to the most current values.
3. The Basic Characteristics can also be changed whenever updated information becomes available.
4. The Calibration data and Reset Values should not be changed for this customised system, except by a specialist HDM-4 user.
5. Update the “Description and Currency” and “Traffic Growth Sets”.



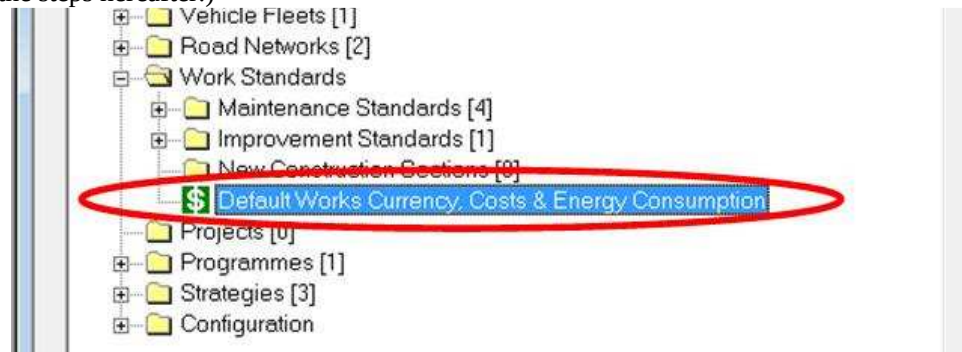
6. Save and close the Vehicle Fleets items once all costs have been updated.

## 12.3. Import the latest Road Network generated in dTIMS.

1. First **rename or delete** the existing “UNRA Network” from the Road Network objects. An import will append records and the resulting road network will thus be a combination of a previous network and the imported network.
2. Click the import button.
3. Select the file **C:\UNRA\_PMS\HDM4\UNRA Network.mdb**, and click next. (This is the file created in Chapter 10)
4. Name of the item = UNRA Network. “Perform data validation” should be checked. Then click on Next.
5. Answer YES to overwrite the existing Road Network, and click next.
6. Click next on Validation step and wait whilst the validation and import processes complete. Click Finish on completion.
7. If errors occur repeat the steps in Chapter 10 and start again with the import of the road network.
8. The UNRA Network should now be opened, verified and edited if needed.
9. Open the road network “UNRA Network”.
10. Click on the “Info” button and make sure the currency is set to **“US Dollars”**.
11. Verify the Asset Valuation information on the road sections is valid. These costs and asset component values will be used in the life cycle cost analysis to calculate asset value.
12. Click “OK” to save the road network info.
13. “Save” and “Close” the “UNRA Network”.

## 12.4. Update Maintenance Standards costs

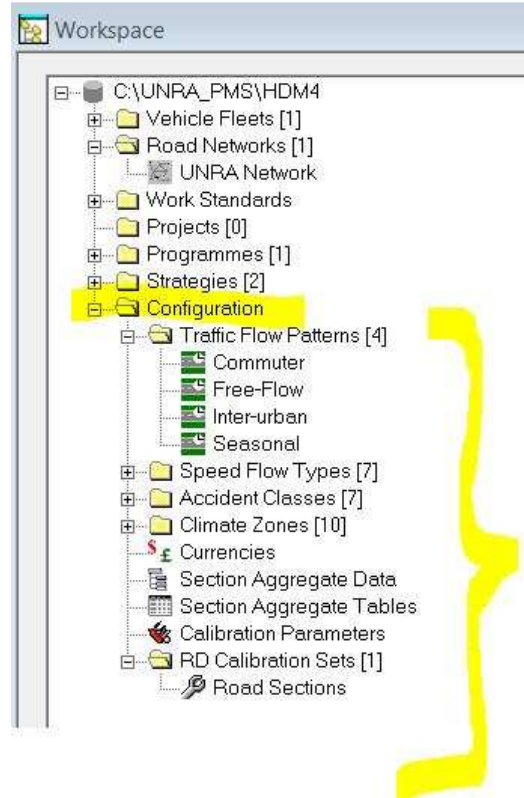
1. Open the item for “Default Works Currency, Costs and Energy Consumption”. Make sure the correct currency is selected, namely “US Dollars” This currency should be same currency that was selected when the vehicle Fleet Currency was customised. Click on the “OK” button to save selected currency. (No costs need entering here because these costs are used when new Works Items are created. The Works Items for UNRA have already been created and will be updated in the steps hereafter.)



2. Four maintenance and one improvement standard have been customised, they are:
  - a. Maint&Rehab Bituminous
  - b. Maintenance Unsealed
  - c. Routine – Bituminous
  - d. Routine – Unsurfaced
  - e. Upgrade to Pave
3. Open each of these standards and update the costs. The advanced user can also change other parameters to enhance/improve the current system.
4. Under the “Asset valuation” tab, enter a realistic “Residual value of new work%”. This percentage will be used to calculate the residual asset value for each road section, based on the “Replacement Cost Value” under the ROAD Network of UNRA.

## 12.5. Configuration

In HDM-4 the objects under the Configuration have been customized with items defining traffic flow, speed flow, calibration factors etc. The road network data of UNRA then points to these items to indicate the characteristics of each road section. The cross-reference between certain fields in the road network (UNRA Network) and the items defined under the Configuration should be exact. Take special care to ensure the spelling and use of capital letters is 100%.

**Figure 17 – Configuration items in HDM-4**

Under the configuration heading, the system has been customized (June 2012) with pre-defined values for UNRA.

The following items have been selected for UNRA's Road Network, these values have been selected in the "Preparation of data in dTIMS".

1. Traffic Flow Patterns: Free-Flow
2. Speed Flow Types and Accident Classes; criteria as follows were used;
  - "Single Lane Road" speed flow type is adopted for all single lane roads.
  - "Four Lane Road" speed flow type is adopted for all four lane roads in UNRA's database.
  - "Intermediate Road" speed flow type is adopted for all two lane roads in UNRA's database if the carriage way width of the road is less than 5.6m.
  - "Two Lane Road" speed flow type is adopted for all two lane roads in UNRA's database if the carriage way width of the road is between 5.6m and 8m.
  - "Two Lane Wide" speed flow type is adopted for all two lane roads in UNRA's database if the carriage way width of the road is greater than 8m.
3. Climate Zone: "Humid Tropical". This indicates the moisture classification as "Humid" and the temperature classification as tropical. This value is stored in the dTIMS data expression: a\_HDM4\_Climate\_Zone.
4. RD Calibration Sets: Road Sections (with items "UNRA AC on Granular", "UNRA Seal on Granular", "UNRA Earth Type A" and "UNRA Gravel Type A")

The advanced user can change the content of the road network (UNRA Network) provided the items are valid in terms of the configuration. E.g. The field/attribute storing Traffic Flow Patterns should store a valid items according the configuration of HDM-4, namely "Commuter", "Free-Flow", "Inter-urban" or "Seasonal".

Once new calibration factors become available, Road Sections under RD Calibration Sets should be updated to the latest values/factors.

The following items are available in UNRA's HDM-4 system.

### 12.5.1. Traffic Flow Patterns

Traffic flow Patterns are used to represent the varying traffic intensities that occurs on roads during the day.

For UNRA, four traffic flow items were configured using the default parameters of the HDM-4 software, see Table 3.

**Table 3: HDM-4 Traffic Flow Patterns available for UNRA**

|                      | Period 1           | Period 2 | Period 3 | Period 4 | Period 5 |
|----------------------|--------------------|----------|----------|----------|----------|
|                      | Hours per year     |          |          |          |          |
|                      | 87.6               | 350.4    | 613.2    | 2978.4   | 4730.4   |
| Traffic Flow Pattern | Percentage of AADT |          |          |          |          |
| Commuter             | 3.05               | 11.33    | 16.55    | 56.26    | 12.81    |
| Seasonal             | 4.25               | 13.24    | 16.6     | 40.32    | 25.59    |
| Inter-urban          | 2.17               | 7.59     | 11.64    | 40.24    | 38.36    |
|                      |                    |          |          |          |          |
|                      | Period 1           |          |          |          |          |
|                      | Hours per year     |          |          |          |          |
|                      | 8760 (all hours)   |          |          |          |          |
| Traffic Flow Pattern | Percentage of AADT |          |          |          |          |
| Free-Flow            | 100                |          |          |          |          |

By default, the “Free-Flow” traffic flow pattern is adopted for all roads in UNRA's database during the process where the HDM-4 data is processed and exported from UNRA's dTIMS database. Any one of the four Traffic Flow Patterns can however be used, to do this the user would have to customise dTIMS to use the other traffic flow patterns. The “Inter-urban” traffic flow pattern should be used for roads with known congestion problems.

The traffic flow patterns for UNRA's road network can be changed in HDM-4 if the information becomes available. To make changes, “Open” one of the four traffic flow patterns in HDM-4 and change the data of the different periods. Do not change the names of the four customized traffic flow patterns.

### 12.5.2. Speed Flow Types

Speed Flow Types are used to represent the capacity characteristics of different road types.

For UNRA, five speed flow types were configured using the default parameters of the HDM-4 software, see Table 4. The speed flow pattern will be selected on the following criteria:

1. “Intermediate Road”, speed flow type is adopted for all two lane roads if the carriage way width of the road is less than 5.6m.
2. “Two Lane Road” speed flow type is adopted for all two lane roads if the carriage way width of the road is between 5.6m and 8m. This was adopted as default for roads with no width.
3. “Two Lane Wide” speed flow type is adopted for all two lane roads if the carriage way width of the road is greater than 8m.
4. “Single Lane Road” speed flow type is adopted for all single lane roads.
5. “Four Lane Road” speed flow type is adopted for all four lane roads.

The speed flow types for UNRA's road network can be changed in HDM-4 if the updated information becomes available from the Traffic Information System. To make changes, "Open" one of the five speed flow types in HDM-4 and change the data. Do not change the names of the five customized speed flow types.

**Table 4: HDM-4 Speed Flow Types available for UNRA**

| Speed flow name   | Number of Lanes | PCSE/lane/hour  |                        |                | JAMSPEED Km/h |
|-------------------|-----------------|-----------------|------------------------|----------------|---------------|
|                   |                 | CAPACITY Q(ult) | FREEFLOW CAPACITY (Qo) | NOMINAL (Qnom) |               |
| Single Lane Road  | 1               | 600             | 0                      | 420            | 10            |
| Two Lane Road     | 2               | 1400            | 140                    | 1260           | 25            |
| Four Lane Road    | 4               | 2000            | 800                    | 1900           | 40            |
| Intermediate Road | 2               | 900             | 0                      | 630            | 20            |
| Two Lane Wide     | 2               | 1600            | 320                    | 1440           | 30            |

### 12.5.3. Accident Classes

Accident Classes are used to represent the accident rate characteristics of different road types. For UNRA, accident classes were configured for five road types using the default parameters of the HDM-4 software, Table 5. The five road types correspond to the five speed flow types defined in the previous paragraph.

Accident rate characteristics are defined in terms of accident rates per 100 million vehicle-km. The accident characteristics of UNRA can be updated in the HDM-4 system once such information becomes available. To make changes, "Open" one of the five accident classes in HDM-4 and change the data. Do not change the names of the five customized accident classes.

**Table 5: HDM-4 Accident Classes available for UNRA**

| Road accident class |
|---------------------|
| Single Lane Road    |
| Two Lane Road       |
| Four Lane Road      |
| Intermediate Road   |
| Two Lane Wide       |

### 12.5.4. Climate Zones

Climate Zones are used to represent the different climatic conditions of a road network. The Climate Zone is divided into two categories; Moisture and Temperature. HDM-4 provides aggregate level parameters for each of the two levels.

For UNRA, the climate zone "Tropical Humid" was adopted for all roads. (Table 6). However, more climate zones are available in HDM-4 that could be used as the default for UNRA's road network. The climate characteristics can be updated in the HDM-4 system if required.

**Table 6: HDM-4 Climate Zones used for UNRA**

|                         | Climate Zone | Tropical Humid |
|-------------------------|--------------|----------------|
| Moisture Classification | MOISTCLASS   | 3 (Humid)      |

|                                 |                |              |
|---------------------------------|----------------|--------------|
| Temperature classification      | TEMPTYPE       | 1 (Tropical) |
| Days temperature > 32 C Deg     | DAYSGT32       | 30           |
| Avg temp range                  | ANNTEMPRG<br>E | 5            |
| Freeze index (C-Days)           | FREEZEIDX      | 0            |
| Moisture Index                  | MOSITIDX       | 40           |
| Mean monthly precipitation (mm) | MMP            | 135          |
| Mean temperature (C Deg)        | MEANTEMP       | 22           |
| Duration of dry season (months) | DRYSEASON      | 3            |
| % Time on snow                  | PCTDS          | 0            |
| % Time on rain                  | PCTDW          | 20           |

#### 12.5.5. Currencies

Currencies comprises the different currencies available when defining the costs and for use in the analysis of the road network. The US Dollar is used in UNRA's HDM-4 system.

#### 12.5.6. Section Aggregate Data and Tables

Section aggregate data and tables are used when the user creates a new section from the Aggregate Data, in the Road Network item.

For UNRA, the road network in HDM-4 is customized and exported from UNRA's dTIMS Database and the section aggregate data and tables are not utilized.

#### 12.5.7. Calibration Parameters

Calibration Parameters are related to Road deterioration and Works Effects Models. The defaults as calibrated HDM-4 were adopted for UNRA's PMS, see the two illustrations below for Bituminous and Unsealed roads respectively.

**Table 7: HDM-4 Standard Calibration Parameters**

The image shows two side-by-side screenshots of the 'Workspace Calibration Parameters' dialog box. The left screenshot is for the 'Bituminous' tab, and the right is for the 'Unsealed' tab. Both tabs have a 'Road Deterioration' section with input fields for 'Wide structural cracking initiated potholes threshold' (20%), 'Ravelling initiated potholes threshold' (30%), and 'Max. thickness of surfacing for plastic flow effects' (100 mm). The 'Work Effects' section includes input fields for 'Area of wide structural cracking that triggers their patching' (20%), '% of wide structural cracks in excess of TRIGACW to be patched' (10%), 'Area of transverse thermal cracking that triggers their patching' (10%), and '% of transverse thermal cracking in excess of TRIGACT to be patched' (20%). Below these is a table for 'Preparatory Works Activity' with columns for 'Reseals' and 'Overlays'. The 'Reseals' column has a dropdown menu with 'Patch Wide Only' selected. The 'Overlays' column has a dropdown menu with 'Nothing' selected. The 'Work Effects' section also includes a 'GRAD' section with input fields for 'Non-motorised grading' (1.4), 'Light motorised grading' (1), and 'Heavy motorised grading' (0.75). At the bottom of each dialog are 'OK', 'Cancel', and 'Apply' buttons. A note at the bottom left of the Bituminous dialog states: 'The percentage at which wide structural cracking initiated potholes arise (ARWpi)'. The Unsealed dialog has a 'Mechanical' section at the bottom.

### 12.5.8. RD Calibration sets

Calibration Sets are sets of calibration coefficients to be defined for a range of pavement types commonly found on the road network being analysed. A Calibration Set is selected for a road network, and a calibration set can contain a number of calibration items for Bituminous, Unsealed and other type roads.

For UNRA, the calibration set “Road Sections” was configured using partly Aurecon’s experience on other road networks in Africa and partly the default factors of HDM-4.

In HDM-4 calibration factors and parameters are used to calibrate and adjust current conditions and future prediction performances of road pavements to represent the reality as closely as possible. Volume 5 of the HDM-4 suite of manuals covers this field of expertise in depth.

The calibration factors adopted for UNRA’s PMS in 2012 are presented in Table 8. These factors should be revised and updated when new calibration factors becomes available. To make changes, “Open” one of the calibration Set “Road sections” and change the data for the two customized “RD Calibration Items” under the “Bituminous” and “Unsealed” items respectively.

**Table 8: HDM-4 calibration for UNRA, 2012**

| Field name | Description                | UNRA Seal on Granular                  | UNRA AC on Granular              | UNRA Gravel Type A | UNRA Earth Type A |
|------------|----------------------------|----------------------------------------|----------------------------------|--------------------|-------------------|
| SURF_CLASS | 0=Bituminous<br>1=Unsealed | 0                                      | 0                                | 1                  | 1                 |
| PAVE_TYPE  |                            | 4 (Surface treatment on Granular base) | 0 (Asphalt mix on Granular base) | 0 (Gravel)         | 1 (Earth)         |
| SURF_MATRL |                            | 8 (Double bituminous surface dressing) | 1 (hot rolled asphalt)           |                    |                   |
| SURF_THICK | Gravel surface thickness   |                                        |                                  | 100                | 0                 |



| Field name | Description                                                    | UNRA Seal<br>on Granular | UNRA AC on<br>Granular | UNRA Gravel<br>Type A | UNRA Earth<br>Type A |
|------------|----------------------------------------------------------------|--------------------------|------------------------|-----------------------|----------------------|
| SURF_D95   | Gravel wearing course:<br>Sieve size of 95% of<br>material     |                          |                        | 21.9                  | 21.9                 |
| SURF_PI    | Gravel wearing course:<br>Plasticity Index                     |                          |                        | 10.1                  | 10.1                 |
| SURF_P02   | Gravel wearing course:<br>% material through<br>2mm sieve      |                          |                        | 51.1                  | 51.1                 |
| SURF_P425  | Gravel wearing course:<br>% material through<br>0.425mm sieve  |                          |                        | 41.6                  | 41.6                 |
| SURF_P075  | Gravel wearing course:<br>% material through<br>0.075mm sieve  |                          |                        | 25.5                  | 25.5                 |
| SUBG_PI    | Subgrade:<br>Plasticity Index                                  |                          |                        | 15                    | 15                   |
| SUBG_P02   | Subgrade:<br>% material through<br>2mm sieve                   |                          |                        | 60                    | 60                   |
| SUBG_P425  | Subgrade:<br>% material through<br>0.425mm sieve               |                          |                        | 40                    | 40                   |
| SUBG_P075  | Subgrade:<br>% material through<br>0.075mm sieve               |                          |                        | 18                    | 18                   |
| SUBG_D95   | Subgrade:<br>Sieve size of 95% of<br>material                  |                          |                        | 13                    | 13                   |
| DEFECTSURF | Construction defects<br>indicator for<br>bituminous surfacings | 1                        | 1                      |                       |                      |
| DEFECTBASE | Construction defects<br>indicator for base                     | 0                        | 0                      |                       |                      |
| CRACK_CRT  | Crack retardation time<br>due to maintenance<br>(years)        | 0                        | 0                      |                       |                      |
| RAVEL_RRF  | Ravelling retardation<br>factor due to<br>maintenance          | 1                        | 1                      |                       |                      |
| ACA_INIT   | Kcia = Calibration factor<br>for all crack initiation          | 0.5                      | 0.5                    |                       |                      |
| ACA_PROG   | Kcpa = Calibration<br>factor for all crack<br>progression      | 0.2                      | 0.2                    |                       |                      |
| ACW_INIT   | Kciw = Calibration<br>factor for wide crack<br>initiation      | 0.5                      | 0.5                    |                       |                      |

| Field name | Description                                                        | UNRA Seal<br>on Granular | UNRA AC on<br>Granular | UNRA Gravel<br>Type A | UNRA Earth<br>Type A |
|------------|--------------------------------------------------------------------|--------------------------|------------------------|-----------------------|----------------------|
| ACW_PROG   | Kcpw = Calibration factor for wide crack initiation                | 0.2                      | 0.2                    |                       |                      |
| ACT_INIT   | Kcit = Calibration factor for transverse thermal crack initiation  | 1                        | 1                      |                       |                      |
| ACT_PROG   | Kcpt = Calibration factor for transverse thermal crack progression | 1                        | 1                      |                       |                      |
| RAVEL_INIT | Kvi = Calibration factor for ravelling initiation                  | 0.5                      | 0.5                    |                       |                      |
| RAVEL_PROG | Kvp = Calibration factor for ravelling progression                 | 0.5                      | 0.5                    |                       |                      |
| PHOLE_KPIC | Kpic = Pothole initiation due to wide structural cracking          | 1                        | 1                      |                       |                      |
| PHOLE_KPIR | Kpic = Pothole initiation due to ravelling                         | 1                        | 1                      |                       |                      |
| PHOLE_PROG | Kpp = Pothole progression factor                                   | 1                        | 1                      |                       |                      |
| EDGEB_PROG | Keb = Calibration factor for edge break progression                | 1                        | 1                      |                       |                      |
| TEXTD_FCTR | Ktd = Calibration factor for texture depth                         | 1                        | 1                      |                       |                      |
| SKIDR_FCTR | Ksfc = Calibration for skid resistance                             | 1                        | 1                      |                       |                      |
| SKIDR_SPED | Ksfcs = Calibration factor for skid resistance speed effects       | 1                        | 1                      |                       |                      |
| RUT_INITDN | Krid = Calibration factor for rut depth initial densification      | 1                        | 1                      |                       |                      |
| RUT_STRUCT | Krst = Calibration factor for structural deformation               | 0.6                      | 0.6                    |                       |                      |
| RUT_PLASTC | Krpd = Calibration factor for plastic deformation                  | 1                        | 1                      |                       |                      |
| PATCH_TIME | Fpat = Interval between pothole patching campaigns, in days.       | 30                       | 30                     |                       |                      |
| DRAINLIFE  | Kdrain = Calibration factor for drain life                         | 1                        | 1                      |                       |                      |

| Field name | Description                                                                   | UNRA Seal on Granular | UNRA AC on Granular | UNRA Gravel Type A | UNRA Earth Type A |
|------------|-------------------------------------------------------------------------------|-----------------------|---------------------|--------------------|-------------------|
| K_SNPk     | KnsPK = Calibration factor for SNPK in roughness model                        | 1                     | 1                   |                    |                   |
| RUT_WEAR   | Krsw = Calibration for surface wear                                           | 1                     | 1                   |                    |                   |
| RUT_KRDS   | Krds = Calibration factor for rut depth standard deviation                    | 1                     | 1                   |                    |                   |
| SNP_RATIO  | Kf = Calibration factor for wet/dry season SNP ratio                          | 1                     | 1                   |                    |                   |
| ENVIR_FCTR | Kgm = Calibration for environment coefficient in roughness model              | 1                     | 1                   |                    |                   |
| ROUGH_KGP  | Hgp = Calibration component for potholing component of roughness model        | 1                     | 1                   |                    |                   |
| ROUGH_KGS  | Kgs = Calibration factor for structural component of roughness model          | 1.0                   | 1.0                 |                    |                   |
| ROUGH_KGC  | Kgc = Calibration coefficient for cracking component of roughness             | 1                     | 1                   |                    |                   |
| ROUGH_KGR  | Kgr = Calibration component of rutting component in roughness model           | 1                     | 1                   |                    |                   |
| STUD_TYRES | Krsw = Calibration factor for surface wear due to vehicles with studded tyres | 0                     | 0                   |                    |                   |
| SALTONROAD | SALT =- Variable for salt on road: TRUE or FALSE                              | FALSE                 | FALSE               | FALSE              | FALSE             |
| DRAINAGE   | Kddf = Calibration factor for drainage factor in SNPK                         | 1                     | 1                   | 0                  | 0                 |
| UNS_KA     | Ka = Calibration factor for roughness effect for grading                      |                       |                     | 1                  | 1                 |
| UNS_KC     | Kc = Calibration factor for roughness progression                             |                       |                     | 1                  | 1                 |
| ROUGH_USER | Roughness limits user defined: TRUE or FALSE                                  |                       |                     | FALSE              | FALSE             |

| Field name | Description                                                  | UNRA Seal on Granular | UNRA AC on Granular | UNRA Gravel Type A | UNRA Earth Type A |
|------------|--------------------------------------------------------------|-----------------------|---------------------|--------------------|-------------------|
| SURFTMLOSS | Kkt = Traffic induced material loss factor of gravel surface |                       |                     | 1                  | 1                 |
| SURFGRVLOS | Kgl = Gravel loss factor of gravel surface                   |                       |                     | 1                  | 1                 |
| MINSURFIRI | Minimum allowable roughness for gravel material (m/km)       |                       |                     | 2.77               | 2.77              |
| MAXSURFIRI | Maximum allowable roughness for gravel material (m/km)       |                       |                     | 22.19              | 22.19             |
| SUBGTMLOSS | Kkt = Traffic induced material loss factor of subgrade       |                       |                     | 1                  | 1                 |
| SUBGGRVLOS | Kgl = Gravel loss factor of subgrade                         |                       |                     | 1                  | 1                 |
| MINSUBGIRI | Minimum allowable roughness for subgrade material (m/km)     |                       |                     | 1.12               | 1.12              |
| MAXSUBGIRI | Maximum allowable roughness for subgrade material (m/km)     |                       |                     | 22.52              | 22.52             |
| NUM_SHLDRS | Number of shoulders                                          | 2                     | 2                   | 2                  | 2                 |
| EDGE_STEP  | Edge break step in mm                                        | 10                    | 10                  |                    |                   |
| NMT_LTYPE  | Non motorised lane type                                      |                       |                     | 3                  | 4                 |

## 12.6. Create and run a “Strategy” analysis

**Strategy Analysis:** *This analysis entails the strategic planning of medium to long term expenditure of a road network. It is a tool to allocate resources, to forecast funding requirements for specified target maintenance standards and to forecast long term network performance under varying levels of funding.*

The first step in the analysis of UNRA’s road network is to run a strategy analysis to determine:

- the medium to long term forecasts of funding requirements,
- the long term network performance under varying levels of funding,
- optimal allocations of funding to the road network and
- road works effects on the candidate project list.

This step produces the results that will enable the PMS user to make a decision on the budget scenario to be investigated if a multi-year works programme is to be generated as a next step from the Programme Analysis.

Follow the steps hereafter to create and run a Strategy analysis:

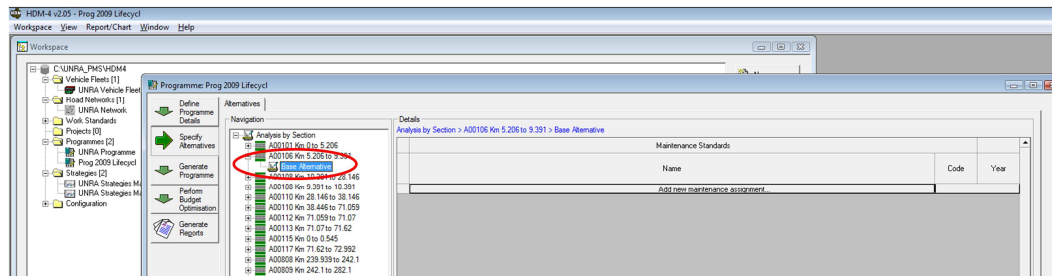
1. Click the new button within UNRA\_PMS\HDM4 workspace and select *Strategies*.

2. Enter an appropriate name, e.g. “UNRA 2011 Strat MaxdIRI”. Use the existing network UNRA Network as the road network. Click OK. (No network matrix is needed). The system will automatically choose “UNRA Vehicle Fleet”.
3. Enter a description of the analysis to be conducted.
4. Under “DEFINE STRATEGY DETAILS”: “**General tab**”:
  - Select an appropriate optimisation method for an analysis.
  - “Maximise dIRI” will maximize the change in roughness for the total road network; it has therefore the objective to maximize the change in condition.
  - “Maximise NPV” has the objective to maximize the Net Present Value of the economic benefits of the road network.
  - “Minimise agency costs” determines the minimum road agency cost necessary to achieve a specified road network condition. This option does not investigate multiple funding levels.
  - Enter the start year and analysis period. The start year should be the SURVEY YEAR+1. E.g. if the condition year in the “UNRA Network” is 2011, the analysis should start at 2012. The analysis period of road networks are typically between 10-20 years.
  - Road network should always be “UNRA Network” with attached vehicle fleet “UNRA Vehicle Fleet”.
  - Select the output currency = “US Dollar”. Verify that the “currencies” are all the same, in this case “US Dollar”.
  - Enter the discount rate, e.g. 8% or 12%.
5. Under the “DEFINE STRATEGY DETAILS”: “Study Sections” tab:
  - Select the road sections to be included in the study. Use the “Select by criteria” button to select all road sections of UNRA’s road network. A sub-network can also be selected if required.
  - Now assign Traffic Growth Sets to all the sections. All sections included in the analysis can be selected before a traffic growth set is assigned.
  - If “Minimise agency costs” is selected, make sure to add the “Target IRI” for each road section.
6. Under the “SPECIFY ALTERNATIVES”: Assign the maintenance and improvement alternatives as follows:
  - Expand the column-width of “Section\_ID” and click on the “Section\_ID” to sort the road sections according to surface type (The Section Id of Bituminous roads are prefixed with “B\_” and the Section Id of Unsealed roads are prefixed with “U\_”).

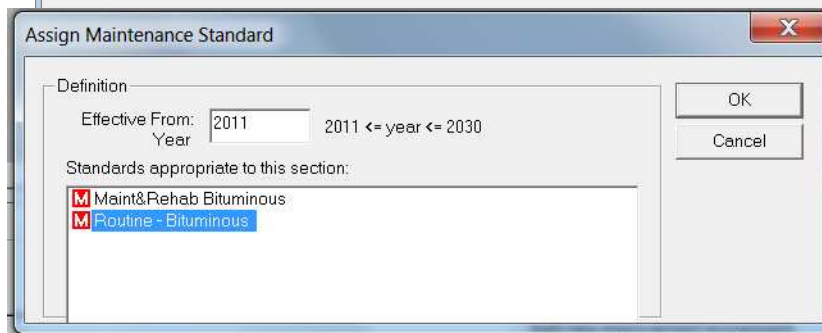
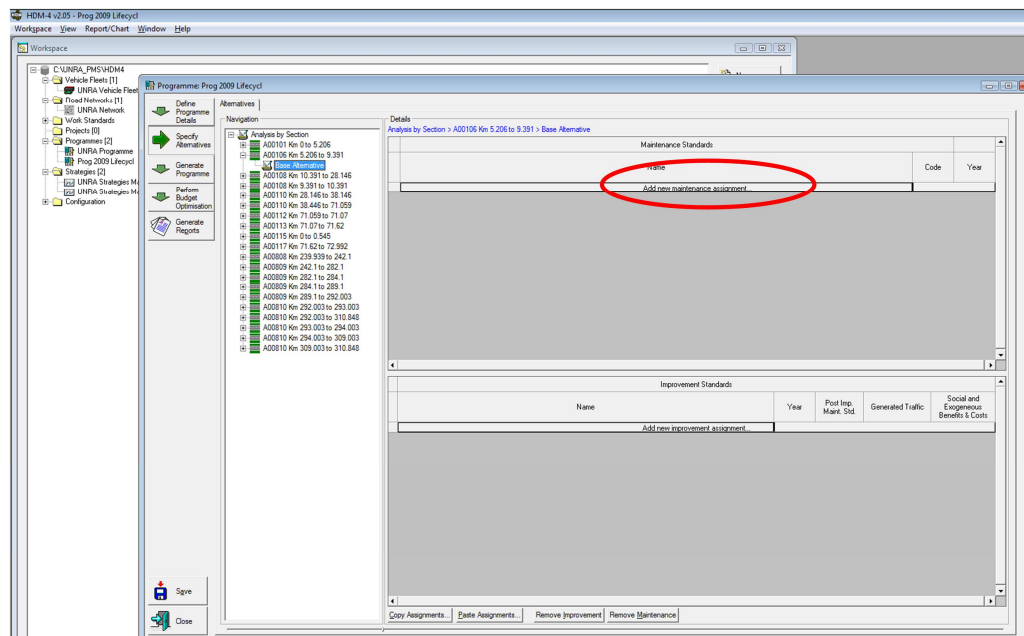
Details  
Analysis by Section

| Section Name              | Section ID         | Number Alternatives |
|---------------------------|--------------------|---------------------|
| A001_02 Km 0 to 0.1       | B_A001_02_0        | 1                   |
| A001_02 Km 0.1 to 0.2     | B_A001_02_0.1      | 1                   |
| A001_02 Km 0.2 to 0.3     | B_A001_02_0.2      | 1                   |
| A001_02 Km 0.3 to 0.4     | B_A001_02_0.3      | 1                   |
| A001_02 Km 0.4 to 0.5     | B_A001_02_0.4      | 1                   |
| A001_02 Km 0.5 to 0.6     | B_A001_02_0.5      | 1                   |
| A001_02 Km 0.6 to 0.7     | B_A001_02_0.6      | 1                   |
| A001_02 Km 0.7 to 0.8     | B_A001_02_0.7      | 1                   |
| A001_02 Km 0.8 to 0.9     | B_A001_02_0.8      | 1                   |
| A001_02 Km 0.9 to 1       | B_A001_02_0.9      | 1                   |
| A001_02 Km 1 to 1.1       | B_A001_02_1        | 1                   |
| A001_02 Km 1.1 to 1.2     | B_A001_02_1.1      | 1                   |
| A001_02 Km 1.2 to 1.3     | B_A001_02_1.2      | 1                   |
| A001_02 Km 1.3 to 1.4     | B_A001_02_1.3      | 1                   |
| A001_02 Km 1.4 to 1.5     | B_A001_02_1.4      | 1                   |
| A001_02 Km 1.5 to 1.6     | B_A001_02_1.5      | 1                   |
| A001_02 Km 1.6 to 1.7     | B_A001_02_1.6      | 1                   |
| A001_02 Km 1.7 to 1.8     | B_A001_02_1.7      | 1                   |
| A001_02 Km 1.8 to 1.9     | B_A001_02_1.8      | 1                   |
| A001_02 Km 1.9 to 2       | B_A001_02_1.9      | 1                   |
| A001_02 Km 2 to 2.1       | B_A001_02_2        | 1                   |
| A002_06 Km 0 to 24.9      | B_A002_06_000000   | 1                   |
| A002_06 Km 24.9 to 39.6   | B_A002_06_024900   | 1                   |
| A002_06 Km 39.6 to 60.9   | B_A002_06_039600   | 1                   |
| A002_06 Km 60.9 to 60.924 | B_A002_06_060900   | 1                   |
| A001_03 Km 1.8 to 1.9     | U_A001_03_1.8_2010 | 1                   |
| A001_03 Km 1.9 to 2       | U_A001_03_1.9_2010 | 1                   |
| A001_03 Km 2 to 2.1       | U_A001_03_2_2010   | 1                   |

- Select the first bituminous road in the list, then select the corresponding section on the left “Analysis by Section” list.
- Now click on the Base Alternative underneath the road id.

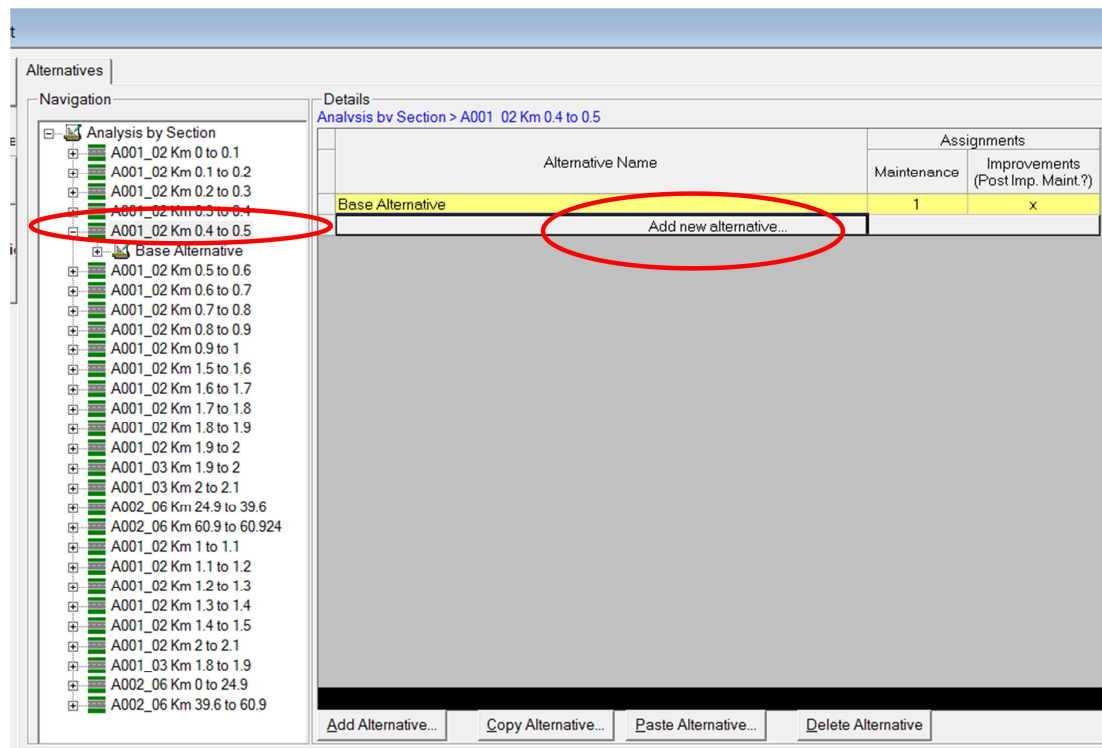


- Add a new maintenance assignment to the Base Alternative for Bituminous roads, namely *Routine – Bituminous*, effective from the first year of the analysis period. This assignment is the Base case alternative where only routine maintenance is done.

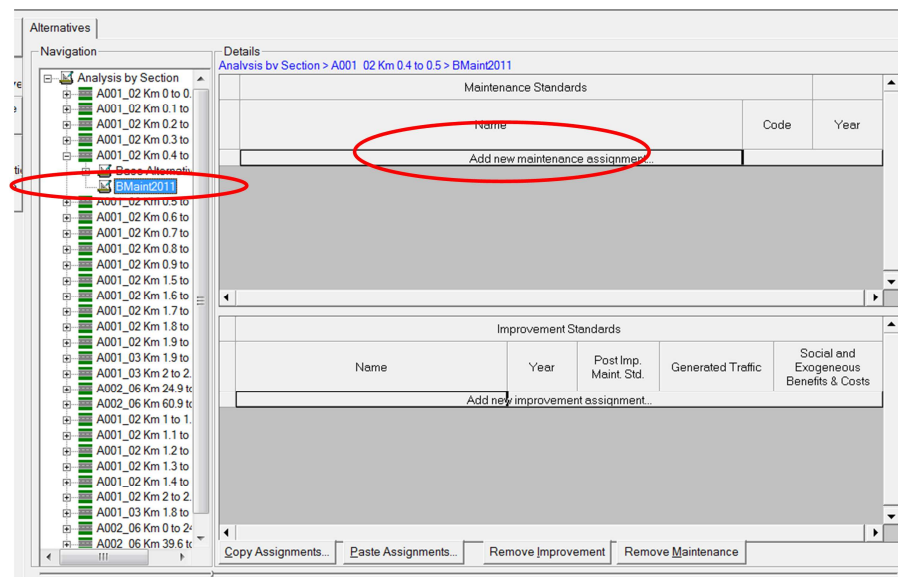


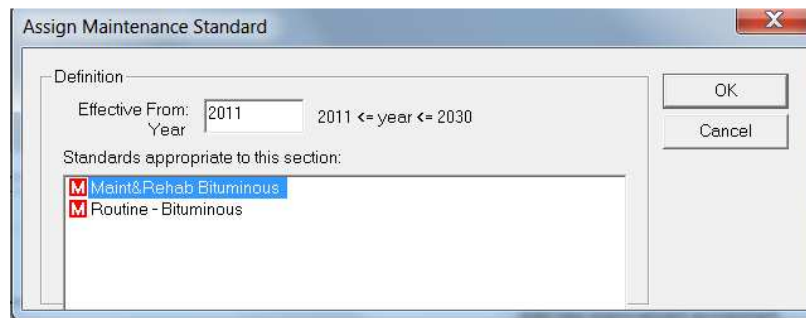
- You will now create several Maintenance alternatives starting from the first year in the analysis period. These maintenance alternatives will be duplicated for the first five, ten or fifteen years of the analysis period to provide the analysis with several works alternatives to choose from during the optimization of constraint budgets.

- Add a new alternative to the selected Bituminous road, and enter a name for the alternative: e.g. BMaint2011, to indicate this is the maintenance alternative starting in 2011 for Bituminous roads. To do this, first click on the Section\_Id in the Navigation window, then click on the tab “Add new alternative”.

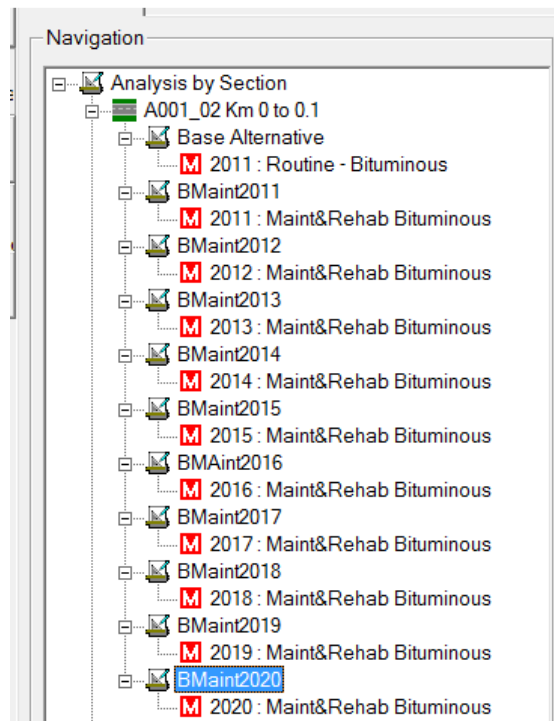


- Click now on the “BMaint2011” underneath the “Base Alternative” id and add a new maintenance assignment to this Alternative for Bituminous roads, namely *Maint&Rehab Bituminous*, effective from the first year of the analysis period. Click Ok to accept.



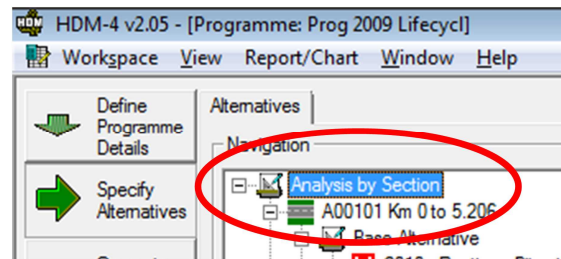


- Two Alternatives have now been created, namely a Base and a Maintenance alternative. In this example “BMaint2011” was created for the first year of the analysis period. Repeat the previous step to create more alternatives for the “Maint&Rehab Bituminous” standard for the years 2 through 10 or 15 of the analysis period. For example the alternatives “BMaint2012”, “BMaint2013”, “BMaint2014”, “BMaint2015”, “BMaint2016”, “BMaint2017”, “BMaint2018”, “BMaint2019 and “BMaint2020” will be created. The section should now have several alternatives assigned to it, see illustration below.

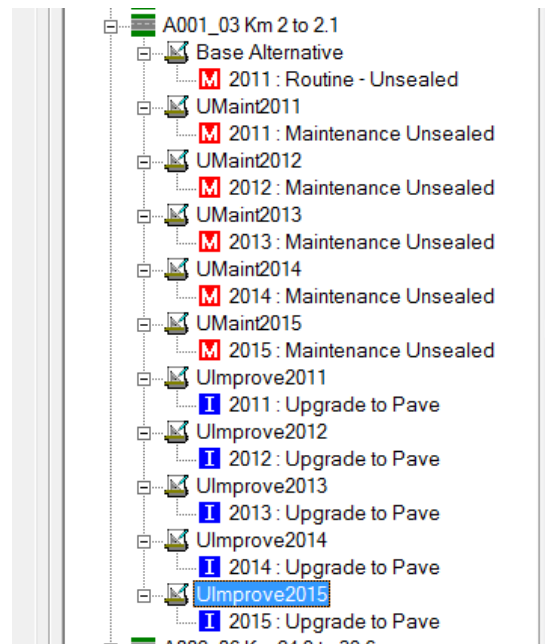


- Once the complete set of alternatives was created for the first section, the entire set will be copied and pasted onto all Bituminous sections.
  - On the navigation window, click on the top line “Analysis by Section” to open the list of sections to be analyzed.





- Use the copy and paste buttons to copy the alternatives of the bituminous section that was just completed to all other bituminous sections. The Section Id of all bituminous roads starts with a B. (Double-click on “Section Id” to sort and group the Bituminous and Unsealed sections together)
  - The number of alternatives should now be the same (6, 11 or 16) for all Bituminous roads.
8. Similar to the procedures for Bituminous roads, add the following alternatives and standards to Unsealed roads. The Section Id of all unsealed roads starts with a U\_.
- Base alternative: Name is “Base Alternative”, and maintenance assignment is “Routine-Unsealed”.
  - Create Maintenance alternatives for the first five, ten or fifteen years of the analysis period: Names are “UMaint2011, UMaint2012, etc.”, and maintenance assignment is “Maintenance Unsealed”.
  - Create Improvement alternatives for the first five, ten or fifteen years of the analysis period: Names are “UImprove2011, UImprove2012, etc.” and the improvement assignment is “Upgrade to Pave”.



- Use the copy and paste buttons to copy the alternatives of the Unsealed section to all other Unsealed sections. The Section Id of all unsealed roads starts with a U.
  - The number of alternatives should now be “11” or more for all Unsealed roads.
9. Use the “Save” to save the work so far.
10. Verify the Number Alternatives for all road sections are valid before continuing.

11. Verify all road sections to be analysed have a traffic growth set assigned before continuing.
12. Under the “GENERATE STRATEGY”: “Perform Run” tab: Change the properties under “Run Setup”:
  - Asset valuation can be included.
  - Choose an appropriate “Run Data Export Directory”. It is recommended that a folder e.g. “HDMSTRATRun\_2011\_October” is created on the analysis folder structure created earlier. (E.g. “C:\UNRA\_PMS\Work\2012LCCA\HDM4Runs” ) The sub-folder can then be deleted if the analysis is abandoned, no longer used or not applicable.
  - In Windows Explorer determine if a RunData.mdb file already exists on the “Run Data Export Directory”; If so **DELETE the file: RunData.mdb**.
  - Click “OK” when done.
13. Click on “START” to begin the analysis.
14. Wait for the analysis to complete!
15. Under the tab “Unconstrained Programme”, the optimal program can be investigated to determine additional budgets to investigate.
16. Under “PERFORM BUDGET OPTIMISATION”: Add the budget scenarios to investigate. Then click the button to “Perform the budget optimization”. Multiple budget scenarios cannot be entered for optimization type “Minimise agency costs”.
17. Reports can now be generated and exported under the “GENERATE REPORTS” tab. The user can investigate and export any of the generated reports, but it is recommended the following reports/graphs are generated and reviewed as a minimum:

| Reports and Graphs to review/investigate the modelling of each road section                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Traffic:</b><br/> <b>AADT for Section Alternatives</b><br/> <b>(Graph):</b><br/>         Use this report to view how traffic (aadt) was grown over the analysis period, for each road section in the analysis set.</p>                                                                                                                                                                                                                                                                                                                           | <p>Section: A001_02 Km 0.9 to 1<br/>         Sensitivity: No Sensitivity Analysis Conducted</p> <p><b>Annual Average Daily Traffic (AADT) for Motorised Vehicles</b></p>                                                                                                                                                            |
| <p><b>Deterioration/Works Effects:</b><br/> <b>Average Roughness by Section</b><br/> <b>(Graph):</b><br/>         Use these graphs to review the pavement performance in terms of roughness for each road section. It is not necessary to open each and every road section, but a few samples can be reviewed to ensure the deterioration and work effects are realistic.<br/>         Changes to all pavement modelling are made under “Configuration: RD Calibrations Sets”.<br/>         Changes to the effects of maintenance/improvements are</p> | <p>Section: A001_02 Km 0 to 1<br/>         Sensitivity: No Sensitivity Analysis Conducted</p> <p>ID: B_A001_02_0<br/>         Rise + Fall: 10.00m/km<br/>         Road Class: Primary or trunk<br/>         Width: 7.00m<br/>         Length: 0.10km<br/>         Curvature: 15.00deg/km</p> <p><b>Average Roughness (m/km)</b></p> |

| Reports and Graphs to review/investigate the modelling of each road section                                                   |                                                                                                                                                                                                                                                                                                                                                                                          |         |                      |               |               |                  |              |                 |              |                  |
|-------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|----------------------|---------------|---------------|------------------|--------------|-----------------|--------------|------------------|
| made under “Works Standards”.                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                          |         |                      |               |               |                  |              |                 |              |                  |
| <b>Deterioration/Works Effects:<br/>Pavement Condition<br/>Summary<br/>Or<br/>Pavement Condition<br/>(per Pavement Type):</b> | <div><div><div><div>Section:</div><div>A001_02 Km 0 to 0.1</div></div><div><div>Alternative:</div><div>Base Alternative</div></div><div><div>Sensitivity:</div><div>No Sensitivity Analysis Conducted</div></div><div><div>Surface Class:</div><div>Bituminous</div></div><div><div>Length:</div><div>0.10km</div></div></div><div>Road Class: <div></div>Width: <div></div></div></div> |         |                      |               |               |                  |              |                 |              |                  |
|                                                                                                                               | Average A                                                                                                                                                                                                                                                                                                                                                                                |         |                      |               |               |                  |              |                 |              |                  |
|                                                                                                                               | Year                                                                                                                                                                                                                                                                                                                                                                                     | MT AADT | ESAL millions /ELANE | IRI bef. m/km | IRI Avg. m/km | All Str Cracks % | Rave-lling % | Edge Break sq.m | Rut Depth mm | No. of Pot-holes |
| 2011                                                                                                                          | 5,796                                                                                                                                                                                                                                                                                                                                                                                    | 0.37    | 6.70                 | 6.35          | 12.50         | 2.51             | 20.89        | 1.17            | 255.89       | 2.84             |
| 2012                                                                                                                          | 5,970                                                                                                                                                                                                                                                                                                                                                                                    | 0.39    | 7.68                 | 7.19          | 15.91         | 6.89             | 56.11        | 1.35            | 329.46       | 2.78             |
| 2013                                                                                                                          | 6,149                                                                                                                                                                                                                                                                                                                                                                                    | 0.40    | 9.07                 | 8.38          | 19.81         | 14.69            | 93.93        | 1.53            | 426.90       | 2.67             |
| 2014                                                                                                                          | 6,334                                                                                                                                                                                                                                                                                                                                                                                    | 0.41    | 11.07                | 10.07         | 24.24         | 26.89            | 135.31       | 1.72            | 556.80       | 2.65             |
| 2015                                                                                                                          | 6,524                                                                                                                                                                                                                                                                                                                                                                                    | 0.42    | 14.01                | 12.54         | 29.22         | 44.38            | 182.38       | 1.92            | 731.13       | 2.63             |
| 2016                                                                                                                          | 6,720                                                                                                                                                                                                                                                                                                                                                                                    | 0.43    | 16.00                | 15.00         | 34.74         | 60.39            | 239.68       | 2.13            | 1,007.50     | 2.49             |
| 2017                                                                                                                          | 6,921                                                                                                                                                                                                                                                                                                                                                                                    | 0.45    | 16.00                | 16.00         | 40.70         | 52.73            | 310.46       | 2.35            | 1,496.22     | 2.33             |
| 2018                                                                                                                          | 7,129                                                                                                                                                                                                                                                                                                                                                                                    | 0.46    | 16.00                | 16.00         | 47.21         | 44.10            | 390.26       | 2.60            | 2,176.54     | 2.28             |
| 2019                                                                                                                          | 7,343                                                                                                                                                                                                                                                                                                                                                                                    | 0.47    | 16.00                | 16.00         | 52.08         | 36.66            | 475.03       | 2.85            | 3,128.01     | 2.24             |
| 2020                                                                                                                          | 7,710                                                                                                                                                                                                                                                                                                                                                                                    | 0.50    | 16.00                | 16.00         | 56.53         | 28.93            | 568.60       | 3.12            | 4,490.51     | 2.06             |
| 2021                                                                                                                          | 8,095                                                                                                                                                                                                                                                                                                                                                                                    | 0.52    | 16.00                | 16.00         | 60.57         | 20.61            | 671.99       | 3.41            | 5,459.74     | 1.85             |
| 2022                                                                                                                          | 8,500                                                                                                                                                                                                                                                                                                                                                                                    | 0.55    | 16.00                | 16.00         | 64.47         | 14.29            | 786.23       | 3.75            | 7,000.00     | 1.84             |
| 2023                                                                                                                          | 8,925                                                                                                                                                                                                                                                                                                                                                                                    | 0.58    | 16.00                | 16.00         | 68.09         | 8.87             | 912.48       | 4.06            | 7,000.00     | 1.86             |
| 2024                                                                                                                          | 9,371                                                                                                                                                                                                                                                                                                                                                                                    | 0.60    | 16.00                | 16.00         | 71.36         | 3.61             | 1,052.03     | 4.40            | 7,000.00     | 1.71             |
| 2025                                                                                                                          | 9,840                                                                                                                                                                                                                                                                                                                                                                                    | 0.64    | 16.00                | 16.00         | 72.77         | 0.00             | 1,206.28     | 4.77            | 7,000.00     | 1.58             |
| 2026                                                                                                                          | 10,332                                                                                                                                                                                                                                                                                                                                                                                   | 0.67    | 16.00                | 16.00         | 72.00         | 0.00             | 1,260.00     | 5.16            | 7,000.00     | 1.69             |

Deterioration/Works Effects:  
Bituminous  
Pavement Condition (Graph)  
Or  
Unsealed Pavement Condition (Graph):

Use these graphs to view the pavements performance values for roughness, cracking, potholes, ravelling, edge break, rutting, structural number and gravel thickness for each maintenance alternative.

Changes to all pavement modelling are made under “Configuration: RD Calibrations Sets”.

Changes to the effects of maintenance/improvements are made under “Works Standards”.

Section:

A001\_02 Km 0.2 to 0.3

Alternative:

Maint Alternative

Sensitivity:

No Sensitivity Analysis Conducted

Length: 0.10kmWidth: 7.00mRise + Fall: 10.00m/kmCurvature: 15.00deg/kmRoad Class: Primary or trunk

Progression of Cracking over time  
(after works values)

% Cracking

2011

2012

2013

2014

2015

2016

2017

2018

2019

2020

2021

2022

2023

2024

2025

2026

2027

2028

2029

2030

■ Total Area of Cracking (ACRA)

Progression of Average Roughness over time

Average Roughness (IRI), m/km

2011

2012

2013

2014

2015

2016

2017

2018

2019

2020

2021

2022

2023

2024

2025

2026

2027

2028

2029

2030

■ Average Roughness

### Reports and Graphs to review/investigate the modelling of each road section

#### Deterioration/Works Effects: Road Works Summary (by Year or by Section):

Lists all works (routine, maintenance and improvements) per road section of per year under the different maintenance alternatives. This information is used to investigate the works and costs for each alternative.

| <b>Section:</b> A001_02 Km 0.1 to 0.2<br><b>Alternative:</b> BMaint2014<br><b>Sensitivity:</b> No Sensitivity Analysis Conducted<br><b>Surface Class:</b> Bituminous<br><b>Length:</b> 0.10km<br><b>Road Class:</b> Primary or trunk<br><b>Width:</b> 7.00m |                    |       |                    |                    |               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-------|--------------------|--------------------|---------------|
| Year                                                                                                                                                                                                                                                        | Description        | Code  | Economic Cost      | Financial Cost     | Work Quantity |
| 2014                                                                                                                                                                                                                                                        | Routine            | ROUT  | 2,210.0            | 2,890.0            | 0.10 km       |
|                                                                                                                                                                                                                                                             | Recon/HeavyRehab   | RECON | 809,200.0          | 1,190,000.0        | 700.00 sq. m  |
| 2015                                                                                                                                                                                                                                                        | Routine            | ROUT  | 2,210.0            | 2,890.0            | 0.10 km       |
|                                                                                                                                                                                                                                                             | Patching           | PATCH | 0.0                | 0.0                | 0.00 sq. m    |
| 2016                                                                                                                                                                                                                                                        | Routine            | ROUT  | 2,210.0            | 2,890.0            | 0.10 km       |
| 2017                                                                                                                                                                                                                                                        | Routine            | ROUT  | 2,210.0            | 2,890.0            | 0.10 km       |
| 2018                                                                                                                                                                                                                                                        | Routine            | ROUT  | 2,210.0            | 2,890.0            | 0.10 km       |
| 2019                                                                                                                                                                                                                                                        | Routine            | ROUT  | 2,210.0            | 2,890.0            | 0.10 km       |
| 2020                                                                                                                                                                                                                                                        | Routine            | ROUT  | 2,210.0            | 2,890.0            | 0.10 km       |
| 2021                                                                                                                                                                                                                                                        | Routine            | ROUT  | 2,210.0            | 2,890.0            | 0.10 km       |
| 2022                                                                                                                                                                                                                                                        | Routine            | ROUT  | 2,210.0            | 2,890.0            | 0.10 km       |
| 2023                                                                                                                                                                                                                                                        | Routine            | ROUT  | 2,210.0            | 2,890.0            | 0.10 km       |
| 2024                                                                                                                                                                                                                                                        | Routine            | ROUT  | 2,210.0            | 2,890.0            | 0.10 km       |
| 2025                                                                                                                                                                                                                                                        | Routine            | ROUT  | 2,210.0            | 2,890.0            | 0.10 km       |
|                                                                                                                                                                                                                                                             | Reseal             | RSEAL | 107,100.0          | 160,650.0          | 700.00 sq. m  |
|                                                                                                                                                                                                                                                             | Prep. Edge Repair  |       | 83,122.3           | 122,238.7          | 71.91 sq. m   |
| 2026                                                                                                                                                                                                                                                        | Routine            | ROUT  | 2,210.0            | 2,890.0            | 0.10 km       |
| 2027                                                                                                                                                                                                                                                        | Routine            | ROUT  | 2,210.0            | 2,890.0            | 0.10 km       |
| 2028                                                                                                                                                                                                                                                        | Routine            | ROUT  | 2,210.0            | 2,890.0            | 0.10 km       |
| 2029                                                                                                                                                                                                                                                        | Routine            | ROUT  | 2,210.0            | 2,890.0            | 0.10 km       |
|                                                                                                                                                                                                                                                             | Overlay/LightRehab | OVERL | 476,000.0          | 714,000.0          | 700.00 sq. m  |
|                                                                                                                                                                                                                                                             | Prep. Edge Repair  |       | 58,208.0           | 85,600.0           | 50.35 sq. m   |
| 2030                                                                                                                                                                                                                                                        | Routine            | ROUT  | 2,210.0            | 2,890.0            | 0.10 km       |
| <b>Total cost for the section:</b>                                                                                                                                                                                                                          |                    |       | <b>1,571,200.3</b> | <b>2,321,618.7</b> |               |

#### Cost Streams and Economic Evaluation: Economic Indicators Summary:

Lists, per section, the economic indicators of each maintenance alternative. This information can be used to investigate the benefits in terms of costs and IRR.

| <b>Section:</b> A001_02 Km 0.5 to 0.6<br><b>Sensitivity:</b> No Sensitivity Analysis Conducted |                                           |                                             |                              |                            |                            |                                     |                          |                          |                               |
|------------------------------------------------------------------------------------------------|-------------------------------------------|---------------------------------------------|------------------------------|----------------------------|----------------------------|-------------------------------------|--------------------------|--------------------------|-------------------------------|
| Alternative                                                                                    | Present Value of Total Agency Costs (RAC) | Present Value of Agency Capital Costs (CAP) | Increase in Agency Costs (C) | Decrease in User Costs (B) | Net Exogenous Benefits (E) | Net Present Value (NPV = B + E - C) | NPV/Cost Ratio (NPV/RAC) | NPV/Cost Ratio (NPV/CAP) | Internal Rate of Return (IRR) |
| Base Alternative                                                                               | 0.072                                     | 0.000                                       | 0.000                        | 0.000                      | 0.000                      | 0.000                               | 0.000                    | 0.000                    | 0.000                         |
| BMaint2011                                                                                     | 1.405                                     | 1.314                                       | 1.370                        | 12.438                     | 0.000                      | 11.068                              | 7.629                    | 8.403                    | 42.7 (1)                      |
| BMaint2012                                                                                     | 1.442                                     | 1.303                                       | 1.370                        | 12.347                     | 0.000                      | 10.976                              | 7.611                    | 8.423                    | 54.1 (1)                      |
| BMaint2013                                                                                     | 1.436                                     | 1.290                                       | 1.364                        | 12.143                     | 0.000                      | 10.779                              | 7.596                    | 8.354                    | 70.0 (1)                      |
| BMaint2014                                                                                     | 1.431                                     | 1.277                                       | 1.359                        | 11.660                     | 0.000                      | 10.306                              | 7.249                    | 8.099                    | 81.4 (2)                      |
| BMaint2015                                                                                     | 1.427                                     | 1.264                                       | 1.355                        | 11.075                     | 0.000                      | 9.720                               | 6.811                    | 7.690                    | 88.3 (2)                      |

Figure in brackets is number of IRR solutions in range -90 to +900

| <b>Section:</b> A001_02 Km 0.6 to 0.7<br><b>Sensitivity:</b> No Sensitivity Analysis Conducted |                                           |                                             |                              |                            |                            |                                     |                          |                          |                               |
|------------------------------------------------------------------------------------------------|-------------------------------------------|---------------------------------------------|------------------------------|----------------------------|----------------------------|-------------------------------------|--------------------------|--------------------------|-------------------------------|
| Alternative                                                                                    | Present Value of Total Agency Costs (RAC) | Present Value of Agency Capital Costs (CAP) | Increase in Agency Costs (C) | Decrease in User Costs (B) | Net Exogenous Benefits (E) | Net Present Value (NPV = B + E - C) | NPV/Cost Ratio (NPV/RAC) | NPV/Cost Ratio (NPV/CAP) | Internal Rate of Return (IRR) |
| Base Alternative                                                                               | 0.072                                     | 0.000                                       | 0.000                        | 0.000                      | 0.000                      | 0.000                               | 0.000                    | 0.000                    | 0.000                         |
| BMaint2011                                                                                     | 0.660                                     | 0.570                                       | 0.589                        | 9.941                      | 0.000                      | 9.352                               | 14.159                   | 16.404                   | 34.6 (1)                      |
| BMaint2012                                                                                     | 0.655                                     | 0.565                                       | 0.584                        | 9.858                      | 0.000                      | 9.275                               | 14.153                   | 16.404                   | 38.3 (1)                      |
| BMaint2013                                                                                     | 1.436                                     | 1.290                                       | 1.364                        | 9.830                      | 0.000                      | 8.466                               | 5.886                    | 6.561                    | 32.2 (1)                      |
| BMaint2014                                                                                     | 1.429                                     | 1.277                                       | 1.357                        | 9.736                      | 0.000                      | 8.379                               | 5.866                    | 6.564                    | 37.2 (2)                      |
| BMaint2015                                                                                     | 1.425                                     | 1.264                                       | 1.353                        | 9.631                      | 0.000                      | 8.278                               | 5.811                    | 6.549                    | 44.3 (2)                      |

Figure in brackets is number of IRR solutions in range -90 to +900

#### Cost Streams and Economic Evaluation: Road Agency and User Cost Streams (Undiscounted):

Lists, per section, the cost streams to the Road Agency and Road Users. Total Transport Cost (an important economic indicator) is given for each maintenance alternative.

| <b>Section:</b> A001_02 Km 0 to 0.1<br><b>Alternative:</b> BMaint2011<br><b>Sensitivity:</b> No Sensitivity Analysis Conducted |                         |              |                              |              |                       |                         |                        |              |                             |                      |               |
|--------------------------------------------------------------------------------------------------------------------------------|-------------------------|--------------|------------------------------|--------------|-----------------------|-------------------------|------------------------|--------------|-----------------------------|----------------------|---------------|
| Sect ID: B_A001_02_0                                                                                                           |                         |              | Road Class: Primary or trunk |              |                       | Curvature: 15.00 deg/km |                        |              |                             |                      |               |
| Length: 0.10 km                                                                                                                |                         |              | Width: 7.00 m                |              |                       | Rise+Fall: 10.00 m/km   |                        |              |                             |                      |               |
| Year                                                                                                                           | Road Agency Costs (RAC) |              |                              |              | Road User Costs (RUC) |                         |                        |              | Net Social / Exogenous Cost | Total Transport Cost |               |
|                                                                                                                                | Capital                 | Recurrent    | Special                      | Total RAC    | MT Vehicle Operation  | MT Travel Time          | NMT Travel & Operation | Accidents    |                             |                      | Total RUC     |
| 2011                                                                                                                           | 0.809                   | 0.002        | 0.000                        | 0.811        | 0.408                 | 0.096                   | 0.000                  | 0.000        | 1.104                       | 0.000                | 1.915         |
| 2012                                                                                                                           | 0.000                   | 0.002        | 0.000                        | 0.002        | 0.367                 | 0.713                   | 0.000                  | 0.000        | 1.080                       | 0.000                | 1.083         |
| 2013                                                                                                                           | 0.000                   | 0.002        | 0.000                        | 0.002        | 0.379                 | 0.735                   | 0.000                  | 0.000        | 1.114                       | 0.000                | 1.116         |
| 2014                                                                                                                           | 0.000                   | 0.002        | 0.000                        | 0.002        | 0.392                 | 0.757                   | 0.000                  | 0.000        | 1.150                       | 0.000                | 1.152         |
| 2015                                                                                                                           | 0.000                   | 0.002        | 0.000                        | 0.002        | 0.406                 | 0.781                   | 0.000                  | 0.000        | 1.187                       | 0.000                | 1.190         |
| 2016                                                                                                                           | 0.000                   | 0.002        | 0.000                        | 0.002        | 0.421                 | 0.805                   | 0.000                  | 0.000        | 1.226                       | 0.000                | 1.228         |
| 2017                                                                                                                           | 0.000                   | 0.002        | 0.000                        | 0.002        | 0.436                 | 0.830                   | 0.000                  | 0.000        | 1.269                       | 0.000                | 1.269         |
| 2018                                                                                                                           | 0.000                   | 0.002        | 0.000                        | 0.002        | 0.452                 | 0.856                   | 0.000                  | 0.000        | 1.308                       | 0.000                | 1.310         |
| 2019                                                                                                                           | 0.000                   | 0.002        | 0.000                        | 0.002        | 0.469                 | 0.883                   | 0.000                  | 0.000        | 1.351                       | 0.000                | 1.354         |
| 2020                                                                                                                           | 0.000                   | 0.002        | 0.000                        | 0.002        | 0.495                 | 0.929                   | 0.000                  | 0.000        | 1.424                       | 0.000                | 1.426         |
| 2021                                                                                                                           | 0.107                   | 0.057        | 0.000                        | 0.164        | 0.524                 | 0.977                   | 0.000                  | 0.000        | 1.501                       | 0.000                | 1.665         |
| 2022                                                                                                                           | 0.000                   | 0.002        | 0.000                        | 0.002        | 0.555                 | 1.029                   | 0.000                  | 0.000        | 1.583                       | 0.000                | 1.586         |
| 2023                                                                                                                           | 0.000                   | 0.002        | 0.000                        | 0.002        | 0.587                 | 1.083                   | 0.000                  | 0.000        | 1.669                       | 0.000                | 1.671         |
| 2024                                                                                                                           | 0.000                   | 0.002        | 0.000                        | 0.002        | 0.621                 | 1.140                   | 0.000                  | 0.000        | 1.760                       | 0.000                | 1.763         |
| 2025                                                                                                                           | 0.000                   | 0.002        | 0.000                        | 0.002        | 0.657                 | 1.200                   | 0.000                  | 0.000        | 1.857                       | 0.000                | 1.860         |
| 2026                                                                                                                           | 0.000                   | 0.002        | 0.000                        | 0.002        | 0.697                 | 1.264                   | 0.000                  | 0.000        | 1.960                       | 0.000                | 1.963         |
| 2027                                                                                                                           | 0.476                   | 0.067        | 0.000                        | 0.543        | 0.739                 | 1.331                   | 0.000                  | 0.000        | 2.070                       | 0.000                | 2.613         |
| 2028                                                                                                                           | 0.000                   | 0.002        | 0.000                        | 0.002        | 0.701                 | 1.399                   | 0.000                  | 0.000        | 2.100                       | 0.000                | 2.102         |
| 2029                                                                                                                           | 0.000                   | 0.002        | 0.000                        | 0.002        | 0.737                 | 1.473                   | 0.000                  | 0.000        | 2.210                       | 0.000                | 2.212         |
| 2030                                                                                                                           | 0.000                   | 0.002        | 0.000                        | 0.002        | 0.774                 | 1.552                   | 0.000                  | 0.000        | 2.326                       | 0.000                | 2.328         |
| <b>Total:</b>                                                                                                                  | <b>1.392</b>            | <b>0.164</b> | <b>0.000</b>                 | <b>1.556</b> | <b>10.819</b>         | <b>20.431</b>           | <b>0.000</b>           | <b>0.000</b> | <b>31.250</b>               | <b>0.000</b>         | <b>32.806</b> |

### Reports and Graphs to investigate and summarise the analysis results for reporting purposes

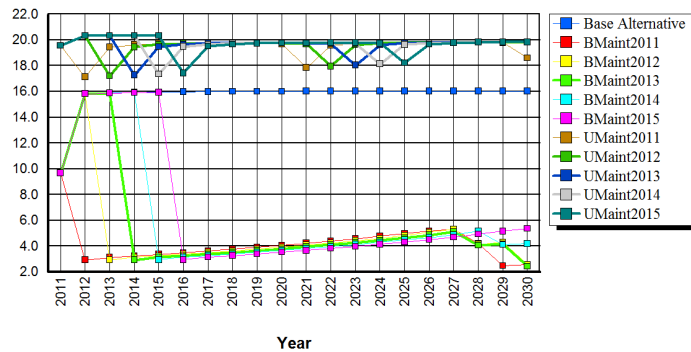
#### Deterioration/Works Effects: Average Roughness by Project (Graph):

Use these graphs to review the average road network pavement performance in terms of roughness (for all sections weighted by length). This graph gives an overview of the expected network performance under each maintenance alternative.

Changes to all pavement modelling are made under “Configuration: RD Calibrations Sets”.

Changes to the effects of maintenance/improvements are made under “Works Standards”.

#### Average Roughness (IRIav) for Project (weighted by section length)



#### Programme and Strategy

##### Analysis:

#### Works Programme Optimised (or Unconstrained) by Section (or Year):

The Works programme for various specific budget scenarios can be generated and exported to Excel files. The results can then be summarised into a graph or table summarising the annual costs and length achieved by each budgets scenario. See sample graphs attached.

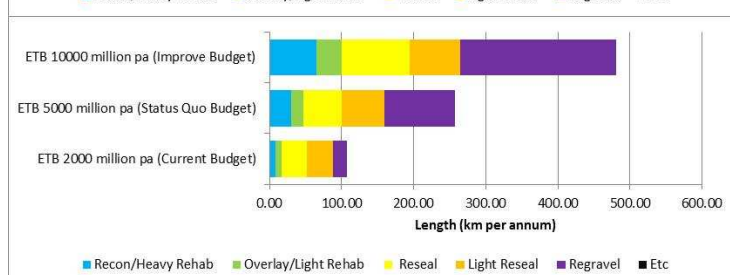
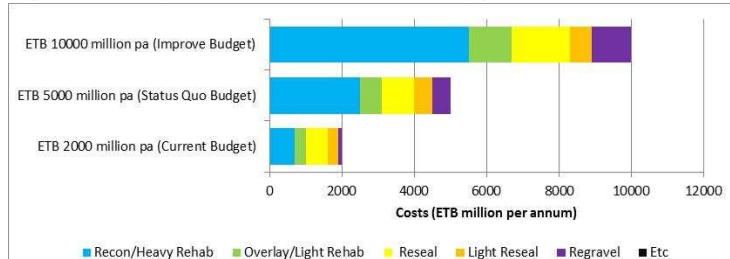
This works programme is also used to compile candidate a multi-year programme with costs. The candidate programme will be used in the next step to prioritise a multi-year works programme under defined budget constraints.

***This programme, BY SECTION, should be exported for display purposes in GIS.***

***Paragraph 12.10 has further details on how to manipulate this Works Programme for importing into UNRA's dTIMS database.***

Budget Scenario: ETB200 mil

| Section                   | Road Class       | Length (km) | AADT  | Surface Class | Year | Work Description   | NPV/CAP | Financial Costs | Cumulative Costs |
|---------------------------|------------------|-------------|-------|---------------|------|--------------------|---------|-----------------|------------------|
| A001_02 Km 0 to 0.1       | Primary or trunk | 0.1         | 5796  | Bituminous    | 2011 | Recon/HeavyRehab   | 7.855   | 1.19            | 1.190            |
| A001_02 Km 0.1 to 0.2     | Primary or trunk | 0.1         | 5796  | Bituminous    | 2011 | Recon/HeavyRehab   | 9.041   | 1.19            | 2.380            |
| A001_02 Km 0.2 to 0.3     | Primary or trunk | 0.1         | 5796  | Bituminous    | 2011 | Recon/HeavyRehab   | 9.222   | 0.95            | 3.332            |
| A001_02 Km 0.3 to 0.4     | Primary or trunk | 0.1         | 5796  | Bituminous    | 2011 | Recon/HeavyRehab   | 9.343   | 0.95            | 4.284            |
| A001_02 Km 0.4 to 0.5     | Primary or trunk | 0.1         | 5796  | Bituminous    | 2011 | Recon/HeavyRehab   | 8.966   | 0.95            | 5.236            |
| A001_02 Km 0.5 to 0.6     | Primary or trunk | 0.1         | 5796  | Bituminous    | 2011 | Recon/HeavyRehab   | 9.174   | 0.95            | 6.188            |
| A001_02 Km 0.6 to 0.7     | Primary or trunk | 0.1         | 5796  | Bituminous    | 2011 | Recon/HeavyRehab   | 9.279   | 149.94          | 156.126          |
| A002_06 Km 60.9 to 60.924 | Primary or trunk | 0.0         | 5150  | Bituminous    | 2011 | Recon/HeavyRehab   | 8.280   | 0.24            | 156.373          |
| A001_02 Km 1.4 to 1.5     | Primary or trunk | 0.1         | 5796  | Bituminous    | 2011 | Recon/HeavyRehab   | 9.035   | 0.95            | 157.325          |
| A001_02 Km 2 to 2.1       | Primary or trunk | 0.1         | 5796  | Bituminous    | 2011 | Recon/HeavyRehab   | 9.697   | 0.95            | 158.277          |
| A001_02 Km 2.1 to 2.2     | Primary or trunk | 0.1         | 5796  | Bituminous    | 2011 | Recon/HeavyRehab   | 9.141   | 1.19            | 159.467          |
| A001_02 Km 0.3 to 0.4     | Primary or trunk | 0.1         | 5796  | Bituminous    | 2011 | Recon/HeavyRehab   | 9.141   | 1.19            | 160.657          |
| A001_02 Km 0.4 to 0.5     | Primary or trunk | 0.1         | 5796  | Bituminous    | 2011 | Recon/HeavyRehab   | 9.172   | 1.19            | 161.847          |
| A001_02 Km 0.5 to 0.6     | Primary or trunk | 0.1         | 5796  | Bituminous    | 2011 | Recon/HeavyRehab   | 8.403   | 1.19            | 163.037          |
| A001_02 Km 0.6 to 0.7     | Primary or trunk | 0.1         | 5796  | Bituminous    | 2011 | Recon/HeavyRehab   | 16.404  | 0.71            | 163.751          |
| A001_02 Km 0.7 to 0.8     | Primary or trunk | 0.1         | 5796  | Bituminous    | 2011 | Recon/HeavyRehab   | 7.763   | 1.19            | 164.941          |
| A001_02 Km 0.8 to 0.9     | Primary or trunk | 0.1         | 5796  | Bituminous    | 2011 | Recon/HeavyRehab   | 8.439   | 1.19            | 166.131          |
| A001_02 Km 1 to 1.1       | Primary or trunk | 0.1         | 5796  | Bituminous    | 2011 | Recon/HeavyRehab   | 16.803  | 0.57            | 166.702          |
| A001_02 Km 1.1 to 1.2     | Primary or trunk | 0.1         | 5796  | Bituminous    | 2011 | Recon/HeavyRehab   | 9.697   | 0.95            | 167.654          |
| A001_02 Km 1.2 to 1.3     | Primary or trunk | 0.1         | 5796  | Bituminous    | 2011 | Recon/HeavyRehab   | 9.696   | 0.95            | 168.606          |
| A001_02 Km 1.3 to 1.4     | Primary or trunk | 0.1         | 5796  | Bituminous    | 2011 | Recon/HeavyRehab   | 9.697   | 0.95            | 169.558          |
| A001_02 Km 0.9 to 1       | Primary or trunk | 0.1         | 5970  | Bituminous    | 2012 | Reseal             | 14.418  | 0.16            | 169.719          |
| A001_02 Km 1.9 to 2       | Primary or trunk | 0.1         | 6149  | Bituminous    | 2013 | Overlay/LightRehab | 13.643  | 0.57            | 170.290          |
| A001_02 Km 0.9 to 1       | Primary or trunk | 0.1         | 7128  | Bituminous    | 2018 | Overlay/LightRehab | 14.418  | 0.71            | 171.004          |
| A001_02 Km 0.2 to 0.3     | Primary or trunk | 0.1         | 10848 | Bituminous    | 2027 | Overlay/LightRehab | 9.141   | 0.71            | 171.718          |
| A001_02 Km 0.3 to 0.4     | Primary or trunk | 0.1         | 10848 | Bituminous    | 2027 | Overlay/LightRehab | 9.141   | 0.71            | 172.432          |



### Reports and Graphs to investigate and summarise the analysis results for reporting purposes

#### Programme and Strategy

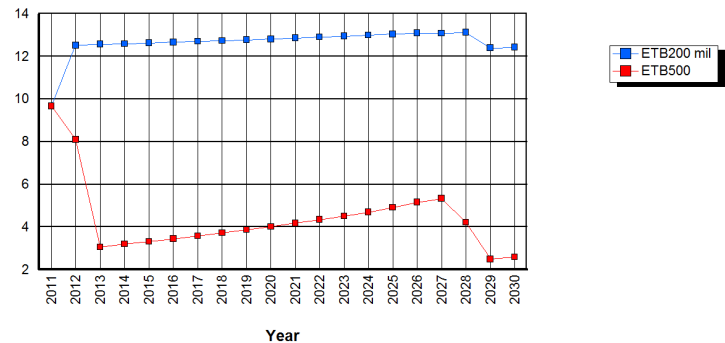
##### Analysis:

#### Roughness: Average for Road Network by Budget Scenario (Graph):

This graph presents the average network condition, in terms of roughness, for each budget scenario investigated. The overall performance of the road network can thus be investigated under different budgets (funding levels).

Bituminous

Annual Average Roughness for each Surface Class of the Optimised Work Programme (weighted by length)



#### Programme and Strategy

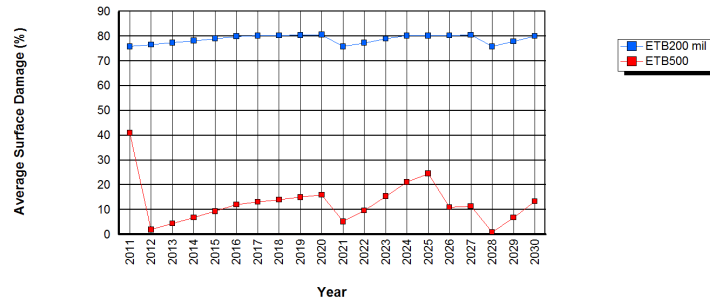
##### Analysis:

#### Pavement Surface Condition Summary by Budget Scenario (Graph):

This graph presents the average network condition, in terms of average surface damage, for each budget scenario investigated, on a link by link basis. The surface condition can thus be investigated under different budgets (funding levels).

Link ID: A002\_06

Annual Average Surface Damage for Link IDs of the Optimised Work Programme (weighted by length)



#### Asset Valuation:

#### Asset Valuation by Section (Graph and Alternative Graph):

##### Asset Value report:

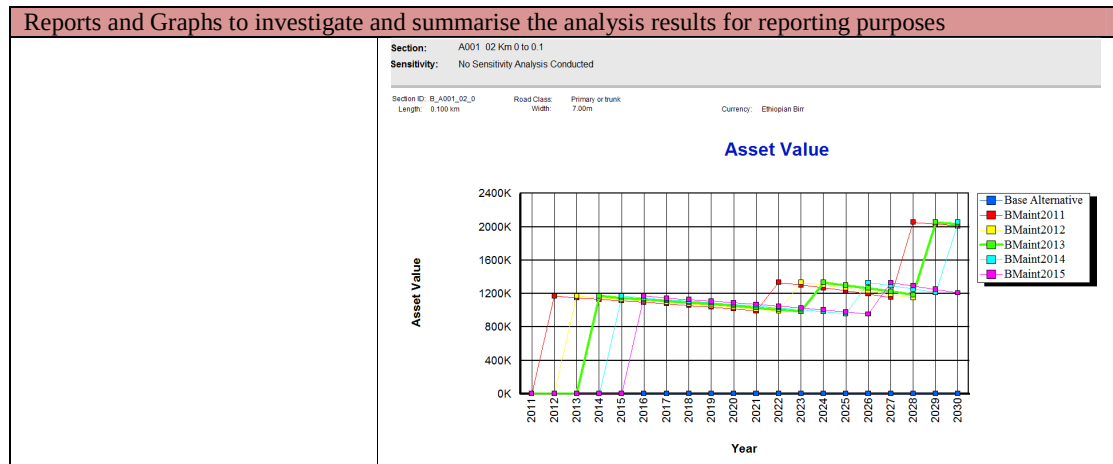
Use these graphs and report to investigate the increase in asset due to expenditure and the asset value change due to different funding investments.

Section: A001\_02 Km 0 to 0.1  
Alternative: BMain2012  
Sensitivity: No Sensitivity Analysis Conducted

Section ID: B\_A001\_02\_0 Road Class: Primary or trunk Length: 0.100 km Width: 7.00m Currency: Ethiopian Birr

#### Asset Value





18. Save and close the analysis on completion.

### 12.7. “Programmes” analysis: Differences between “Multi-Year Forward Programme” and “Life-cycle” analysis type

Programme Analysis: A list of candidate projects can be generated from the Strategy Analysis or it can be based on some criteria, e.g. reseal when pavement surface damage is more than 20%. The objective of a Programme Analysis is to prioritize a long list of candidate road projects into a multi-year or one-year working programme under defined budget constraints.

The set of investment options is user-defined and does not include the entire road network; the analysis is therefore not true optimisation as is the case with the Strategy analysis in HDM-4.

| Difference between analysis types in Programmes Analysis                                                           |                                                                                                                                                                                                                                                                                               |
|--------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| “Life-cycle”analysis                                                                                               | “Multi-Year Forward Programme”analysis                                                                                                                                                                                                                                                        |
| For the development of an unconstrained work programme or a constrained works programme under budget optimisation. | For the development of a one or multi-year forward programme. The analysis is typical a simplification of the life cycle comparing the alternatives to a) either assign the road works in the budget period or b) postpone the road works until after the first year after the budget period. |
| Maintenance and improvement assignments are selected over a period of 10-15 years.                                 |                                                                                                                                                                                                                                                                                               |
| Analysis period = 10 – 15 years                                                                                    | Analysis period = 1-5 years                                                                                                                                                                                                                                                                   |

### 12.8. Create and run a “Programmes” analysis with “Multi-Year Forward Programme” analysis type

During this step the candidate projects are refined to prepare a multi-year rolling works programme, subject to resource constraints. If there is insufficient budget, works are post-poned and not included in the programme.

1. Click the new button within the workspace and select *Programme*.
2. Enter an appropriate name, e.g. “UNRA 2011 Prog MYF” and select *UNRA Network* as the road network. Click OK.
3. Under “DEFINE PROGRAMME DETAILS”: “General tab”:
  - The type of analysis = Multi-Year Forward Programme.
  - Enter the start year and analysis period. Period is typically between 3 and 5 years.



- Road network should always be UNRA Network.
  - Select the output currency = US Dollars.
  - Enter the discount rate, e.g. 10%.
4. Under the “DEFINE PROGRAMME DETAILS”: “Study Sections” tab:
    - Select the road sections to be included in the study. Include only the candidate projects to be investigated for inclusion in the works programme
    - Assign Traffic Growth Sets to the selected sections.
  5. Under the “SPECIFY ALTERNATIVES”: Assign the maintenance and improvement alternatives as follows:
    - Assign “MaintB” and “MaintU” as the maintenance alternative to “Bituminous” and “Unsealed” roads respectively.
    - Assign “UPGRAD” as the improvement alternative to “Unsealed” roads with alternative to upgrade-to-pave . Maintenance after Improvement should be “MaintB” to maintain the upgraded road.

| Work Programming        |               |                  |             |             |                               |
|-------------------------|---------------|------------------|-------------|-------------|-------------------------------|
| Section                 | Surface class | Road class       | Maintenance | Improvement | Maintenance after Improvement |
| A001_02 Km 0.4 to 0.5   | Bituminous    | Primary or trunk | MaintB      |             |                               |
| A001_02 Km 0.5 to 0.6   | Bituminous    | Primary or trunk | MaintB      |             |                               |
| A001_02 Km 0.7 to 0.8   | Bituminous    | Primary or trunk | MaintB      |             |                               |
| A001_02 Km 1 to 1.1     | Bituminous    | Primary or trunk | MaintB      |             |                               |
| A001_02 Km 1.5 to 1.6   | Bituminous    | Primary or trunk | MaintB      |             |                               |
| A001_03 Km 1.9 to 2     | Unsealed      | Primary or trunk | MaintU      | UPGRAD      | MaintB                        |
| A001_03 Km 2 to 2.1     | Unsealed      | Primary or trunk | MaintU      | UPGRAD      | MaintB                        |
| A002_06 Km 0 to 24.9    | Bituminous    | Primary or trunk | MaintB      |             |                               |
| A002_06 Km 24.9 to 39.6 | Bituminous    | Primary or trunk | MaintB      |             |                               |

6. Under the GENERATE PROGRAMME: Perform Run tab: Change the properties under Run Setup button if needed, then start the analysis.
  - Asset valuation can be included and
  - Choose an appropriate “Run Data Export Directory”. It is recommended that a folder e.g. “HDMMULTYRRun\_2011\_October” is created on the analysis folder structure created earlier. (E.g. “C:\UNRA\_PMS\Work\2012LCCA\HDM4Runs” ) The sub-folder can then be deleted if the analysis is abandoned, no longer used or not applicable.
  - In Windows Explorer determine if a RunData.mdb file already exists on the “Run Data Export Directory”; If so **DELETE the file: RunData.mdb**.
  - Click “OK” when done.
7. Under PERFORM BUDGET OPTIMISATION: Add/customise the required budget/s and perform the budget optimisation.
8. Reports can now be generated and exported under the GENERATE REPORTS tab.
  - The same reports can be generated as per the “Strategy Analysis” in the previous sub-chapter.
  - The most significant report here will be under **Programme and Strategy Analysis: Works Programme Optimised** indicating the selected projects for the rolling works programme. Generate this report and export to Excel for further manipulation.
9. Save and close the analysis on completion.



## 12.9. Create and run a “Programmes” analysis with “Life-cycle” analysis type

During this step the candidate projects are refined to prepare a multi-year rolling works programme, subject to resource constraints. If there is insufficient budget, works are post-poned and not included in the programme.

1. Click the new button within the workspace and select *Programme*.
2. Enter an appropriate name, e.g. “UNRA 2011 Prog LC” and select *UNRA Network* as the road network. Click OK.
3. Under “DEFINE PROGRAMME DETAILS”: “General tab”:
  - The type of analysis = Life cycle
  - Enter the start year and analysis period. Period is typically between 10 and 15 years.
  - Road network should always be UNRA Network.
  - Select the output currency = US Dollars.
  - Enter the discount rate, e.g. 10%.
4. Under the “DEFINE STRATEGY DETAILS”: “Study Sections” tab:
  - According to the HDM-4 Manuals; the candidate projects identified during the strategy analysis should be further investigated here (if necessary). Select only the road sections to be included in this study.
  - Now assign Traffic Growth Sets to all the sections. All sections included in the analysis can be selected before a traffic growth set is assigned.
5. The rest of the input and result information are the same as the parameters specified under the “Strategy Analysis”. Use the same steps as followed in paragraph 12.6 to complete this analysis.

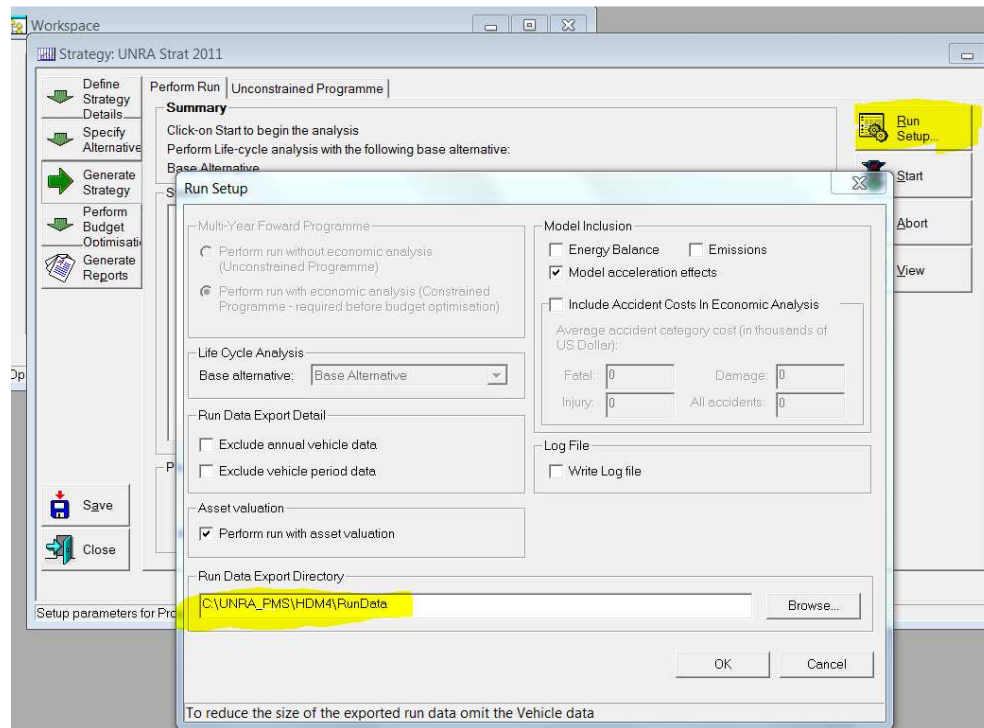
## 12.10. Manipulating the Results for Importing into dTIMS

The output from the Works Programme in HDM-4 can be manipulated for importing into UNRA’s dTIMS database and to calculate costs, etc.

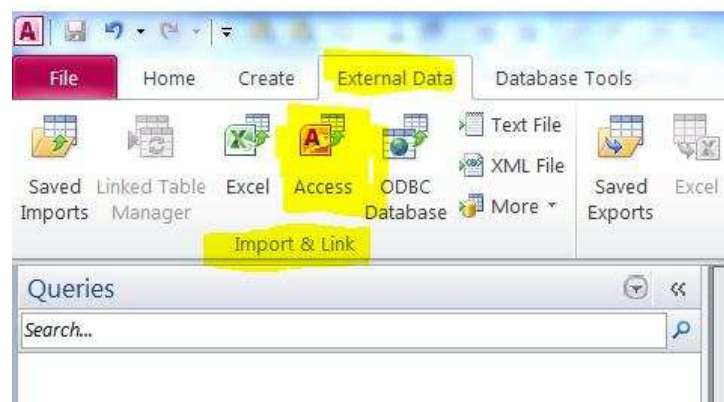
Investigate the attributes of the perspective: **Results\_HDM4** in UNRA’s dTIMS database to comprehend the data needs of the HDM4 results. The attributes are also detailed below in a table.

Many methods can be used to populate the Results\_HDM4 perspective in dTIMS with an HDM4 Works Programme. The method below is one of many methods the user can adopt to prepare the Results\_HDM4 perspective with data. Alternatively the Works Programme can be exported from HDM4 and macros/formulae in Excel can be used to manipulate the content.

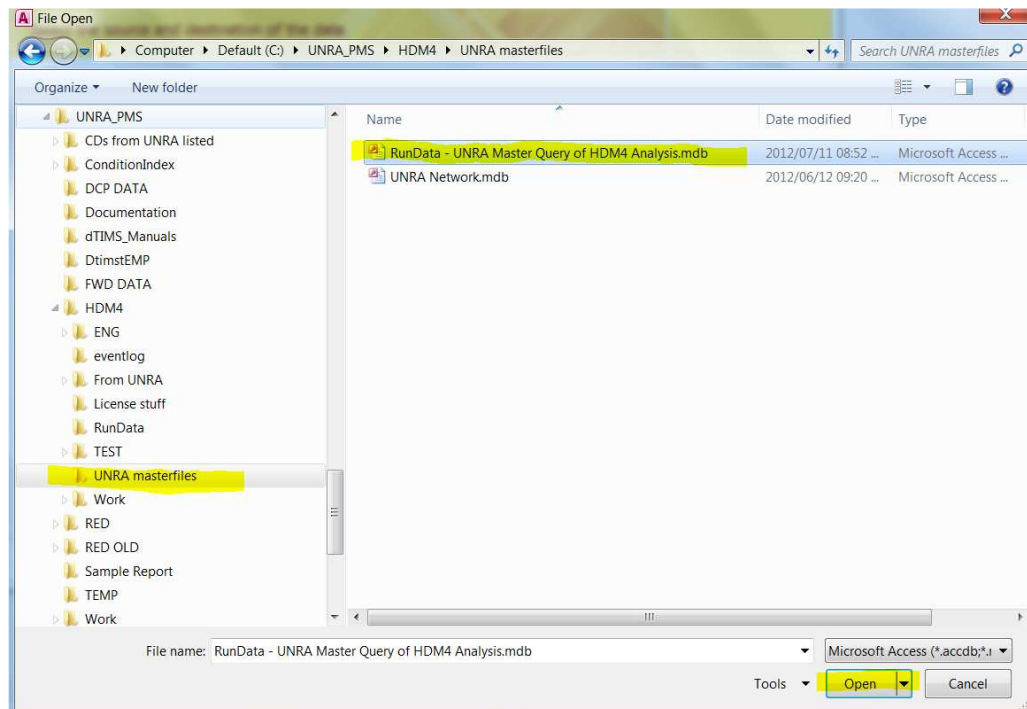
1. Follow the steps hereafter to prepare the data generated by HDM-4 for further use.
2. Locate the file “RunData.mdb” that is associated with the analysis you have just completed. It should reside on the sub-directory of “C:\UNRA\_PMS\Work” created during the “GENERATE STRATEGIES” phase.
3. Open this file: RunData.mdb. The screen below shows where to search for the location of this file.
4. Make sure you are accessing the correct mdb file, there could be a number of these files on your hard drive. For every analysis you run in HDM-4 a file by name “RunData.mdb” is created.



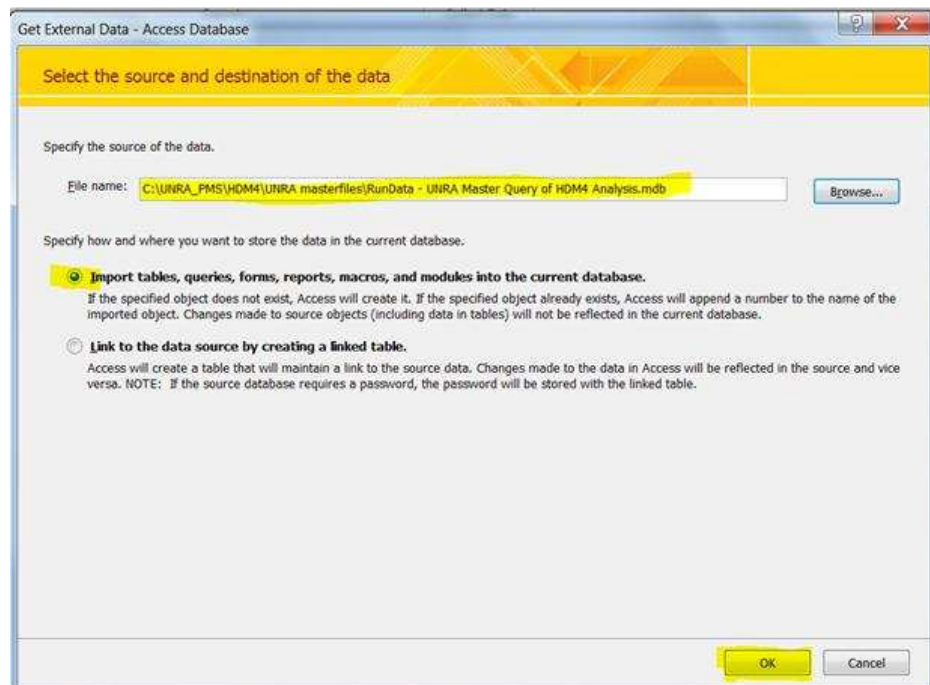
5. Go to the “Queries” object, there should not be any queries, but if there are, DELETE ALL.
6. You will now import the queries customised for UNRA into your RunData.mdb file. The master file **“RunData - UNRA Master Query of HDM4 Analysis.mdb”** on **“C:\UNRA\_PMS\HDM4\UNRA masterfiles”** contains the queries. You do NOT need to open the master mdb file.
7. In Access (“RunData.mdb” file), under “External data” , “Import & Link”, click on the Access tab. See screen shot below.



8. Browse to the file **RunData - UNRA Master Query of HDM4 Analysis.mdb** on **“C:\UNRA\_PMS\HDM4\UNRA masterfiles”**.



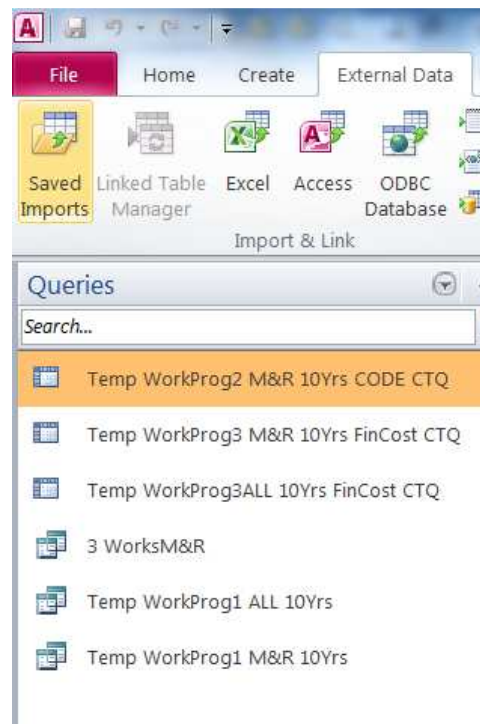
9. Select the first bullet and click OK.



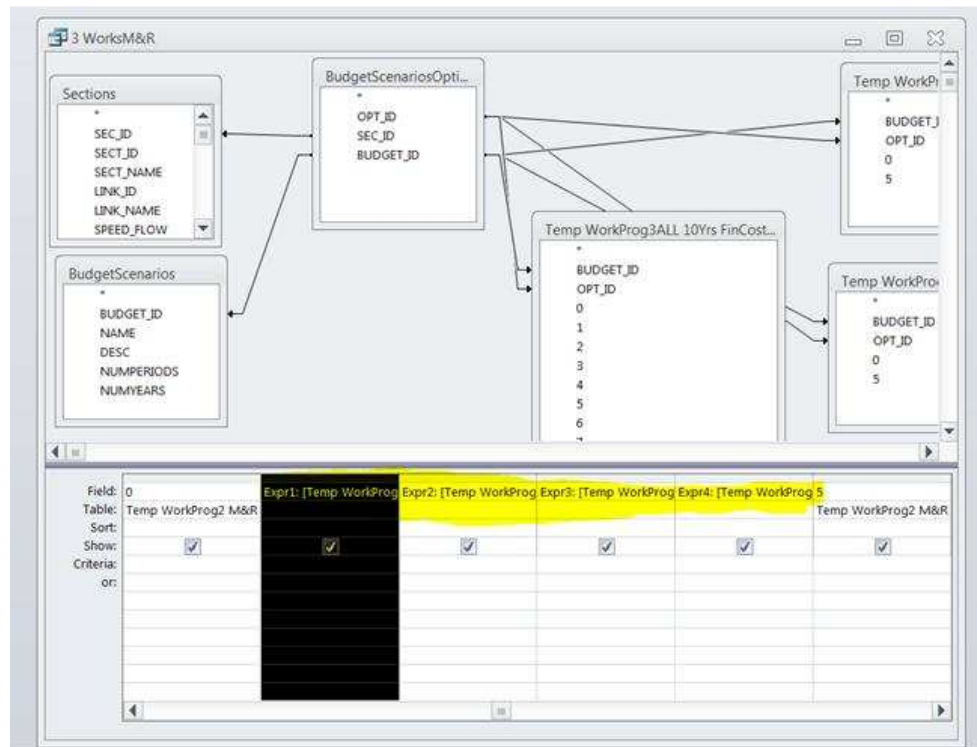
10. Go to “Queries”, “Select All” and click OK. Wait for the import to complete! Close the command box once the import is done!



11. Step 7: The “RunData.mdb” should now contain a number queries as the example below. The query “3 Works M&R” produces data to be used for the compilation of a Works programme.



12. Open the Design View of the Query “3 WorksM&R”.
13. Delete all the columns that have “Expr:..” in the first row (Field). These columns represent years that have no data and the query will not work if these are not deleted.



14. Now Open(Execute) the Query “3 WorksM&R”. (If the query do not open due to errors, open the Design View again and check for more columns with no data – Expr... in Fields row).
15. Copy the Query results to Excel and manipulate the results for input into the perspective **Results\_HDM4** in UNRA’s dTIMS database.
16. Study the sample file; **Sample for HDM4 Results Import.xlsx**, on “C:\UNRA\_PMS\HDM4\UNRA masterfiles” to understand further manipulation on the HDM4 Works programme for import into dTIMS. See sheet **Import Results\_HDM4**. – **Do not use this file, make a copy of it for manipulation.**
17. The following attributes are needed for Results\_HDM4 perspective.

| COLUMN | HEADING/<br>ATTRIBUTE NAME | COMMENT                                                                                                         |
|--------|----------------------------|-----------------------------------------------------------------------------------------------------------------|
| A      | f_IKey                     | This should start at 1 and continue with increments of 1.                                                       |
| B      | Road                       | Road names, to match 100% with the road names stored in UNRA’s dTIMS database.                                  |
| C      | From                       | Offset in km to indicate the start of the road segment.                                                         |
| D      | From_Description           | Not used, keep blank.                                                                                           |
| E      | To                         | Offset in km to indicate the start of the road segment.                                                         |
| F      | To_Description             | Not used, keep blank.                                                                                           |
| G      | ElementID                  | Unique id of road segment: Combination of Road Name, Start Offset in metres, Year and Short budget description. |

| COLUMN | HEADING/<br>ATTRIBUTE NAME | COMMENT                                                                                                 |
|--------|----------------------------|---------------------------------------------------------------------------------------------------------|
| H      | Length                     | Length of road segment in km.                                                                           |
| I      | Analysis_StartYR           | Enter the start year of the analysis period in HDM4.                                                    |
| J      | BudgetName                 | Enter a name to identify the budget scenario (max 19 characters)                                        |
| K      | MRCostYr01                 | Cost of the Recurrent Maintenance, Rehabilitation or Upgrading intervention. (In US\$ million). Year 1  |
| L      | MRCostYr02                 | Cost of the Recurrent Maintenance, Rehabilitation or Upgrading intervention. (In US\$ million). Year 2  |
| M      | MRCostYr03                 | Cost of the Recurrent Maintenance, Rehabilitation or Upgrading intervention. (In US\$ million). Year 3  |
| N      | MRCostYr04                 | Cost of the Recurrent Maintenance, Rehabilitation or Upgrading intervention. (In US\$ million). Year 4  |
| O      | MRCostYr05                 | Cost of the Recurrent Maintenance, Rehabilitation or Upgrading intervention. (In US\$ million). Year 5  |
| P      | MRCostYr06                 | Cost of the Recurrent Maintenance, Rehabilitation or Upgrading intervention. (In US\$ million). Year 6  |
| Q      | MRCostYr07                 | Cost of the Recurrent Maintenance, Rehabilitation or Upgrading intervention. (In US\$ million). Year 7  |
| R      | MRCostYr08                 | Cost of the Recurrent Maintenance, Rehabilitation or Upgrading intervention. (In US\$ million). Year 8  |
| S      | MRCostYr09                 | Cost of the Recurrent Maintenance, Rehabilitation or Upgrading intervention. (In US\$ million). Year 9  |
| T      | MRCostYr10                 | Cost of the Recurrent Maintenance, Rehabilitation or Upgrading intervention. (In US\$ million). Year 10 |
| U      | MRTreatmentYr01            | Name of the Recurrent Maintenance, Rehabilitation or Upgrading intervention. Year 1                     |
| V      | MRTreatmentYr02            | Name of the Recurrent Maintenance, Rehabilitation or Upgrading intervention. Year 2                     |
| W      | MRTreatmentYr03            | Name of the Recurrent Maintenance, Rehabilitation or Upgrading intervention. Year 3                     |
| X      | MRTreatmentYr04            | Name of the Recurrent Maintenance, Rehabilitation or Upgrading intervention. Year 4                     |
| Y      | MRTreatmentYr05            | Name of the Recurrent Maintenance, Rehabilitation or Upgrading intervention. Year 5                     |
| Z      | MRTreatmentYr06            | Name of the Recurrent Maintenance, Rehabilitation or Upgrading intervention. Year 6                     |
| AA     | MRTreatmentYr07            | Name of the Recurrent Maintenance, Rehabilitation or Upgrading intervention. Year 7                     |
| AB     | MRTreatmentYr08            | Name of the Recurrent Maintenance, Rehabilitation or Upgrading intervention. Year 8                     |
| AC     | MRTreatmentYr09            | Name of the Recurrent Maintenance, Rehabilitation or Upgrading intervention. Year 9                     |
| AD     | MRTreatmentYr10            | Name of the Recurrent Maintenance, Rehabilitation or Upgrading intervention. Year 10                    |

| COLUMN | HEADING/<br>ATTRIBUTE NAME | COMMENT                                                     |
|--------|----------------------------|-------------------------------------------------------------|
| AE     | RoutCostYr01               | Cost of the Routine Maintenance. (In US\$ million). Year 1  |
| AF     | RoutCostYr02               | Cost of the Routine Maintenance. (In US\$ million). Year 2  |
| AG     | RoutCostYr03               | Cost of the Routine Maintenance. (In US\$ million). Year 3  |
| AH     | RoutCostYr04               | Cost of the Routine Maintenance. (In US\$ million). Year 4  |
| AI     | RoutCostYr05               | Cost of the Routine Maintenance. (In US\$ million). Year 5  |
| AJ     | RoutCostYr06               | Cost of the Routine Maintenance. (In US\$ million). Year 6  |
| AK     | RoutCostYr07               | Cost of the Routine Maintenance. (In US\$ million). Year 7  |
| AL     | RoutCostYr08               | Cost of the Routine Maintenance. (In US\$ million). Year 8  |
| AM     | RoutCostYr09               | Cost of the Routine Maintenance. (In US\$ million). Year 9  |
| AN     | RoutCostYr10               | Cost of the Routine Maintenance. (In US\$ million). Year 10 |

18. Make sure the data of each column for each road segment is valid and agrees with the attribute properties expected in the perspective “Results\_HDM4” in UNRA’s dTIMS database.
19. The perspective “Results\_HDM4” is a Historic perspective, thus many analyses can be stored for the same road/road segment, provided the ElementID on each element is unique. For this reason, the year and budget name should be part of the ElementID.
20. The format of the work sheet **Import Results\_HDM4** can directly be imported into dTIMS. If all rows have not been imported into dTIMS, check the error file for errors, otherwise import the Excel sheet into an Access (mdb) file and import into dTIMS. Make sure the LRM is set to “Roads”.

### 13. Running a LCCA in RED software

For detailed instructions on the RED software consult the file: **RED Road User Guide and Case Studies V3.2.pdf**<sup>3</sup> residing on C:\UNRA\_PMS\RED\RED Manuals.

The RED system customised for UNRA consists of the following files:

- **UNRA RED - HDM-4 VOC models.xls** – file containing the latest VOC models. This file should have been updated prior to this step, refer to chapter 5 for detail procedures on how to update RED with the latest VOC information. The steps to follow will read the necessary vehicle operating costs from this file.

<sup>3</sup> Before continuing: the user must be familiar with the document: Roads Economic Decision Model, Software User Guide and Case Studies. It is SSATP Working Paper No 78 of Sub-Saharan Africa Policy Program, July 2004. This manual can be found on the directory C:\UNRA\_PMS\RED\RED Manuals

- **UNRA RED - Program.xls** –file that will be used when a road network or sub-network is analysed during this step.
- **UNRA RED - Main.xls** – file that will be used for project level analysis of a specific road section or road.
- **UNRA RED - Data from dTIMS.xls** – file that was exported by dTIMS, containing formulae to customise the road network/sub-network into the correct format before the network is pasted into UNRA RED – Program.xls. The file was exported from dTIMS during chapter 10.
- **UNRA RED - Risk.xls** –file that will be used for risk analysis.

### 13.1. Preparation of data and input parameters for Program Analysis in RED

Follow the steps hereafter to prepare the data and complete an economic evaluation of low-volume rural roads in the RED software.

1. Open the file **UNRA RED – Data from dTIMS.xls** residing on C:\UNRA\_PMS\RED.
2. Open the sheet **From\_dTIMS1**: This sheet contains all the links of UNRA’s road network exported from dTIMS. Only 285 records can be analysed at a time. It is the responsibility of the user to scrutinize the file and delete the unnecessary link-records from this sheet, until a maximum of 285 records remain. Only records from row:2 up to row:286 will thus be used. Complete the following steps in the **From\_dTIMS1** sheet:
  - a) Sort the data ascending according to the first two columns: Road and E\_RED\_From.
  - b) Use Excel Filter and delete the rows that will not be analysed in RED. These will typically be roads with higher traffic ( $MT\_AADT > 200$ ) and paved roads ( $SURF\_CLASS = 0$ ).
  - c) Make sure there are not more than 285 rows of data in the sheet **From\_dTIMS1**. These are the roads (rows) that will be analysed.
  - d) Remove the filter from this sheet. During the next step only rows that are filtered will be copied across to the next sheet for analysis.
3. Open the sheet **From\_dTIMS**: Click the button to copy the first 285 records from the sheet: **From\_dTIMS1** to this sheet. (The user can also do this step manually, make sure the row headings does not change and make sure the data in rows 2 through 285 are valid.)
4. Open the first sheet **RED Parameters**: This step simplifies the customization of the link-records for an analysis. Populate all the yellow-shaded cells with input values; remember all costs should be updated prior to an analysis. These values will be used to customise the link-records for a program analysis in RED. View also the file **UNRA RED - Main.xls**, as this file has specific details to road type codes, vehicle class descriptions, etc. The software assumes seven (7) alternatives will be analysed, if any alternative is not used it is recommended that the details are entered as follows:



| Description                                          | Not used |
|------------------------------------------------------|----------|
| RoadType(X/Y/Z)                                      | X        |
| Roughness(IRI)                                       | 99       |
| InvestmentDurationin<br>Years(0/1/2/3)               | 1        |
| Percent ofInvestment Costs in Year<br>1(%)           | 100      |
| Percent ofInvestment Costs in Year<br>2(%)           | 0        |
| Percent ofInvestment Costs in Year<br>3(%)           | 0        |
| FinancialInvestmentCost('000\$/km)                   | 0        |
| FixedFinancialMaint.<br>Costs('000\$/km/year)        | 999999   |
| VariableFinancialMaint.<br>Costs('000\$/km/year/ADT) | 999999   |
| AccidentsRate(# per 100 million<br>vehicle-km)       | 999999   |
| PercentWithFatality(%)                               | 11%      |
| PercentWithInjury(%)                                 | 17%      |
| PercentWithDamage Only(%)                            | 72%      |

5. Open the sheet **RED Final**: Rows 11 through 295 contain the link-records as populated from the data exported from dTIMS and the information on the **RED Parameters** sheet. The user can make any changes he/she deems necessary here. Keep in mind that this sheet has been customised with formulae; any hard-coded changes will wipe-out the customised formulae. Save the file to keep all changes made. Then select the rows to be analysed and copied.
6. Open the file **UNRA RED – Program.xls** residing on C:\UNRA\_PMS\RED. Open the sheet: **RED Inputs**.
  - e) Delete the data in rows 11 downwards to the end.
  - f) Use Alt&Tab to go back to the file **UNRA RED – Data from dTIMS.xls** and copy the data from Rows: 11 to 295 (or a portion thereof), Columns “A” to “IU”.
  - g) Use Alt&Tab to go back to the file **UNRA RED – Program.xls** and paste **AS VALUES** the selected data at position A11.
7. Save and close the two Excel files that were opened during the previous steps to ensure all changes are kept.
8. Open the file **UNRA RED – Program.xls** residing on C:\UNRA\_PMS\RED. Open the sheet **RED Inputs**.
9. This file is now ready for an analysis.
  - a) Use the manual **RED Road User Guide and Case Studies V3.2.pdf** and verify each column of the data in the sheet **RED Inputs**. Scroll down all the rows to be analysed and inspect the data to be within the acceptable range of each data item. Pay attention to the headings and verify costs entered agree with the unit specified; e.g. ‘000/km or /’000/km/year, etc.
  - b) The name in cell B2 should be **UNRA RED - Main.xls**. Click on the **Execute RED Model** button to start the program analysis.
  - c) The Case Study on the Program Module (pages 17 to 26) in the RED Manual contains details on the results from a Program Analysis.
  - d) Investigate these results and repeat the processes in this paragraph until you have achieved a suitable program for the sub network/list of roads analysed.

- e) Select a Budget Scenario (Unconstrained or one of the Constraint Budgets) that agrees with your objective. The data/graphs/tables can then be exported from the RED spreadsheet to reports/documents/slide shows for presentation.
10. If errors occur during the analysis stage, make the costs of the “Not Used” alternative smaller, or replace the “Not Used” alternative with valid maintenance action or reduce the number of records in the “RED Input” sheet.

## 13.2. Manipulate Results for Import into dTIMS database

The output from the file: UNRA RED – Program.xls, can be manipulated to import the recommended/selected Works programme into UNRA’s dTIMS database and to calculate costs, etc.

Investigate the attributes of the perspective: **Results\_RED** in UNRA’s dTIMS database to comprehend the data needs of the RED results. The attributes are also detailed in the table below.

The method below is one of many methods the user can adopt to prepare the Results\_RED perspective with data. Alternatively the user can prepare his/her own set of processes to manipulate the results of the RED analysis for import.

1. Make sure you import only relevant analyses that require storage, otherwise the UNRA’s dTIMS database will grow with unnecessary data.
2. The worksheet “**EXPORTtoDTIMS**” in the file **UNRA RED – Program.xls**, has been customized to tailor the data from one of the budget scenarios. This worksheet name is shaded in red.
3. It is recommended that the file **UNRA RED – Program.xls**, is copied under another name
4. Follow the steps indicated in the first 8 rows; then use the data and headings from rows 11 onwards to import the results into UNRA’s dTIMS database or to manipulate the results for reporting. The following are of importance:
  - a) Make sure the spelling of the worksheet name in step 3 agrees 100% with worksheet name of the budget scenario you have selected for importing.
  - b) The Year and Budget Description in steps 6 and 7 will be used to identify each RED analysis. The Year and Budget Description are used to identify the different analyses. These should be unique for each set of data that are prepared for importing.
  - c) Cell A12: This should always start at 41 and continue with increments of one in column A. The number 41 indicates the start row number in the road segments in the budget-scenario-worksheet.
  - d) Do not change the content of the values in cells K10 to AX10: These cells store the column-positions of the Capital and Recurrent costs in the worksheets “Unconstrained Budget”, “Budget Constraint A”, “Budget Constraint B”, etc. (if the content is changed, the formulae will extract incorrect data from the optimized budget worksheet).
5. The following columns were customised in the work sheet “**EXPORTtoDTIMS**”:

| COLUMN | HEADING/<br>ATTRIBUTE NAME | COMMENT                                                                        |
|--------|----------------------------|--------------------------------------------------------------------------------|
| A      | f_Key                      | This should always start at 41 and continue with increments of 1.              |
| B      | Road                       | Road names, to match 100% with the road names stored in UNRA’s dTIMS database. |
| C      | From                       | Offset in km to indicate the start of the road segment.                        |
| D      | From_Description           | Not used, keep blank.                                                          |

| COLUMN         | HEADING/<br>ATTRIBUTE NAME | COMMENT                                                                                                                             |
|----------------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <i>E</i>       | To                         | Offset in km to indicate the start of the road segment.                                                                             |
| <i>F</i>       | To_Description             | Not used, keep blank.                                                                                                               |
| <i>G</i>       | ElementID                  | Unique id of road segment: Combination of Road Name, Start Offset in metres, Year and Short budget descripton.                      |
| <i>H</i>       | Length                     | Length of road segment in km.                                                                                                       |
| <i>I</i>       | Analysis_StartYR           | Dependant on values entered in previous step.                                                                                       |
| <i>J</i>       | Budget_Name                | Dependant on values entered in previous step.                                                                                       |
| <i>K - AD</i>  | CAPCostYrNN                | Where NN = 01 to 20, indicating the year in the 20-year analysis period.<br><br>Stores the Capital Cost per year, in USD million.   |
| <i>AE - AX</i> | RECCostYrNN                | Where NN = 01 to 20, indicating the year in the 20-year analysis period.<br><br>Stores the Recurrent Cost per year, in USD million. |
| <i>AY</i>      | RecurrentAnnualCost        | Recurrent cost as an annualised value in USD million.                                                                               |
| <i>AZ</i>      | TreatmentName              | Name of the selected Treatment Alternative.                                                                                         |
| <i>BA</i>      | TreatmentNPVEconCost       | Net present values of expenditure, USD million.                                                                                     |
| <i>BB</i>      | TreatmentRoughness         | Average roughness of the road segment (IRI), for the selected treatment.                                                            |

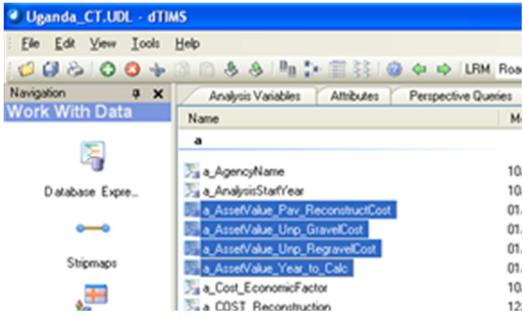
6. Make sure the data of each column for each road segment is valid and agrees with the attribute properties expected in the perspective “Results\_RED” in UNRA’s dTIMS database.
7. The data from rows 12 onwards could now be imported in the perspective “Results\_RED”. The following method is recommended:
8. Export the data view of the dTIMS perspective Results\_RED to an Access table. Clear the Access table of all data and paste the data of the work sheet “**EXPORTtoDTIMS**” (columns 12 onwards) into the Access table. Import the Access table into perspective “Results\_RED”.
9. The perspective “Results\_RED” is a Historic perspective, thus many analyses can be stored for the same road/road segment, provided the ElementID on each element is unique. For this reason, the year and budget name should be part of the ElementID.

## 14. Calculating Asset Value

### 14.1. Current Asset Value

The RMS User Manual (chapter “Reporting in dTIMS”) describes the steps to be followed after surveyed data was imported into the Visuals perspective in dTIMS. These steps include the calculation of Asset Value for each road segment surveyed, based the condition of the road segment and unit costs for the reconstruction of paved roads and gravelling/regravelling of unpaved roads.

If asset values in the Visuals perspective should be recalculated, follow the steps hereafter:

| Update Asset Values for a Specific Year                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Option a (using dTIMS WF)                                                                                                                                                                                                                                                                       | Option b (using dTIMS)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Use the workflow <i>Asset Value calculation</i> in dTIMS WF.                                                                                                                                                                                                                                    | Perform the calculation of asset values in dTIMS.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <p>The re/calculation of asset values is done for a specific year. If more than one year's values need calculated, you will need to run the workflow more than once.</p> <p>Make sure the unit costs correspond to the monetary value for the year for which asset values are recalculated.</p> | <p>The re/calculation of asset values is done for a specific year. If more than one year's values need calculated, you will need to repeat the steps below more than once.</p> <p>Update the unit costs. These are found in the Database Expressions listed below. All costs are defined in US Dollars.</p> <p>Make sure the unit costs correspond to the monetary value for the year for which asset values are recalculated.</p> <p>Update the following Database Expressions:</p> <p>a_AssetValue_Year_to_Calc with the year for which Asset Value is to be recalculated.</p> <p>a_AssetValue_Pav_ReconstructCost: Cost of a reconstruction on paved road, per sq meter</p> <p>a_AssetValue_Unp_GravelCost: Cost to upgrade an earth road to a gravel road, per km</p> <p>a_AssetValue_Unp_RegravelCost: Cost to regavel an existing gravel road, per km</p>  <p>Thereafter the Formula Transformations are run; they (re)calculates the Asset Value based on the above-defined unit costs and a specific year, prompted from the user. All of these cost data are stored in the Visuals perspective along with the calculated Asset Value, the cost basis is thus clear on which the asset value has been calculated, and to enable comparison within dTIMS of historical asset value data.</p> <p>To recalculate the asset values in the Visuals perspective, execute the Formula Transformations,</p> |

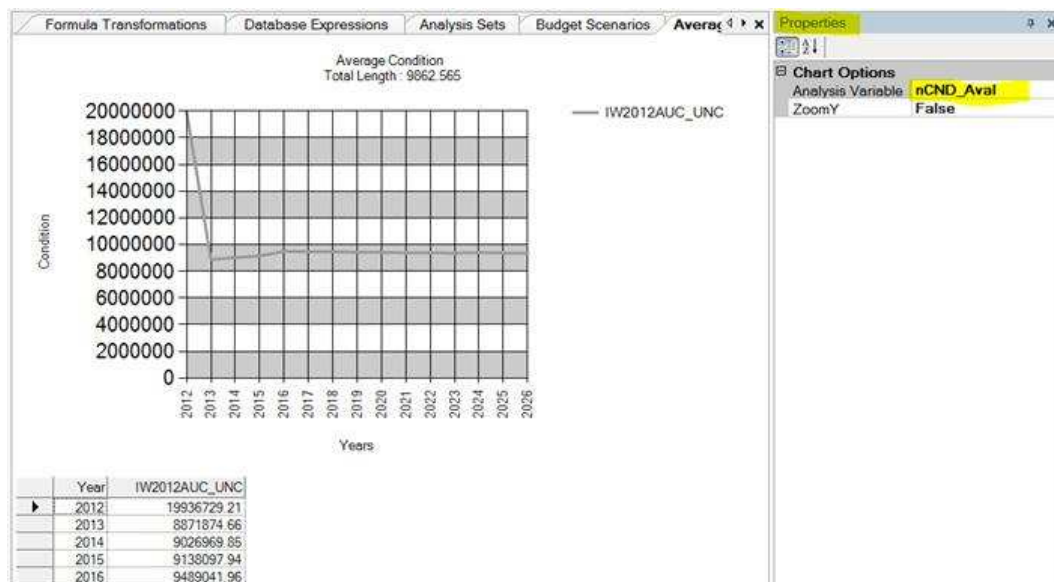
| Update Asset Values for a Specific Year                      |                                                                                                                                                                                                                                                         |
|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Option a (using dTIMS WF)                                    | Option b (using dTIMS)                                                                                                                                                                                                                                  |
| Use the workflow <i>Asset Value calculation</i> in dTIMS WF. | Perform the calculation of asset values in dTIMS.                                                                                                                                                                                                       |
|                                                              | in the order below: <ol style="list-style-type: none"> <li>1. FT_Visuals_Pav_ReconCost</li> <li>2. FT_Visuals_Unp_GravCost</li> <li>3. FT_Visuals_Unp_RegravCost</li> <li>4. FT_Visuals_AssetValue_Pav</li> <li>5. FT_Visuals_AssetValue_Unp</li> </ol> |

## 14.2. Future Predicted Asset value

The dTIMS LCCA calculated the future predicted asset value, on each road segments that was analysed. The predicted asset values are associated with a funding level and the specific fund allocation between the different intervention treatments.

To view future predicted asset values:

1. Go to **Budget Scenarios** under **Analyze & Report**.
2. Select a single or multiple budget scenarios.
3. From the **Tasks Pane** select the item: **Average Condition** under **Charts that Compare Budget Scenarios**.
4. The system will automatically open the Average Condition (nCND\_CIND) graph and table. Change this to display the Asset Values using the analysis variable nCND\_Aval. See example below.



5. The Grid Data and Chart Image can now be copied into a document and/or work file (such as Excel) for further manipulation/

## 15. Presentation of Analyses

The presentations of a life cycle cost analysis are usually accompanied by a detailed. The content of such a report is determined by the requirements of the roads authority. It is recommended that the following aspects are addressed in such a report:

1. An introduction to the road authority and information regarding surveys, policies, departmental practices, software and systems, etc.
2. A description of the road network in terms of road length, districts and traffic.
3. A description of the current condition of paved and unpaved roads separately, according to the most recent survey data.
4. Background to the analysis that was conducted: unit costs, budgets, maintenance policies.
5. The consequences of the current funding level that will be adopted for the maintenance, rehabilitations and upgrading of roads.
6. The consequences of increased funding scenarios or alternative maintenance policies.

A Life Cycle Cost Needs Analysis Report was compiled for UNRA during 2009; based on the 2009 Phase I condition data. This report can be found on “C:\UNRA\_PMS\Sample report” and can be used as an example. The outline of this report is only a guide; it is up to the PMS personnel, responsible for UNRA’s PMS, to produce this report and to determine the appropriate content thereof.

The sub-paragraphs hereafter can be used as guide for extracting and preparing data for such a LCC Needs Analysis Report. The graphs shown here are not related to UNRA’s road network, they are merely examples.

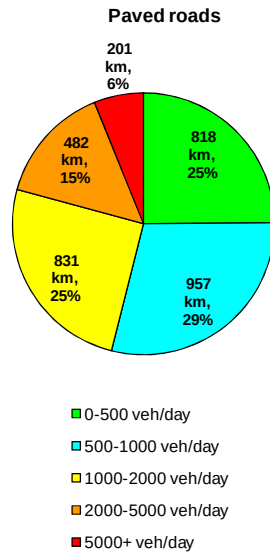
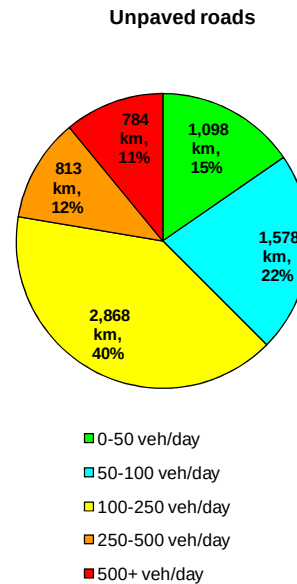
### 15.1. Network information extracted from dTIMS

#### 15.1.1. Network length

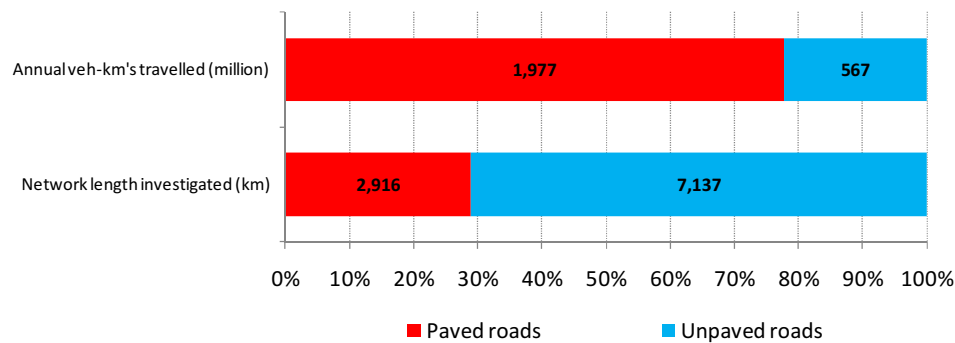
- The road network length is determined from the Roads perspective in UNRA’s dTIMS database. Export the Data Sheet View of the Roads perspective and total the LENGTH attribute. The road network length can also be summarized for the different road classed found in the Roads perspective.
- The Data Sheet View of any other perspective can be exported and used further to summarise the road network length, e.g. ADMIN\_CLASS or ADMIN\_NAME from the Administration perspective.
- Alternatively a query can be created or an existing query can be used to export the necessary attributes for summarizing the network length.

#### 15.1.2. Traffic

- Traffic information should be exported from the Data Sheet View of the Traffic perspective in UNRA’s dTIMS database.
- Summarize the traffic data into traffic load categories and prepare tables and/or graphs, see example below.

**Traffic distribution on paved roads, 2008/2009****Traffic distribution on unpaved roads, 2008,2009**

- Calculate the annual vehicle-kilometers driven on the roads network of UNRA. Annualised vehicle-kilometers are the product of AADT and Road Length and 365 (days).

**Network length versus vehicle-kilometres travelled, 2008/2009****15.1.3. Distress ratings**

Categorise the condition of paved and unpaved roads separately into tables and graphs. It is recommended that the colour scheme below is used to present the distress conditions:

- For paved roads ratings are ■ very poor, ■ poor, ■ fair, ■ good and ■ very good. These ratings can indirectly be related to the proportion (length or percentage) of the paved road network with ■ none, ■ isolated, ■ moderate, ■ warning and ■ severe problems.
- For unpaved roads the rating scores are done on a scale of 1 to 5 where 1 is ■ acceptable (very good), and 5 is ■ severe (very poor). These ratings can indirectly be related to the proportion (length or percentage) of the unpaved road network with ■ none, ■ isolated, ■ moderate, ■ warning and ■ severe problems.

Condition information should be exported from the Data Sheet View of the Visuals perspective in UNRA's dTIMS database. The data can then be filtered for a specific year. Alternatively a Time Dependent Query can be created and executed to produce the most recent visual assessment information on each road segment.

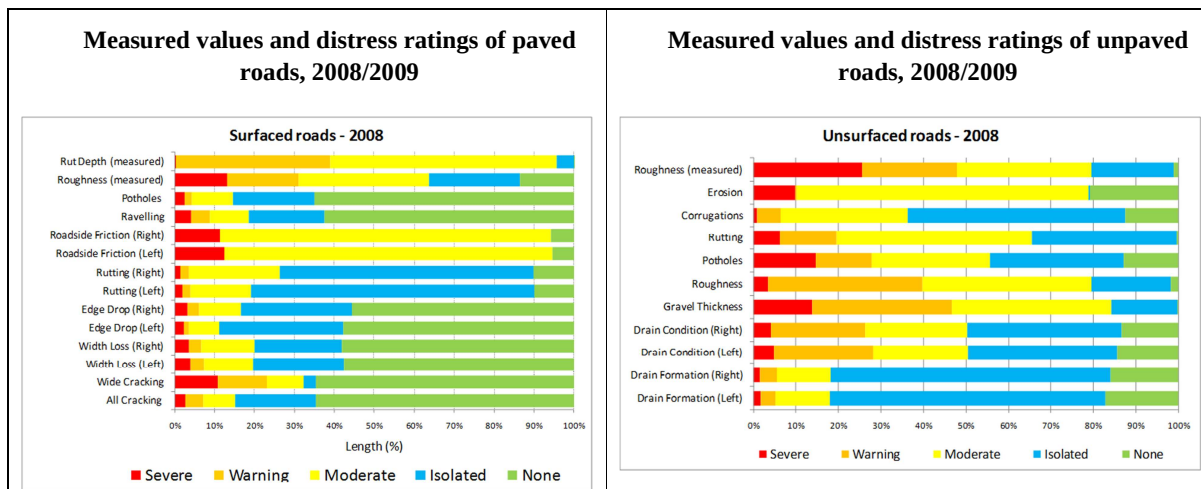
Use the distress categories below to prepare distress distribution graphs, examples of such graphs follow after the tables below:

| <b>Distress</b>                                                                                                                           | <b>Attribute in Visuals perspective</b> | <b>Data range describing each problem category – paved roads</b> |                       |                        |                        |                         |
|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|------------------------------------------------------------------|-----------------------|------------------------|------------------------|-------------------------|
|                                                                                                                                           |                                         | <b>Severe / Very Poor</b>                                        | <b>Warning / Poor</b> | <b>Moderate / Fair</b> | <b>Isolated / Good</b> | <b>None / Very Good</b> |
| Visual Condition Index (VCI) %                                                                                                            | VCI                                     | 0-29%                                                            | 30-49%                | 50-69%                 | 70-84%                 | 85%+                    |
| All Cracking                                                                                                                              | CO_ALLCRCKS                             | 5                                                                | 4                     | 3                      | 2                      | 1                       |
| Wide Cracking                                                                                                                             | CO_WIDECRACKS                           | 5                                                                | 4                     | 3                      | 2                      | 1                       |
| Width Loss (Left)                                                                                                                         | CL_WIDTHLOSS                            | 5                                                                | 4                     | 3                      | 2                      | 1                       |
| Width Loss (Right)                                                                                                                        | CR_WIDTHLOSS                            | 5                                                                | 4                     | 3                      | 2                      | 1                       |
| Edge Drop (Left)                                                                                                                          | CL_EDGEDROP                             | 5                                                                | 4                     | 3                      | 2                      | 1                       |
| Edge Drop (Right)                                                                                                                         | CR_EDGEDROP                             | 5                                                                | 4                     | 3                      | 2                      | 1                       |
| Rutting (Left)                                                                                                                            | CL_RUTTING                              | 5                                                                | 4                     | 3                      | 2                      | 1                       |
| Rutting (Right)                                                                                                                           | CR_RUTTING                              | 5                                                                | 4                     | 3                      | 2                      | 1                       |
| Roadside Friction (Left)                                                                                                                  | CL_RDFRICTION                           | 5                                                                | 4                     | 3                      | 2                      | 1                       |
| Roadside Friction (Right)                                                                                                                 | CR_RDFRICTION                           | 5                                                                | 4                     | 3                      | 2                      | 1                       |
| Ravelling                                                                                                                                 | CO_RAVELLING                            | 5                                                                | 4                     | 3                      | 2                      | 1                       |
| Potholes (Number)                                                                                                                         | CO_POTHOLES                             | 101 – 300                                                        | 51-100                | 11 - 50                | 1 - 10                 | 0                       |
| Roughness (measured)                                                                                                                      | Roughness_Measured*                     | > 6.5 IRI                                                        | 4.5 – 6.5 IRI         | 3 – 4.5 IRI            | 2 – 3 IRI              | <= 2 IRI                |
| Rut Depth (measured)                                                                                                                      | Rutting_Measured*                       | >20 mm                                                           | 10 – 20 mm            | 5 – 10 mm              | 2 – 5 mm               | <= 2 mm                 |
| * The measured roughness and rut depth values can also be extracted from the ROUGHNESS and RUTTING perspectives in UNRA's dTIMS database. |                                         |                                                                  |                       |                        |                        |                         |

| <b>Distress</b>                | <b>Attribute in Visuals perspective</b> | <b>Data range describing each problem category – unpaved roads</b> |                       |                        |                        |                         |
|--------------------------------|-----------------------------------------|--------------------------------------------------------------------|-----------------------|------------------------|------------------------|-------------------------|
|                                |                                         | <b>Severe / Very Poor</b>                                          | <b>Warning / Poor</b> | <b>Moderate / Fair</b> | <b>Isolated / Good</b> | <b>None / Very Good</b> |
| Visual Condition Index (VCI) % | VCI                                     | 0-29%                                                              | 30-49%                | 50-69%                 | 70-84%                 | 85%+                    |
| Drain Formation                | L_Drain_Formatio                        | 5                                                                  | 4                     | 3                      | 2                      | 1                       |



|                         |                             |         |             |            |           |          |
|-------------------------|-----------------------------|---------|-------------|------------|-----------|----------|
| (Left)                  | $n_{UNP}$                   |         |             |            |           |          |
| Drain Formation (Right) | $R_{Drain\_Formation\_UNP}$ | 5       | 4           | 3          | 2         | 1        |
| Drain Condition (Left)  | $L_{Drain\_Cond\_UNP}$      | 5       | 4           | 3          | 2         | 1        |
| Drain Condition (Right) | $R_{Drain\_Cond\_UNP}$      | 5       | 4           | 3          | 2         | 1        |
| Gravel Thickness        | $C_{GRAVEL\_THICK\_UNP}$    | 5       | 4           | 3          | 2         | 1        |
| Roughness               | $C_{ROUGHNESS\_UNP}$        | 5       | 4           | 3          | 2         | 1        |
| Potholes                | $C_{POTHOLE\_UNP}$          | 5       | 4           | 3          | 2         | 1        |
| Rutting                 | $C_{RUTTING\_UNP}$          | 5       | 4           | 3          | 2         | 1        |
| Corrugations            | $C_{CORRUGATIONS\_UNP}$     | 5       | 4           | 3          | 2         | 1        |
| Erosion                 | $C_{EROSION\_GULLIES\_UNP}$ | 5       | 4           | 3          | 2         | 1        |
| Roughness (measured)    | $Roughness\_Measured^*$     | >16 IRI | 12 - 16 IRI | 8 - 12 IRI | 4 - 8 IRI | <= 4 IRI |



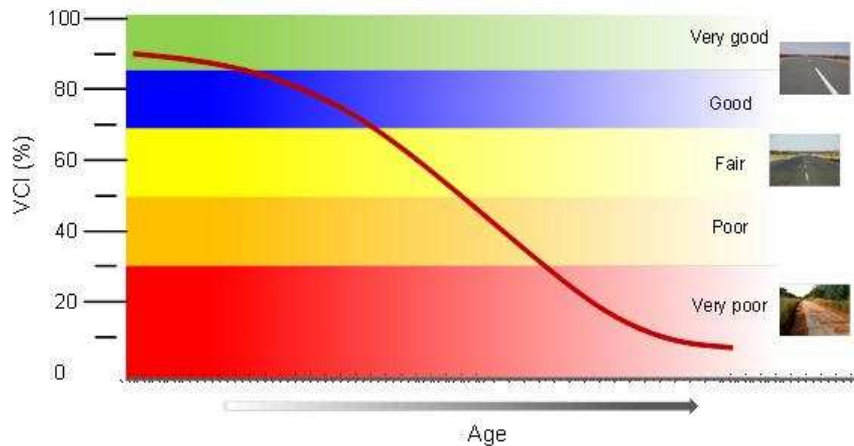
#### 15.1.4. Visual Condition Index (VCI)

Describe the current and historic condition of paved and unpaved roads networks in terms of the following two measures:

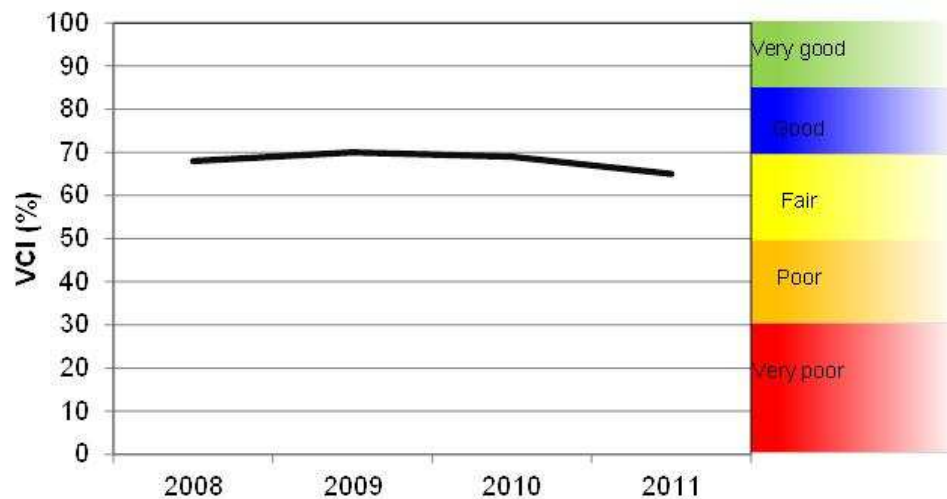
- The average network VCI. This value presents the overall condition of the paved and unpaved road networks separately or combined. The formula to calculate this value is:
  - Avg Network VCI =  $\sum_{i=1}^n (VCI_i * Length_i) / \sum_{i=1}^n (Length_i)$
  - And n = number of records in road network or sub-network

- o A graph such as the one below can then be prepared to show the current and historic conditions over time. The categories shown on the graph are used to explain the overall condition category of the road network or sub-network.

### The VCI scale and typical road deterioration



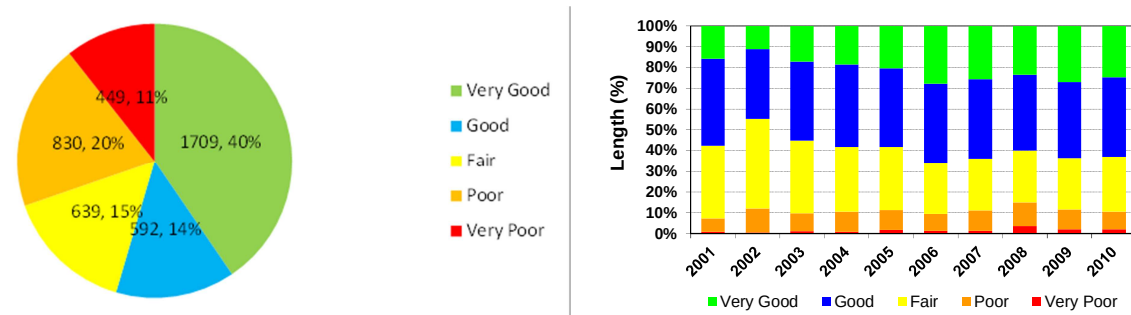
### A typical VCI graph



- The distribution of road network length between the different condition categories; very poor, poor, fair, good and very good. The current index values can be compared to historic data to determine a trend for the network's performance. Take note of the percentage of roads in a poor to very poor condition as these have huge future financial implications to users due to extra road user costs and the road authority due to the high costs associated with rehabilitation. A condition distribution graph, showing the length/percentage road in a very poor, poor, fair, good and very good condition, can also be done per region to investigate which region is worse.
  - o The categories adopted are:
  - o Very Good = 85% to 100%
  - o Good = 70% to 84%
  - o Fair = 50% to 69%
  - o Poor = 30% to 49%

- o Very Poor = 0% to 29%

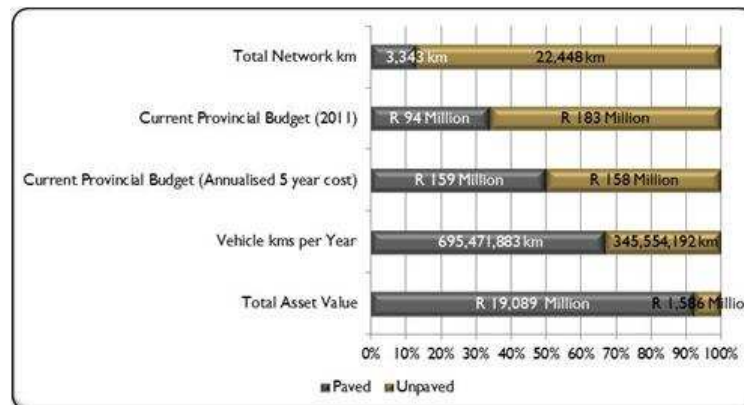
### Typical condition distribution graphs, current and historic



#### 15.1.5. Asset Value

Current asset value of the paved and unpaved road networks can be extracted from UNRA dTIMS database. Use one of the methods below:

- The Perspective Query: PQ\_Road\_Asset\_Value will export the Visual perspective with the asset values for all years of visual assessment data. The user can then filter the appropriate data.
- Create a perspective query, to sum the asset values from the visuals perspective onto the elements of Roads perspective. Add a filter to make sure the correct year's data is transformed from the Visuals perspective onto the elements of Roads perspective.
- The asset values can then be summed and presented with other information from UNRA's database, see example below.



#### 15.2. Background of the LCCA

Include information describing the input parameters, cost and any other information influenced the life cycle cost analysis that was done.

Refer to the document on "C:\UNRA\_PMS\Sample report" for an example of such a section. Make sure to describe the budget levels and maintenance policies that were investigated, as well as unit costs.

#### 15.3. Vehicle operating costs

Include a chapter to describe the vehicle operating costs. Document the input values and VOC models as it was calculated in chapter 5.

## 15.4. Consequences of various budget scenarios

The results of the three LCCA tools are described in chapters 11, 11.3 and 13. Use the sample report on “C:\UNRA\_PMS\Sample report” as a guide and manipulate the output data from analyses into formats suitable for your report.

## 16. Presentation of Results in GIS

The presentation of the LCCA results in GIS is detailed in UNRA’s RMS User Manual.

### 16.1. dTIMS

One of the outputs of a dTIMS analysis is a Maintenance and Rehabilitation (M&R) program based on a given budget, either constrained or unconstrained. In the output of a dTIMS, the M&R report is referred to as the Construction Program.

After a budget scenario has been optimised; go to the **Analyze & Report/Budget Scenarios** tab.

This procedure exports the selected strategy on each road section for the selected budget scenario to a file (Excel, Access, etc.) for further manipulation into a file GIS.

1. Select one budget scenario at a time, by clicking on the name. Select **Construction Programmes** from the task menu.
2. The options can be changed as needed but it is recommended that the report is done by section, all treatments are included, at least 5 years of treatments are included and the values of the DataView **V\_AnalMaster\_For\_ConstructionProgramme** are included.
3. The table below describes the output fields/columns of the exported Construction Programme.

| Column/Field Heading | Description of content                                                                                                   |
|----------------------|--------------------------------------------------------------------------------------------------------------------------|
| Analysis Set         | Name of analysis set that was executed in dTIMS.                                                                         |
| BudScen              | Name of budget scenario that was optimised in dTIMS.                                                                     |
| Year                 | Year of the first treatment that was selected for the road segment under the funding constraints of the budget scenario. |
| ElementID            | Element id of the road segment in the perspective: AnalMaster.                                                           |
| Length               | Length in kilometres of the road segment in the perspective: AnalMaster.                                                 |
| Road                 | Road name of the road segment in the perspective: AnalMaster.                                                            |
| From_Add             | Start km of the road segment in the perspective: AnalMaster.                                                             |
| From_Offset          | Same as previous, ignore this.                                                                                           |
| From_Desc            | Not used.                                                                                                                |
| To_Add               | End km of the road segment in the perspective: AnalMaster.                                                               |
| To_Offset            | Same as previous, ignore this.                                                                                           |
| To_Desc              | Not used.                                                                                                                |
| ElementKey           | Not used.                                                                                                                |
| Crk_Crka             | For paved roads: Percentage carriageway area with all cracks.                                                            |
| Crk_Crkw             | For paved roads: Percentage carriageway area with wide cracks.                                                           |
| Inv_Region           | Region name.                                                                                                             |
| Inv_Station          | Station name.                                                                                                            |
| Inv_FromDesc         | Description at start of road section in AnalMaster.                                                                      |
| Inv_ToDesc           | Description at end of road section in AnalMaster.                                                                        |
| Inv_Link             | Link name.                                                                                                               |
| Inv_Terrain          | Terrain code for Flat, Rolling or Mountainous .                                                                          |
| Inv_Width            | Carriageway width of road in metres.                                                                                     |
| Pot_PotNo            | For paved roads: No of 0.1m <sup>2</sup> potholes per km.                                                                |

| Column/Field Heading | Description of content                                                                                                                                                                              |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pot_Ptch             | For paved roads: No data are recorded for this (Percentage area patched).                                                                                                                           |
| Rgh_Irir             | For paved roads: Average roughness on the road segment from measured values, in IRI (m/km).                                                                                                         |
| Rgh_Year             | For paved roads: Year in which roughness was measured.                                                                                                                                              |
| Rut_Ruttm            | For paved roads: Average rut depth of the road segment in mm.                                                                                                                                       |
| Rvl_Ravl             | For paved roads: Percentage area of carriage way with ravelling.                                                                                                                                    |
| Struc_Pavetype       | Pavement type: F = Flexible (paved), U = Unsealed (unpaved).                                                                                                                                        |
| Struc_PaveYear       | For paved roads: Year of construction or rehabilitation of structural pavement layers. 1900 if no data is available.                                                                                |
| Struc_SNC            | For paved roads: Modified structural number from falling weight deflections, for HDM-4 modelling.                                                                                                   |
| Struc_SNP            | For paved roads: Adjusted structural number from falling weight deflections, for HDM-4 modelling. (This will be the same as the previous column).                                                   |
| Struc_SurfType       | Surface type:<br>FA = Asphalt on paved road<br>FC = Chip and spray on paved road<br>FS = Slurry on paved road<br>UG = Unsealed road with gravel wearing course<br>UE = Earth road<br>US = Sand road |
| Struc_SurfYear       | For paved roads: Year of most recent surfacing. 1900 if no data is available.                                                                                                                       |
| Traf_AADT            | Average annual daily traffic in both directions for the road segment.                                                                                                                               |
| Traf_AADH_Split      | Percentage of heavy vehicle on the road segment.                                                                                                                                                    |
| Traf_AADT_Year       | Year of traffic data.                                                                                                                                                                               |
| Unp_GravThick        | For unpaved roads: Average gravel thickness of road segment, in mm.                                                                                                                                 |
| VCI                  | Average Visual Condition Index (%) for the road segment, derived from the visual assessment distresses. Ranges from 100% (very good), to 0% (very poor).                                            |
| Treatment            | Name of the first treatment that was selected for the road segment under the funding constraints of the budget scenario.<br><br>This is not necessarily the optimum treatment.                      |
| Cost                 | Cost of the first treatment that was selected for the road segment under the funding constraints of the budget scenario.                                                                            |

- Use the file created in the previous steps and follow the instructions of the paragraph “Displaying LCCA Data in the GIS Client” in chapter 10 of the RMS User Manual to display the results in GIS.

## 16.2. HDM-4

One of the outputs of a HDM-4 analysis is a Maintenance and Rehabilitation (M&R) program based on a given budget, either constrained or unconstrained. In the output of HDM-4 (Generate Reports tab), the M&R report is referred to as the Works Program.

After a Strategy or Programme analysis in HDM4 has completed; export the “Works programme optimised by Section” under the GENERATE REPORTS tab to an Excel file. Follow the instructions of the paragraph “Displaying HDM-4 Data in the GIS Client” in chapter 10 of the RMS User Manual to display the results in GIS.

### 16.3. RED

After the “Execute RED Model” process has executed and completed in the file: “UNRA RED - Program.xls”; go to the sheet; “Recommended Programme” or go to any of the “Budget Constraint sheets (A – E)”. Use the output and headings from rows 39 onwards and manipulate the file for display purposes in GIS. Follow the instructions of the paragraph “Displaying RED Data in the GIS Client” in chapter 10 of the RMS User Manual.

## 17. Customisation of Models and Treatments in dTIMS

### 17.1. Performance Models

In dTIMS, performance models are entered as Analysis Variables. The customization can thus be viewed under Analysis Variables.

| Table 9 – Pavement performance models details 1, as customized in dTIMS |                                                                                                                                                       |                                                  |
|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| Name                                                                    | Description                                                                                                                                           | Model for Paved, Unpaved or Both pavement types? |
| cDAV_Material                                                           | Material type – not used for UNRA                                                                                                                     | Unpaved                                          |
| Comp_Poth_Block<br>(not used for UNRA)                                  | Not used for UNRA                                                                                                                                     | Paved                                            |
| Comp_Regravel_Block<br>(not used for UNRA)                              | Not used for UNRA                                                                                                                                     | Unpaved                                          |
| Comp_Roughness_Block<br>(not used for UNRA)                             | Not used for UNRA                                                                                                                                     | Paved                                            |
| Comp_SealAgeBlock<br>(not used for UNRA)                                | Not used for UNRA                                                                                                                                     | Paved                                            |
| nAGE_Pave                                                               | Age of pavement layers in years (Age since construction, reconstruction, rehabilitation for paved roads)                                              | Paved                                            |
| nAGE_Seal                                                               | Age of surfacing in years (age of most recent surfacing on paved roads)                                                                               | Paved                                            |
| nCND_AgeCrka                                                            | Theoretical age of surfacing according to all cracking% in years                                                                                      | Paved                                            |
| nCND_AgeCrkw                                                            | Theoretical age of surfacing according to wide cracking% in years                                                                                     | Paved                                            |
| nCND_AgeRavl                                                            | Theoretical age of surfacing according to ravelling % in years                                                                                        | Paved                                            |
| nCND_Aval                                                               | Asset Value of the element, for the entire length of the element in the AnalMaster perspective.                                                       | Both                                             |
| nCND_Cind                                                               | Model for the condition index (%), initially obtained from VCI, for future predictions a combined index from the HDM-4 performance prediction models. | Both                                             |
| nCND_Crka                                                               | All cracking model (% carriageway area)                                                                                                               | Paved                                            |
| nCND_Crkw                                                               | Wide cracking model (% carriageway area)                                                                                                              | Paved                                            |
| nCND_D95                                                                | Unpaved model for size of sieve retaining 95% of material                                                                                             | Unpaved                                          |
| nCND_GravelThickness                                                    | Unpaved model for predicting gravel thickness due to gravel loss on unsurfaced roads                                                                  | Unpaved                                          |
| nCND_Edge                                                               | Edge break model for paved roads. M3/km                                                                                                               | Paved                                            |
| nCND_Irir                                                               | Model for Roughness Progression (IRI)                                                                                                                 | Both                                             |
| nCND_LabMax                                                             | Unpaved model for maximum particle size (mm)                                                                                                          | Unpaved                                          |
| nCND_P075                                                               | Unpaved model for % material passing 0.075mm sieve                                                                                                    | Unpaved                                          |
| nCND_P2                                                                 | Unpaved model for % material passing 2mm sieve                                                                                                        | Unpaved                                          |
| nCND_P26                                                                | Unpaved model for % material passing 26mm sieve                                                                                                       | Unpaved                                          |
| nCND_P425                                                               | Unpaved model for % material passing 0.425mm sieve                                                                                                    | Unpaved                                          |

**Table 9 – Pavement performance models details 1, as customized in dTIMS**

| Name                    | Description                                                                                                                                      | Model for Paved, Unpaved or Both pavement types? |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| nCND_PI                 | Unpaved model for plasticity index of gravel material.                                                                                           | Unpaved                                          |
| nCND_PL                 | Unpaved model for plastic limit of gravel material                                                                                               | Unpaved                                          |
| nCND_Poth               | Pothole model (number 0.1m2 potholes / km)                                                                                                       | Paved                                            |
| nCND_Ptch               | Pothole patching model (% carriageway area)                                                                                                      | Paved                                            |
| nCND_RAVL               | Ravelling model (% carriageway area)                                                                                                             | Paved                                            |
| nCND_Rough_AfterBlade   | Unpaved model for roughness after blading, QI                                                                                                    | Unpaved                                          |
| nCND_Rough_BeforeBlade  | Unpaved model for roughness before blading, QI                                                                                                   | Unpaved                                          |
| nCND_RoughMax           | Unpaved model for maximum roughness (QI)                                                                                                         | Unpaved                                          |
| nCND_RoughMin           | Unpaved model for minimum roughness (QI)                                                                                                         | Unpaved                                          |
| nCND_RoughTemp          | Temporary roughness used for calculations for gravel roads, QI.                                                                                  | Unpaved                                          |
| nCND_Rutm               | Rut depth model (mm)                                                                                                                             | Paved                                            |
| nCND_Ruts               | Rut depth standard deviation (mm)                                                                                                                | Paved                                            |
| nCND_SMG                | Unpaved model for SMG calculation from material properties                                                                                       | Unpaved                                          |
| nCND_Snp                | Adjusted structural number model                                                                                                                 | Paved                                            |
| nCND_TNYear             | Technical needs year (for internal use)                                                                                                          | Both                                             |
| nCPV_TNYear             | Technical needs benefit function (for internal use)                                                                                              | Both                                             |
| nCST_AgencyAnnEconomic  | Annual Agency Economic Cost (Not NPV) (US\$)                                                                                                     | Both                                             |
| nCST_AgencyAnnFinancial | Annual Agency Financial Cost (Not NPV) (US\$)                                                                                                    | Both                                             |
| nDAV_BladeFreq          | Unpaved model for blade frequency in days                                                                                                        | Unpaved                                          |
| nDAV_MatQual            | Unpaved material quality, reset to 1 after regravelling (1 to 5)                                                                                 | Unpaved                                          |
| nDAV_Pavetype           | Model for pavement type, F(lexible), U(npaved)                                                                                                   | Both                                             |
| nDAV_Pcra               | Model for Previous Cracking (% all cracking before most recent surfacing)                                                                        | Paved                                            |
| nDAV_Pcrw               | Model for Previous Cracking (% wide cracking before most recent surfacing)                                                                       | Paved                                            |
| nDAV_Surfetype          | Model for surfacing type, FS(slurry), FA(asphalt), FC(chip and spray), UE(earth), UG(gravel)                                                     | Both                                             |
| nDAV_Taci               | Time to all crack initiation (in years)                                                                                                          | Paved                                            |
| nDAV_Trvi               | Time to ravelling initiation (in years)                                                                                                          | Paved                                            |
| nDAV_Twci               | Time to wide crack initiation (in years)                                                                                                         | Paved                                            |
| nPV_AUC                 | Present value of area under the condition curve – Compilation variable. Internal field for benefit function to maximise condition.               | Both                                             |
| nPV_EconAgency          | Net present value of economic agency cost at end of analysis period (US\$)                                                                       | Both                                             |
| nPV_FinAgency           | Net present value of financial agency cost at end of analysis period (US\$)                                                                      | Both                                             |
| nPV_TTC_AUC_UPGRADES    | TTC for use when benefit function is AUC to discard “Upgrade to paved roads” that are not economical in terms of VOC savings and costs. (US\$)   | Both                                             |
| nPV_TTC_Benefit         | (Total Transportation Cost of Do Nothing Option) minus (Total transportation Cost of the alternative) = Benefit in term of NPV. Discounted costs | Both                                             |
| nTRF_Aadt               | Average annual daily traffic                                                                                                                     | Both                                             |
| nTRF_CumEsals           | Cummulative no of equivalent standard axles loads                                                                                                | Both                                             |
| nTRF_Esal               | # Equivalent standard axles loads, per year                                                                                                      | Both                                             |
| nTRF_Yax                | Total number of all axles, million per lane                                                                                                      | Both                                             |
| nTTC_Annual             | Sum of VOC and Agency Cost per annum, for use to calculate Benefit of PV                                                                         | Both                                             |

**Table 9 – Pavement performance models details 1, as customized in dTIMS**

| Name        | Description                                                                                                                        | Model for Paved, Unpaved or Both pavement types? |
|-------------|------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
|             | TTC at the end of the analysis period (US\$)                                                                                       |                                                  |
| nVOC_Annual | Model for annual vehicle operating costs , Not discounted. Sum of all vehicle classes. – Given as Annual VOC (US\$) per kilometre. | Both                                             |

**Table 10 – Pavement performance models details 1, as customized in dTIMS**

| Name                                        | Shift | Keep  | Reset | Type        | Slope | DiscardFilt       | PropAtt                   |
|---------------------------------------------|-------|-------|-------|-------------|-------|-------------------|---------------------------|
| cDAV_Material                               | FALSE | FALSE | TRUE  | Dynamic     | Down  | (none)            | AnalMaster->Unp_Material  |
| Comp_Poth_Block<br>(not used for UNRA)      | FALSE | TRUE  | FALSE | Compilation | Down  | bFBlock_Poth      | AnalMaster->Pot_PotNo     |
| Comp_Regravel_Block<br>(not used for UNRA)  | FALSE | TRUE  | FALSE | Compilation | Down  | (none)            | AnalMaster->Unp_GravThick |
| Comp_Roughness_Block<br>(not used for UNRA) | FALSE | TRUE  | FALSE | Compilation | Down  | bFBlock_Roughness | AnalMaster->Rgh_Irir      |
| Comp_SealAgeBlock<br>(not used for UNRA)    | FALSE | TRUE  | FALSE | Compilation | Down  | bFBlock_SealAge   | AnalMaster->nShort        |
| nAGE_Pave                                   | FALSE | TRUE  | TRUE  | Annual      | Up    | (none)            | AnalMaster->nShort        |
| nAGE_Seal                                   | FALSE | TRUE  | TRUE  | Annual      | Up    | (none)            | AnalMaster->nShort        |
| nCND_AgeCrka                                | FALSE | FALSE | TRUE  | Annual      | Up    | (none)            | AnalMaster->nShort        |
| nCND_AgeCrkw                                | FALSE | FALSE | TRUE  | Annual      | Up    | (none)            | AnalMaster->nShort        |
| nCND_AgeRavl                                | FALSE | FALSE | TRUE  | Annual      | Up    | (none)            | AnalMaster->nShort        |
| nCND_Aval                                   | FALSE | TRUE  | TRUE  | Annual      | Down  | (none)            | AnalMaster->nDummyDouble  |
| nCND_Cind                                   | FALSE | TRUE  | TRUE  | Annual      | Down  | (none)            | AnalMaster->nPercent      |
| nCND_Crka                                   | FALSE | TRUE  | TRUE  | Annual      | Up    | (none)            | AnalMaster->Crk_Crka      |
| nCND_Crkw                                   | FALSE | TRUE  | TRUE  | Annual      | Up    | (none)            | AnalMaster->Crk_Crkw      |
| nCND_D95                                    | FALSE | FALSE | TRUE  | Annual      | Down  | (none)            | AnalMaster->nSingle       |
| nCND_Edge                                   | FALSE | TRUE  | TRUE  | Annual      | Up    | (none)            | AnalMaster->nSingle       |
| nCND_GravelThickness                        | FALSE | TRUE  | TRUE  | Annual      | Down  | (none)            | AnalMaster->Unp_GravThick |
| nCND_Irir                                   | FALSE | TRUE  | TRUE  | Annual      | Up    | (none)            | AnalMaster->Rgh_Irir      |
| nCND_LabMax                                 | FALSE | FALSE | TRUE  | Annual      | Down  | (none)            | AnalMaster->nSingle       |
| nCND_P075                                   | FALSE | FALSE | TRUE  | Annual      | Down  | (none)            | AnalMaster->nSingle       |
| nCND_P2                                     | FALSE | FALSE | TRUE  | Annual      | Down  | (none)            | AnalMaster->nSingle       |
| nCND_P26                                    | FALSE | FALSE | TRUE  | Annual      | Down  | (none)            | AnalMaster->nSingle       |
| nCND_P425                                   | FALSE | FALSE | TRUE  | Annual      | Down  | (none)            | AnalMaster->nSingle       |
| nCND_PL                                     | FALSE | FALSE | TRUE  | Annual      | Down  | (none)            | AnalMaster->nSingle       |
| nCND_Poth                                   | FALSE | TRUE  | TRUE  | Annual      | Up    | (none)            | AnalMaster->Pot_PotNo     |
| nCND_Ptch                                   | FALSE | TRUE  | TRUE  | Dynamic     | Down  | (none)            | AnalMaster->Pot_Ptch      |
| nCND_RAVL                                   | FALSE | TRUE  | TRUE  | Annual      | Up    | (none)            | AnalMaster->Rvl_Ravl      |



**Table 10 – Pavement performance models details 1, as customized in dTIMS**

| Name                    | Shift | Keep  | Reset | Type        | Slope | DiscardFilter                 | PropAtt                     |
|-------------------------|-------|-------|-------|-------------|-------|-------------------------------|-----------------------------|
| nCND_Rough_AfterBlade   | FALSE | FALSE | TRUE  | Annual      | Down  | (none)                        | AnalMaster->nSingle         |
| nCND_Rough_BeforeBlade  | FALSE | FALSE | TRUE  | Annual      | Down  | (none)                        | AnalMaster->nSingle         |
| nCND_RoughMax           | FALSE | FALSE | TRUE  | Annual      | Down  | (none)                        | AnalMaster->nSingle         |
| nCND_RoughMin           | FALSE | FALSE | TRUE  | Annual      | Down  | (none)                        | AnalMaster->nSingle         |
| nCND_RoughTemp          | FALSE | FALSE | FALSE | Annual      | Down  | (none)                        | AnalMaster->nSingle         |
| nCND_Rutm               | FALSE | TRUE  | TRUE  | Annual      | Up    | (none)                        | AnalMaster->Rut_Rutm        |
| nCND_Ruts               | FALSE | FALSE | TRUE  | Dynamic     | Down  | (none)                        | AnalMaster->Rut_Ruts        |
| nCND_SMG                | FALSE | FALSE | TRUE  | Annual      | Down  | (none)                        | AnalMaster->nSingle         |
| nCND_Snp                | FALSE | TRUE  | TRUE  | Annual      | Up    | (none)                        | AnalMaster->Struc_SNP       |
| nCND_TNYear             | FALSE | TRUE  | TRUE  | Annual      | Up    | (none)                        | AnalMaster->nShort          |
| nCST_AgencyAnnEconomic  | FALSE | TRUE  | TRUE  | Annual      | Up    | (none)                        | AnalMaster->nDummyDouble    |
| nCST_AgencyAnnFinancial | FALSE | TRUE  | TRUE  | Annual      | Up    | (none)                        | AnalMaster->nDummyDouble    |
| nDAV_BladeFreq          | FALSE | TRUE  | TRUE  | Dynamic     | Down  | (none)                        | AnalMaster->nZeroToThousand |
| nDAV_MatQual            | FALSE | FALSE | TRUE  | Dynamic     | Down  | (none)                        | AnalMaster->Unp_GravQual    |
| nDAV_Pavetype           | FALSE | FALSE | TRUE  | Dynamic     | Down  | (none)                        | AnalMaster->Struc_Pavetype  |
| nDAV_Pcra               | FALSE | FALSE | TRUE  | Dynamic     | Down  | (none)                        | AnalMaster->Crk_Pcra        |
| nDAV_Pcrw               | FALSE | FALSE | TRUE  | Dynamic     | Down  | (none)                        | AnalMaster->Crk_Pcrw        |
| nDAV_Surfetype          | FALSE | FALSE | TRUE  | Dynamic     | Down  | (none)                        | AnalMaster->Struc_SurfType  |
| nDAV_Taci               | FALSE | TRUE  | TRUE  | Dynamic     | Down  | (none)                        | AnalMaster->nShort          |
| nDAV_Trvl               | FALSE | FALSE | TRUE  | Dynamic     | Down  | (none)                        | AnalMaster->nShort          |
| nDAV_Twci               | FALSE | FALSE | TRUE  | Dynamic     | Down  | (none)                        | AnalMaster->nShort          |
| nPV_AUC                 | FALSE | TRUE  | FALSE | Compilation | Up    | (none)                        | AnalMaster->nLong           |
| nPV_EconAgency          | FALSE | TRUE  | FALSE | Compilation | Down  | (none)                        | AnalMaster->nDummyDouble    |
| nPV_FinAgency           | FALSE | FALSE | FALSE | Compilation | Down  | (none)                        | AnalMaster->nDummyDouble    |
| nPV_TTC_AUC_UPGRADES    | FALSE | TRUE  | FALSE | Compilation | Down  | bFBlock_TTC_Negative_Upgrades | AnalMaster->nDummyDouble    |
| nPV_TTC_Benefit         | FALSE | TRUE  | FALSE | Compilation | Down  | bFBlock_TCC_Negative          | AnalMaster->nDummyDouble    |
| nTRF_Aadt               | FALSE | TRUE  | FALSE | Annual      | Up    | (none)                        | AnalMaster->Traf_AADT       |
| nTRF_CumEsals           | FALSE | FALSE | TRUE  | Annual      | Up    | (none)                        | AnalMaster->nDummyDouble    |
| nTRF_Esal               | FALSE | TRUE  | FALSE | Annual      | Up    | (none)                        | AnalMaster->nDummyDouble    |
| nTRF_Yax                | FALSE | FALSE | FALSE | Annual      | Up    | (none)                        | AnalMaster->nDummyDouble    |
| nTTC_Annual             | FALSE | TRUE  | TRUE  | Annual      | Up    | (none)                        | AnalMaster-                 |

**Table 10 – Pavement performance models details 1, as customized in dTIMS**

| Name        | Shift | Keep | Reset | Type   | Slope | DiscardFilt | PropAtt                  |
|-------------|-------|------|-------|--------|-------|-------------|--------------------------|
|             |       |      |       |        |       |             | >nDummyDouble            |
| nVOC_Annual | FALSE | TRUE | TRUE  | Annual | Up    | (none)      | AnalMaster->nDummyDouble |

**Table 11 – Pavement performance models details 2, as customized in dTIMS**

| Name                 | InitFrom                   | Filter                | CurveExp              |
|----------------------|----------------------------|-----------------------|-----------------------|
| cDAV_Material        | AnalMaster->Unp_Material   |                       |                       |
| Comp_Poth_Block      |                            | bF_Always             | nCBlock_Poth          |
| Comp_Regravel_Block  |                            | bF_Always             | nCBlock_Regravel      |
| Comp_Roughness_Block |                            | bF_Always             | nCBlock_Roughness     |
| Comp_SealAgeBlock    |                            | bF_Always             | nCBlock_SealAge       |
| nAGE_Pave            | nCAGE_Pave_Yr1             | bFPavement_F          | nCAGE_Pave            |
| nAGE_Pave            | nCAGE_Pave_Yr1             | bFPavement_U          | nZero                 |
| nAGE_Seal            | nCAGE_Seal_Yr1             | bFPavement_F          | nCAGE_Seal            |
| nAGE_Seal            | nCAGE_Seal_Yr1             | bFPavement_U          | nZero                 |
| nCND_AgeCrka         | nCCND_Crka_AgeYr1          | bFPavement_U          | nZero                 |
| nCND_AgeCrka         | nCCND_Crka_AgeYr1          | bFPavement_F          | nCCND_Crka_Age        |
| nCND_AgeCrkw         | nCCND_Crkw_AgeYr1          | bFPavement_F          | nCCND_Crkw_Age        |
| nCND_AgeCrkw         | nCCND_Crkw_AgeYr1          | bFPavement_U          | nZero                 |
| nCND_AgeRavl         | nCCND_Ravl_AgeYr1          | bFPavement_F          | nCCND_Ravl_Age        |
| nCND_AgeRavl         | nCCND_Ravl_AgeYr1          | bFPavement_U          | nZero                 |
| nCND_Aval            | AnalMaster->Inv_AssetValue | bF_Always             | nCCND_Aval            |
| nCND_Cind            | nCCND_Cind_YR1             | bFPavement_F          | nCCND_CIND_F          |
| nCND_Cind            | nCCND_Cind_YR1             | bFPavement_U          | nCCND_CIND_G          |
| nCND_Crka            | nCCND_Crka_Yr1             | bFCND_Crka_FC_50to100 | nCCND_FC_Crka_50to100 |
| nCND_Crka            | nCCND_Crka_Yr1             | bFCND_Crka100         | nHundred              |
| nCND_Crka            | nCCND_Crka_Yr1             | bFPavement_U          | nZero                 |
| nCND_Crka            | nCCND_Crka_Yr1             | bFCND_Crka_FC_0to50   | nCCND_FC_Crka_0to50   |
| nCND_Crka            | nCCND_Crka_Yr1             | bFCND_Crka_FA_0to50   | nCCND_FA_Crka_0to50   |
| nCND_Crka            | nCCND_Crka_Yr1             | bFCND_Crka0           | nZero                 |
| nCND_Crka            | nCCND_Crka_Yr1             | bFCND_Crka_FA_50to100 | nCCND_FA_Crka_50to100 |
| nCND_Crkw            | nCCND_Crkw_Yr1             | bFCND_Crkw0           | nZero                 |
| nCND_Crkw            | nCCND_Crkw_Yr1             | bFCND_Crkw_FA_0to50   | nCCND_FA_Crkw_0to50   |
| nCND_Crkw            | nCCND_Crkw_Yr1             | bFCND_Crkw_FC_0to50   | nCCND_FC_Crkw_0to50   |
| nCND_Crkw            | nCCND_Crkw_Yr1             | bFCND_Crkw_FA_50to100 | nCCND_FA_Crkw_50to100 |
| nCND_Crkw            | nCCND_Crkw_Yr1             | bFCND_Crkw_FC_50to100 | nCCND_FC_Crkw_50to100 |
| nCND_Crkw            | nCCND_Crkw_Yr1             | bFCND_Crkw100         | nHundred              |
| nCND_Crkw            | nCCND_Crkw_Yr1             | bFPavement_U          | nZero                 |
| nCND_D95             | nCCND_D95                  | bFPavement_F          | nZero                 |
| nCND_D95             | nCCND_D95                  | bFPavement_U          | nCCND_D95             |
| nCND_Edge            | nCND_Edge_YR1              | bFPavement_F          | nCCND_Edgebreak       |

**Table 11 – Pavement performance models details 2, as customized in dTIMS**

| Name                   | InitFrom             | Filter             | CurveExp            |
|------------------------|----------------------|--------------------|---------------------|
| nCND_Edge              | nCND_Edge_Yr1        | bFPavement_U       | nZero               |
| nCND_GravelThickness   | nCCND_GravelThickYr1 | bFPavement_U       | nCCND_GravelThick_U |
| nCND_GravelThickness   | nCCND_GravelThickYr1 | bFPavement_F       | nCCND_GravelThick_F |
| nCND_Irir              | nCCND_IrirYr1        | bFPavement_F       | nCCnd_Irir_F        |
| nCND_Irir              | nCCND_IrirYr1        | bFPavement_U       | nCCnd_Irir_G        |
| nCND_LabMax            | nCCND_LabMax         | bFPavement_U       | nCCND_LabMax        |
| nCND_LabMax            | nCCND_LabMax         | bFPavement_F       | nZero               |
| nCND_P075              | nCCND_P075           | bFPavement_U       | nCCND_P075          |
| nCND_P075              | nCCND_P075           | bFPavement_F       | nZero               |
| nCND_P2                | nCCND_P2             | bFPavement_U       | nCCND_P2            |
| nCND_P2                | nCCND_P2             | bFPavement_F       | nZero               |
| nCND_P26               | nCCND_P26            | bFPavement_U       | nCCND_P26           |
| nCND_P26               | nCCND_P26            | bFPavement_F       | nZero               |
| nCND_P425              | nCCND_P425           | bFPavement_F       | nZero               |
| nCND_P425              | nCCND_P425           | bFPavement_U       | nCCND_P425          |
| nCND_PI                | nCCND_PI             | bFPavement_U       | nCCND_PI            |
| nCND_PI                | nCCND_PI             | bFPavement_F       | nZero               |
| nCND_PL                | nCCND_PL             | bFPavement_U       | nCCND_PL            |
| nCND_PL                | nCCND_PL             | bFPavement_F       | nZero               |
| nCND_Poth              | nCCND_Poth_Yr1       | bFPavement_F       | nCCND_Poth          |
| nCND_Poth              | nCCND_Poth_Yr1       | bFPavement_U       | nZero               |
| nCND_Ptch              | nCCND_Ptch_Yr1       |                    |                     |
| nCND_RAVL              | nCCND_Ravl_Yr1       | bFCND_Ravl100      | nHundred            |
| nCND_RAVL              | nCCND_Ravl_Yr1       | bFPavement_U       | nZero               |
| nCND_RAVL              | nCCND_Ravl_Yr1       | bFCND_Ravl_50to100 | nCCND_Ravl_50TO100  |
| nCND_RAVL              | nCCND_Ravl_Yr1       | bFCND_Ravl_0to50   | nCCND_Ravl_0to50    |
| nCND_RAVL              | nCCND_Ravl_Yr1       | bFCND_Ravl0        | nZero               |
| nCND_Rough_AfterBlade  | nCCND_RoughHigh      | bFPavement_F       | nZero               |
| nCND_Rough_AfterBlade  | nCCND_RoughHigh      | bFPavement_U       | nCCND_RoughHigh     |
| nCND_Rough_BeforeBlade | nCCND_RoughHigh      | bFPavement_U       | nCCND_RoughHigh     |
| nCND_Rough_BeforeBlade | nCCND_RoughHigh      | bFPavement_F       | nZero               |
| nCND_RoughMax          | nCCND_RoughnessMax   | bFPavement_U       | nCCND_RoughnessMax  |
| nCND_RoughMax          | nCCND_RoughnessMax   | bFPavement_F       | nZero               |
| nCND_RoughMin          | nCCND_RoughnessMin   | bFPavement_U       | nCCND_RoughnessMin  |
| nCND_RoughMin          | nCCND_RoughnessMin   | bFPavement_F       | nZero               |
| nCND_RoughTemp         | nThreeHundred        | bFPavement_U       | nCCND_RoughTemp     |
| nCND_RoughTemp         | nThreeHundred        | bFPavement_F       | nZero               |
| nCND_Rutm              | nCCND_Rutm_Yr1       | bFPavement_F       | nCCND_Rutm          |
| nCND_Rutm              | nCCND_Rutm_Yr1       | bFPavement_U       | nZero               |
| nCND_Ruts              | nCCND_RutsYr1        |                    |                     |
| nCND_SMG               | nCCND_SMG            | bFPavement_U       | nCCND_SMG           |
| nCND_SMG               | nCCND_SMG            | bFPavement_F       | nZero               |

**Table 11 – Pavement performance models details 2, as customized in dTIMS**

| Name                    | InitFrom                   | Filter               | CurveExp            |
|-------------------------|----------------------------|----------------------|---------------------|
| nCND_Snp                | nCCND_SNPYr1               | bFPavement_F         | nCCND_Snp           |
| nCND_Snp                | nCCND_SNPYr1               | bFPavement_U         | nZero               |
| nCND_TNYear             | n30                        | bF_Always            | n30                 |
| nCPV_TNYears            |                            | bF_Always            | nCPV_TNYear         |
| nCST_AgencyAnnEconomic  | nZero                      | bF_Always            | nZero               |
| nCST_AgencyAnnFinancial | nZero                      | bF_Always            | nZero               |
| nDAV_BladeFreq          | nCDAV_BladeFreq            |                      |                     |
| nDAV_MatQual            | AnalMaster->Unp_GravQual   |                      |                     |
| nDAV_Pavetype           | AnalMaster->Struc_Pavetype |                      |                     |
| nDAV_Pcra               | nCDAV_PcraYr1              |                      |                     |
| nDAV_Pcrw               | nCDAV_PcrwYr1              |                      |                     |
| nDAV_Surftype           | AnalMaster->Struc_SurfType |                      |                     |
| nDAV_Taci               | nCDAV_TaciYr1              |                      |                     |
| nDAV_Trvl               | nCDAV_TrvlYr1              |                      |                     |
| nDAV_Twci               | nCDAV_TwciYr1              |                      |                     |
| nPV_AUC                 |                            | bF_Always            | nCPV_AUC            |
| nPV_EconAgency          |                            | bF_Always            | nCPV_EconAgencyCost |
| nPV_FinAgency           |                            | bF_Always            | nCPV_FinAgencyCost  |
| nPV_TTC_AUC_UPGRADES    |                            | bF_Always            | nOne                |
| nPV_TTC_Benefit         |                            | bF_Always            | nCPV_TTC_Benefit    |
| nTRF_Aadt               | nCTRF_AADT_Yr1             | bF_Always            | nCTRF_AADT          |
| nTRF_CumEsals           | nCTRF_CumEsalYr1           | bFPavement_F         | nCTRF_CumEsal       |
| nTRF_CumEsals           | nCTRF_CumEsalYr1           | bFPavement_U         | nZero               |
| nTRF_Esal               | nCTRF_EsalYr1              | bFPavement_F         | nCTRF_Esal          |
| nTRF_Esal               | nCTRF_EsalYr1              | bFPavement_U         | nZero               |
| nTRF_Yax                | nCTRF_Yax_YR1              | bF_Use_AM_VehSplits  | nCTRF_Yax_AM        |
| nTRF_Yax                | nCTRF_Yax_YR1              | bF_Use_DEF_VehSplits | nCTRF_Yax_DEF       |
| nTRF_Yax                | nCTRF_Yax_YR1              | bFPavement_U         | nZero               |
| nTTC_Annual             | nCTTC_Annual               | bF_Always            | nCTTC_Annual        |
| nVOC_Annual             | nCVOC_Annual               | bF_Always            | nCVOC_Annual        |

**Table 12 – Expressions for the curves and filters of the performance models, as customized in dTIMS**

| Type     | Name              | TheExpression                                                                                            |
|----------|-------------------|----------------------------------------------------------------------------------------------------------|
| CurveExp | nCBlock_Poth      | GET4CAV_MAX(nCND_Poth)                                                                                   |
| CurveExp | nCBlock_Regravel  | GET_ANALVAR_4_YR(nCND_GravelThickness,15)                                                                |
| CurveExp | nCBlock_Roughness | GET4CAV_MAX(nCND_Irir)                                                                                   |
| CurveExp | nCBlock_SealAge   | MAX(MAX(MAX(GET_TRTYR('Reseal'),GET_TRTYR('HeavyRehab')),GET_TRTYR('LightRehab')),GET_TRTYR('FogSpray')) |
| CurveExp | nCAGE_Pave        | nAGE_Pave + 1.0                                                                                          |
| CurveExp | nZero             | 0.0                                                                                                      |
| CurveExp | nCAGE_Seal        | nAGE_Seal + 1.0                                                                                          |
| CurveExp | nZero             | 0.0                                                                                                      |
| CurveExp | nCCND_Crka_Age    | nCND_AgeCrka + 1.0                                                                                       |
| CurveExp | nZero             | 0.0                                                                                                      |
| CurveExp | nCCND_Crkw_Age    | nCND_AgeCrkw + 1.0                                                                                       |
| CurveExp | nZero             | 0.0                                                                                                      |

**Table 12 – Expressions for the curves and filters of the performance models, as customized in dTIMS**

| Type     | Name                  | TheExpression                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CurveExp | nCCND_Ravl_Age        | nCND_AgeRavl + 1.0                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| CurveExp | nZero                 | 0.0                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| CurveExp | nCCND_Aval            | IF(nDAV_Pavetype = 'F',(a_AssetValue_Pav_ReconstructCost*0.4 + (a_AssetValue_Pav_ReconstructCost * 0.6 * nCND_Cind / 100.0)) * AnalMaster->Length * 1000.0 * AnalMaster->Inv_Width, ((a_CostGravel - a_CostRegravel) + (a_CostRegravel * MIN(nCND_GravelThickness,a_HDM4_GravThickMax) / a_HDM4_GravThickMax)) * AnalMaster->Length)                                                                                                                        |
| CurveExp | nCCND_CIND_F          | -6.875 + 0.0104 * (100.0 - 15.0 / 40.0 * MIN(40.0,nCND_Crka) - 27.5 / 30.0 * MIN(30.0,nCND_Crkw) - 20.0 / 300.0 * MIN(300.0,nCND_Poth) - 17.5 / 25.0 * MIN(25.0,nCND_Rutm) - 20.0 / 9.0 * MAX(0.0,nCND_Irir - 1.0)) ** 2.0                                                                                                                                                                                                                                  |
| CurveExp | nCCND_CIND_G          | (MIN(a_HDM4_GravThickMax,nCND_GravelThickness) / a_HDM4_GravThickMax * 100.0) * (0.0012*YR**2.0 - 0.0663 * YR + 0.9996 )                                                                                                                                                                                                                                                                                                                                    |
| CurveExp | nCCND_Edgebreak       | nCND_Edge + IF(AnalMaster->Inv_Width >= a_HDM4_CWmax,0.0, (a_HDM4_KEB * IF(Left(AnalMaster->Struc_SurfType,2) = "FA",25,75) * (max(min(max(0.2+2.65 * AnalMaster->Inv_Width, (a_HDM4_CWmax-AnalMaster->Inv_Width)/10),1),0)) * (nTRF_Aadt** 2)*AnalMaster->Edgebreak_EStep * ((0.9*a_HDM4_SPEED_LIM)/10)* (0.2 + (AnalMaster->Inv_MMP/1)/1000000 ))                                                                                                         |
| CurveExp | nCCND_FA_Crka_0to50   | nCND_Crka + a_HDM4_KCP * ((1.0-0.12*a_HDM4_CRK_CRT)/1.0) * 1.0 * ((1.0*1.84*0.45*1.0 + MIN(nCND_Crka,(100.0-nCND_Crka))**0.45)**(1.0/0.45) - MIN(nCND_Crka,100.0-nCND_Crka))                                                                                                                                                                                                                                                                                |
| CurveExp | nCCND_FC_Crka_0to50   | nCND_Crka + a_HDM4_KCP * ((1.0-0.12*a_HDM4_CRK_CRT)/1.0) * 1.0 * ((1.0*1.76*0.32*1.0 + MIN(nCND_Crka,(100.0-nCND_Crka))**0.32)**(1.0/0.32) - MIN(nCND_Crka,100.0-nCND_Crka))                                                                                                                                                                                                                                                                                |
| CurveExp | nCCND_FA_Crka_50to100 | nCND_Crka + a_HDM4_KCP * ((1.0-0.12*a_HDM4_CRK_CRT)/1.0) * -1.0 * ((-1.0*1.84*0.45*1.0 + MIN(nCND_Crka,(100.0-nCND_Crka))**0.45)**(1.0/0.45) - MIN(nCND_Crka,100.0-nCND_Crka))                                                                                                                                                                                                                                                                              |
| CurveExp | nCCND_FC_Crka_50to100 | nCND_Crka + a_HDM4_KCP * ((1.0-0.12*a_HDM4_CRK_CRT)/1.0) * -1.0 * ((-1.0*1.76*0.32*1.0 + MIN(nCND_Crka,(100.0-nCND_Crka))**0.32)**(1.0/0.32) - MIN(nCND_Crka,100.0-nCND_Crka))                                                                                                                                                                                                                                                                              |
| CurveExp | nHundred              | 100.0                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| CurveExp | nZero                 | 0.0                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| CurveExp | nCCND_FA_Crkw_0to50   | nCND_Crkw + a_HDM4_KCP * ((1.0-0.12*a_HDM4_CRK_CRT)/1.0) * 1.0 * ((1.0*1.84*0.45*1.0 + MIN(nCND_Crkw,(100.0-nCND_Crkw))**0.45)**(1.0/0.45) - MIN(nCND_Crkw,100.0-nCND_Crkw))                                                                                                                                                                                                                                                                                |
| CurveExp | nCCND_FC_Crkw_0to50   | nCND_Crkw + a_HDM4_KCP * ((1.0-0.12*a_HDM4_CRK_CRT)/1.0) * 1.0 * ((1.0*1.76*0.32*1.0 + MIN(nCND_Crkw,(100.0-nCND_Crkw))**0.32)**(1.0/0.32) - MIN(nCND_Crkw,100.0-nCND_Crkw))                                                                                                                                                                                                                                                                                |
| CurveExp | nCCND_FA_Crkw_50to100 | nCND_Crkw + a_HDM4_KCP * ((1.0-0.12*a_HDM4_CRK_CRT)/1.0) * -1.0 * ((-1.0*1.84*0.45*1.0 + MIN(nCND_Crkw,(100.0-nCND_Crkw))**0.45)**(1.0/0.45) - MIN(nCND_Crkw,100.0-nCND_Crkw))                                                                                                                                                                                                                                                                              |
| CurveExp | nCCND_FC_Crkw_50to100 | nCND_Crkw + a_HDM4_KCP * ((1.0-0.12*a_HDM4_CRK_CRT)/1.0) * -1.0 * ((-1.0*1.76*0.32*1.0 + MIN(nCND_Crkw,(100.0-nCND_Crkw))**0.32)**(1.0/0.32) - MIN(nCND_Crkw,100.0-nCND_Crkw))                                                                                                                                                                                                                                                                              |
| CurveExp | nHundred              | 100.0                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| CurveExp | nCCND_D95             | IF(AnalMaster->Struc_Pavetype <> 'U',0.0,IF(nCND_GravelThickness > 0.0,XTAB(MatProperties,AnalMaster->Unp_GravQual,5.0,34.0),XTAB(MatProperties,0.0,5.0,100.0)))                                                                                                                                                                                                                                                                                            |
| CurveExp | nCCND_GravelThick_U   | nCND_GravelThickness - 3.65 * (3.46 + 0.246 * IF(AnalMaster->Inv_MMP > 0.0,AnalMaster->Inv_MMP,a_HDM4_MMP) * AnalMaster->Unp_RF + MAX(0.0,0.022 + 0.969 * AnalMaster->Unp_C / 57300.0 + 0.00342 * IF(AnalMaster->Inv_MMP > 0.0,AnalMaster->Inv_MMP,a_HDM4_MMP) * nCND_P075 - 0.0092 * IF(AnalMaster->Inv_MMP > 0.0,AnalMaster->Inv_MMP,a_HDM4_MMP) * nCND_PI - 0.101 * IF(AnalMaster->Inv_MMP > 0.0,AnalMaster->Inv_MMP,a_HDM4_MMP)) * MAX(nTRF_Aadt,10.0)) |
| CurveExp | nCCND_GravelThick_F   | nCND_GravelThickness                                                                                                                                                                                                                                                                                                                                                                                                                                        |

**Table 12 – Expressions for the curves and filters of the performance models, as customized in dTIMS**

| Type     | Name               | TheExpression                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CurveExp | nCCnd_Irir_F       | nCND_Irir + a_HDM4_KGP * ((134.0 * EXP(a_HDM4_m_ENVIRONCOEFF * a_HDM4_KGM * nAGE_Pave) * ((1.0 + nCND_Snp) ** (-5.0)) * nTRF_Esal/1000000.0) + (0.0066 * (nCND_Crka - GET_ANALVAR_4_YR(nCND_Crka, YR-1.0))) + (0.088 * (nCND_Rutm - GET_ANALVAR_4_YR(nCND_Rutm, YR-1.0))) + (0.00019 * (2.0 - MAX(MIN(0.25 * (AnalMaster->Inv_Width-3.0), 1.0), 0.0) * MAX((1.0 - nTRF_Aadt/5000.0), 0.0)) * (nCND_Poth ** 1.5 - GET_ANALVAR_4_YR(nCND_Poth, YR-1.0) ** 1.5))) + (a_HDM4_m_ENVIRONCOEFF * a_HDM4_KGM * nCND_Irir) |
| CurveExp | nCCnd_Irir_G       | IF(nCND_GravelThickness > 0.0, IF(nDAV_BladeFreq = 30.0, 3.3886 * EXP(0.0023 * MIN(400.0, nTRF_Aadt)), IF(nDAV_BladeFreq = 60.0, 4.4327 * EXP(0.0026 * MIN(400.0, nTRF_Aadt)), 30.0)), IF(nDAV_BladeFreq = 30.0, 6.9329 * EXP(0.0027 * MIN(400.0, nTRF_Aadt)), IF(nDAV_BladeFreq = 60.0, 6.8584 * (MIN(400.0, nTRF_Aadt) ** 0.1909), 30.0)))                                                                                                                                                                      |
| CurveExp | nCCND_LabMax       | IF(AnalMaster->Struc_Pavetype <> 'U', 0.0, IF(nCND_GravelThickness > 0.0, XTAB(MatProperties, AnalMaster->Unp_GravQual, 6.0, 53.0), XTAB(MatProperties, 0.0, 6.0, 110.0)))                                                                                                                                                                                                                                                                                                                                        |
| CurveExp | nCCND_P075         | IF(AnalMaster->Struc_Pavetype <> 'U', 0.0, IF(nCND_GravelThickness > 0.0, XTAB(MatProperties, AnalMaster->Unp_GravQual, 4.0, 12.0), XTAB(MatProperties, 0.0, 4.0, 10.0)))                                                                                                                                                                                                                                                                                                                                         |
| CurveExp | nCCND_P2           | IF(AnalMaster->Struc_Pavetype <> 'U', 0.0, IF(nCND_GravelThickness > 0.0, XTAB(MatProperties, AnalMaster->Unp_GravQual, 2.0, 40.0), XTAB(MatProperties, 0.0, 2.0, 30.0)))                                                                                                                                                                                                                                                                                                                                         |
| CurveExp | nCCND_P26          | IF(AnalMaster->Struc_Pavetype <> 'U', 0.0, IF(nCND_GravelThickness > 0.0, XTAB(MatProperties, AnalMaster->Unp_GravQual, 1.0, 82.0), XTAB(MatProperties, 0.0, 1.0, 70.0)))                                                                                                                                                                                                                                                                                                                                         |
| CurveExp | nCCND_P425         | IF(AnalMaster->Struc_Pavetype <> 'U', 0.0, IF(nCND_GravelThickness > 0.0, XTAB(MatProperties, AnalMaster->Unp_GravQual, 3.0, 18.0), XTAB(MatProperties, 0.0, 3.0, 20.0)))                                                                                                                                                                                                                                                                                                                                         |
| CurveExp | nCCND_PI           | IF(AnalMaster->Struc_Pavetype <> 'U', 0.0, IF(nCND_GravelThickness > 0.0, XTAB(MatProperties, AnalMaster->Unp_GravQual, 8.0, 12.0), XTAB(MatProperties, 0.0, 8.0, 20.0)))                                                                                                                                                                                                                                                                                                                                         |
| CurveExp | nCCND_PL           | IF(AnalMaster->Struc_Pavetype <> 'U', 0.0, IF(nCND_GravelThickness > 0.0, XTAB(MatProperties, AnalMaster->Unp_GravQual, 7.0, 18.0), XTAB(MatProperties, 0.0, 7.0, 15.0)))                                                                                                                                                                                                                                                                                                                                         |
| CurveExp | nCCND_Poth         | nCND_Poth + nCCND_Poth_Enlargemnt + nCCND_Poth_WideCrack + nCCND_Poth_Ravelling                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| CurveExp | nCCND_Ravl_0to50   | nCND_RAVL + (a_HDM4_KVP/a_HDM4_Ravel_RRF) * 1.0 * (1.0 * (0.3 + 1.5 * nTRF_Yax) * 0.352)                                                                                                                                                                                                                                                                                                                                                                                                                          |
| CurveExp | nCCND_Ravl_50TO100 | nCND_RAVL + (a_HDM4_KVP/a_HDM4_Ravel_RRF) * (100.0 - nCND_RAVL)                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| CurveExp | nHundred           | 100.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| CurveExp | nCCND_RoughHigh    | IF(AnalMaster->Struc_Pavetype <> 'U', 0.0, 25.0 * 13.0)                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| CurveExp | nCCND_RoughnessMax | IF(AnalMaster->Struc_Pavetype <> 'U', 0.0, MAX(279.0 - 421.0 * (0.5 - IF(nCND_P425 = 0.0, 1.0, nCND_P075 / nCND_P425)) ** 2.0 + 0.22 * AnalMaster->Unp_C - 9.93 * AnalMaster->Unp_RF * IF(AnalMaster->Inv_MMP > 0.0, AnalMaster->Inv_MMP, a_HDM4_MMP), 150.0))                                                                                                                                                                                                                                                    |
| CurveExp | nCCND_RoughnessMin | IF(AnalMaster->Struc_Pavetype <> 'U', 0.0, MAX(10.0, MIN(100.0, 4.69 * nCND_D95 * (1.0 - 2.78 * nCND_SMG))))                                                                                                                                                                                                                                                                                                                                                                                                      |
| CurveExp | nCCND_RoughTemp    | nCND_Rough_BeforeBlade                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| CurveExp | nCCND_Rutm         | nCND_Rutm + a_HDM4_KRP * (44950.0 * nCND_Snp ** (-1.14) * ((nTRF_Esal/1000000.0) ** 0.11) * IF(AnalMaster->Struc_COMP > 80.0 And AnalMaster->Struc_COMP < 120.0, AnalMaster->Struc_COMP, 95.0) ** (-2.3)) + IF(nCND_Crka > 0.0, a_HDM4_KRP * (0.0000248 * nCND_Snp ** (-0.84) * (nTRF_Esal / 1000000.0) ** 0.14 * IF(AnalMaster->Inv_MMP > 0.0, (AnalMaster->Inv_MMP ** 1.07), (a_HDM4_MMP ** 1.07)) * (nCND_Crka ** 1.11)), 0.0)                                                                                 |
| CurveExp | nCCND_SMG          | IF(AnalMaster->Struc_Pavetype <> 'U', 0.0, MIN(MIN((IF(nCND_D95 > 4.0, (LOG(nCND_P2 / 95.0) / LOG(2.0 / nCND_D95)), IF(nCND_D95 > 1.0, (LOG(nCND_P425 / 95.0) / LOG(0.425 / nCND_D95)), 0.3)) + IF(nCND_D95 > 1.0, (LOG(nCND_P425 / 95.0) / LOG(0.425 / nCND_D95)), 0.3)) + IF(nCND_D95 > 4.0, (LOG(nCND_P075 / 95.0) / LOG(0.075 / nCND_D95)), 0.3)) / 3.0, 1.0 - (IF(nCND_D95                                                                                                                                   |

**Table 12 – Expressions for the curves and filters of the performance models, as customized in dTIMS**

| Type     | Name                | TheExpression                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                     | $> 4.0, (\text{LOG}(\text{nCND\_P2} / 95.0) / \text{LOG}(2.0 / \text{nCND\_D95})), \text{IF}(\text{nCND\_D95} > 1.0, (\text{LOG}(\text{nCND\_P425} / 95.0) / \text{LOG}(0.425 / \text{nCND\_D95})), 0.3) + \text{IF}(\text{nCND\_D95} > 1.0, (\text{LOG}(\text{nCND\_P425} / 95.0) / \text{LOG}(0.425 / \text{nCND\_D95})), 0.3) + \text{IF}(\text{nCND\_D95} > 4.0, (\text{LOG}(\text{nCND\_P075} / 95.0) / \text{LOG}(0.075 / \text{nCND\_D95})), 0.3) / (3.0, 0.36))$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| CurveExp | nCCND_Snp           | $\text{nCND\_Snp} - 0.0000758 * (\text{MIN}(63.0, \text{nCND\_Crka}) * \text{a\_HDM4\_HSNEW} + \text{MAX}(\text{MIN}(\text{nCND\_Crka} - (0.62 * \text{nDAV\_Pcra} + 0.39 * \text{nDAV\_Pcrw}), 40.0), 0.0) * \text{a\_HDM4\_HSOLD})$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| CurveExp | n30                 | 30.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| CurveExp | nCPV_TNYear         | $\text{GET4CAV\_PVDIFF}(\text{nCND\_TNYear}, 31, \text{nTRF\_Aadt}, 0)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| CurveExp | nCPV_AUC            | $\text{GET4CAV\_PVDIFF}(\text{nCND\_Cind}, 0, \text{nTRF\_Aadt}, 1)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| CurveExp | nCPV_EconAgencyCost | $\text{GET4CAV\_PV}(\text{nCST\_AgencyAnnEconomic})$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| CurveExp | nCPV_FinAgencyCost  | $\text{GET4CAV\_PV}(\text{nCST\_AgencyAnnFinancial})$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| CurveExp | nOne                | 1.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| CurveExp | nCPV_TTC_Benefit    | $\text{GET4CAV\_PVDIFF}(\text{nTTC\_Annual}, 9999999999, \text{nTRF\_Aadt}, 0) + \text{IF}(\text{AnalMaster} \rightarrow \text{Com\_Year} > 0.0 \text{ And } \text{IS\_COMMITTED}(), \text{AnalMaster} \rightarrow \text{Com\_ExBen}, 0.0)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| CurveExp | nCTRF_AADT          | $\text{IF}(\text{AnalMaster} \rightarrow \text{Traf\_Growth} > -1.0, \text{nTRF\_Aadt} * (1.0 + \text{AnalMaster} \rightarrow \text{Traf\_Growth} / 100.0), \text{nTRF\_Aadt} * (1.0 + \text{a\_Traf\_Growth} / 100.0))$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| CurveExp | nCTRF_CumEsal       | $\text{nTRF\_CumEsals} + \text{nTRF\_Esal}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| CurveExp | nCTRF_Esal          | $\text{IF}(\text{nDAV\_Pavetype} = 'U', 0.0, (\text{AnalMaster} \rightarrow \text{Traf\_Veh1\_Esals} + \text{AnalMaster} \rightarrow \text{Traf\_Veh2\_Esals} + \text{AnalMaster} \rightarrow \text{traf\_Veh3\_Esals} + \text{AnalMaster} \rightarrow \text{Traf\_Veh4\_Esals} + \text{AnalMaster} \rightarrow \text{Traf\_Veh5\_Esals} + \text{AnalMaster} \rightarrow \text{Traf\_Veh6\_Esals} + \text{AnalMaster} \rightarrow \text{Traf\_Veh7\_Esals} + \text{AnalMaster} \rightarrow \text{Traf\_Veh8\_Esals} + \text{AnalMaster} \rightarrow \text{Traf\_Veh9\_Esals}) * ((1.0 + (\text{a\_Traf\_Growth} / 100.0)) * (\text{YR} - 1.0)))$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| CurveExp | nCTRF_Yax_AM        | $\text{IF}(\text{nDAV\_Pavetype} < 'F', 0.0, \text{nTRF\_Aadt} * ((\text{AnalMaster} \rightarrow \text{Traf\_Veh1\_Split} / 100.0 * \text{IF}(\text{AnalMaster} \rightarrow \text{Traf\_Veh1\_Axles} > -1.0, \text{AnalMaster} \rightarrow \text{Traf\_Veh1\_Axles}, \text{a\_Traf\_Veh1\_Axles})) + (\text{AnalMaster} \rightarrow \text{Traf\_Veh2\_split} / 100.0 * \text{IF}(\text{AnalMaster} \rightarrow \text{Traf\_Veh2\_Axles} > -1.0, \text{AnalMaster} \rightarrow \text{Traf\_Veh2\_Axles}, \text{a\_Traf\_Veh2\_Axles})) + (\text{AnalMaster} \rightarrow \text{Traf\_Veh3\_Split} / 100.0 * \text{IF}(\text{AnalMaster} \rightarrow \text{Traf\_Veh3\_Axles} > -1.0, \text{AnalMaster} \rightarrow \text{Traf\_Veh3\_Axles}, \text{a\_Traf\_Veh3\_Axles})) + (\text{AnalMaster} \rightarrow \text{Traf\_Veh4\_Split} / 100.0 * \text{IF}(\text{AnalMaster} \rightarrow \text{Traf\_Veh4\_Axles} > -1.0, \text{AnalMaster} \rightarrow \text{Traf\_Veh4\_Axles}, \text{a\_Traf\_Veh4\_Axles})) + (\text{AnalMaster} \rightarrow \text{Traf\_Veh5\_Split} / 100.0 * \text{IF}(\text{AnalMaster} \rightarrow \text{Traf\_Veh5\_Axles} > -1.0, \text{AnalMaster} \rightarrow \text{Traf\_Veh5\_Axles}, \text{a\_Traf\_Veh5\_Axles})) + (\text{AnalMaster} \rightarrow \text{Traf\_Veh6\_Split} / 100.0 * \text{IF}(\text{AnalMaster} \rightarrow \text{Traf\_Veh6\_Axles} > -1.0, \text{AnalMaster} \rightarrow \text{Traf\_Veh6\_Axles}, \text{a\_Traf\_Veh6\_Axles})) + (\text{AnalMaster} \rightarrow \text{Traf\_Veh7\_Split} / 100.0 * \text{IF}(\text{AnalMaster} \rightarrow \text{Traf\_Veh7\_Axles} > -1.0, \text{AnalMaster} \rightarrow \text{Traf\_Veh7\_Axles}, \text{a\_Traf\_Veh7\_Axles})) + (\text{AnalMaster} \rightarrow \text{Traf\_Veh8\_Split} / 100.0 * \text{IF}(\text{AnalMaster} \rightarrow \text{traf\_Veh8\_Axles} > -1.0, \text{AnalMaster} \rightarrow \text{traf\_Veh8\_Axles}, \text{a\_Traf\_Veh8\_Axles})) + (\text{AnalMaster} \rightarrow \text{Traf\_Veh9\_Split} / 100.0 * \text{IF}(\text{AnalMaster} \rightarrow \text{Traf\_Veh9\_Axles} > -1.0, \text{AnalMaster} \rightarrow \text{Traf\_Veh9\_Axles}, \text{a\_Traf\_Veh9\_Axles}))) * 365.0 * \text{AnalMaster} \rightarrow \text{Traf\_Lane\_Max} / 100.0 / 1000000.0)$ |
| CurveExp | nCTRF_Yax_DEF       | $\text{IF}(\text{nDAV\_Pavetype} < 'F', 0.0, \text{nTRF\_Aadt} * ((\text{a\_Traf\_Veh1\_Split} / 100.0 * \text{IF}(\text{AnalMaster} \rightarrow \text{Traf\_Veh1\_Axles} > -1.0, \text{AnalMaster} \rightarrow \text{Traf\_Veh1\_Axles}, \text{a\_Traf\_Veh1\_Axles})) + (\text{a\_Traf\_Veh2\_Split} / 100.0 * \text{IF}(\text{AnalMaster} \rightarrow \text{Traf\_Veh2\_Axles} > -1.0, \text{AnalMaster} \rightarrow \text{Traf\_Veh2\_Axles}, \text{a\_Traf\_Veh2\_Axles})) + (\text{a\_Traf\_Veh3\_Split} / 100.0 * \text{IF}(\text{AnalMaster} \rightarrow \text{Traf\_Veh3\_Axles} > -1.0, \text{AnalMaster} \rightarrow \text{Traf\_Veh3\_Axles}, \text{a\_Traf\_Veh3\_Axles})) + (\text{a\_Traf\_Veh4\_Split} / 100.0 * \text{IF}(\text{a\_Traf\_Veh4\_Axles} > -1.0, \text{AnalMaster} \rightarrow \text{Traf\_Veh4\_Axles}, \text{a\_Traf\_Veh4\_Axles})) + (\text{a\_Traf\_Veh5\_Split} / 100.0 * \text{IF}(\text{AnalMaster} \rightarrow \text{Traf\_Veh5\_Axles} > -1.0, \text{AnalMaster} \rightarrow \text{Traf\_Veh5\_Axles}, \text{a\_Traf\_Veh5\_Axles})) + (\text{a\_Traf\_Veh6\_Split} / 100.0 * \text{IF}(\text{AnalMaster} \rightarrow \text{Traf\_Veh6\_Axles} > -1.0, \text{AnalMaster} \rightarrow \text{Traf\_Veh6\_Axles}, \text{a\_Traf\_Veh6\_Axles})) + (\text{a\_Traf\_Veh7\_Split} / 100.0 * \text{IF}(\text{AnalMaster} \rightarrow \text{Traf\_Veh7\_Axles} > -1.0, \text{AnalMaster} \rightarrow \text{Traf\_Veh7\_Axles}, \text{a\_Traf\_Veh7\_Axles})) + (\text{a\_Traf\_Veh8\_Split} / 100.0 * \text{IF}(\text{AnalMaster} \rightarrow \text{traf\_Veh8\_Axles} > -1.0, \text{AnalMaster} \rightarrow \text{traf\_Veh8\_Axles}, \text{a\_Traf\_Veh8\_Axles})) + (\text{a\_Traf\_Veh9\_Split} / 100.0 * \text{IF}(\text{AnalMaster} \rightarrow \text{Traf\_Veh9\_Axles} > -1.0, \text{AnalMaster} \rightarrow \text{Traf\_Veh9\_Axles}, \text{a\_Traf\_Veh9\_Axles}))) * 365.0 * \text{AnalMaster} \rightarrow \text{Traf\_Lane\_Max} / 100.0 / 1000000.0)$                                                                                                                                                                                                                                                                               |
| CurveExp | nCTTC_Annual        | $\text{nVOC\_Annual} + \text{GST\_COST\_E} / \text{GET\_LENGTH}(\text{AnalMaster})$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

**Table 12 – Expressions for the curves and filters of the performance models, as customized in dTIMS**

| Type     | Name                  | TheExpression                                                                                                                        |
|----------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| CurveExp | nCVOC_Annual          | nCVOC_Type01 + nCVOC_Type02 + nCVOC_Type03 + nCVOC_Type04 + nCVOC_Type05 + nCVOC_Type06 + nCVOC_Type07 + nCVOC_Type08 + nCVOC_Type09 |
| Filter   | bF_Always             | TRUE                                                                                                                                 |
| Filter   | bFPavement_F          | nDAV_Pavetype = 'F'                                                                                                                  |
| Filter   | bFPavement_U          | nDAV_Pavetype = 'U'                                                                                                                  |
| Filter   | bFCND_Crka0           | nDAV_Pavetype = 'F' And nCND_AgeCrka <= nDAV_Taci                                                                                    |
| Filter   | bFCND_Crka_FA_0to50   | nDAV_Pavetype = 'F' And LEFT(nDAV_Surftype,2.0) = 'FA' And nCND_AgeCrka > nDAV_Taci And nCND_Crka < 50.0                             |
| Filter   | bFCND_Crka_FC_0to50   | nDAV_Pavetype = 'F' And LEFT(nDAV_Surftype,2.0) <> 'FA' And nCND_AgeCrka > nDAV_Taci And nCND_Crka < 50.0                            |
| Filter   | bFCND_Crka_FA_50to100 | nDAV_Pavetype = 'F' And LEFT(nDAV_Surftype,2.0) = 'FA' And nCND_AgeCrka > nDAV_Taci And nCND_Crka >= 50.0 And nCND_Crka < 100.0      |
| Filter   | bFCND_Crka_FC_50to100 | nDAV_Pavetype = 'F' And LEFT(nDAV_Surftype,2.0) <> 'FA' And nCND_AgeCrka > nDAV_Taci And nCND_Crka >= 50.0 And nCND_Crka < 100.0     |
| Filter   | bFCND_Crka100         | nDAV_Pavetype = 'F' And nCND_Crka >= 100.0                                                                                           |
| Filter   | bFCND_Crkw0           | nDAV_Pavetype = 'F' And nCND_AgeCrkw <= nDAV_Twci                                                                                    |
| Filter   | bFCND_Crkw_FA_0to50   | nDAV_Pavetype = 'F' And LEFT(nDAV_Surftype,2.0) = 'FA' And nCND_AgeCrkw > nDAV_Twci And nCND_Crkw < 50.0                             |
| Filter   | bFCND_Crkw_FC_0to50   | nDAV_Pavetype = 'F' And LEFT(nDAV_Surftype,2.0) <> 'FA' And nCND_AgeCrkw > nDAV_Twci And nCND_Crkw < 50.0                            |
| Filter   | bFCND_Crkw_FA_50to100 | nDAV_Pavetype = 'F' And LEFT(nDAV_Surftype,2.0) = 'FA' And nCND_AgeCrkw > nDAV_Twci And nCND_Crkw >= 50.0 And nCND_Crkw < 100.0      |
| Filter   | bFCND_Crkw_FC_50to100 | nDAV_Pavetype = 'F' And LEFT(nDAV_Surftype,2.0) <> 'FA' And nCND_AgeCrkw > nDAV_Twci And nCND_Crkw >= 50.0 And nCND_Crkw < 100.0     |
| Filter   | bFCND_Crkw100         | nDAV_Pavetype = 'F' And nCND_Crkw >= 100.0                                                                                           |
| Filter   | bFCND_Ravl0           | nDAV_Pavetype = 'F' And nCND_AgeRavl <= nDAV_Trvl                                                                                    |
| Filter   | bFCND_Ravl_0to50      | nDAV_Pavetype = 'F' And nCND_AgeRavl > nDAV_Trvl And nCND_RAVL < 50.0                                                                |
| Filter   | bFCND_Ravl_50to100    | nDAV_Pavetype = 'F' And nCND_AgeRavl > nDAV_Trvl And nCND_RAVL >= 50.0 And nCND_RAVL < 100.0                                         |
| Filter   | bFCND_Ravl100         | nDAV_Pavetype = 'F' And nCND_RAVL >= 100.0                                                                                           |
| Filter   | bFPavement_U          | nDAV_Pavetype = 'U'                                                                                                                  |
| Filter   | bF_Use_AM_VehSplits   | nDAV_Pavetype = 'F' And nCCAV_SumOfVehTypes >= 96.0 And nCCAV_SumOfVehTypes <= 104.0                                                 |
| Filter   | bF_Use_DEF_VehSplits  | nDAV_Pavetype = 'F' And nCCAV_SumOfVehTypes <> 100.0                                                                                 |

## 17.2. Treatments and triggers

**Table 13 – Treatments and treatments costs in dTIMS**

| Treatment  | Description              | Cost type | Expression name | Expression                                                                                                                    |
|------------|--------------------------|-----------|-----------------|-------------------------------------------------------------------------------------------------------------------------------|
| Crack_Seal | Crack Seal               | Economic  | nCTEC_CrackSeal | (nCND_Crka / 100.0) * GET_LENGTH(AnalMaster) * 1000.0 * AnalMaster->Inv_Width * a_CostCrackSeal * 2.0 * a_Cost_EconomicFactor |
|            |                          | Financial | nCTFC_CrackSeal | (nCND_Crka / 100.0) * GET_LENGTH(AnalMaster) * 1000.0 * AnalMaster->Inv_Width * a_CostCrackSeal * 2.0                         |
| FogSpray   | Fogspray minor treatment | Economic  | nCTEC_FogSpray  | GET_LENGTH(AnalMaster) * 1000.0 * AnalMaster->Inv_Width * a_CostFogSpray * a_Cost_EconomicFactor                              |
|            |                          | Financial | nCTFC_FogSpray  | GET_LENGTH(AnalMaster) * 1000.0 * AnalMaster->Inv_Width * a_CostFogSpray                                                      |



**Table 13 – Treatments and treatments costs in dTIMS**

| Treatment    | Description                             | Cost type | Expression name    | Expression                                                                                                         |
|--------------|-----------------------------------------|-----------|--------------------|--------------------------------------------------------------------------------------------------------------------|
| HeavyRehab   | Heavy Rehabilitation                    | Economic  | nCTEC_Heavy_Rehab  | GET_LENGTH(AnalMaster) * 1000.0 * AnalMaster->Inv_Width * a_CostHeavyRehab * a_Cost_EconomicFactor                 |
|              |                                         | Financial | nCTFC_Heavy_rehab  | GET_LENGTH(AnalMaster) * 1000.0 * AnalMaster->Inv_Width * a_CostHeavyRehab                                         |
| LightRehab   | Light Rehabilitation                    | Economic  | nCTEC_Light_Rehab  | GET_LENGTH(AnalMaster) * 1000.0 * AnalMaster->Inv_Width * a_CostLightRehab * a_Cost_EconomicFactor                 |
|              |                                         | Financial | nCTFC_Light_Rehab  | GET_LENGTH(AnalMaster) * 1000.0 * AnalMaster->Inv_Width * a_CostLightRehab                                         |
| Patching     | Patching of all Potholes > 0.2%         | Economic  | nCTEC_Patching     | a_CostPatching * nCND_Poth * 0.1 * a_Cost_EconomicFactor                                                           |
|              |                                         | Financial | nCTFC_Patching     | nCND_Poth * 0.1 * a_CostPatching                                                                                   |
| Regravel     | Regravelling                            | Economic  | nCTEC_Gravel       | GET_LENGTH(AnalMaster) * IF(AnalMaster->Struc_SurfType = 'UG',a_CostRegravel,a_CostGravel) * a_Cost_EconomicFactor |
|              |                                         | Financial | nCTFC_Gravel       | GET_LENGTH(AnalMaster) * IF(AnalMaster->Struc_SurfType = 'UG',a_CostRegravel,a_CostGravel)                         |
| Reseal       | Reseal                                  | Economic  | nCTEC_Seal         | GET_LENGTH(AnalMaster) * 1000.0 * AnalMaster->Inv_Width * a_CostSeal * a_Cost_EconomicFactor                       |
|              |                                         | Financial | nCTFC_Seal         | GET_LENGTH(AnalMaster) * 1000.0 * AnalMaster->Inv_Width * a_CostSeal                                               |
| Reseal_Light | Reseal light                            | Economic  | nCTEC_Reseal_Light | GET_LENGTH(AnalMaster) * 1000.0 * AnalMaster->Inv_Width * a_Cost_Reseal_Light * a_Cost_EconomicFactor              |
|              |                                         | Financial | nCTFC_Reseal_Light | GET_LENGTH(AnalMaster) * 1000.0 * AnalMaster->Inv_Width * a_Cost_Reseal_Light                                      |
| UpgradePave  | Upgravel gravel road to paved standards | Economic  | nCTEC_UpgradePave  | GET_LENGTH(AnalMaster) * IF(nTRF_Aadt > 500.0,a_CostPaveUpgradeHigh,a_CostPaveUpgradeLow) * a_Cost_EconomicFactor  |
|              |                                         | Financial | nCTFC_UpgradePave  | GET_LENGTH(AnalMaster) * IF(nTRF_Aadt > 500.0,a_CostPaveUpgradeHigh,a_CostPaveUpgradeLow)                          |

**Table 14 – Treatment reset values in dTIMS**

| Treatment | Analysis variable (Performance parameter) | Expression name of reset value | Reset expression   |
|-----------|-------------------------------------------|--------------------------------|--------------------|
| FogSpray  | nCND_AgeCrka                              | nCTR_Fogspray_AgeCrka          | nCND_AgeCrka - 1.0 |
| FogSpray  | nCND_AgeCrkw                              | nCTR_FogSpray_AgeCrkw          | nCND_AgeCrkw - 1.0 |
| FogSpray  | nCND_AgeRavl                              | nCTR_FogSpray_AgeRavl          | nCND_AgeRavl - 1.0 |
| FogSpray  | nCND_Crka                                 | nZero                          | 0.0                |
| FogSpray  | nCND_Crkw                                 | nZero                          | 0.0                |
| FogSpray  | nCND_RAVL                                 | nZero                          | 0.0                |
| FogSpray  | nCND_Poth                                 | nZero                          | 0.0                |
| FogSpray  | nCND_Rutm                                 | nCTR_Fogspray_Rutm             | 0.99 * nCND_Rutm   |
| FogSpray  | nCND_Ruts                                 | nCTR_All_Ruts                  | nCND_Rutm * 0.5    |
| FogSpray  | nCND_Irir                                 | nCTR_Fogspray_Irir             | nCND_Irir * 0.99   |

Table 14 – Treatment reset values in dTIMS

| Treatment  | Analysis variable<br>(Performance<br>parameter) | Expression name of reset<br>value | Reset expression                                                                                                                                                                                                                                                                                                                                           |
|------------|-------------------------------------------------|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FogSpray   | nCND_Aval                                       | nCCND_Aval                        | IF(nDAV_Pavetype =<br>'F',(a_AssetValue_Pav_ReconstructCost*0.4 +<br>(a_AssetValue_Pav_ReconstructCost * 0.6 *<br>nCND_Cind / 100.0)) * AnalMaster->Length * 1000.0<br>* AnalMaster->Inv_Width, ((a_CostGravel -<br>a_CostRegravel) + (a_CostRegravel *<br>MIN(nCND_GravelThickness,a_HDM4_GravThickMa<br>x) / a_HDM4_GravThickMax)) * AnalMaster->Length) |
| FogSpray   | nCND_TNYear                                     | nCTR_All_TN_Year                  | YR                                                                                                                                                                                                                                                                                                                                                         |
| FogSpray   | nCST_AgencyAnnEco<br>nomic                      | nCCST_AgencyEconCost              | GST_COST_E / GET_LENGTH(AnalMaster)                                                                                                                                                                                                                                                                                                                        |
| FogSpray   | nCST_AgencyAnnFin<br>ancial                     | nCCST_AgencyFinCost               | GST_COST_F / GET_LENGTH(AnalMaster)                                                                                                                                                                                                                                                                                                                        |
| FogSpray   | nVOC_Annual                                     | nCVOC_Annual                      | nCVOC_Type01 + nCVOC_Type02 + nCVOC_Type03<br>+ nCVOC_Type04 + nCVOC_Type05 +<br>nCVOC_Type06 + nCVOC_Type07 + nCVOC_Type08<br>+ nCVOC_Type09                                                                                                                                                                                                              |
| FogSpray   | nTTC_Annual                                     | nCTTC_Annual                      | nVOC_Annual + GST_COST_E /<br>GET_LENGTH(AnalMaster)                                                                                                                                                                                                                                                                                                       |
| FogSpray   | nCND_Cind                                       | nCCND_CIND_F                      | -6.875 + 0.0104 * (100.0 - 15.0 / 40.0 *<br>MIN(40.0,nCND_Crka) - 27.5 / 30.0 *<br>MIN(30.0,nCND_Crkw) - 20.0 / 300.0 *<br>MIN(300.0,nCND_Poth) - 17.5 / 25.0 *<br>MIN(25.0,nCND_Rutm) - 20.0 / 9.0 *<br>MAX(0.0,nCND_Irir - 1.0)) ** 2.0                                                                                                                  |
| HeavyRehab | nDAV_Surftype                                   | cCTR_Rehab_Surftype               | 'FA'                                                                                                                                                                                                                                                                                                                                                       |
| HeavyRehab | nAGE_Pave                                       | nZero                             | 0.0                                                                                                                                                                                                                                                                                                                                                        |
| HeavyRehab | nAGE_Seal                                       | nZero                             | 0.0                                                                                                                                                                                                                                                                                                                                                        |
| HeavyRehab | nTRF_CumEsals                                   | nZero                             | 0.0                                                                                                                                                                                                                                                                                                                                                        |
| HeavyRehab | nCND_Snp                                        | nCTR_HeavyRehab_SNC               | nCND_Snp + 0.014 * 15.0 + 0.005 * 120.0                                                                                                                                                                                                                                                                                                                    |
| HeavyRehab | nDAV_Pcra                                       | nZero                             | 0.0                                                                                                                                                                                                                                                                                                                                                        |
| HeavyRehab | nDAV_Pcrw                                       | nZero                             | 0.0                                                                                                                                                                                                                                                                                                                                                        |
| HeavyRehab | nDAV_Taci                                       | nCTR_All_Taci                     | MAX(1.0,13.21 * a_HDM4_KCI * EXP(-20.7 *<br>(nTRF_Esal / 1000000.0) / nCND_Snp / nCND_Snp) *<br>MAX(0.0,(1.0 - nDAV_Pcrw / 20.0)))                                                                                                                                                                                                                         |
| HeavyRehab | nDAV_Twci                                       | nCTR_All_Twci                     | MAX(2.66 + 0.88 * nDAV_Taci,1.16 * nDAV_Taci)                                                                                                                                                                                                                                                                                                              |
| HeavyRehab | nDAV_Trvl                                       | nCTR_All_Trvl                     | a_HDM4_KVI * 10.5 * EXP(-0.156 * nTRF_Yax)                                                                                                                                                                                                                                                                                                                 |
| HeavyRehab | nCND_AgeCrka                                    | nZero                             | 0.0                                                                                                                                                                                                                                                                                                                                                        |
| HeavyRehab | nCND_AgeRavl                                    | nZero                             | 0.0                                                                                                                                                                                                                                                                                                                                                        |
| HeavyRehab | nCND_AgeCrkw                                    | nZero                             | 0.0                                                                                                                                                                                                                                                                                                                                                        |
| HeavyRehab | nCND_Crka                                       | nZero                             | 0.0                                                                                                                                                                                                                                                                                                                                                        |
| HeavyRehab | nCND_Crkw                                       | nZero                             | 0.0                                                                                                                                                                                                                                                                                                                                                        |
| HeavyRehab | nCND_RAVL                                       | nZero                             | 0.0                                                                                                                                                                                                                                                                                                                                                        |
| HeavyRehab | nCND_Poth                                       | nZero                             | 0.0                                                                                                                                                                                                                                                                                                                                                        |
| HeavyRehab | nCND_Ptch                                       | nZero                             | 0.0                                                                                                                                                                                                                                                                                                                                                        |
| HeavyRehab | nCND_Rutm                                       | nOne                              | 1.0                                                                                                                                                                                                                                                                                                                                                        |
| HeavyRehab | nCND_Ruts                                       | nCTR_All_Ruts                     | nCND_Rutm * 0.5                                                                                                                                                                                                                                                                                                                                            |
| HeavyRehab | nCND_Irir                                       | nCTR_HeavyRehab_Irir              | 1.4                                                                                                                                                                                                                                                                                                                                                        |
| HeavyRehab | nCND_Aval                                       | nCCND_Aval                        | IF(nDAV_Pavetype =<br>'F',(a_AssetValue_Pav_ReconstructCost*0.4 +<br>(a_AssetValue_Pav_ReconstructCost * 0.6 *<br>nCND_Cind / 100.0)) * AnalMaster->Length * 1000.0<br>* AnalMaster->Inv_Width, ((a_CostGravel -<br>a_CostRegravel) + (a_CostRegravel *<br>MIN(nCND_GravelThickness,a_HDM4_GravThickMa<br>x) / a_HDM4_GravThickMax)) * AnalMaster->Length) |

Table 14 – Treatment reset values in dTIMS

| Treatment  | Analysis variable (Performance parameter) | Expression name of reset value | Reset expression                                                                                                                                                                                                                                                                                                                     |
|------------|-------------------------------------------|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HeavyRehab | nCND_TNYear                               | nCTR_All_TN_Year               | YR                                                                                                                                                                                                                                                                                                                                   |
| HeavyRehab | nCST_AgencyAnnFinancial                   | nCCST_AgencyFinCost            | GST_COST_F / GET_LENGTH(AnalMaster)                                                                                                                                                                                                                                                                                                  |
| HeavyRehab | nCST_AgencyAnnEconomic                    | nCCST_AgencyEconCost           | GST_COST_E / GET_LENGTH(AnalMaster)                                                                                                                                                                                                                                                                                                  |
| HeavyRehab | nVOC_Annual                               | nCVOC_Annual                   | nCVOC_Type01 + nCVOC_Type02 + nCVOC_Type03 + nCVOC_Type04 + nCVOC_Type05 + nCVOC_Type06 + nCVOC_Type07 + nCVOC_Type08 + nCVOC_Type09                                                                                                                                                                                                 |
| HeavyRehab | nTTC_Annual                               | nCTTC_Annual                   | nVOC_Annual + GST_COST_E / GET_LENGTH(AnalMaster)                                                                                                                                                                                                                                                                                    |
| HeavyRehab | nCND_Cind                                 | nCCND_CIND_F                   | -6.875 + 0.0104 * (100.0 - 15.0 / 40.0 * MIN(40.0,nCND_Crka) - 27.5 / 30.0 * MIN(30.0,nCND_Crkw) - 20.0 / 300.0 * MIN(300.0,nCND_Poth) - 17.5 / 25.0 * MIN(25.0,nCND_Rutm) - 20.0 / 9.0 * MAX(0.0,nCND_Irir - 1.0)) ** 2.0                                                                                                           |
| LightRehab | nDAV_Surftype                             | cCTR_Rehab_Surftype            | 'FA'                                                                                                                                                                                                                                                                                                                                 |
| LightRehab | nAGE_Pave                                 | nZero                          | 0.0                                                                                                                                                                                                                                                                                                                                  |
| LightRehab | nAGE_Seal                                 | nZero                          | 0.0                                                                                                                                                                                                                                                                                                                                  |
| LightRehab | nTRF_CumEsals                             | nZero                          | 0.0                                                                                                                                                                                                                                                                                                                                  |
| LightRehab | nCND_Snp                                  | nCTR_LightRehab_SNC            | nCND_Snp + 0.014 * 15.0 + 0.005 * 80.0                                                                                                                                                                                                                                                                                               |
| LightRehab | nDAV_Pcra                                 | nZero                          | 0.0                                                                                                                                                                                                                                                                                                                                  |
| LightRehab | nDAV_Pcrw                                 | nZero                          | 0.0                                                                                                                                                                                                                                                                                                                                  |
| LightRehab | nDAV_Taci                                 | nCTR_All_Taci                  | MAX(1.0,13.21 * a_HDM4_KCI * EXP(-20.7 * (nTRF_Esal / 1000000.0) / nCND_Snp / nCND_Snp) * MAX(0.0,(1.0 - nDAV_Pcrw / 20.0)))                                                                                                                                                                                                         |
| LightRehab | nDAV_Twci                                 | nCTR_All_Twci                  | MAX(2.66 + 0.88 * nDAV_Taci,1.16 * nDAV_Taci)                                                                                                                                                                                                                                                                                        |
| LightRehab | nDAV_Trvi                                 | nCTR_All_Trvi                  | a_HDM4_KVI * 10.5 * EXP(-0.156 * nTRF_Yax)                                                                                                                                                                                                                                                                                           |
| LightRehab | nCND_AgeCrka                              | nZero                          | 0.0                                                                                                                                                                                                                                                                                                                                  |
| LightRehab | nCND_AgeRavl                              | nZero                          | 0.0                                                                                                                                                                                                                                                                                                                                  |
| LightRehab | nCND_AgeCrkw                              | nZero                          | 0.0                                                                                                                                                                                                                                                                                                                                  |
| LightRehab | nCND_Crka                                 | nZero                          | 0.0                                                                                                                                                                                                                                                                                                                                  |
| LightRehab | nCND_Crkw                                 | nZero                          | 0.0                                                                                                                                                                                                                                                                                                                                  |
| LightRehab | nCND_RAVL                                 | nZero                          | 0.0                                                                                                                                                                                                                                                                                                                                  |
| LightRehab | nCND_Poth                                 | nZero                          | 0.0                                                                                                                                                                                                                                                                                                                                  |
| LightRehab | nCND_Ptch                                 | nZero                          | 0.0                                                                                                                                                                                                                                                                                                                                  |
| LightRehab | nCND_Rutm                                 | nOne                           | 1.0                                                                                                                                                                                                                                                                                                                                  |
| LightRehab | nCND_Ruts                                 | nCTR_All_Ruts                  | nCND_Rutm * 0.5                                                                                                                                                                                                                                                                                                                      |
| LightRehab | nCND_Irir                                 | nCTR_LightRehab_Irir           | 1.5                                                                                                                                                                                                                                                                                                                                  |
| LightRehab | nCND_Aval                                 | nCCND_Aval                     | IF(nDAV_Pavetype = 'F',(a_AssetValue_Pav_ReconstructCost*0.4 + (a_AssetValue_Pav_ReconstructCost * 0.6 * nCND_Cind / 100.0)) * AnalMaster->Length * 1000.0 * AnalMaster->Inv_Width, ((a_CostGravel - a_CostRegravel) + (a_CostRegravel * MIN(nCND_GravelThickness,a_HDM4_GravThickMax) / a_HDM4_GravThickMax)) * AnalMaster->Length) |
| LightRehab | nCND_TNYear                               | nCTR_All_TN_Year               | YR                                                                                                                                                                                                                                                                                                                                   |
| LightRehab | nCST_AgencyAnnEconomic                    | nCCST_AgencyEconCost           | GST_COST_E / GET_LENGTH(AnalMaster)                                                                                                                                                                                                                                                                                                  |
| LightRehab | nCST_AgencyAnnFinancial                   | nCCST_AgencyFinCost            | GST_COST_F / GET_LENGTH(AnalMaster)                                                                                                                                                                                                                                                                                                  |

Table 14 – Treatment reset values in dTIMS

| Treatment  | Analysis variable<br>(Performance<br>parameter) | Expression name of reset<br>value | Reset expression                                                                                                                                                                                                                                                                                                                                           |
|------------|-------------------------------------------------|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LightRehab | nVOC_Annual                                     | nCVOC_Annual                      | nCVOC_Type01 + nCVOC_Type02 + nCVOC_Type03<br>+ nCVOC_Type04 + nCVOC_Type05 +<br>nCVOC_Type06 + nCVOC_Type07 + nCVOC_Type08<br>+ nCVOC_Type09                                                                                                                                                                                                              |
| LightRehab | nTTC_Annual                                     | nCTTC_Annual                      | nVOC_Annual + GST_COST_E /<br>GET_LENGTH(AnalMaster)                                                                                                                                                                                                                                                                                                       |
| LightRehab | nCND_Cind                                       | nCCND_CIND_F                      | -6.875 + 0.0104 * (100.0 - 15.0 / 40.0 *<br>MIN(40.0,nCND_Crka) - 27.5 / 30.0 *<br>MIN(30.0,nCND_Crkw) - 20.0 / 300.0 *<br>MIN(300.0,nCND_Poth) - 17.5 / 25.0 *<br>MIN(25.0,nCND_Rutm) - 20.0 / 9.0 *<br>MAX(0.0,nCND_Irir - 1.0)) ** 2.0                                                                                                                  |
| Patching   | nCND_Ptch                                       | nCTR_Patching_Ptch                | nCND_Ptch + nCND_Poth                                                                                                                                                                                                                                                                                                                                      |
| Patching   | nCND_Poth                                       | nZero                             | 0.0                                                                                                                                                                                                                                                                                                                                                        |
| Patching   | nCND_Irir                                       | nCCnd_Irir_F                      | nCND_Irir + a_HDM4_KGP * ((134.0 *<br>EXP(a_HDM4_m_ENVIRONCOEFF *<br>a_HDM4_KGM * nAGE_Pave)*(1.0+nCND_Snp)**<br>(-5.0)) * nTRF_Esal/1000000.0) +<br>(0.0066*(nCND_Crka-<br>GET_ANALVAR_4_YR(nCND_Crka,YR-1.0))) +<br>(0.088*(nCND_Rutm -<br>GET_ANALVAR_4_YR(nCND_Rutm,YR-1.0))) + (                                                                      |
| Patching   | nCND_Aval                                       | nCCND_Aval                        | IF(nDAV_Pavetype =<br>'F',(a_AssetValue_Pav_ReconstructCost*0.4 +<br>(a_AssetValue_Pav_ReconstructCost * 0.6 *<br>nCND_Cind / 100.0)) * AnalMaster->Length * 1000.0<br>* AnalMaster->Inv_Width, ((a_CostGravel -<br>a_CostRegravel) + (a_CostRegravel *<br>MIN(nCND_GravelThickness,a_HDM4_GravThickMa<br>x) / a_HDM4_GravThickMax)) * AnalMaster->Length) |
| Patching   | nCST_AgencyAnnEco<br>nomic                      | nCCST_AgencyEconCost              | GST_COST_E / GET_LENGTH(AnalMaster)                                                                                                                                                                                                                                                                                                                        |
| Patching   | nCST_AgencyAnnFin<br>ancial                     | nCCST_AgencyFinCost               | GST_COST_F / GET_LENGTH(AnalMaster)                                                                                                                                                                                                                                                                                                                        |
| Patching   | nVOC_Annual                                     | nCVOC_Annual                      | nCVOC_Type01 + nCVOC_Type02 + nCVOC_Type03<br>+ nCVOC_Type04 + nCVOC_Type05 +<br>nCVOC_Type06 + nCVOC_Type07 + nCVOC_Type08<br>+ nCVOC_Type09                                                                                                                                                                                                              |
| Patching   | nTTC_Annual                                     | nCTTC_Annual                      | nVOC_Annual + GST_COST_E /<br>GET_LENGTH(AnalMaster)                                                                                                                                                                                                                                                                                                       |
| Patching   | nCND_Cind                                       | nCCND_CIND_F                      | -6.875 + 0.0104 * (100.0 - 15.0 / 40.0 *<br>MIN(40.0,nCND_Crka) - 27.5 / 30.0 *<br>MIN(30.0,nCND_Crkw) - 20.0 / 300.0 *<br>MIN(300.0,nCND_Poth) - 17.5 / 25.0 *<br>MIN(25.0,nCND_Rutm) - 20.0 / 9.0 *<br>MAX(0.0,nCND_Irir - 1.0)) ** 2.0                                                                                                                  |
| Regravel   | nDAV_Surftype                                   | cCTR_Grav_Surftype                | 'UG'                                                                                                                                                                                                                                                                                                                                                       |
| Regravel   | nDAV_MatQual                                    | nCTR_GravMatQual                  | 1.0                                                                                                                                                                                                                                                                                                                                                        |
| Regravel   | nCND_GravelThickne<br>ss                        | nCTR_GravThickness                | nCND_GravelThickness + a_HDM4_GravThickMax                                                                                                                                                                                                                                                                                                                 |
| Regravel   | nCND_P26                                        | nCCND_P26                         | IF(AnalMaster->Struc_Pavetype <><br>'U',0.0,IF(nCND_GravelThickness ><br>0.0,XTAB(MatProperties,AnalMaster-<br>>Unp_GravQual,1.0,82.0),XTAB(MatProperties,0.0,1.0<br>,70.0)))                                                                                                                                                                              |

Table 14 – Treatment reset values in dTIMS

| Treatment | Analysis variable<br>(Performance parameter) | Expression name of reset value | Reset expression                                                                                                                                                                                                                                                |
|-----------|----------------------------------------------|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Regravel  | nCND_P2                                      | nCCND_P2                       | IF(AnalMaster->Struc_Pavetype <> 'U',0.0,IF(nCND_GravelThickness > 0.0,XTAB(MatProperties,AnalMaster->Unp_GravQual,2.0,40.0),XTAB(MatProperties,0.0,2.0,30.0)))                                                                                                 |
| Regravel  | nCND_P425                                    | nCCND_P425                     | IF(AnalMaster->Struc_Pavetype <> 'U',0.0,IF(nCND_GravelThickness > 0.0,XTAB(MatProperties,AnalMaster->Unp_GravQual,3.0,18.0),XTAB(MatProperties,0.0,3.0,20.0)))                                                                                                 |
| Regravel  | nCND_P075                                    | nCCND_P075                     | IF(AnalMaster->Struc_Pavetype <> 'U',0.0,IF(nCND_GravelThickness > 0.0,XTAB(MatProperties,AnalMaster->Unp_GravQual,4.0,12.0),XTAB(MatProperties,0.0,4.0,10.0)))                                                                                                 |
| Regravel  | nCND_D95                                     | nCCND_D95                      | IF(AnalMaster->Struc_Pavetype <> 'U',0.0,IF(nCND_GravelThickness > 0.0,XTAB(MatProperties,AnalMaster->Unp_GravQual,5.0,34.0),XTAB(MatProperties,0.0,5.0,100.0)))                                                                                                |
| Regravel  | nCND_LabMax                                  | nCCND_LabMax                   | IF(AnalMaster->Struc_Pavetype <> 'U',0.0,IF(nCND_GravelThickness > 0.0,XTAB(MatProperties,AnalMaster->Unp_GravQual,6.0,53.0),XTAB(MatProperties,0.0,6.0,110.0)))                                                                                                |
| Regravel  | nCND_PL                                      | nCCND_PL                       | IF(AnalMaster->Struc_Pavetype <> 'U',0.0,IF(nCND_GravelThickness > 0.0,XTAB(MatProperties,AnalMaster->Unp_GravQual,7.0,18.0),XTAB(MatProperties,0.0,7.0,15.0)))                                                                                                 |
| Regravel  | nCND_Irir                                    | nCCnd_Irir_G                   | IF(nCND_GravelThickness > 0.0,IF(nDAV_BladeFreq = 30.0,3.3886 * EXP(0.0023 * MIN(400.0,nTRF_Aadt)),IF(nDAV_BladeFreq = 60.0,4.4327 * EXP(0.0026 * MIN(400.0,nTRF_Aadt)),30.0)),IF(nDAV_BladeFreq = 30.0,6.9329 * EXP(0.0027 * MIN(400.0,nTRF_Aadt)),IF(nDAV_Bla |
| Regravel  | nCND_TNYear                                  | nCTR_RegravelTN                | IF(GET_LASTMAJTRT() = 'Regravel' and YR > 1.0,30.0,YR)                                                                                                                                                                                                          |
| Regravel  | nCST_AgencyAnnEconomic                       | nCCST_AgencyEconCost           | GST_COST_E / GET_LENGTH(AnalMaster)                                                                                                                                                                                                                             |
| Regravel  | nCST_AgencyAnnFinancial                      | nCCST_AgencyFinCost            | GST_COST_F / GET_LENGTH(AnalMaster)                                                                                                                                                                                                                             |
| Regravel  | nVOC_Annual                                  | nCVOC_Annual                   | nCVOC_Type01 + nCVOC_Type02 + nCVOC_Type03 + nCVOC_Type04 + nCVOC_Type05 + nCVOC_Type06 + nCVOC_Type07 + nCVOC_Type08 + nCVOC_Type09                                                                                                                            |
| Regravel  | nTTC_Annual                                  | nCTTC_Annual                   | nVOC_Annual + GST_COST_E / GET_LENGTH(AnalMaster)                                                                                                                                                                                                               |
| Regravel  | nCND_Cind                                    | nCCND_CIND_G                   | (MIN(a_HDM4_GravThickMax,nCND_GravelThickness) / a_HDM4_GravThickMax * 100.0) * (0.0012*YR**2.0 - 0.0663 * YR + 0.9996 )                                                                                                                                        |

Table 14 – Treatment reset values in dTIMS

| Treatment    | Analysis variable<br>(Performance parameter) | Expression name of reset value | Reset expression                                                                                                                                                                                                                                                                                                                     |
|--------------|----------------------------------------------|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Regravel     | nCND_Aval                                    | nCCND_Aval                     | IF(nDAV_Pavetype = 'F',(a_AssetValue_Pav_ReconstructCost*0.4 + (a_AssetValue_Pav_ReconstructCost * 0.6 * nCND_Cind / 100.0)) * AnalMaster->Length * 1000.0 * AnalMaster->Inv_Width, ((a_CostGravel - a_CostRegravel) + (a_CostRegravel * MIN(nCND_GravelThickness,a_HDM4_GravThickMax) / a_HDM4_GravThickMax)) * AnalMaster->Length) |
| Reseal       | nDAV_SurfType                                | cCTR_Seal_SurfType             | 'FC'                                                                                                                                                                                                                                                                                                                                 |
| Reseal       | nAGE_Seal                                    | nZero                          | 0.0                                                                                                                                                                                                                                                                                                                                  |
| Reseal       | nCND_Snp                                     | nCTR_Seal_Snc                  | nCND_Snp + 0.014 * 15.0                                                                                                                                                                                                                                                                                                              |
| Reseal       | nDAV_Pcra                                    | nCTR_Seal_PCRA                 | nCND_Crka                                                                                                                                                                                                                                                                                                                            |
| Reseal       | nDAV_Pcrw                                    | nCTR_Seal_PCRW                 | nCND_Crkw                                                                                                                                                                                                                                                                                                                            |
| Reseal       | nDAV_Taci                                    | nCTR_All_Taci                  | MAX(1.0,13.21 * a_HDM4_KCI * EXP(-20.7 * (nTRF_Esal / 1000000.0) / nCND_Snp / nCND_Snp) * MAX(0.0,(1.0 - nDAV_Pcrw / 20.0)))                                                                                                                                                                                                         |
| Reseal       | nDAV_Twci                                    | nCTR_All_Twci                  | MAX(2.66 + 0.88 * nDAV_Taci,1.16 * nDAV_Taci)                                                                                                                                                                                                                                                                                        |
| Reseal       | nDAV_Trvl                                    | nCTR_All_Trvl                  | a_HDM4_KVI * 10.5 * EXP(-0.156 * nTRF_Yax)                                                                                                                                                                                                                                                                                           |
| Reseal       | nCND_AgeCrka                                 | nZero                          | 0.0                                                                                                                                                                                                                                                                                                                                  |
| Reseal       | nCND_AgeCrkw                                 | nZero                          | 0.0                                                                                                                                                                                                                                                                                                                                  |
| Reseal       | nCND_AgeRavl                                 | nZero                          | 0.0                                                                                                                                                                                                                                                                                                                                  |
| Reseal       | nCND_Crka                                    | nZero                          | 0.0                                                                                                                                                                                                                                                                                                                                  |
| Reseal       | nCND_Crkw                                    | nZero                          | 0.0                                                                                                                                                                                                                                                                                                                                  |
| Reseal       | nCND_RAVL                                    | nZero                          | 0.0                                                                                                                                                                                                                                                                                                                                  |
| Reseal       | nCND_Poth                                    | nZero                          | 0.0                                                                                                                                                                                                                                                                                                                                  |
| Reseal       | nCND_Ptch                                    | nZero                          | 0.0                                                                                                                                                                                                                                                                                                                                  |
| Reseal       | nCND_Rutm                                    | nCTR_Seal_Rutm                 | 0.95 * nCND_Rutm                                                                                                                                                                                                                                                                                                                     |
| Reseal       | nCND_Ruts                                    | nCTR_All_Ruts                  | nCND_Rutm * 0.5                                                                                                                                                                                                                                                                                                                      |
| Reseal       | nCND_Irir                                    | nCTR_Seal_Irir                 | nCND_Irir * 0.95                                                                                                                                                                                                                                                                                                                     |
| Reseal       | nCND_TNYear                                  | nCTR_All_TN_Year               | YR                                                                                                                                                                                                                                                                                                                                   |
| Reseal       | nCND_Aval                                    | nCCND_Aval                     | IF(nDAV_Pavetype = 'F',(a_AssetValue_Pav_ReconstructCost*0.4 + (a_AssetValue_Pav_ReconstructCost * 0.6 * nCND_Cind / 100.0)) * AnalMaster->Length * 1000.0 * AnalMaster->Inv_Width, ((a_CostGravel - a_CostRegravel) + (a_CostRegravel * MIN(nCND_GravelThickness,a_HDM4_GravThickMax) / a_HDM4_GravThickMax)) * AnalMaster->Length) |
| Reseal       | nCST_AgencyAnnEconomic                       | nCCST_AgencyEconCost           | GST_COST_E / GET_LENGTH(AnalMaster)                                                                                                                                                                                                                                                                                                  |
| Reseal       | nCST_AgencyAnnFinancial                      | nCCST_AgencyFinCost            | GST_COST_F / GET_LENGTH(AnalMaster)                                                                                                                                                                                                                                                                                                  |
| Reseal       | nVOC_Annual                                  | nCVOC_Annual                   | nCVOC_Type01 + nCVOC_Type02 + nCVOC_Type03 + nCVOC_Type04 + nCVOC_Type05 + nCVOC_Type06 + nCVOC_Type07 + nCVOC_Type08 + nCVOC_Type09                                                                                                                                                                                                 |
| Reseal       | nTTC_Annual                                  | nCTTC_Annual                   | nVOC_Annual + GST_COST_E / GET_LENGTH(AnalMaster)                                                                                                                                                                                                                                                                                    |
| Reseal       | nCND_Cind                                    | nCCND_CIND_F                   | -9.875 + 0.0104 * (100.0 - 15.0 / 40.0 * MIN(40.0,nCND_Crka) - 27.5 / 30.0 * MIN(30.0,nCND_Crkw) - 20.0 / 300.0 * MIN(300.0,nCND_Poth) - 17.5 / 20.0 * MIN(20.0,nCND_Rutm) - 20.0 / 9.0 * MAX(0.0,nCND_Irir - 1.0)) ** 2.0                                                                                                           |
| Reseal_Light | nDAV_SurfType                                | cCTR_Seal_SurfType             | 'FC'                                                                                                                                                                                                                                                                                                                                 |
| Reseal_Light | nAGE_Seal                                    | nZero                          | 0.0                                                                                                                                                                                                                                                                                                                                  |

Table 14 – Treatment reset values in dTIMS

| Treatment    | Analysis variable<br>(Performance parameter) | Expression name of reset value | Reset expression                                                                                                                                                                                                                                                                                                                     |
|--------------|----------------------------------------------|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reseal_Light | nCND_Snp                                     | nCTR_Seal_Snc                  | nCND_Snp + 0.014 * 15.0                                                                                                                                                                                                                                                                                                              |
| Reseal_Light | nDAV_Pcra                                    | nCTR_Seal_PCRA                 | nCND_Crka                                                                                                                                                                                                                                                                                                                            |
| Reseal_Light | nDAV_Pcrw                                    | nCTR_Seal_PCRW                 | nCND_Crkw                                                                                                                                                                                                                                                                                                                            |
| Reseal_Light | nDAV_Taci                                    | nCTR_All_Taci                  | MAX(1.0,13.21 * a_HDM4_KCI * EXP(-20.7 * (nTRF_Esal / 1000000.0) / nCND_Snp / nCND_Snp) * MAX(0.0,(1.0 - nDAV_Pcrw / 20.0)))                                                                                                                                                                                                         |
| Reseal_Light | nDAV_Twci                                    | nCTR_All_Twci                  | MAX(2.66 + 0.88 * nDAV_Taci,1.16 * nDAV_Taci)                                                                                                                                                                                                                                                                                        |
| Reseal_Light | nDAV_Trvl                                    | nCTR_All_Trvl                  | a_HDM4_KVI * 10.5 * EXP(-0.156 * nTRF_Yax)                                                                                                                                                                                                                                                                                           |
| Reseal_Light | nCND_AgeCrka                                 | nZero                          | 0.0                                                                                                                                                                                                                                                                                                                                  |
| Reseal_Light | nCND_AgeCrkw                                 | nZero                          | 0.0                                                                                                                                                                                                                                                                                                                                  |
| Reseal_Light | nCND_AgeRavl                                 | nZero                          | 0.0                                                                                                                                                                                                                                                                                                                                  |
| Reseal_Light | nCND_Crka                                    | nZero                          | 0.0                                                                                                                                                                                                                                                                                                                                  |
| Reseal_Light | nCND_Crkw                                    | nZero                          | 0.0                                                                                                                                                                                                                                                                                                                                  |
| Reseal_Light | nCND_RAVL                                    | nZero                          | 0.0                                                                                                                                                                                                                                                                                                                                  |
| Reseal_Light | nCND_Poth                                    | nZero                          | 0.0                                                                                                                                                                                                                                                                                                                                  |
| Reseal_Light | nCND_Ptch                                    | nZero                          | 0.0                                                                                                                                                                                                                                                                                                                                  |
| Reseal_Light | nCND_Rutm                                    | nCTR_Seal_Rutm                 | 0.95 * nCND_Rutm                                                                                                                                                                                                                                                                                                                     |
| Reseal_Light | nCND_Ruts                                    | nCTR_All_Ruts                  | nCND_Rutm * 0.5                                                                                                                                                                                                                                                                                                                      |
| Reseal_Light | nCND_Irir                                    | nCTR_Seal_Irir                 | nCND_Irir * 0.95                                                                                                                                                                                                                                                                                                                     |
| Reseal_Light | nCND_TNYear                                  | nCTR_All_TN_Year               | YR                                                                                                                                                                                                                                                                                                                                   |
| Reseal_Light | nCND_Aval                                    | nCCND_Aval                     | IF(nDAV_Pavetype = 'F',(a_AssetValue_Pav_ReconstructCost*0.4 + (a_AssetValue_Pav_ReconstructCost * 0.6 * nCND_Cind / 100.0)) * AnalMaster->Length * 1000.0 * AnalMaster->Inv_Width, ((a_CostGravel - a_CostRegravel) + (a_CostRegravel * MIN(nCND_GravelThickness,a_HDM4_GravThickMax) / a_HDM4_GravThickMax)) * AnalMaster->Length) |
| Reseal_Light | nCST_AgencyAnnEconomic                       | nCCST_AgencyEconCost           | GST_COST_E / GET_LENGTH(AnalMaster)                                                                                                                                                                                                                                                                                                  |
| Reseal_Light | nCST_AgencyAnnFinancial                      | nCCST_AgencyFinCost            | GST_COST_F / GET_LENGTH(AnalMaster)                                                                                                                                                                                                                                                                                                  |
| Reseal_Light | nVOC_Annual                                  | nCVOC_Annual                   | nCVOC_Type01 + nCVOC_Type02 + nCVOC_Type03 + nCVOC_Type04 + nCVOC_Type05 + nCVOC_Type06 + nCVOC_Type07 + nCVOC_Type08 + nCVOC_Type09                                                                                                                                                                                                 |
| Reseal_Light | nTTC_Annual                                  | nCTTC_Annual                   | nVOC_Annual + GST_COST_E / GET_LENGTH(AnalMaster)                                                                                                                                                                                                                                                                                    |
| Reseal_Light | nCND_Cind                                    | nCCND_CIND_F                   | -9.875 + 0.0104 * (100.0 - 15.0 / 40.0 * MIN(40.0,nCND_Crka) - 27.5 / 30.0 * MIN(30.0,nCND_Crkw) - 20.0 / 300.0 * MIN(300.0,nCND_Poth) - 17.5 / 20.0 * MIN(20.0,nCND_Rutm) - 20.0 / 9.0 * MAX(0.0,nCND_Irir - 1.0)) ** 2.0                                                                                                           |
| UpgradePave  | nDAV_Pavetype                                | cCTR_Upgrade_Pavetype          | 'F'                                                                                                                                                                                                                                                                                                                                  |
| UpgradePave  | nDAV_SurfType                                | cCTR_Seal_SurfType             | 'FC'                                                                                                                                                                                                                                                                                                                                 |
| UpgradePave  | nAGE_Pave                                    | nZero                          | 0.0                                                                                                                                                                                                                                                                                                                                  |
| UpgradePave  | nAGE_Seal                                    | nZero                          | 0.0                                                                                                                                                                                                                                                                                                                                  |
| UpgradePave  | nCND_Snp                                     | nCTR_Upgrade_SNP               | 3.5                                                                                                                                                                                                                                                                                                                                  |
| UpgradePave  | nDAV_Pcra                                    | nZero                          | 0.0                                                                                                                                                                                                                                                                                                                                  |
| UpgradePave  | nDAV_Pcrw                                    | nZero                          | 0.0                                                                                                                                                                                                                                                                                                                                  |
| UpgradePave  | nDAV_Taci                                    | nCTR_All_Taci                  | MAX(1.0,13.21 * a_HDM4_KCI * EXP(-20.7 * (nTRF_Esal / 1000000.0) / nCND_Snp / nCND_Snp) * MAX(0.0,(1.0 - nDAV_Pcrw / 20.0)))                                                                                                                                                                                                         |
| UpgradePave  | nDAV_Twci                                    | nCTR_All_Twci                  | MAX(2.66 + 0.88 * nDAV_Taci,1.16 * nDAV_Taci)                                                                                                                                                                                                                                                                                        |

Table 14 – Treatment reset values in dTIMS

| Treatment   | Analysis variable<br>(Performance<br>parameter) | Expression name of reset<br>value | Reset expression                                                                                                                                                                                                                                                                                                                                                                             |
|-------------|-------------------------------------------------|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UpgradePave | nDAV_Trvl                                       | nCTR_All_Trvl                     | $a\_HDM4\_KVI * 10.5 * EXP(-0.156 * nTRF\_Yax)$                                                                                                                                                                                                                                                                                                                                              |
| UpgradePave | nCND_AgeCrka                                    | nZero                             | 0.0                                                                                                                                                                                                                                                                                                                                                                                          |
| UpgradePave | nCND_AgeCrkw                                    | nZero                             | 0.0                                                                                                                                                                                                                                                                                                                                                                                          |
| UpgradePave | nCND_AgeRavl                                    | nZero                             | 0.0                                                                                                                                                                                                                                                                                                                                                                                          |
| UpgradePave | nCND_Crka                                       | nZero                             | 0.0                                                                                                                                                                                                                                                                                                                                                                                          |
| UpgradePave | nCND_Crkw                                       | nZero                             | 0.0                                                                                                                                                                                                                                                                                                                                                                                          |
| UpgradePave | nCND_RAVL                                       | nZero                             | 0.0                                                                                                                                                                                                                                                                                                                                                                                          |
| UpgradePave | nCND_Poth                                       | nZero                             | 0.0                                                                                                                                                                                                                                                                                                                                                                                          |
| UpgradePave | nCND_Ptch                                       | nZero                             | 0.0                                                                                                                                                                                                                                                                                                                                                                                          |
| UpgradePave | nCND_Rutm                                       | nOne                              | 1.0                                                                                                                                                                                                                                                                                                                                                                                          |
| UpgradePave | nCND_Ruts                                       | nCTR_All_Ruts                     | $nCND\_Rutm * 0.5$                                                                                                                                                                                                                                                                                                                                                                           |
| UpgradePave | nCND_GravelThickne<br>ss                        | nCTR_Upgrade_GravelThic<br>k      | 250.0                                                                                                                                                                                                                                                                                                                                                                                        |
| UpgradePave | nDAV_BladeFreq                                  | nOne                              | 1.0                                                                                                                                                                                                                                                                                                                                                                                          |
| UpgradePave | nCND_Irir                                       | nCTR_Upgrade_IRI                  | 1.4                                                                                                                                                                                                                                                                                                                                                                                          |
| UpgradePave | nCND_Aval                                       | nCCND_Aval                        | $IF(nDAV\_Pavetype = 'F', (a\_AssetValue\_Pav\_ReconstructCost * 0.4 + (a\_AssetValue\_Pav\_ReconstructCost * 0.6 * nCND\_Cind / 100.0)) * AnalMaster \rightarrow Length * 1000.0 * AnalMaster \rightarrow Inv\_Width, ((a\_CostGravel - a\_CostRegravel) + (a\_CostRegravel * MIN(nCND\_GravelThickness, a\_HDM4\_GravThickMax) / a\_HDM4\_GravThickMax)) * AnalMaster \rightarrow Length)$ |
| UpgradePave | nCND_TNYear                                     | nCTR_All_TN_Year                  | YR                                                                                                                                                                                                                                                                                                                                                                                           |
| UpgradePave | nCST_AgencyAnnEco<br>nomic                      | nCCST_AgencyEconCost              | $GST\_COST\_E / GET\_LENGTH(AnalMaster)$                                                                                                                                                                                                                                                                                                                                                     |
| UpgradePave | nCST_AgencyAnnFin<br>ancial                     | nCCST_AgencyFinCost               | $GST\_COST\_F / GET\_LENGTH(AnalMaster)$                                                                                                                                                                                                                                                                                                                                                     |
| UpgradePave | nVOC_Annual                                     | nCVOC_Annual                      | $nCVOC\_Type01 + nCVOC\_Type02 + nCVOC\_Type03 + nCVOC\_Type04 + nCVOC\_Type05 + nCVOC\_Type06 + nCVOC\_Type07 + nCVOC\_Type08 + nCVOC\_Type09$                                                                                                                                                                                                                                              |
| UpgradePave | nTTC_Annual                                     | nCTTC_Annual                      | $nVOC\_Annual + GST\_COST\_E / GET\_LENGTH(AnalMaster)$                                                                                                                                                                                                                                                                                                                                      |
| UpgradePave | nCND_Cind                                       | nCCND_CIND_F                      | $-9.875 + 0.0104 * (100.0 - 15.0 / 40.0 * MIN(40.0, nCND\_Crka) - 27.5 / 30.0 * MIN(30.0, nCND\_Crkw) - 20.0 / 300.0 * MIN(300.0, nCND\_Poth) - 17.5 / 20.0 * MIN(20.0, nCND\_Rutm) - 20.0 / 9.0 * MAX(0.0, nCND\_Irir - 1.0)) ** 2.0$                                                                                                                                                       |



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## *Appendix A: dTIMS WF – Workflow: Installation and Operation*

---

# Installation of dTIMS WF

The complete set of installation files can be found on C:\UNRA\_PMS\Installation Files\dTIMS WF.

In internet explorer navigate to <http://www.dtimes.net/wf/v8.6/launch.htm>

## Deighton Associates Limited dTIMS wf

**Note: Running version 8.6 of Workflow will upgrade your dTIMS CT v8.5 database. This upgrade is permanent.**

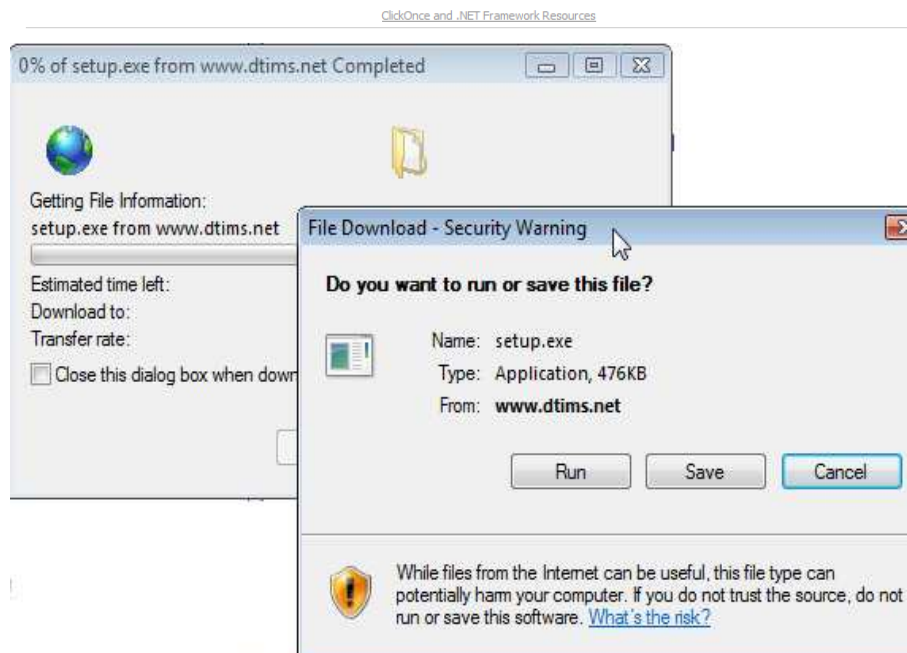
Click [here](#) to open version 8.6

The following prerequisites are required:

- Windows Installer 3.1
- Windows Installer 4.5
- Microsoft .NET Framework 4 (x86 and x64)
- Deighton Trust Fix for WF

If these components are not already installed, you can [Install](#) them now.

Click the “Install”  
link



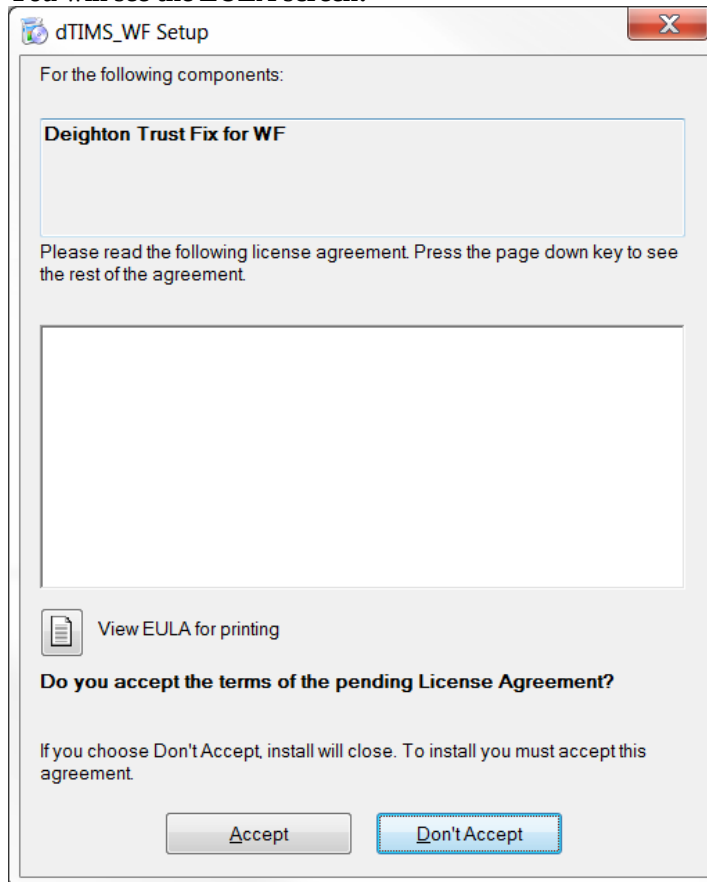
Click the “Run”  
button

You will see the following warning message:



Click the “Run” button

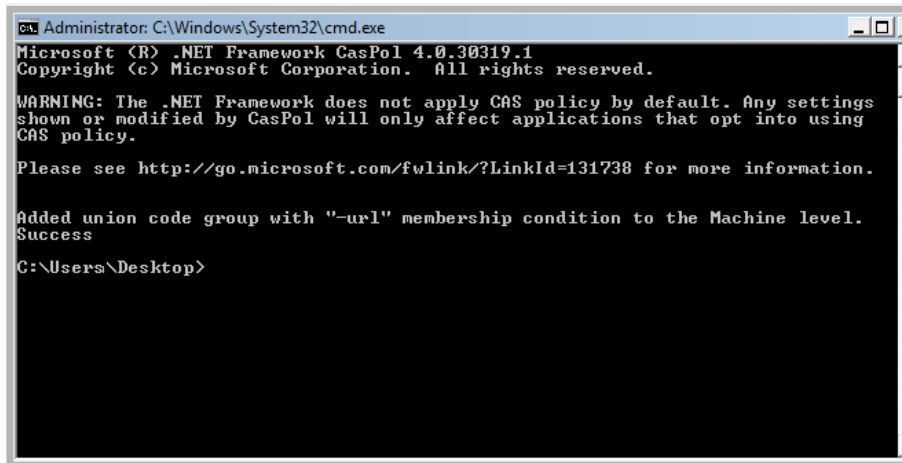
**You will see the EULA screen:**



Click “Accept”

**A windows screen will open and remain open, type the word `exit` {enter} (no spaces, no quotation marks) to close the window.**

Type **exit**



```
C:\Windows\System32\cmd.exe
Microsoft (R) .NET Framework CasPol 4.0.30319.1
Copyright (c) Microsoft Corporation. All rights reserved.

WARNING: The .NET Framework does not apply CAS policy by default. Any settings
shown or modified by CasPol will only affect applications that opt into using
CAS policy.

Please see http://go.microsoft.com/fwlink/?LinkId=131738 for more information.

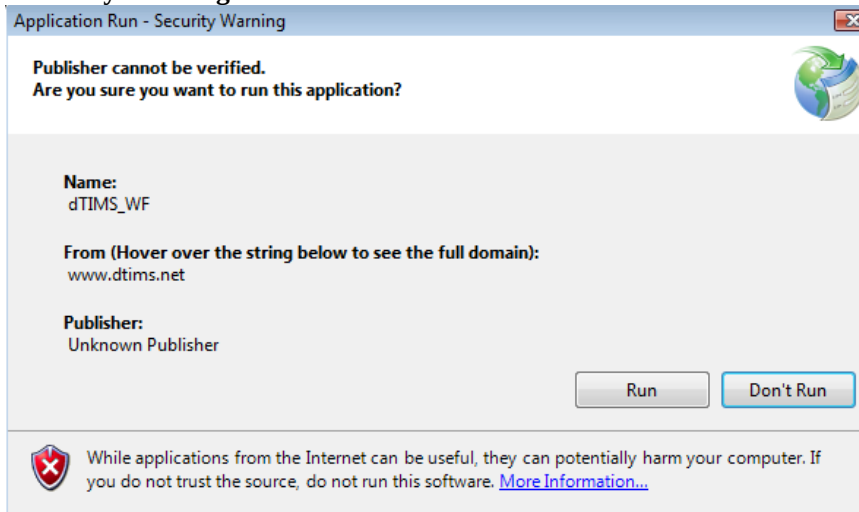
Added union code group with "-url" membership condition to the Machine level.
Success

C:\Users\Desktop>
```

You may see the WF login screen in which case you are ready to use dTIMS wf

Go to [WF login screen](#)

Dependent upon your security settings, you may see the following screens: “Security Warning” screen:



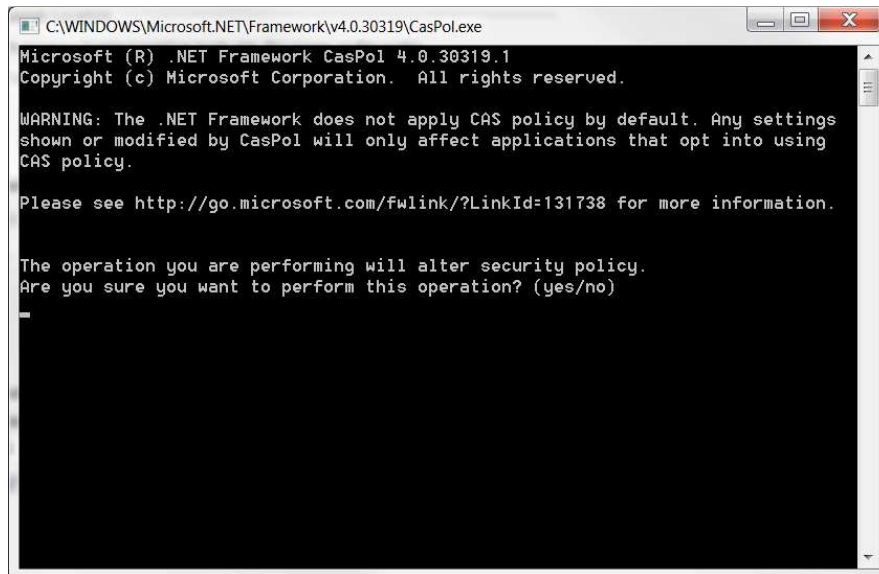
Click “Run”

“User Account Control” prompt



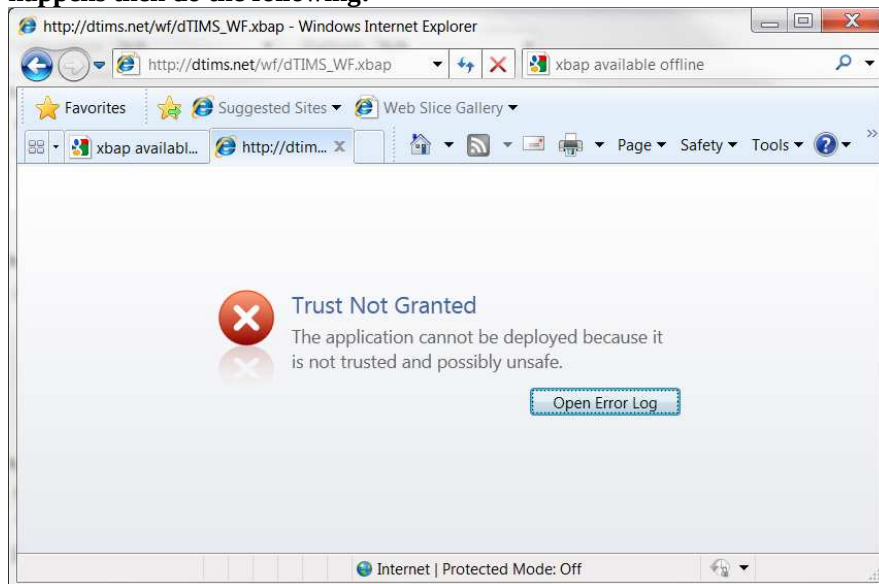
Click “Yes”

Now you may need to minimize the Internet Explorer window in order to see the following screen:



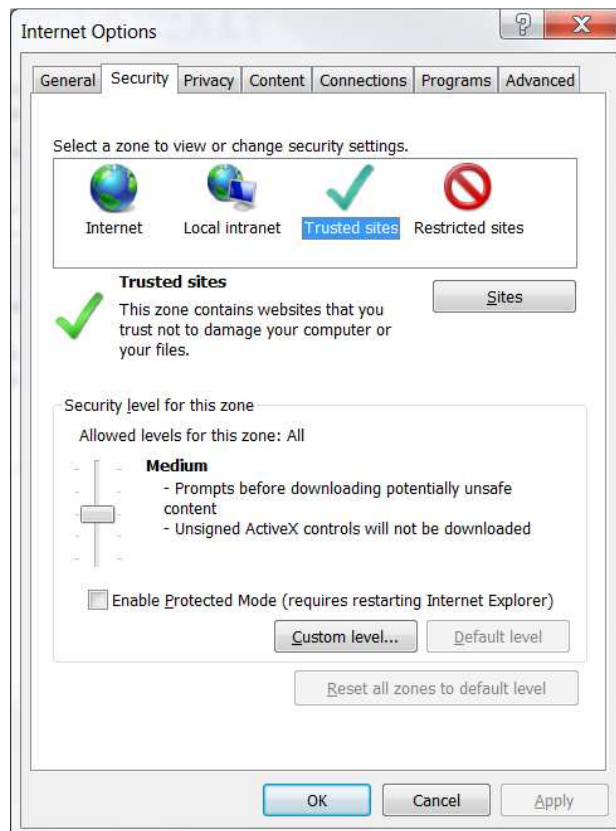
When you see this prompt enter: "Yes" without the quotes and hit enter. Check for errors on this screen and if none, just close it.

You may see a screen telling you that "trust is not granted". If this happens then do the following:



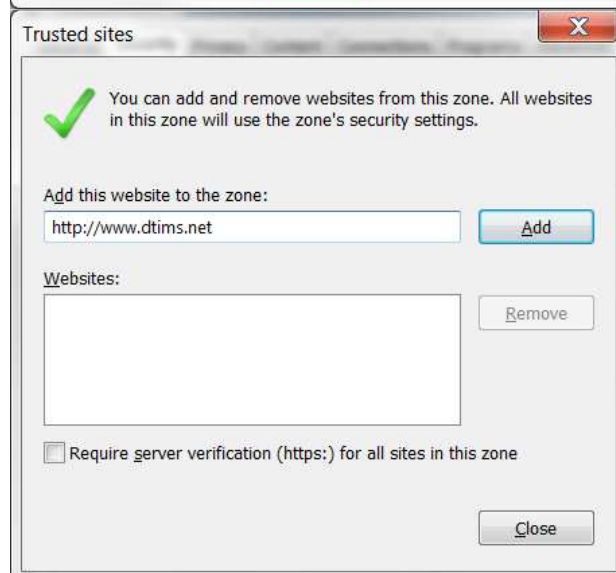
Open Internet Explorer -> Tools -> Internet Options

Choose the security tab



Click the green check mark for “Trusted Sites”

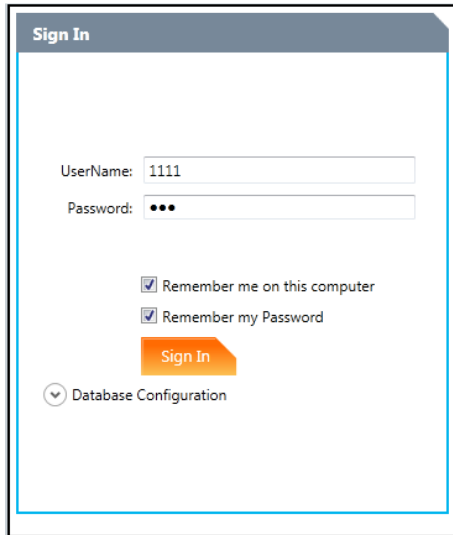
Click the “Sites” button



Type “<http://www.dtimes.net>” in the “Add this website to the zone:” textbox  
Ensure that the “Require server verification (https:) for all sites in this zone” checkbox is **NOT** checked  
Click the “Add” button.

Click Close  
Click the OK button

Hit refresh in IE or Open Internet Explorer and navigate to <http://www.dtimes.net/wf/publish.htm> again and now click the “launch” link



Sign In

UserName: 1111

Password: ●●●

☒ Remember me on this computer

☒ Remember my Password

Sign In

▼ Database Configuration

**You should now be able to pick an database udl and use dTIMS wf.**

### Subsequent Usage

Click the <http://www.dtimes.net/wf/v8.6/launch.htm> “launch” link to run the application

# Execution of dTIMS WF

In internet explorer navigate to <http://www.dtimes.net/wf/v8.6/launch.htm>

## Workflows Navigation Page

### UNRA's Customised Workflows to perform a LCCA in dTIMS

Workflow to perform all the steps in the WF, select the WF **Step ALL for UNRA LCCA**, and click on **START** to execute.

Alternatively, the workflows can be selected on-by-one and executed (started) sequentially (according to the step numbers).

UNRA's dTIMS WF was customised with a set of instruction files residing on the drive C:\UNRA\_PMS\WF. These files contain guidelines and instructions to assist the LCCA user and can be obtained from the CD of UNRA's RMS prepared by Aurecon during October 2012.

## Datasheet View Navigation Page

Follow the instructions in the file C:\UNRA\_PMS\Installation Files\dTIMS WF\Manuals Installation Guides\ *dTIMS wf Version 1\_0.pdf*.

There are also examples in the file C:\UNRA\_PMS\Installation Files\dTIMS WF\Manuals Installation Guides\ *Workflow\_wf\_examples.pdf*

## Review and Adjust Navigation Page

Follow the instructions in the file C:\UNRA\_PMS\Installation Files\dTIMS WF\Manuals Installation Guides\ *dTIMS wf Version 1\_0.pdf*.

There are also examples in the file C:\UNRA\_PMS\Installation Files\dTIMS WF\Manuals Installation Guides\ *Workflow\_wf\_examples.pdf*



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## *Appendix B: dTIMS MD – Management Dashboard: Installation and Operation*

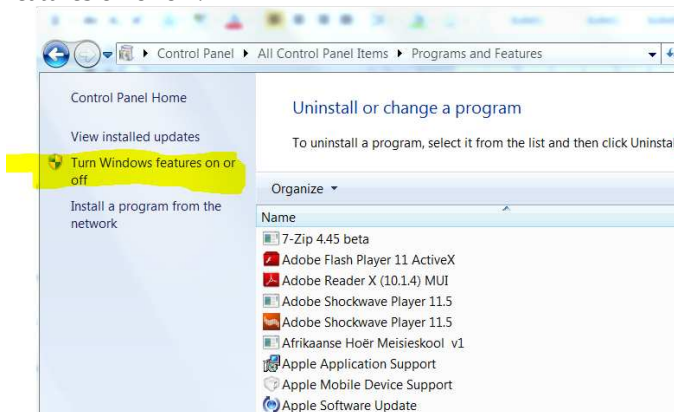
---

# Installation of dTIMS MD

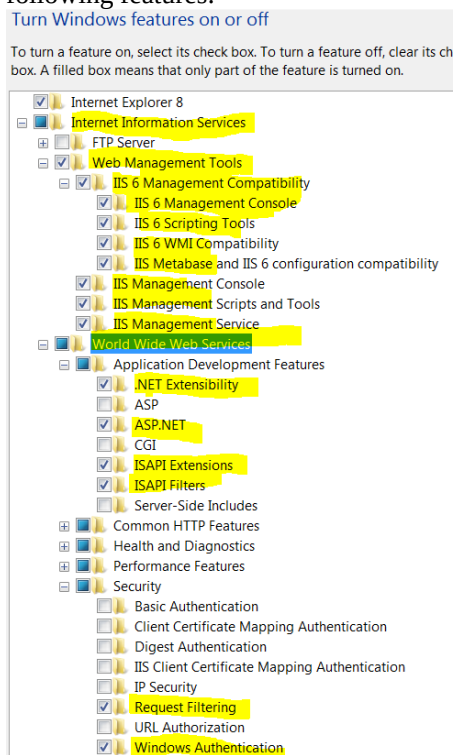
The complete set of installation files can be found on C:\UNRA\_PMS\Installation Files\dTIMS MD.

## dTIMS MD Setup

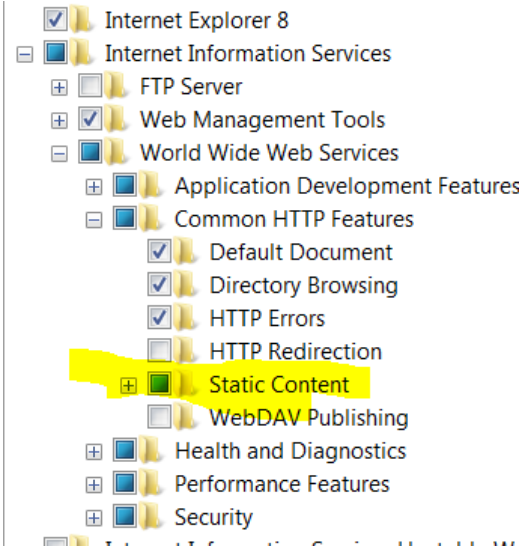
1. IIS must be installed with .NET Extensibility and ASP.NET options.
  - a. In Control panel, got to Programs and Features, then select the option Turn Windows features on or off.



- b. Under Internet Information Services, Web Management Tools check (select) the following features:



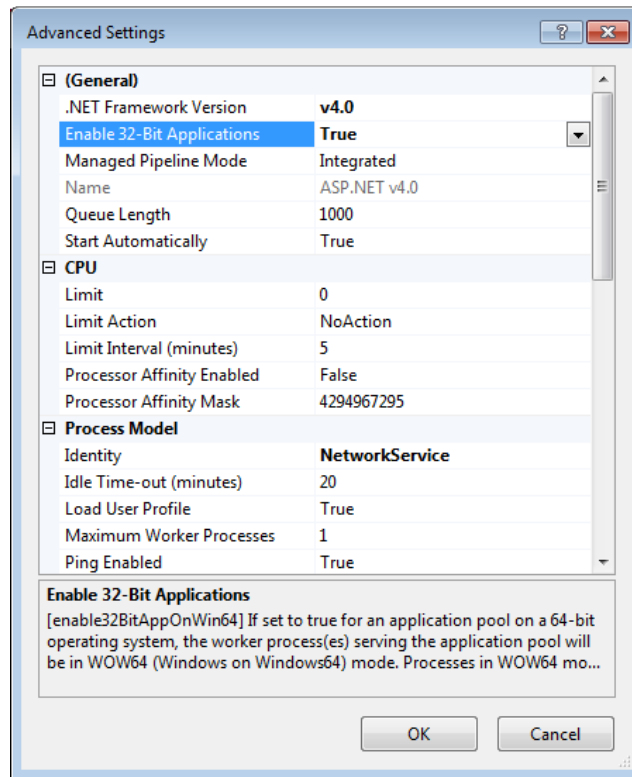
And Static Content under Common HTTP Features.



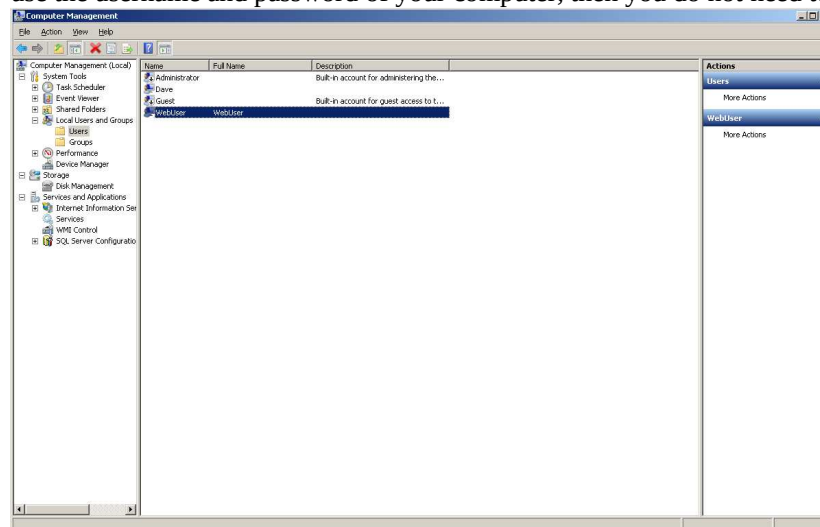
- c. Click on OK to start the IIS and ASP.NET installation process. This could take a while.
- d. Close Control Panel when done.
- e. Check now if all is ok: Open your browser and navigate to <http://localhost>.
- f. If you see the display below then very thing is ok



2. .NET Framework 4.0 must be installed and can be downloaded [here](#).
3. Application Pool Preparation – Open the IIS manager and navigate to the application pools area using the navigation tree. Select the “ASP.NET v4.0” AppPool. In the actions menu on the right, click “Advanced Settings”. Set the “Enable 32 bit applications” property to true.



4. Using the local users and groups console create a user. This is the user that dTIMS MD will run as. For the purposes of this document we will call this user “WebUser”. Alternatively you should use the username and password of your computer, then you do not need to create this username.



5. Copy the dTIMS MD application files to a folder under the Inetpub folder. This folder was created by the IIS setup. (Usually C:\Inetpub\wwwroot\dTIMSmd)
6. In the IIS manager, select the folder where the dTIMS MD application files have been copied, right click and select “Convert to Application”. Click “Ok” on the resulting dialog to accept default options.
7. In the actions menu on the right, click “Advanced Settings”. Set the “Application Pool” property to “ASP.NET v4.0” AppPool.

8. Open the “web.config” file located in the dTIMS MD application folder using notepad. Locate the following entry “<identity impersonate>”. Edit this line to include the login information for your own username and password OR for the “WebUser” created in step 4. This line will now look like this.

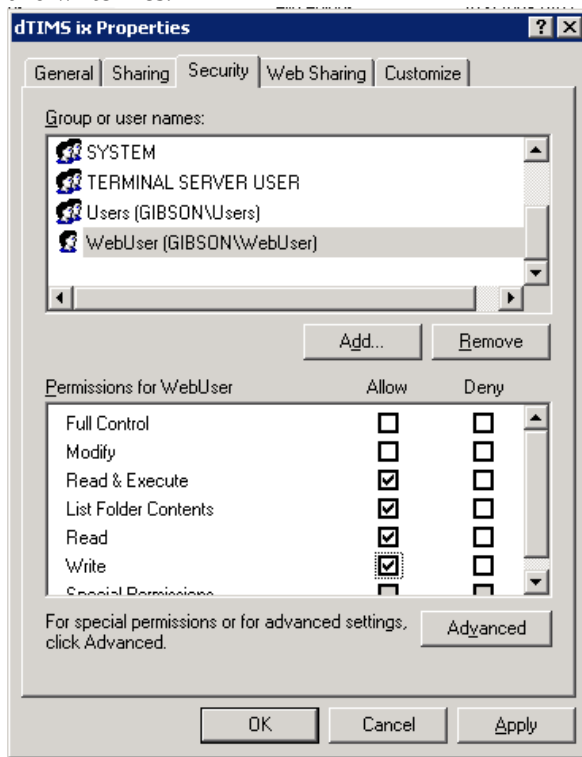
<identity impersonate="true " userName="**your username**" password="**your password**"/>  
OR

<identity impersonate="true " userName="WebUser" password="your password"/>

Save changes to Web.Config.

**Remember, any changes to your password should be changed in the “web.config” file, otherwise MD will not work.**

9. Close Internet Information Services Manager.
10. Open windows explorer and navigate to the folder where the dTIMS MD application files have been copied. Open the properties of the dTIMSmd Folder.
11. Select the security tab and select your username allowing yourself to read and write files.
12. If “Webuser” was screated, select the security tab and add “WebUser” allowing this user to read and write files.



13. Give yourself (or “WebUser”) read and write access to the temporary ASP.NET files by performing the same operation as above. The default location of this folder is C:\WINDOWS\Microsoft.NET\Framework\v2.0.50727\Temporary ASP.NET Files.
14. dTIMS MD Connects to an enterprise database. It does this utilizing a “udl” file. This file defines which database on which server should be utilized. See creating udl files in dTIMS documentation. Create a udl file in a folder other than the dTIMS MD folder. – This should have been done already by your administrator.
15. Ensure that yourself ( or WebUser ) has both Read and write access to this folder as dTIMS MD will write temporary files in this folder for data caching purposes.
16. Install the HASP Network license manager on any server in your network that is visible to the server that hosts dTIMS MD. In some network configurations a Nethasp.ini file may need to be configured. This ini file should be placed in the Windows\System folder of the server that will host dTIMS MD. . – This should have been done already by your administrator.

17. Open a browser and navigate to <http://localhost/dTIMSmd/ContentConfig.aspx>
18. Set the database path to the full path and filename created in step 14.
19. Save your configuration.
20. Close your browser and reopen navigating to  
<http://localhost/dTIMSmd/Login.aspx?Connect=DBPath>
21. Login using a valid dTIMS user name and password. See your dTIMS database administrator to obtain valid credentials.

# Execution of dTIMS MD

## Configuration

The configuration is only needed when the shape file has changed or when dTIMS MD is configured for the very first time.

In internet explorer navigate to <http://localhost/ContentConfig.aspx>

The following display will open:

**md**  
dTIMS<sup>®</sup> MANAGEMENT DASHBOARD

Dashboard Data Sheet View Work With Data Strategic Analysis Module Cross Asset ROI Construction Program Links Edit Configuration Help Logout

**Site Configuration**  
dTIMS MD Version 1.0.4248.14485

**Database Path:** s:\Microsoft SQL Server\IMSSQL10\_50.SQL2008\IMSSQL\DATA\UgandaCTOnServer.UDL

**Custom Links:**

Name:

URL:

|                                             | Name             | URL                                               |
|---------------------------------------------|------------------|---------------------------------------------------|
| <a href="#">Edit</a> <a href="#">Delete</a> | ILAP             | http://www.ilap.com.mx/ilaplow_archivos/frame.htm |
| <a href="#">Edit</a> <a href="#">Delete</a> | Deighton Website | http://deighton.com                               |

**Mapping Configuration**

☒ Use dTIMS MD Maps

**Map Units:** Meter

**Base Layer Shapefile:** C:\UNRA\_PMS\GIS\ROADS\_26Sep2012\ROADS.shp

**Map Projection:** WGS 84 / UTM zone 36N

**Base Layer Element ID:** ROAD

**Shp layer field containing the 'From Measure':**

**Online Map Overlay:**

☒ Google  
☐ Yahoo  
☐ Microsoft Virtual Earth  
☐ None

**Online Map Options:**

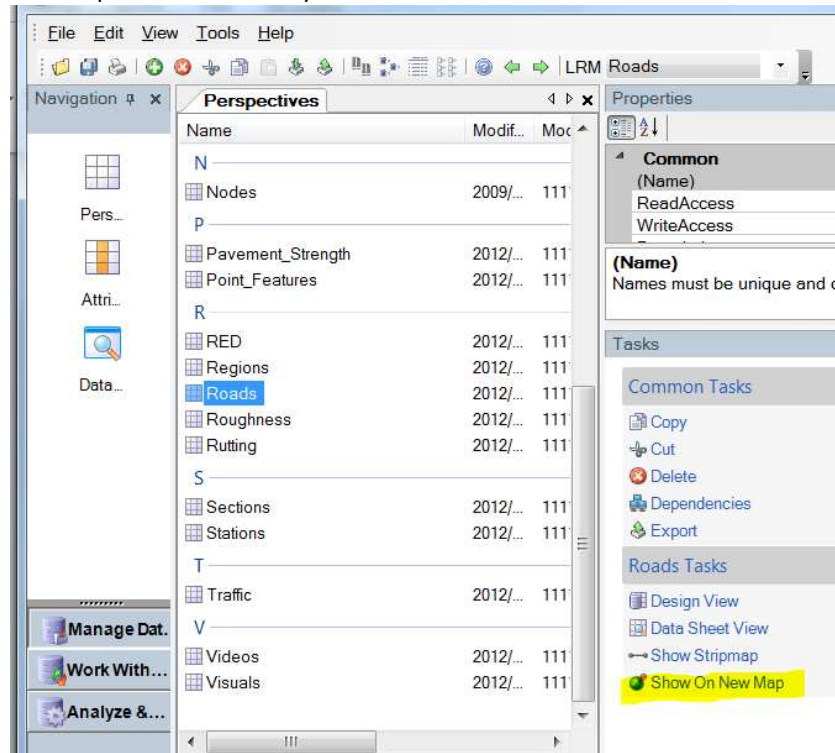
☐ Normal  
☒ Satellite

Use the Help function (top right of the display) to get detailed information on the operation of the software from the developer (Deighton).

The following notes are important for the configuration of UNRA's MD:

- Database path: A file with extension .UDL should have been created on your computer. See the installation guide for Microsoft SQL Server on C:\UNRA\_PMS\Installation Files\dTIMS for further details on how to connect the UDL file with the dTIMS database.
- Base layer Shapefile: This should have been created by your GIS department.
  - A base map in the form of a map document (.shp) contains a layer that defines UNRA's network, namely the ROADS perspective in dTIMS. This base map will be indexed to the ROADS field that is used as the unique key in dTIMS to defines roads.
  - Make sure the path points to the correct shp file, and fille in the Map projection as per the screen above.

- o The shape file should contain an attribute ROAD that has exactly the same road names as those in UNRA's dTIMS database.
- o When a new map document has been created, ask your dTIMS administrator to complete the following steps before continuing in DTIMS MD:
  - First delete the existing shape files (with .shp and .idx files from the computer), and replace the files with the new set of shape file. MAKE SURE THE .IDX FILE DOES NOT EXIST.
  - In dTIMS (CT), select the task "Show on new map" for the ROADS perspective. The select the correct/new .shp file for the ROADS perspective. This will create a new file .idx on the same directory as the .shp file.
  - The shape file is now ready for access in dTIMS MD and dTIMS WF.



- Save when done.

### Navigation in dTIMS MD

In internet explorer navigate to <http://localhost/ContentDBoardV.aspx>.

Below is a print-out of the Help function to help you navigate through the Dashboard.



### Overview:

The **md** dTIMS MANAGEMENT DASHBOARD is designed for management individuals who are responsible for deciphering, reviewing and reporting using information amassed and stored in the dTIMS CT Oracle or SQL database.



This easy to use web browser, read-only analysis reporting tool provides an overview of all assets in the network using clear charts, visual maps and concise reports.

The July 2011 release boasts:

- A multi level increase to .Net framework 4.0 which improves overall performance.
- A new feature ROI (Return on Investment) allows the user to test the effects to Condition of applying additional budget, or conversely, testing the effects to budget of increasing condition.

The request is sent to the database for data and the most up-to-date information will be retrieved each time the request is made to view or review the map, charts, Stripmaps etc. While gathering information from the database an update indicator will be shown in one of two animated ways



A permanently displayed menu contains the options available to you to view and review data. There are five viewing options, each able to display your data in different formats for ease of use.

The Dashboard offers an overview of the network in terms of its overall condition and costs to maintain the condition.

Datasheet View provides a view of the data in both a map and list of elements and the attributes of those elements in a text format.

This document contains information on:

- [Logging into md](#)
- [Navigation in md](#)
- [What's on the Dashboard](#)
  - ✚ [Map](#)
  - ✚ [Charts](#)
- [Configuring your Dashboard](#)
- [Looking at the actual data in Datasheet View](#)
  - ✚ [Selecting data and showing it on the map](#)
- [Viewing pre-configured queries and reports in Work with Data](#)
- [Strategic Analysis Module's what-if Scenarios](#)
- [Reviewing results of additional funding on Cross Asset](#)
- [Getting Results of "What If" scenarios with ROI](#)
- [Reporting and viewing the recommended Construction Program](#)
- [Linking to other web sites and web applications](#)
  - ✚ [Setting up new links](#)

### Logging into md dTIMS MANAGEMENT DASHBOARD:

Each user must login to be given access to the Management Dashboard. This has been set up by your Information Technology group who are responsible for giving access and linking to the database. The user will connect using their company browser (i.e.: Windows Internet Explorer) to the Management Dashboard site for login. Login consists of a User Name and Password. The user name and password will be supplied to you by your IT staff. Use the **Logout** option when finished your session.

- The password is case sensitive
- Select the option to **Remember me next time** to avoid entry of the password each time you log in.
- Logout is on the dashboard menu
- Logout saves the Asset & Funding Category selections and Dashboard Configuration
- Logout does not save the dashboard layout or the zoom level
- More than one person can be logged into the site at the same time.
- Depending upon the server setup, you may have a session expiry on the management dashboard login.
- Expiry is usually set at about 20 minutes. After 20 minutes of inactivity, the session will expire and return to the Login screen.

### Navigation in md:

**md dTIMS MANAGEMENT DASHBOARD** is a single document application which means that only one tab can be open at a time, when another tab is selected, the first one will be replaced.

The default on open is the Dashboard, which gives you visuals using maps (if configured) and charts. The Management Dashboard is made up of seven different tabs to organize the data being viewed. In keeping with all web applications, options are available by selection with the mouse or with the tab and enter keys.

#### Navigating in the web page display area:

As used with all web applications, use the slider bars to move the display up/down left/right in the workspace



Note: in Data Sheet View, use the scroll bar to see data off the screen. In reports, the report is moved down the display area; use the scroll bar to move back to the top of the page.

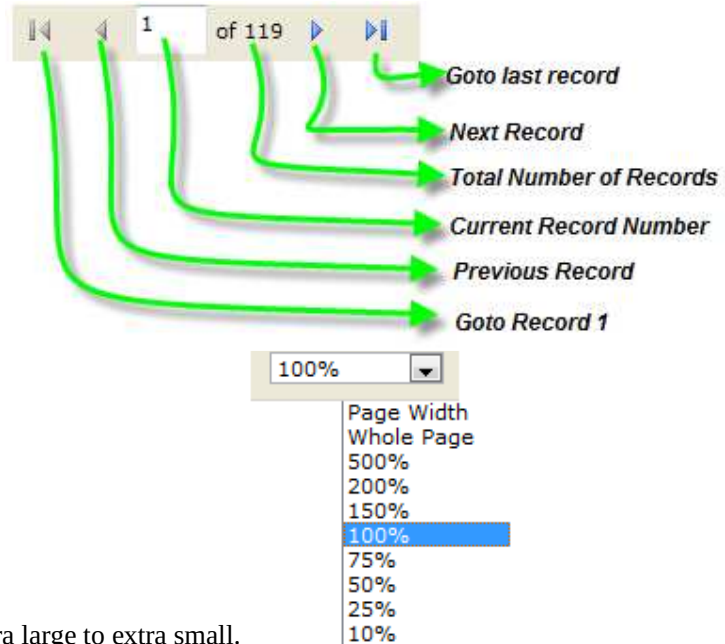
A drop-down  field contains a list to offer a choice

Clicking  will link to a function or option

Hovering over the button  will invoke another window

Navigation within reports:

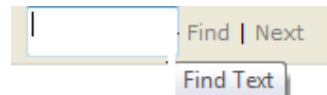
Moving within the records



Zoom by varying percentages from extra large to extra small.

Search for text

- Enter the value you're looking for (not column or case specific)



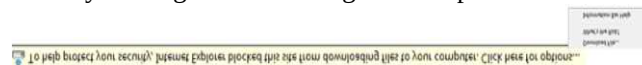
- Click *Find*
- Click *Next*

- Ctrl | F

- Will stop at the first occurrence of the text
- Will stop at the next occurrence of the text until all entries have been found and will then report that the entire file has been searched.
- Will invoke the *Find on this Page* option at the top of the web banner

Exporting Data

Depending upon the Internet Explorer settings, at login time you may receive a security message in the message banner prior to downloading files.



Select the **Download File...** option to continue.

*This interruption will, unfortunately, reset your current function; you will have to make your request for download a second time.*

Navigating within the Map:Panning Tool

- Use the mouse to click and hold the



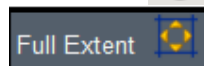
directional indicator to pan the map in one of the four directions.

- Clicking the center globe will give you a world map.



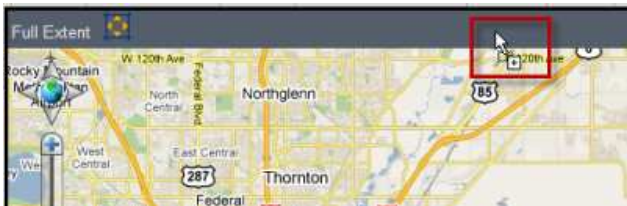
#### Zooming Tool

- Zoom in and out using the slider bar
- You can also zoom a section using the **shift / left mouse** box selector



#### Map Extent

- Will reset the map to its full extent, eliminating any zoom level and returning the display to the default



#### Using the Mouse for Navigation within the Map Display:

Move map in *Datasheet View* by clicking on header to drag and drop map window to another screen location.

left mouse

left mouse

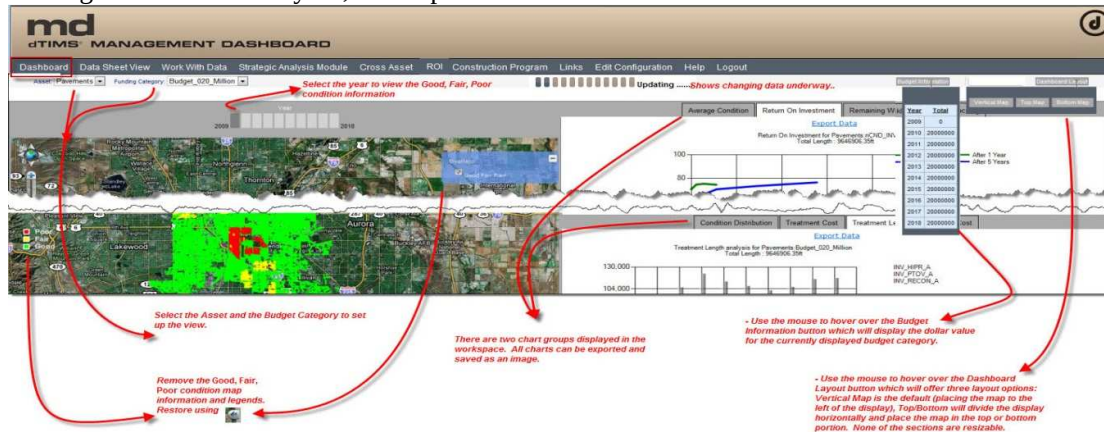
wheel

Shift + left mouse

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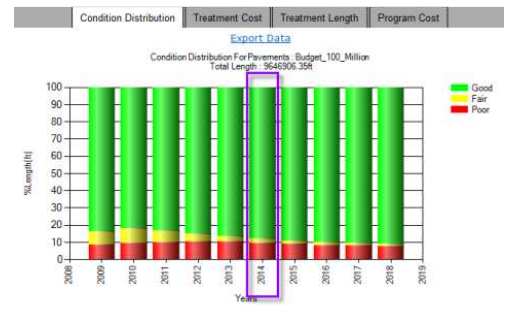
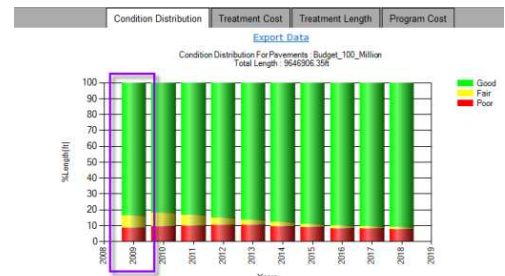
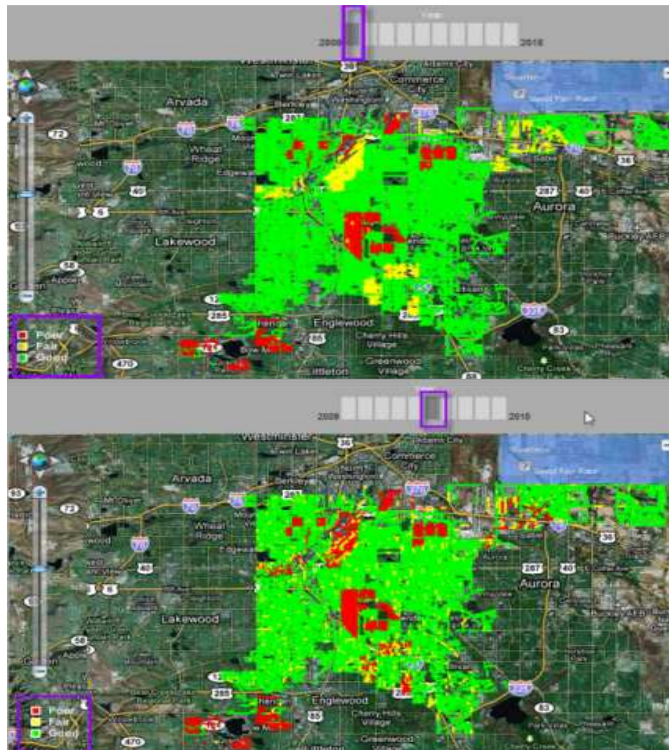
## What's on the Dashboard:

The default dashboard contains the map, and two groups of charts. None of the charts can be closed or moved. None of the charts or the map can be resized. The one change that can be made in the display is on the Dashboard Layout where you can change the position of the map on the md workspace. Each time a change is made to the layout, the map will reset the zoom level.



## The Map:

A very informative **Condition** update is shown on the map data. The example used in the documentation includes three categories Good, Fair, Poor. The agency's own condition categories will be reported on their system. The information recorded on the Condition Distribution graph coincides with the map visual for the selected year. This information gives a constantly updating visual of the condition of the network during the selected year of the analysis period.



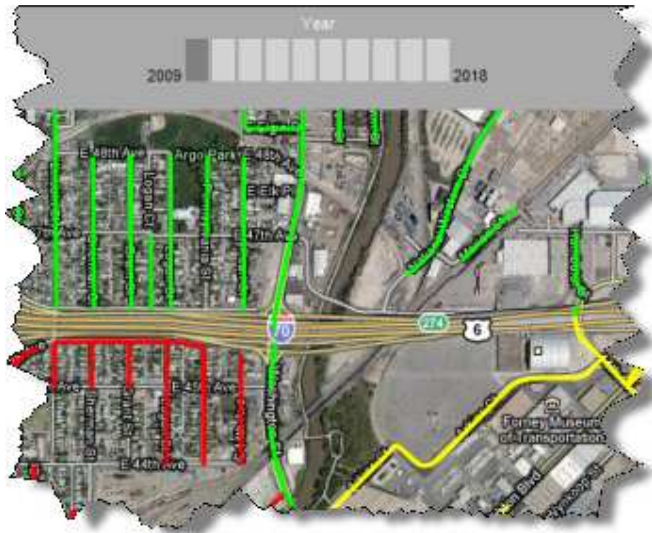
Of course the entire network can be difficult to visualize.

Therefore, a more effective use of the map and its information is to make a small area selection and visualize the condition changes over the years.

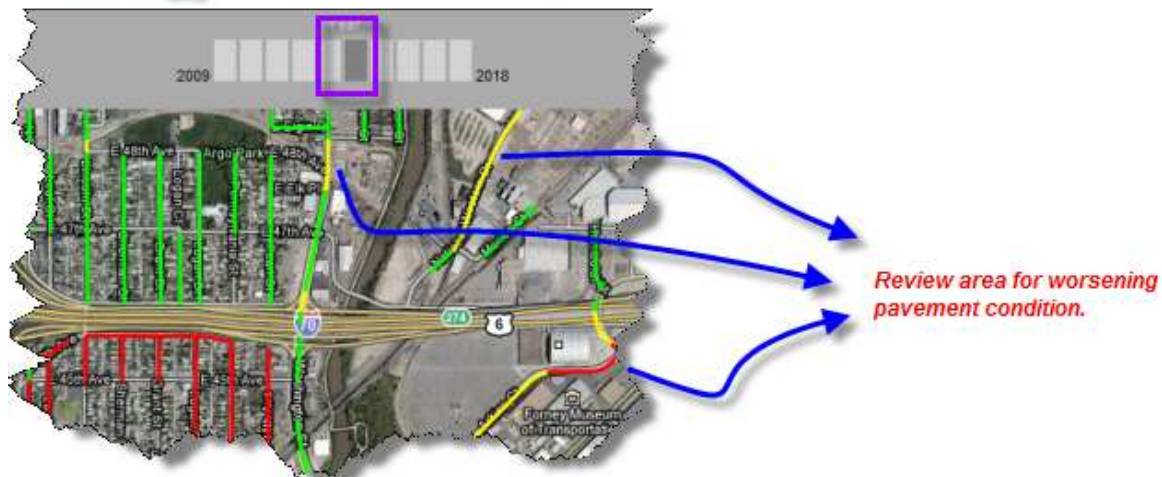
- Make a small area selection by using **shift | left-mouse draw**



- This example shows the condition of this selection for the year 2009 and its portion is representative of the corresponding year column in the Condition Distribution Chart for the entire network.



- Change the year to view the change to the area after five years



### The Charts:

There are two groups of charts. None of the charts can be closed, moved or resized. Each chart's data can be exported to an excel file. You can **change the name and save location by using the *Save File*** option in the download window. Each chart image can be saved to an image file. To invoke the save image, right-click on the chart and ***Save Picture As...***

Also on the right-click menu are options to **print** or **email** the chart image. The print and email options do not produce .gif or .bmp files and therefore will not work for Average Condition, Length in Backlog, Treatment Costs/Length, Program Costs.

Charts that compare Budget Scenarios such as *Average Condition*, *Return on Investment*, *Remaining Work* and *Length in Backlog* require that all Budget Scenarios are optimized. When one or more Budget Scenarios are not optimized, the charts will not be drawn.

When optimization within the database is required, a message will be displayed in the MD application citing '*at least one Budget Scenario needs to be optimized*'.

| Chart Name        | Description                                                                                                 | Excel Export Files                                                                                            |
|-------------------|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Average Condition | Shows how the average network impact performance index changes over the analysis period for the all Funding | <b>AvgCondExport.xls</b><br>MS Excel file contains the Analysis Year and Budget value for each of the Funding |

|                        |                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                         |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        | Categories. The average shown is the weighted average taken at the end of the analysis year over all elements.                                                                                                                                                                                                                                                                                                        | Categories.                                                                                                                                                             |
| Return on Investment   | Shows cost over fifteen years, shown after 1 year, 5 years and 10 years of investment using data from all Funding categories.                                                                                                                                                                                                                                                                                         | <b>ROIExport.xls</b><br>MS Excel file contains the Analysis year, cost and condition for each of the Funding Categories.                                                |
| Remaining Work         | Based on the two poorest condition levels in the network, shows plot of how much length would remain in inferior elements vs. how much money you would spend after so many years of investment. All Funding Categories are used.                                                                                                                                                                                      | <b>RemWorkExport.xls</b><br>MS Excel file contains the Analysis year, Funding Category, Cost and Length.                                                                |
| Length in Backlog      | Based on the two poorest condition levels in the network, shows plot of how much length would remain in the inferior elements vs. how the years of investment in the analysis period. All Funding Categories are used.                                                                                                                                                                                                | <b>BacklogExport.xls</b><br>MS Excel file contains the Analysis year and length values for each budget category.                                                        |
| Condition Distribution | Used to produce the map in its Condition visual, this graph shows the percent length of elements in each condition category by analysis year.                                                                                                                                                                                                                                                                         | <b>CondDistExport.xls</b><br>MS Excel file contains the Analysis year and length values for each of the condition categories.                                           |
| Treatment Cost         | A visual of how the total costs of the construction program are distributed to each type of treatment by analysis year. It is useful for viewing how the distribution of costs for the treatments changes over the analysis period. It is also useful to help compare how the distribution of treatments changes as the money gets tighter for different funding categories. All Funding Categories are used.         | <b>TrtCostExport.xls</b><br>MS Excel file contains the Analysis year, Treatments found in the construction program and the cost values associated with the treatments.  |
| Treatment Length       | Shows a plot of the total length of elements in the construction program receiving each type of treatment by analysis year. This chart is used to see the ways in which the distribution of lengths is affected by different treatment changes over the analysis period. It is also useful to help you compare how the distribution of treatments changes as the money gets tighter for different funding categories. | <b>TrtLenExport.xls</b><br>MS Excel file contains the Analysis year, Treatments found in the construction program and the length values associated with the treatments. |
| Program Costs          | Shows the distribution of program cost by analysis year and by Budget Category. The graph is useful for viewing the relative amount spent in each budget category.                                                                                                                                                                                                                                                    | <b>ProgCostExport.xls</b><br>MS Excel file contains the Analysis year, Budget Categories and associated costs.                                                          |

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## Configuring your Dashboard:

This tab helps you to set up your application configuration to identify and link to your map file and also to link to the database. Use the Custom Links section to add, edit or delete links to other applications or web pages.

**Note:** The Database Path refers to the location of the database on the server. This is usually set up by the IT department and does not require changes unless specifically requested.

**md**  
dTIMS MANAGEMENT DASHBOARD

Dashboard Data Sheet View Work With Data Strategic Analysis Module Cross Asset ROI Construction Program Links **Edit Configuration** Help Logout

Site Configuration: dTIMS MD Version 1.0.4168.15317 *dTIMS md version number*

Database Path: *Path linking to the database's .udl file*

Custom Links:

| Name             | URL                                               |
|------------------|---------------------------------------------------|
| ILAP             | http://www.ilap.com.mx/ilaplow_archivos/frame.htm |
| Deighton Website | http://deighton.com                               |

Mapping Configuration: ☒ Use dTIMS MD Maps

Map Units: Feet

Base Layer Shapefile: *A Base map in the form of a map document (.shp) contains a layer that defines the network's Base perspective. This base map will be indexed to a field which exists in dTIMS (such as Road).*

Map Projection: *Map Projection which relates the shape file to the satellite map file for your network area. This will be supplied with the shapefile.*

Base Layer Element ID: *Base Layer ElementID identifies the name of the field being used as the index to the map (such as Road).*

Shp layer field containing the 'From Measure': *For Polyline M type base maps, define the Shape layer field containing 'From Measure' entry. If none, select blank entry.*

Online Map Overlay: ☒ Bing ☐ None

Online Map Options: ☒ Normal ☐ Satellite ☐ Hybrid

Excel Link Path: *Path linking to the excel file (normally same as database path) to hold the exported Construction Program in a formatted report with charting*

**Save** *Save changes to the configuration file.*

**Note:** dTIMS CT uses the Bing Online Map Overlay which is equivalent to Microsoft Virtual Earth. Both selections are compatible.

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## Data Sheet View:

This tab offers a map of the network and an option to drill down to the actual data that is stored on each of the elements in the network.

The display in this tab consists of a data grid and a map. The map is moveable in this tab by clicking on the gray top banner and dropping within the workspace. The map is not resizable.

While in the Data Sheet View tab, you can show the values of the attributes on the map as it pertains to the network. You accomplish this using the **Map Data** option where you can select the attribute, apply color and show it on the map display.

When we talk about viewing the data in text format, it is best to have an idea of the setup of the database. You have the option of viewing by Perspective or by Data View.

A **Perspective** categorizes information on top of the base network road. It could be information on condition, maintenance or inventory such as signage, curbs, alleys etc as it relates to the base network road, stored in attributes and recorded as a record indexed by a base Road and ElementID.

A **Data View** is a method of picking and choosing the attribute data fields you actually want to see within the perspective.

**LRM** is Location Reference Method relates locations to one another. The objective is to identify virtually any spot along the asset. To accommodate this we use known points and measure distances from them. Location referencing results in an address consisting of a know point identifier, plus or minus, then a distance from the know point. An example of this is how we



would keep track of landmarks on the road. We know that for example, *landmarkA* is at *point1* + 30 feet. If you are unsure which LRM you should use, then use the Base Perspective (Roads, Base).

What's on the Data Sheet View Display:

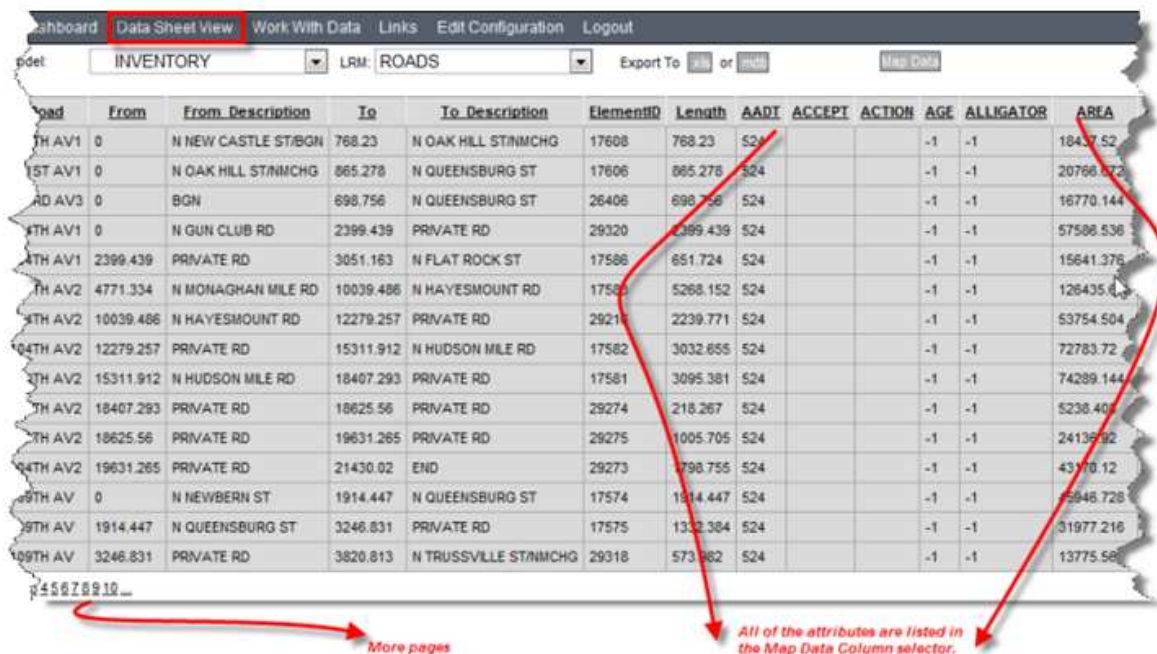


Drop-down list shows all available Perspectives and Data Views in alphabetical order. Select the one which will control the attribute fields in the Data Sheet view.

Select one of the Section or Point type perspectives listed

Export the Data Sheet View text to an external Excel or Access file.

Using **Open Export file** may open the file beneath the MD application. You may need to look in your task bar for the open Office application.



| Road     | From      | From Description    | To        | To Description       | ElementID | Length   | AADI | ACCEPT | ACTION | AGE | ALLIGATOR | AREA      |
|----------|-----------|---------------------|-----------|----------------------|-----------|----------|------|--------|--------|-----|-----------|-----------|
| 1TH AV1  | 0         | N NEW CASTLE ST/BGN | 768.23    | N OAK HILL ST/MCHG   | 17608     | 768.23   | 524  |        |        | -1  | -1        | 18437.52  |
| 1ST AV1  | 0         | N OAK HILL ST/MCHG  | 865.278   | N QUEENSBURG ST      | 17606     | 865.278  | 524  |        |        | -1  | -1        | 20766.72  |
| RD AV3   | 0         | BGN                 | 698.756   | N QUEENSBURG ST      | 26406     | 698.756  | 524  |        |        | -1  | -1        | 16770.144 |
| 4TH AV1  | 0         | N GUN CLUB RD       | 2399.439  | PRIVATE RD           | 29320     | 2399.439 | 524  |        |        | -1  | -1        | 57586.536 |
| 4TH AV1  | 2399.439  | PRIVATE RD          | 3051.163  | N FLAT ROCK ST       | 17586     | 651.724  | 524  |        |        | -1  | -1        | 15641.376 |
| 7TH AV2  | 4771.334  | N MONAGHAN MILE RD  | 10039.486 | N HAYESMOUNT RD      | 17583     | 5268.152 | 524  |        |        | -1  | -1        | 126435.6  |
| 5TH AV2  | 10039.486 | N HAYESMOUNT RD     | 12279.257 | PRIVATE RD           | 29218     | 2239.771 | 524  |        |        | -1  | -1        | 53754.504 |
| 64TH AV2 | 12279.257 | PRIVATE RD          | 15311.912 | N HUDSON MILE RD     | 17582     | 3032.655 | 524  |        |        | -1  | -1        | 72783.72  |
| 7TH AV2  | 15311.912 | N HUDSON MILE RD    | 18407.293 | PRIVATE RD           | 17581     | 3095.381 | 524  |        |        | -1  | -1        | 74289.144 |
| 7TH AV2  | 18407.293 | PRIVATE RD          | 18625.56  | PRIVATE RD           | 29274     | 218.267  | 524  |        |        | -1  | -1        | 5238.40   |
| 7TH AV2  | 18625.56  | PRIVATE RD          | 19631.265 | PRIVATE RD           | 29275     | 1005.705 | 524  |        |        | -1  | -1        | 24139.92  |
| 64TH AV2 | 19631.265 | PRIVATE RD          | 21430.02  | END                  | 29273     | 1798.755 | 524  |        |        | -1  | -1        | 43170.12  |
| 9TH AV   | 0         | N NEWBERN ST        | 1914.447  | N QUEENSBURG ST      | 17574     | 1914.447 | 524  |        |        | -1  | -1        | 5848.726  |
| 9TH AV   | 1914.447  | N QUEENSBURG ST     | 3246.831  | PRIVATE RD           | 17575     | 1332.384 | 524  |        |        | -1  | -1        | 31977.216 |
| 109TH AV | 3246.831  | PRIVATE RD          | 3820.813  | N TRUSSVILLE ST/MCHG | 29318     | 573.982  | 524  |        |        | -1  | -1        | 13775.58  |

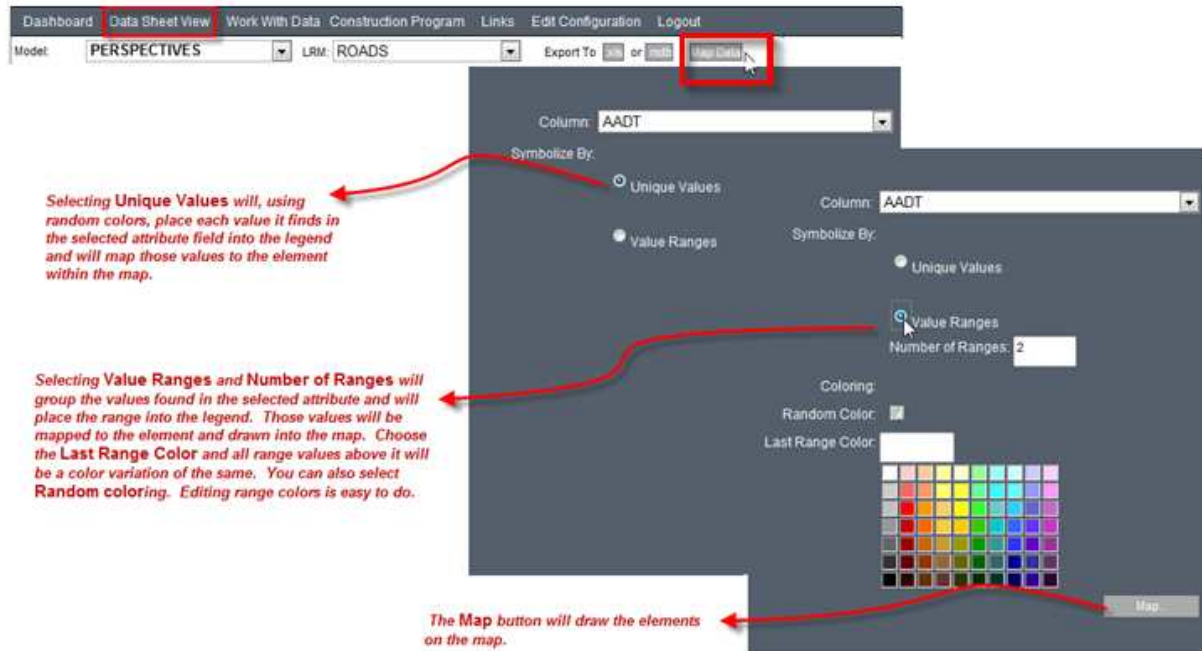
More pages

All of the attributes are listed in the Map Data Column selector.

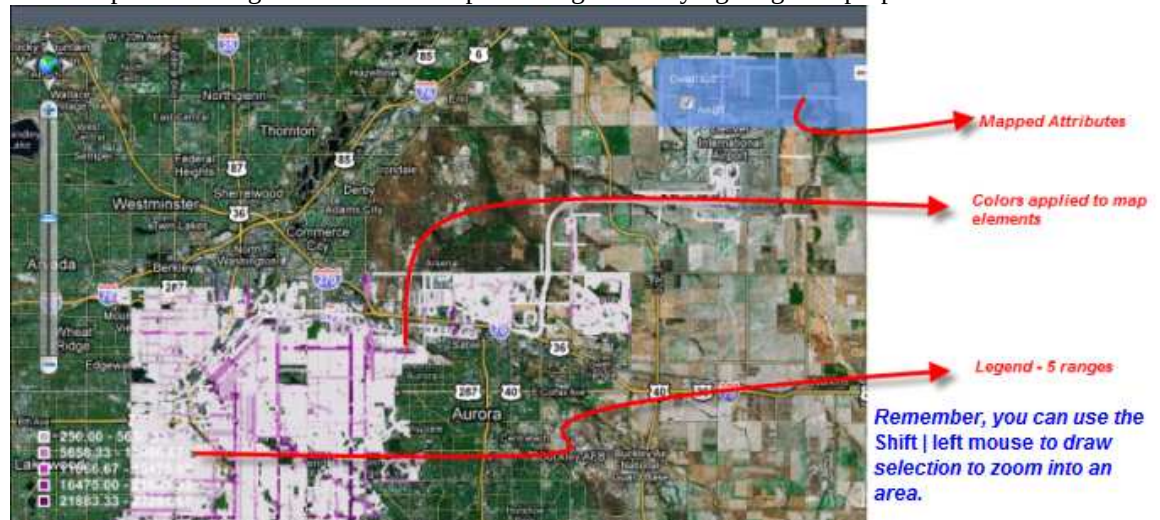
Use the web page scroll bar to move data in the worksheet to the right.

Adding Attributes and their values to the Map:

Update the map using the **Map Data** option.



An example of writing the AADT the map in 5 ranges of varying ranges of purple.



### Editing the Range Values and the Color:

When using ranges, you have to determine the number of ranges to use and then let the application write to the map before you can make any changes. You may want to control the colors and range values on the map. You edit the values using the **Change Map Layer Style** option which is available only after a map has already been drawn.

Map Data Change Map Layer Style

**Displays the current ranges and their colors. Use the Edit Link to make changes.**

| #                    | Range Min | Range Max | Color   |
|----------------------|-----------|-----------|---------|
| <a href="#">Edit</a> | 250       | 4885.714  | #E8E3E8 |
| <a href="#">Edit</a> | 4885.714  | 9521.429  | #D6C9D6 |
| <a href="#">Edit</a> | 9521.429  | 14157.14  | #CSAEC5 |
| <a href="#">Edit</a> | 14157.14  | 18792.86  | #B692B6 |
| <a href="#">Edit</a> | 18792.86  | 23428.57  |         |
| <a href="#">Edit</a> | 23428.57  | 28064.29  |         |

**The first range minimum can be changed. The default value shows you the data's actual minimum. You can assign your own minimum value. No other minimum value is editable.**

**Click into the range you are changing and type the values you want to use. Let's say from 250 to 350 and the color you want to use is blue.**

**Be sure to Update before you leave the row, else it will cancel the change.**

| #                                             | Range Min | Range Max | Color   |
|-----------------------------------------------|-----------|-----------|---------|
| <a href="#">Update</a> <a href="#">Cancel</a> | 250       | 350       | #0000FF |
| <a href="#">Edit</a>                          | 4885.714  | 9521.42   |         |
| <a href="#">Edit</a>                          | 9521.429  | 14157.1   |         |
| <a href="#">Edit</a>                          | 14157.14  | 18792.8   |         |
| <a href="#">Edit</a>                          | 18792.86  | 23428.5   |         |
| <a href="#">Edit</a>                          | 23428.57  | 28064.29  | #985898 |

**Update Map**

**Notice that the next minimum range will default to start at the same value as the previous maximum value. These values must be contiguous. Edit the next range maximum values and colors until done.**

| #                                             | Range Min | Range Max | Color   |
|-----------------------------------------------|-----------|-----------|---------|
| <a href="#">Edit</a>                          | 250       | 350       | #0000FF |
| <a href="#">Update</a> <a href="#">Cancel</a> | 350       | 9521.429  | #D6C9D6 |
| <a href="#">Edit</a>                          | 9521.429  | 14157.14  | #CSAEC5 |
| <a href="#">Edit</a>                          | 14157.14  | 18792.86  | #B692B6 |
| <a href="#">Edit</a>                          | 18792.86  | 23428.57  | #A874A8 |
| <a href="#">Edit</a>                          | 23428.57  | 28064.29  | #985898 |

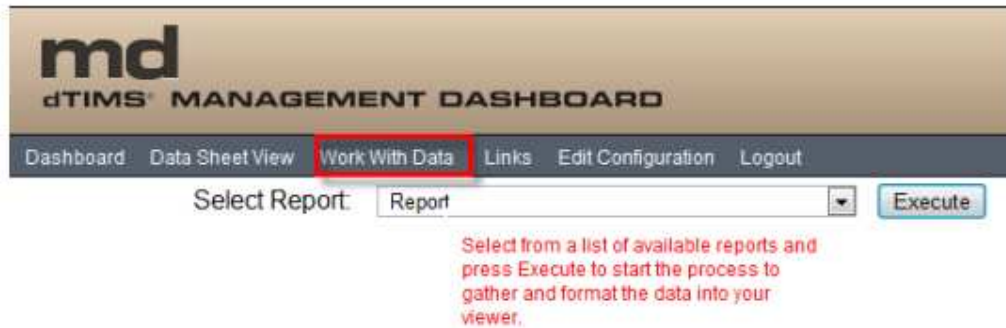
**Update Map**

**When editing is complete, use the Update Map to redraw.**

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## Work with Data:

Designed for the display and review of data reports in the form of predesigned queries, this tab offers a report viewer, search tool and export to Microsoft Excel or to Acrobat .pdf file.



Select Report:

RSL\_Report

▼

Execute

1

of 8

100%

▼

Find | Next

Select a format

▼

Export

Perspective queries

Run time: 2/26/2010 4:31:15 PM

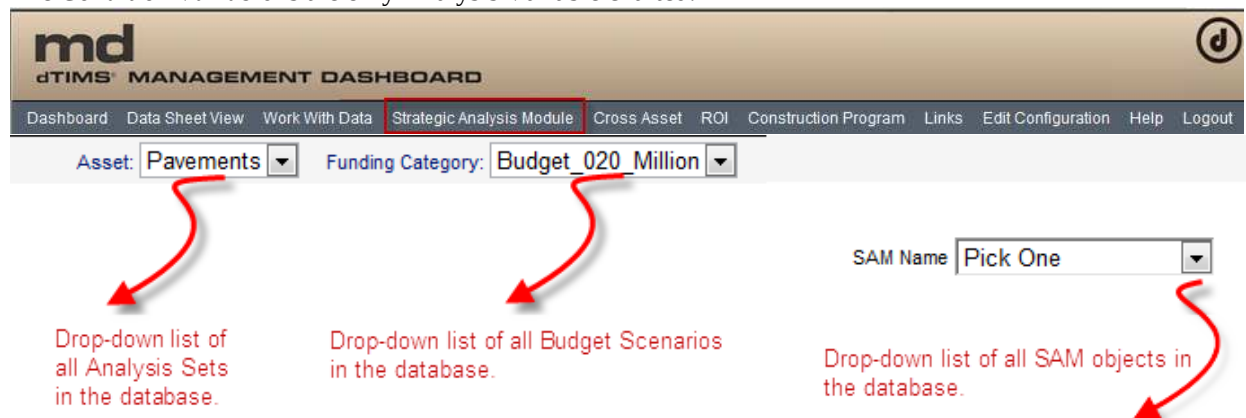
| Element Key | Elem_Id | AREA         | ACT    | Key 00 | Key 01 | Key 02 | RS |
|-------------|---------|--------------|--------|--------|--------|--------|----|
| 2935        | NE024   | 2,159,683.52 | 125.23 | 22.64  | 0.00   | 0.00   |    |

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## Strategic Analysis Module (SAM):

The SAM tab allows the user to view and review the different Budget Scenario groups and see the impact to the network condition by adding investment to the selected Budget Scenarios.

Intervals are pre-defined in the SAM object and the graphic will be redrawn for each position on the slider bar. The value box will also update with the corresponding budget amount. Should you want to define your own budget value, simply type the value into the box (as long as it falls within the range set by the minimum and maximum values on the slider bar) and depress the Show button to draw the graph. The *Condition Variable* is the only Analysis Variable charted.





125,000.00

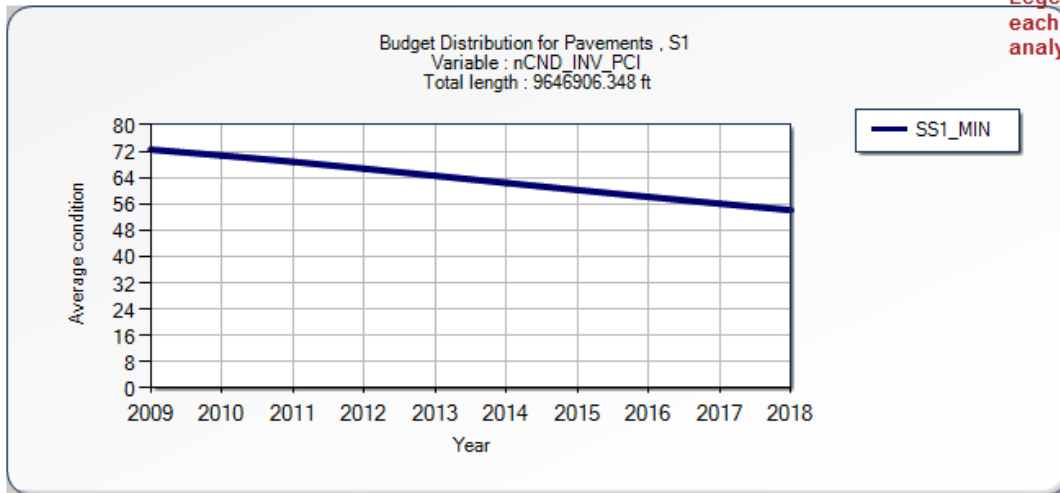
Show

Click slider bar to graph the condition for each interval in the analysis.

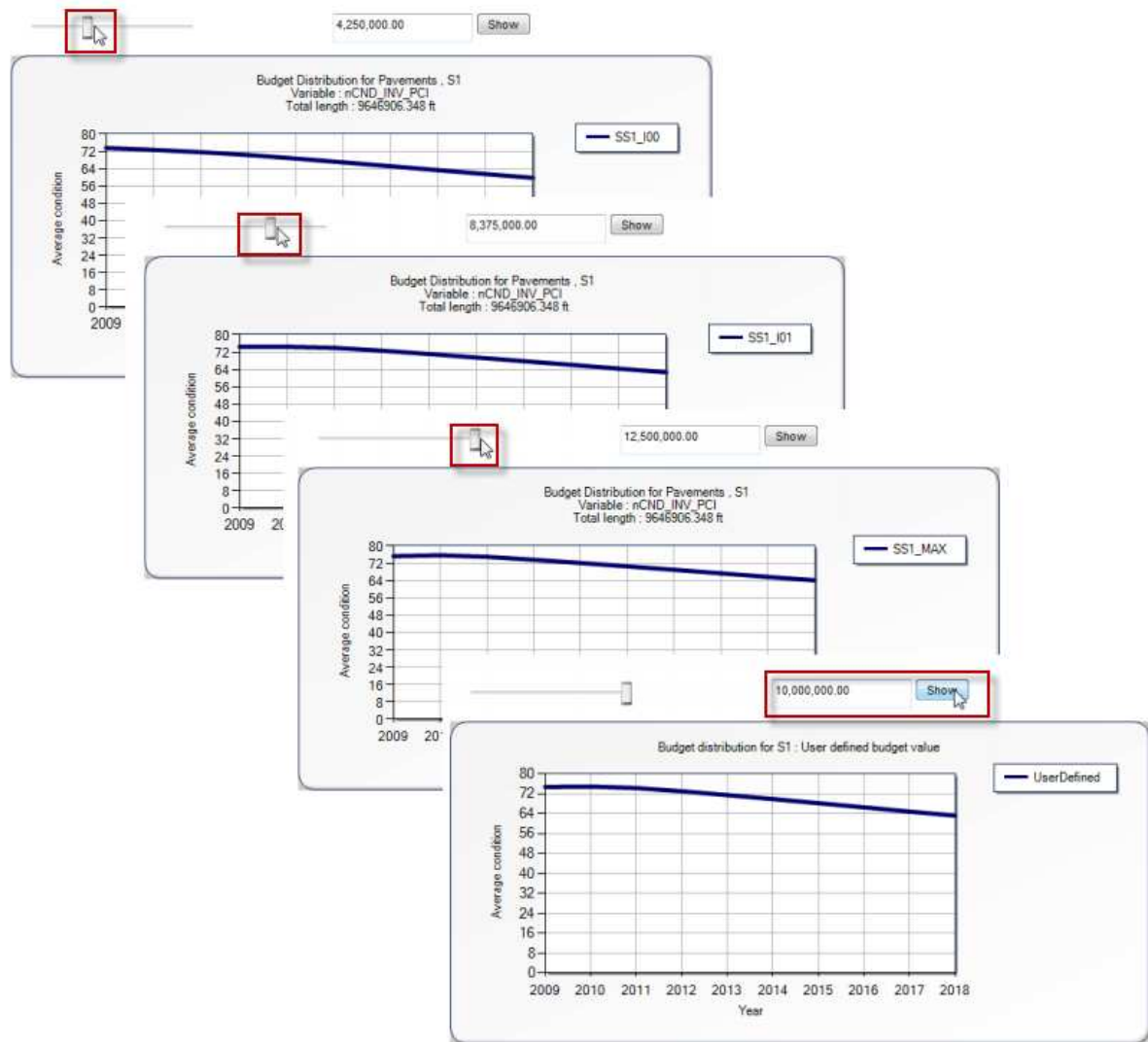
Value increases for each interval in the analysis.

Type a value into the value box and click Show to see condition for a certain

Legend changes for each interval in the analysis.







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### Cross Asset Analysis & Optimization:

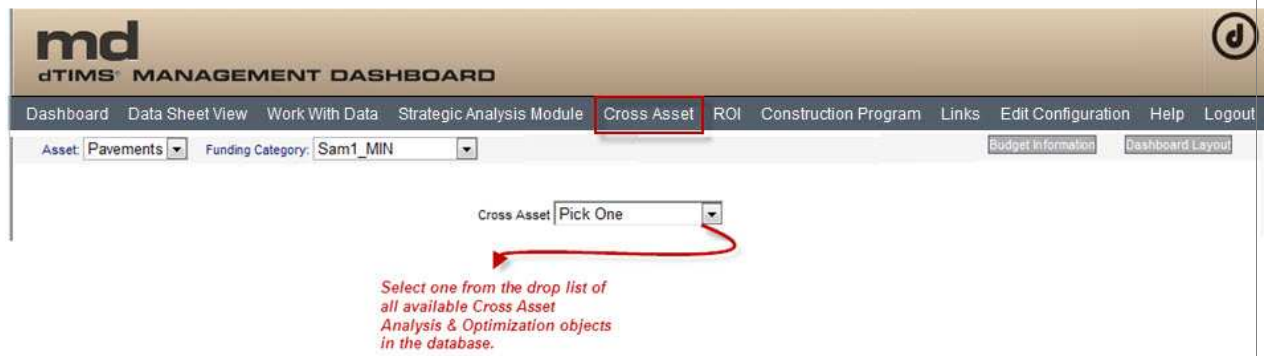
Provides a function to enable the user to perform, review and report economic trade-off analysis between different assets, subsets of assets or sub-sets of a network. The purpose is to return an optimum maintenance and rehabilitation program across multiple assets or across subsets of the same asset.

The Cross Asset Analysis and Optimization object will be used to specify the parameters that control the analysis which will include the assets being analyzed, the budgets available, the benefit and cost fields as well as the type of optimization to be completed.

Additional funding will be optionally added by year.

Optimization constraints such as Allow DoNothing, Committed Strategies, Multipass will be taken from the individually selected Budget Scenarios.

The object will be used to execute the analysis, review and adjust the results, and provide access to the summary reports and graphs.

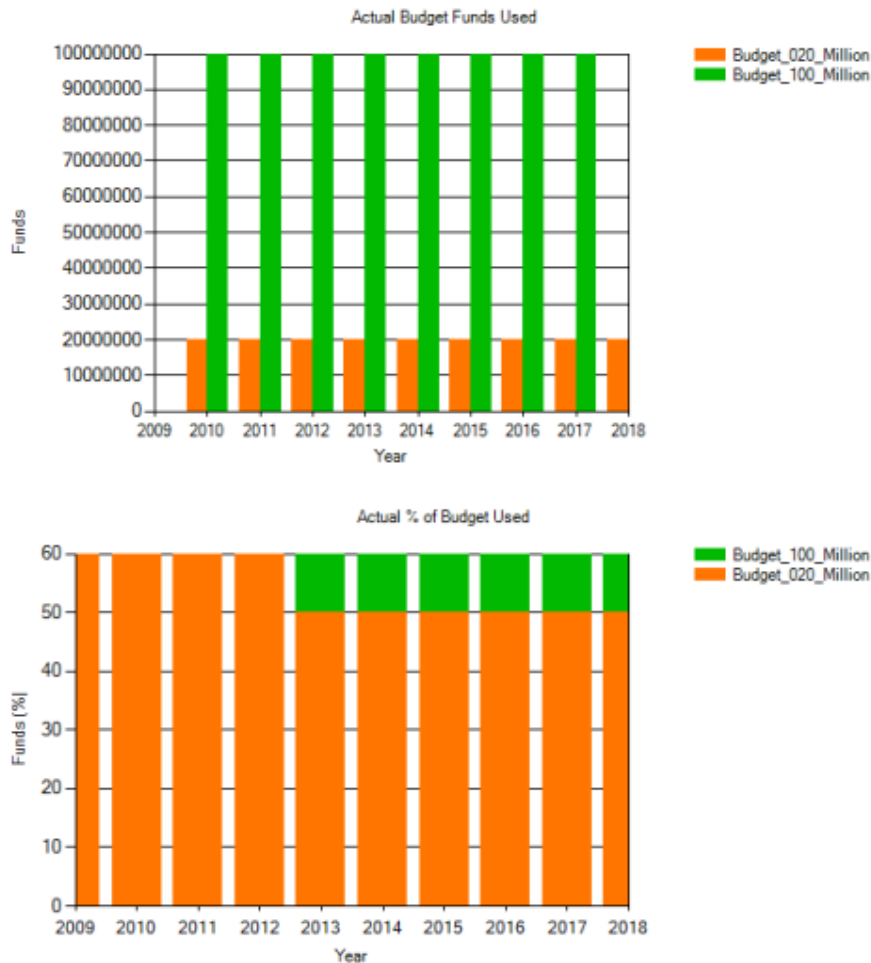


The results of the Cross Asset are shown in two parts. The first is a data grid showing the selected Budget Scenarios and both budgeted dollars and additional dollars, as well as their percentages used. The second part of the report is one or two charts. The default chart is the Budget Amount. Use the *Pick One* option to change the default chart. Use the second *Pick One* to add one of the percentage charts on demand.

- Use the Pick One option to clear the grid or the charts
- Use the Export Data to export the grid data to an Excel file
- Use the right-click Print or Print Preview for the report.
- Click into the chart and use the left mouse click to click and hold a zoom portion
- Use the reset zoom ↶ arrow on both axes to return the display to 100%

| Cross Asset |                    | CAAO1            | Actual Budget Funds       |                          |                          |                              |                         | Actual % of Budget      |                                     |                        |
|-------------|--------------------|------------------|---------------------------|--------------------------|--------------------------|------------------------------|-------------------------|-------------------------|-------------------------------------|------------------------|
| Year        | Budget Scenario    | Budget Amount    | Max % of Additional Funds | Total Additional Funding | Actual Budget Funds Used | Actual Additional Funds Used | Actual Total Funds Used | Actual % of Budget Used | Actual % of Additional Funding Used | Actual % of Total Used |
| 2009        | Budget_020_Million | \$0.00           | 75                        | \$100,000.00             | \$0.00                   | \$0.00                       | \$0.00                  | NaN                     | 0.00                                | 0.00                   |
| 2009        | Budget_100_Million | \$0.00           | 25                        | \$100,000.00             | \$0.00                   | \$0.00                       | \$0.00                  | NaN                     | 0.00                                | 0.00                   |
| 2010        | Budget_020_Million | \$20,000,000.00  | 75                        | \$100,000.00             | \$19,997,259.15          | \$0.00                       | \$19,997,259.15         | 99.99                   | 0.00                                | 99.49                  |
| 2010        | Budget_100_Million | \$100,000,000.00 | 25                        | \$100,000.00             | \$99,980,395.51          | \$0.00                       | \$99,980,395.51         | 99.98                   | 0.00                                | 99.88                  |
| 2011        | Budget_020_Million | \$20,000,000.00  | 75                        | \$100,000.00             | \$19,992,229.50          | \$0.00                       | \$19,992,229.50         | 99.96                   | 0.00                                | 99.46                  |
| 2011        | Budget_100_Million | \$100,000,000.00 | 25                        | \$100,000.00             | \$99,988,842.42          | \$0.00                       | \$99,988,842.42         | 99.99                   | 0.00                                | 99.89                  |
| 2012        | Budget_020_Million | \$20,000,000.00  | 75                        | \$100,000.00             | \$19,993,345.24          | \$0.00                       | \$19,993,345.24         | 99.97                   | 0.00                                | 99.47                  |
| 2012        | Budget_100_Million | \$100,000,000.00 | 25                        | \$100,000.00             | \$99,973,625.57          | \$0.00                       | \$99,973,625.57         | 99.97                   | 0.00                                | 99.87                  |
| 2013        | Budget_020_Million | \$20,000,000.00  | 50                        | \$100,000.00             | \$19,992,951.48          | \$0.00                       | \$19,992,951.48         | 99.96                   | 0.00                                | 99.47                  |
| 2013        | Budget_100_Million | \$100,000,000.00 | 50                        | \$100,000.00             | \$99,978,962.12          | \$0.00                       | \$99,978,962.12         | 99.98                   | 0.00                                | 99.88                  |
| 2014        | Budget_020_Million | \$20,000,000.00  | 50                        | \$100,000.00             | \$19,991,464.04          | \$0.00                       | \$19,991,464.04         | 99.96                   | 0.00                                | 99.46                  |
| 2014        | Budget_100_Million | \$100,000,000.00 | 50                        | \$100,000.00             | \$99,994,829.11          | \$0.00                       | \$99,994,829.11         | 99.99                   | 0.00                                | 99.89                  |
| 2015        | Budget_020_Million | \$20,000,000.00  | 50                        | \$100,000.00             | \$19,996,462.47          | \$0.00                       | \$19,996,462.47         | 99.98                   | 0.00                                | 99.48                  |
| 2015        | Budget_100_Million | \$100,000,000.00 | 50                        | \$100,000.00             | \$99,987,515.14          | \$0.00                       | \$99,987,515.14         | 99.99                   | 0.00                                | 99.89                  |
| 2016        | Budget_020_Million | \$20,000,000.00  | 50                        | \$100,000.00             | \$19,999,123.54          | \$0.00                       | \$19,999,123.54         | 100.00                  | 0.00                                | 99.50                  |
| 2016        | Budget_100_Million | \$100,000,000.00 | 50                        | \$100,000.00             | \$99,985,464.74          | \$0.00                       | \$99,985,464.74         | 99.99                   | 0.00                                | 99.89                  |
| 2017        | Budget_020_Million | \$20,000,000.00  | 50                        | \$100,000.00             | \$19,995,654.46          | \$0.00                       | \$19,995,654.46         | 99.98                   | 0.00                                | 99.48                  |
| 2017        | Budget_100_Million | \$100,000,000.00 | 50                        | \$100,000.00             | \$99,962,508.31          | \$0.00                       | \$99,962,508.31         | 99.96                   | 0.00                                | 99.86                  |
| 2018        | Budget_020_Million | \$20,000,000.00  | 50                        | \$100,000.00             | \$19,997,003.51          | \$0.00                       | \$19,997,003.51         | 99.99                   | 0.00                                | 99.49                  |
| 2018        | Budget_100_Million | \$100,000,000.00 | 50                        | \$100,000.00             | \$99,981,873.00          | \$0.00                       | \$99,981,873.00         | 99.98                   | 0.00                                | 99.88                  |

[Export Data](#)



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## ROI (Return on Investment Tool):

Designed for management staff charged with making budgetary program schedules, the Return on Investment Tool (ROI) provides the user with the answers to the three elusive questions:

1. If I want to maintain the current condition, how much money will I need to spend?
2. If I want a specified return, how much money do I need to spend?
3. If I spend 'x' dollars, what is the return?

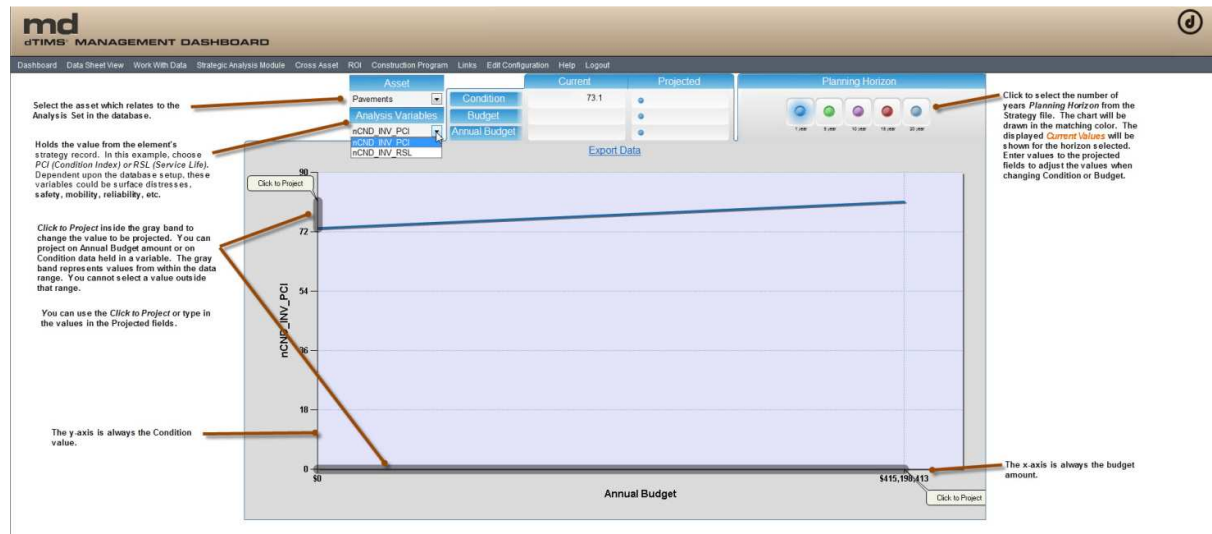
In order to enter the new condition value or the new budget value, the form offers two areas to change the projection. You can click into the gray band on either of the axis to populate the **Projected** area or you can type the text value into the **Projected** area. Once entered, the chart will be refreshed, showing the new values.

**Notes:** Be careful to click into the gray area. Clicking the axis line is ineffectual.

If you click out of or type a value out of the range of data (gray band) then the **top data value** will automatically be returned.

Refresh a blank display by keying a value in the Projected Condition or Projected Annual Budget field.





The following are examples to display the use of the Condition and Budget values.

**Question 1:** *If I want to maintain the current condition, how much money will I need to spend?*

*Let's look at maintaining a condition of 73.1 over the next 10 years. How much money should I budget to do that?*

Type 73.1 to the *Projected* field and click the *Planning Horizon* to 10 years.

The results show that over ten years a reduced total budget investment of \$659,963.978 would be required to keep the Condition Index at par.



*Let's take a look at the budget for one year when we want to kick up the condition index to 80.*

**Question 2:** *If I want a specified return, how much money do I need to spend?*

Click the Condition (PCI) up to 80

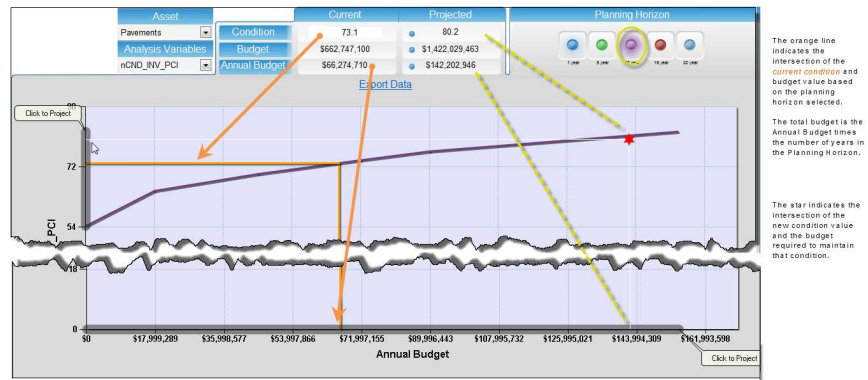
The projected values will update and graph is redrawn to cross the index and budget values.



*Let's look at the same example but this time, use a longer planning horizon.*

Select the **Planning Horizon** of 10 years and the Condition of 80

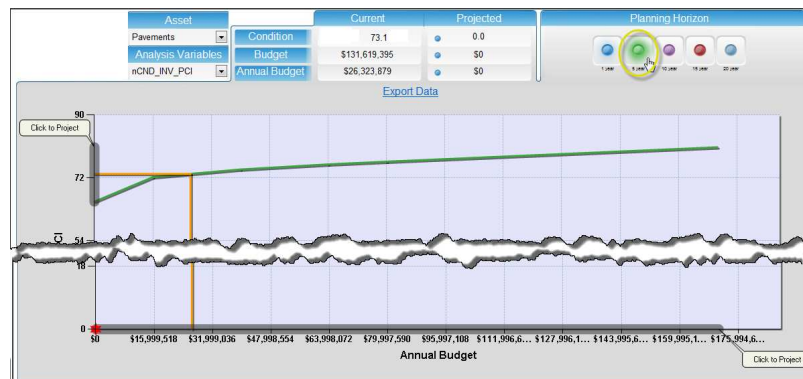
The results show that if I want to maintain a Condition Index of 80, I will have to increase the annual budget by \$75,928,236.



*What if you want to test the condition results if you spend more money over 5 years?*  
**Question 3:** If I spend \$40 million each year for 5 years, what is the return?

Open ROI and select the 5 year horizon

You can see that an investment of \$26 million over 5 years provides a PCI of 73. Pretty good, but what if you spend \$40 million?



Click into the gray band on the Annual Budget to select \$40 million

Both **Current** intersection and **Projected** intersection is displayed.

The results show the asset's condition increasing by 1.3 points by adding funds of \$14,005,654.

Do the results warrant the investment?



Not easy to click into the right spot to round out the value?

Just type the value to the Annual Budget field in the **Projected** area.

Let's look at the results of spending \$50 million per year

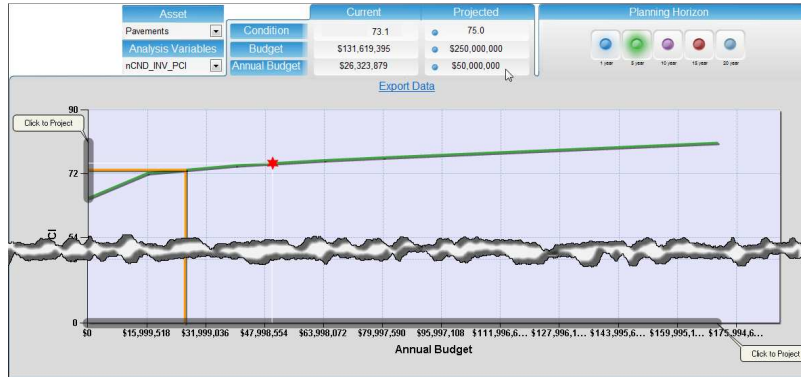


The total budget now shows the \$50 million times the number of years in the *Planning Horizon*.

Both **Current** intersection and  projected intersection is displayed.

The results show the asset's condition increasing by 1.9 points by adding funds of \$23,676,121.

Do the results warrant the investment?



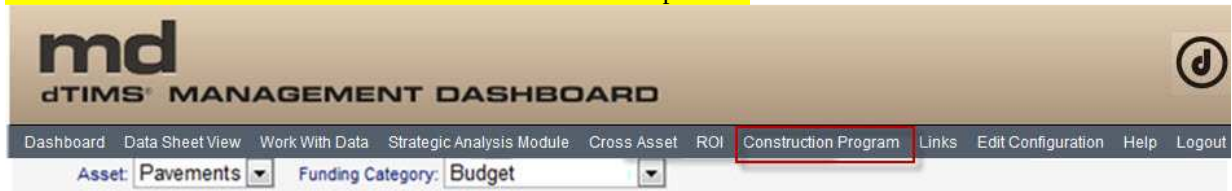
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## Construction Program:

The Construction Program is compiled from the strategy list of all selected treatments and their costs sorted by element identifier (Section) or by Year (Year). Element information is listed for both reports and in the case of a Data View request, all attributes selected in the Data View will also be included. This tab offers a report viewer, search tool, export and print function. You can also view the Budget

dollars for the selected Funding category using the [Budget Information](#) button.

**Note:** When exporting your construction program, please use the **Save** option to name and create the external file and **Print** from the external Excel or Acrobat .pdf file.



Choose the asset and Funding Category from the drop down list. To close the report and return to this window, change the Funding Category or the Asset.

[Export to Excel](#)

Report Type

☒ By Section
 ☐ By Year

Years to Include

Start Year: 2009
 End Year: 2025

Treatments to include

☒ Treatment Name1  
☒ Treatment Name2  
☒ Treatment Name3  
☒ Treatment Name4  
☒ Treatment Name5

Include Values From This Data View

(none)

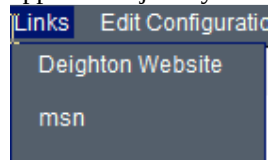
[View report](#)

|                                    |                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Export to Excel                    | When this link is followed, the data will be exported to a custom Excel application MDReports.xls which formats the data, charts by Year, by Section, by Budget Scenario and summaries. More than just a flat file and based on a pivot table, this format allows you to use the options in Excel to sort, filter and reorganize data.<br><br>See operations instructions in the Excel worksheet. |
| Report Type                        | Radio button selection to sort Treatments Report by Section or by Year.                                                                                                                                                                                                                                                                                                                           |
| Years to Include                   | Start Year and End Year provide the option to run the report for a specified number of years rather than the defaulted entire period covered by the selected strategy.                                                                                                                                                                                                                            |
| Treatments to include              | All available treatments are listed.<br>Select <input checked="" type="checkbox"/> and clear <input type="checkbox"/> by double clicking.                                                                                                                                                                                                                                                         |
| Include Values from this Data View | This list is made up of all Data Views which contain at least one attribute from the Master Perspective.                                                                                                                                                                                                                                                                                          |
| View Report                        | Displayed in the workspace, the construction report contains all element information fields as well as the Treatment name and Treatment cost. If the Data View is named, then all fields listed in the Data View will be included. This file can also be exported to Excel or pdf file.                                                                                                           |

| Year | Road        | From_Add | From_Desc | To_Add | To_Desc     | ElementID | Length | Treatment | Cost        |
|------|-------------|----------|-----------|--------|-------------|-----------|--------|-----------|-------------|
| 2018 | Main Street | 0.00     |           | 333.00 |             | 1000      | 333.00 | INV_PT    | \$68,262.50 |
| 2015 | King Street | 0.00     |           | 407.70 | S NEWTON ST | 10000     | 407.70 | INV_PT    | \$66,861.00 |

### Links:

The Links tab is a list of currently linked web site addresses which will open another web site or web application just by clicking onto the link.



The maintenance of these links is handled in the *Edit [Configuration](#)* tab under *Custom Links*.

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